GATE OVERFLOW

Volume 1

All previous year NIELIT CSE Questions

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GATE Overflow Team
led by

Arjun Suresh

Computer Science & IT

QnA

Chat

FB Group

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1 Algorithms (89)

1.0.1 NIELIT 2016 DEC Scientist B (IT) - Section B: 7

What is the type of the algorithm used in solving the 4 Queens problem?

A. Greedy
B. Branch and Bound
C. Dynamic
D. Backtracking

nielit2016dec-scientistb-it algorithms

Answer key

1.1 Algorithm Design (13)

1.1.1 Algorithm Design: NIELIT 2021 Dec Scientist A - Section B: 71

The Highest Lower Bound on the number of Comparisons in the worst case for comparison-based sorting order of :

A. n
B. n^2
C. $\mathrm{n} \log \mathrm{n}$
D. $\mathrm{n} \log 2\mathrm{n}$

nielit2021dec-scientista sorting asymptotic-notations algorithm-design algorithms

Answer key

1.1.2 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 116

Which notation gives the lower bound of a function ?

A. Θ – notation
B. O – notation
C. Ω – notation
D. None of the these

nielit2021dec-scientistb asymptotic-notations algorithm-design

Answer key

1.1.3 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 55

100 elements can be sorted in 100 sec using bubble sort. In 400 sec, approximately ____________ elements can be sorted.

A. 100
B. 200
C. 300
D. 400

nielit2021dec-scientistb sorting time-complexity algorithm-design

Answer key

1.1.4 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 6

A sorting algorithm which passes through a list to exchange the first element with smallest element in the remaining elements is known as____________.

A. Insertion sort
B. Selection sort
C. Heap sort
D. Bubble sort

nielit-2021-it-dec-scientistb sorting algorithm-design data-structures

Answer key

1.1.5 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 61

The number of swaps required to sort the array

$8, 22, 7, 9, 31, 19, 5, 13$

In ascending order using bubble sort is :

A. 10 B. 9 C. 13 D. 14

nielit-2021-it-dec-scientistb nielit2021dec-scientistb data-structures sorting algorithm-design

Answer key

1.1.6 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 75

The time complexity of heap sort in worst case is :

A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$ D. $O(n^2)$

nielit-2021-it-dec-scientistb time-complexity sorting binary-heap algorithm-design data-structures

1.1.7 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 78

Which of the following sorting algorithms has the lowest worst-case complexity ?

A. Merge sort
B. Bubble sort
C. Quick sort
D. Insertion sort

nielit2021dec-scientistb sorting algorithm-design time-complexity data-structures

Answer key

1.1.8 Algorithm Design: NIELIT 2022 April Scientist B | Section B | Question: 105

What is the time complexity of the following recursive function?

```
int ComputFun(int n)
{
   if(n<=2)
      return 1;
   else
       return (DoSomething(floor(sqrt(n)))+n);
}
```

A. $\Theta(n)$
B. $\Theta(\log n)$
C. $\Theta(n \log n)$
D. $\Theta(\log \log n)$

nielit2022apr-scientistb recurrence-relation time-complexity programming-in-c algorithm-design

1.1.9 Algorithm Design: NIELIT Scientist B 2020 November: 100

Consider the algorithm that solves problems of size n by recursively solving two sub problems of size $n-1$ and then combining the solutions in constant time. Then the running time of the algorithm would be:

A. $O(n)$ B. $O(\log n)$ C. $O(n \log n)$ D. $O(n^2)$

nielit-scb-2020 time-complexity recurrence-relation algorithm-design asymptotic-notations

Answer key☟

1.1.10 Algorithm Design: NIELIT Scientist B 2020 November: 44

Which sorting algorithm sorts by moving the current data element past the already sorted values and repeatedly interchanges it with the preceding value until it is in its correct place?

A. Insertion sort
B. Internal sort
C. External sort
D. Radix sort

nielit-scb-2020 sorting data-structures algorithm-design

Answer key☟

1.1.11 Algorithm Design: NIELIT Scientist B 2020 November: 52

The best running time is defined as/obtained as/by:

A. the least or smallest of all the running times the algorithm takes, on inputs of a particular size
B. an input that requires maximum computations or resources
C. averaging the different running times for all inputs of a particular kind
D. none of the options

nielit-scb-2020 algorithm-design time-complexity asymptotic-notations

Answer key☟

1.1.12 Algorithm Design: NIELIT Scientist B 2020 November: 77

The recurrence relation $T(n) = 7T(n/7) + n$ has the solution:

A. $O(n)$ B. $O(logn)$ C. $O(nlog(n))$ D. $O(n^2)$

nielit-scb-2020 recurrence-relation asymptotic-notations algorithm-design

Answer key☟

1.1.13 Algorithm Design: NIELIT Scientist B 2020 November: 91

What is the advantage of bubble sort over other sorting techniques?

A. It is faster
B. Consumes less memory
C. Detects whether the input is already sorted
D. All of the options

nielit-scb-2020 sorting algorithm-design data-structures

Answer key

1.2 Algorithm Design Techniques (2)

1.2.1 Algorithm Design Techniques: NIELIT 2016 DEC Scientist B (CS) - Section B: 7

The Knapsack problem belongs to which domain of problems?

A. Optimization B. NP complete C. Linear Solution D. Sorting

nielit2016dec-scientistb-cs algorithms knapsack-problem algorithm-design-techniques

Answer key

1.2.2 Algorithm Design Techniques: NIELIT 2018-74

Dijkstra's algorithm is based on

A. Greedy approach
B. Dynamic programming
C. Backtracking paradigm
D. Divide and conquer paradigm

nielit-2018 algorithms algorithm-design-techniques

Answer key

1.3 Array (2)

1.3.1 Array: NIELIT 2016 MAR Scientist C - Section C: 53

An element in an array X is called a leader if it is greater than all elements to the right of it in X. The best algorithm to find all leaders in an array

A. solves it in linear time using a left to right pass of the array
B. solves in linear time using a right to left pass of the array
C. solves it using divide and conquer in time $\theta(n \log n)$
D. solves it in time $\theta(n^2)$

nielit2016mar-scientistc algorithms array

Answer key

1.3.2 Array: NIELIT 2022 April Scientist B | Section B | Question: 86

A Young tableau is a $2\mathrm{D}$ array of integers increasing from left to right and from top to bottom. Any unfilled entries are marked with ∞, and hence there cannot be any entry to the right of, or below a ∞. The following Young tableau consists of unique entries.

$$\begin{array}{cccc} \mathbf{1} & 2 & 5 & 14 \\ 3 & 4 & 6 & 23 \\ 10 & 12 & 18 & 25 \\ 31 & \infty & \infty & \infty \end{array}$$

When an element is removed from a Young tableau, other elements should be moved into its place so that the resulting table is still a Young tableau (unfilled entries may be filled with a ∞). The minimum number of entries (other than 1) to be shifted, to remove 1 from the given Young tableau is _____.

A. 2 B. 5 C. 6 D. 18

nielit2022apr-scientistb data-structures sorting array

Answer key☟

1.4 Asymptotic Notations (2)

1.4.1 Asymptotic Notations: NIELIT 2016 MAR Scientist C - Section C: 18

Two alternative package A and B are available for processing a database having 10^k records. Package A requires $0.0001n^2$ time units and package B requires $10n\log_{10} n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

A. 12 B. 10 C. 6 D. 5

nielit2016mar-scientistc algorithms asymptotic-notations

Answer key☟

1.4.2 Asymptotic Notations: NIELIT 2017 July Scientist B (IT) - Section B: 59

What is the running time of the following function (specified as a function of the input value)?

```
void Function(int n){
    int i=1;
    int s=1;
    while(s<=n){
        i++;
        s=s+i;
    }
}
```

A. $O(n)$ B. $O(n^2)$ C. $O(1)$ D. $O(\sqrt{n})$

nielit2017july-scientistb-it algorithms time-complexity asymptotic-notations

Answer key☟

1.5 Backtracking (2)

1.5.1 Backtracking: NIELIT 2017 DEC Scientific Assistant A - Section B: 22

Which type of algorithm is used to solve the "8 Queens" problem ?

A. Greedy B. Dynamic C. Divide and conquer D. Backtracking

nielit2017dec-assistanta algorithms backtracking easy

Answer key

1.5.2 Backtracking: NIELIT 2021 Dec Scientist A - Section B: 85

In what manner is a state-space tree for a backtracking algorithm constructed ?

A. Breadth-first search
B. Twice around the tree
C. Depth-first search
D. Nearest neighbour first

nielit2021dec-scientista algorithm-design backtracking data-structures

Answer key

1.6 Binary Search (2)

1.6.1 Binary Search: NIELIT 2021 Dec Scientist A - Section B: 88

The recurrence relation for binary search algorithm is :

A. $\mathrm{T(n) = 2T\ (n/2) + O\ (1)}$
B. $\mathrm{T(n) = 2T\ (n/2) + O\ (n)}$
C. $\mathrm{T(n) = T\ (n/2) + O\ (1)}$
D. $\mathrm{T(n) = T\ (n/2) + O\ (n)}$

nielit2021dec-scientista recurrence-relation binary-search algorithm-design data-structures asymptotic-notations

Answer key

1.6.2 Binary Search: NIELIT Scientist B 2020 November: 101

The program written for binary search, calculates the midpoint of the span as $\mathrm{mid :=(Low{+}High)/2}$. The program works well if the number of elements in the list is small (about $32,000$) but it behaves abnormally when the number of elements is large. This can be avoided by performing the calculation as:

A. $\mathrm{mid :=(High\text{-}Low)/2{+}Low}$
B. $\mathrm{mid :=(High\text{-}Low{+}1)/2}$
C. $\mathrm{mid :=(High\text{-}Low)/2}$
D. $\mathrm{mid :=(High{+}Low)/2}$

nielit-scb-2020 programming-in-c data-structures binary-search algorithms

Answer key

1.7 Binary Search Tree (1)

1.7.1 Binary Search Tree: NIELIT 2021 Dec Scientist A - Section B: 61

Let $\mathrm{T(n)}$ be the number of different binary search trees on n distinct elements-then $\mathrm{T(n)} = \sum_{k=1}^{n} \mathrm{T(K\text{-}1)}\ T(x)$ where x is:

A. $\mathrm{n-k+1}$
B. $\mathrm{n-k}$
C. $\mathrm{n-k-1}$
D. $\mathrm{n-k-2}$

nielit2021dec-scientista binary-search-tree recurrence-relation data-structures algorithms

Answer key

1.8 Cryptography (1)

1.8.1 Cryptography: NIELIT Scientist B 2020 November: 79

Which of the following property is related to a cryptographic hash functions?

A. One way Function
B. Inversible
C. Non-Deterministic
D. All of the options

nielit-scb-2020 cryptography

Answer key☟

1.9 Dijkstras Algorithm (2)

1.9.1 Dijkstras Algorithm: NIELIT 2021 Dec Scientist B - Section B: 56

The maximum number of times the decrease key operation performed in Dijkstra's algorithm will be equal to _______________.

A. Total number of vertices
B. Total number of edges
C. Number of vertices -1
D. Number of edges -1

nielit2021dec-scientistb dijkstras-algorithm graph-algorithms algorithms

Answer key☟

1.9.2 Dijkstras Algorithm: NIELIT 2021 Dec Scientist B - Section B: 66

Dijkstra's Algorithm cannot be applied on ______________ .

A. Directed and weighted graphs
B. Graphs having negative weight function
C. Unweighted graph
D. Undirected and unweighted graphs

nielit2021dec-scientistb graph-algorithms dijkstras-algorithm algorithms

Answer key☟

1.10 Divide and Conquer (3)

1.10.1 Divide and Conquer: NIELIT 2016 DEC Scientist B (IT) - Section B: 52

Find the odd one out

A. Merge Sort
B. TVSP Problem
C. Knapsack Problem
D. OBST Problem

nielit2016dec-scientistb-it algorithms easy dynamic-programming divide-and-conquer

Answer key☟

1.10.2 Divide and Conquer: NIELIT 2017 DEC Scientific Assistant A - Section B: 39

Merge sort uses :

A. Divide-and-conquer
B. Backtracking
C. Heuristic approach
D. Greedy approach

nielit2017dec-assistanta algorithms sorting merge-sort divide-and-conquer

Answer key☟

1.10.3 Divide and Conquer: NIELIT Scientific Assistant A 2020 November: 63

What is the product of following matrix using Strassen's matrix multiplication algorithm?

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \qquad B = \begin{bmatrix} 8 & 4 \\ 6 & 2 \end{bmatrix}$$

A. $C_{11} = 80; C_{12} = 07; C_{21} = 15; C_{22} = 34$
B. $C_{11} = 82; C_{12} = 26; C_{21} = 10; C_{22} = 34$
C. $C_{11} = 15; C_{12} = 07; C_{21} = 80; C_{22} = 34$
D. $C_{11} = 26; C_{12} = 10; C_{21} = 82; C_{22} = 34$

nielit-sta-2020 algorithms divide-and-conquer

Answer key☟

1.11 Dynamic Programming (6)

1.11.1 Dynamic Programming: NIELIT 2017 July Scientist B (CS) - Section B: 39

Which of the following standard algorithms is not Dynamic Programming based?

A. Bellman-Ford Algorithm for single source shortest path
B. Floyd Warshall Algorithm for all pairs shortest paths
C. $0-1$ Knapsack problem
D. Prim's Minimum Spanning Tree

nielit2017july-scientistb-cs algorithms easy dynamic-programming

Answer key☟

1.11.2 Dynamic Programming: NIELIT 2021 Dec Scientist A - Section B: 68

Given the following characteristics :

i. Optimal substructure
ii. Overlapping subproblems
iii. Memorization
iv. Decrease and conquer

Dynamic programming has the following characteristics :

A. (i), (ii), (iv)
B. (i), (ii), (iii)
C. (ii), (iii), (iv)
D. (i), (iii), (iv)

nielit2021dec-scientista algorithm-design dynamic-programming data-structures algorithms

Answer key☟

1.11.3 Dynamic Programming: NIELIT 2021 Dec Scientist A - Section B: 95

The time complexity of solving the Longest Common Subsequence problem using Dynamic Programming is : (m and n are lengths of subsequences)

A. O(m.n)
B. O(m+n)
C. O(log m.n)
D. O(m/n)

nielit2021dec-scientista dynamic-programming time-complexity algorithm-design data-structures

Answer key

1.11.4 Dynamic Programming: NIELIT 2021 Dec Scientist B - Section B: 12

Which of the following methods can be used to solve the Knapsack problem?

A. Brute force algorithm
B. Recursion
C. Dynamic programming
D. Brute force, Recursion, and Dynamic Programming

nielit-2021-it-dec-scientistb algorithm-design dynamic-programming recursion algorithms

Answer key

1.11.5 Dynamic Programming: NIELIT 2021 Dec Scientist B - Section B: 16

In dynamic programming approach the optimum solution is calculated in the following way:

A. Divide and conquer
B. Top up fashion
C. Bottom-up approach
D. Mixed approach

nielit-2021-it-dec-scientistb dynamic-programming algorithm-design programming-in-c data-structures

Answer key

1.11.6 Dynamic Programming: NIELIT Scientific Assistant A 2020 November: 105

Assembly line scheduling and Longest Common Subsequence problems are an example of _______.

A. Dynamic Programming
B. Greedy Algorithms
C. Greedy Algorithms and Dynamic Programming respectively
D. Dynamic Programming and Branch and Bound respectively

nielit-sta-2020 algorithms dynamic-programming

Answer key

1.12 Graph Algorithms (3)

1.12.1 Graph Algorithms: NIELIT 2017 July Scientist B (CS) - Section B: 42

Let G be a graph with n vertices and m edges.What is the tightest upper bound on the running time of Depth First Search of G, when G is represented using adjacency matrix?

A. $O(n)$
B. $O(m+n)$
C. $O(n^2)$
D. $O(mn)$

nielit2017july-scientistb-cs algorithms graph-algorithms

Answer key

1.12.2 Graph Algorithms: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 29

Which of the following algorithm solve the all-pair shortest path problem?

A. Dijakstra's algorithm
B. Floyd's algorithm
C. Prim's algorithm
D. Warshall's algorithm

nielit2017oct-assistanta-cs algorithms graph-algorithms

Answer key☟

1.12.3 Graph Algorithms: NIELIT Scientific Assistant A 2020 November: 79

Which of the following algorithms can be used to most efficiently find whether a cycle is present in a given graph?

A. Prim's Minimum Spanning Tree Algorithm
B. Breadth First Search
C. Depth First Search
D. Kruskal's Minimum Spanning Tree Algorithm

nielit-sta-2020 algorithms graph-algorithms

Answer key☟

1.13 Hashing (5)

1.13.1 Hashing: NIELIT 2016 MAR Scientist B - Section C: 17

A hash function f defined as $f(\text{key}) = \text{key} \bmod 7$, with linear probing, insert the keys $37, 38, 72, 48, 98, 11, 56$ into a table indexed from 11 will be stored in the location

A. 3
B. 4
C. 5
D. 6

nielit2016mar-scientistb algorithms hashing

Answer key☟

1.13.2 Hashing: NIELIT 2016 MAR Scientist B - Section C: 20

A hash table has space for 100 records. Then the probability of collision before the table is 10% full, is

A. 0.45
B. 0.5
C. 0.3
D. 0.34 (approximately)

nielit2016mar-scientistb algorithms hashing

Answer key☟

1.13.3 Hashing: NIELIT 2017 DEC Scientific Assistant A - Section B: 36

The average search time of hashing, with linear probing will be less if the load factor :

A. is far less than 1
B. equals 1

C. is far greater than 1

D. none of the options

nielit2017dec-assistanta algorithms hashing

Answer key☟

1.13.4 Hashing: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 36

The average search time for hashing with linear probing will be less if the load factor

A. Is far less than one

B. Equals one

C. Is far greater than one

D. None of the above

nielit2017oct-assistanta-it algorithms hashing

Answer key☟

1.13.5 Hashing: NIELIT Scientist B 2020 November: 55

In which of the following hash functions, do consecutive keys map to consecutive hash values?

A. Division method

B. Multiplication method

C. Folding method

D. Mid-square method

nielit-scb-2020 hashing data-structures

Answer key☟

1.14 Heapify (1)

1.14.1 Heapify: NIELIT 2022 April Scientist B | Section B | Question: 96

Suppose we are sorting an array of eight integers using heapsort, and we have just finished some heapify (either maxheapify or minheapify) operations. The array now looks like this:
$16 \ 14 \ 15 \ 10 \ 12 \ 27 \ 28$
How many heapify operations have been performed on root of heap ?

A. 1

B. 2

C. 3 or 4

D. 5 or 6

nielit2022apr-scientistb sorting data-structures binary-heap heapify algorithm-design

1.15 Knapsack Problem (2)

1.15.1 Knapsack Problem: NIELIT 2017 July Scientist B (IT) - Section B: 60

$0/1$-Knapsack is a well known problem where, it is desired to get the maximum total profit by placing n items (each item is having some weight and associated profit) into a knapsack of capacity W. The table given below shows the weights and associated profits for 5 items, where one unit of each item is available to you. It is also given that the knapsack capacity W is 8. If the given $0/1$ knapsack problem is solved using Dynamic Programming, which one of the following will be maximum earned profit by placing the items into the knapsack of capacity 8.

Item#	Weight	Associated Profit
1	1	3
2	2	5
3	4	9
4	5	11
5	8	18

A. 19 B. 18 C. 17 D. 20

nielit2017july-scientistb-it algorithms greedy-algorithms knapsack-problem

Answer key☟

1.15.2 Knapsack Problem: NIELIT Scientific Assistant A 2020 November: 64

Which of the following is a correct time complexity to solve the $0/1$ knapsack problem where n and w represents the number of items and capacity of knapsack respectively?

A. $O(n)$ B. $O(w)$ C. $O(nw)$ D. $O(n+w)$

nielit-sta-2020 algorithms dynamic-programming easy knapsack-problem time-complexity

Answer key☟

1.16 Linear Search (1)

1.16.1 Linear Search: NIELIT 2021 Dec Scientist A - Section B: 117

Worst case scenario in case of linear search algorithm is _______________ .

A. Item is somewhere in the middle of the array.
B. Item is not in the array at all.
C. Item is the last element in the array.
D. Item is the last element in the array or is not there at all.

nielit2021dec-scientista data-structures algorithms linear-search time-complexity

Answer key☟

1.17 Master Theorem (3)

1.17.1 Master Theorem: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 4

The running time of an algorithm $T(n)$, where $'n'$ is the input size , is given by
$T(n) = 8T(n/2) + qn$, if $n > 1$
$= p$, if $n = 1$
Where p, q are constants. The order of this algorithm is

A. n^2 B. n^n C. n^3 D. n

nielit2017oct-assistanta-cs algorithms recurrence-relation time-complexity master-theorem

Answer key☟

1.17.2 Master Theorem: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 14

The running time of an algorithm $T(n)$, where $'n'$ is the input size , is given by
$T(n) = 8T(n/2) + qn$, if $n > 1$
$= p$, if $n = 1$
Where p, q are constants. The order of this algorithm is

A. n^2 B. n^n C. n^3 D. n

nielit2017oct-assistanta-it algorithms recurrence-relation time-complexity master-theorem

1.17.3 Master Theorem: NIELIT 2021 Dec Scientist B - Section B: 69

In the recurrence relation
$\mathrm{T}(n) = 0.5 * \mathrm{T}(n/2) + 1/n$, which case of Master Theorem is suitable?

A. Case 1
B. Master Theorem not applicable in this situation
C. Case 2
D. Case 3

nielit-2021-it-dec-scientistb recurrence-relation master-theorem algorithm-design

Answer key☟

1.18 Matrix Chain Ordering (2)

1.18.1 Matrix Chain Ordering: NIELIT 2017 July Scientist B (CS) - Section B: 41

Four Matrices M_1, M_2, M_3 and M_4 of dimensions $p \times q$, $q \times r$, $r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example, when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of multiplications is $pqr + rst + prt$. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is $pqr + prs + pst$.

If $p = 10, q = 100, r = 20, s = 5$ and $t = 80$, then the number of scalar multiplications needed is

A. 248000 B. 44000 C. 19000 D. 25000

nielit2017july-scientistb-cs algorithms dynamic-programming matrix-chain-ordering

1.18.2 Matrix Chain Ordering: NIELIT 2021 Dec Scientist B - Section B: 71

The number of operations in matrix multiplication $\mathrm{M1, M2, M3, M4}$ and $\mathrm{M5}$ of sizes $5 \times 10, 10 \times 100, 100 \times 2, 2 \times 20$ and 20×50 respectively will be:

A. 5830 B. 4600 C. 6900 D. 12890

nielit-2021-it-dec-scientistb algorithms dynamic-programming matrix-chain-ordering

Answer key☟

1.19 Merge Sort (4)

1.19.1 Merge Sort: NIELIT 2017 DEC Scientific Assistant A - Section B: 53

Given two sorted list of size 'm' and 'n' respectively. The number of comparisons needed in the worst case by the merge sort algorithm will be :

A. $m*n$ B. minimum of m, n C. maximum of m, n D. $m+n-1$

nielit2017dec-assistanta algorithms sorting merge-sort

Answer key☟

1.19.2 Merge Sort: NIELIT 2021 Dec Scientist B - Section B: 23

Merge Sort divides the input list in:

A. N equal parts
B. Three equal parts
C. Two parts which may not be equal
D. N parts which may not be equal

nielit-2021-it-dec-scientistb merge-sort sorting algorithm-design data-structures

Answer key☟

1.19.3 Merge Sort: NIELIT 2021 Dec Scientist B - Section B: 63

Division in merge sort algorithm is based on which approach:

A. Parallel B. Random C. Interactive D. Recursive

nielit-2021-it-dec-scientistb merge-sort algorithm-design recursion data-structures sorting

1.19.4 Merge Sort: NIELIT Scientist B 2020 November: 113

If one uses straight two-way merge sort algorithm to sort the following elements in ascending order $20, 47, 15, 8, 9, 4, 40, 30, 12, 17$ then the order of these elements after the second pass of the algorithm is:

A. $8, 9, 15, 20, 47, 4, 12, 17, 30, 40$
B. $8, 15, 20, 47, 4, 9, 30, 40, 12, 17$
C. $15, 20, 47, 4, 8, 9, 12, 30, 40, 17$
D. $4, 8, 9, 15, 20, 47, 12, 17, 30, 40$

nielit-scb-2020 sorting merge-sort data-structures algorithm-design

Answer key☟

1.20 Minimum Spanning Tree (2)

1.20.1 Minimum Spanning Tree: NIELIT 2016 DEC Scientist B (IT) - Section B: 27

Complexity of Kruskal's algorithm for finding minimum spanning tree of an undirected graph containing n vertices and m edges if the edges are sorted is:

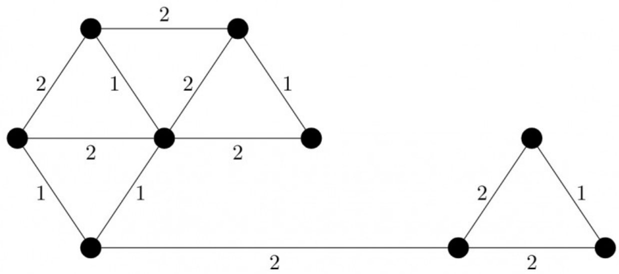
A. $O(mn)$ B. $O(m)$ C. $O(m+n)$ D. $O(n)$

nielit2016dec-scientistb-it algorithms minimum-spanning-tree time-complexity

Answer key

1.20.2 Minimum Spanning Tree: NIELIT 2022 April Scientist B | Section B | Question: 81

The number of distinct minimum spanning trees for the weighted graph below is ________.



A. 4 B. 5 C. 6 D. 7

nielit2022apr-scientistb minimum-spanning-tree graph-algorithms algorithms

1.21 Quick Sort (4)

1.21.1 Quick Sort: NIELIT 2016 DEC Scientist B (CS) - Section B: 12

The running time of Quick sort algorithm depends heavily on the selection of:

A. No. of inputs
B. Arrangement of elements in an array
C. Size of elements
D. Pivot Element

nielit2016dec-scientistb-cs algorithms sorting quick-sort

Answer key

1.21.2 Quick Sort: NIELIT 2021 Dec Scientist B - Section B: 43

Write Recurrence of Quick Sort in worst case.

A. $\mathrm{T(n) = T\ (n\text{-}1) + 1}$
B. $\mathrm{T(n) = T\ (n\text{-}1) + n}$
C. $\mathrm{T(n) = 2T\ (n\text{-}1) + n}$
D. $\mathrm{T(n) = \ T\ (n/3) + T\ (2n/2) + n}$

nielit2021dec-scientistb recurrence-relation quick-sort sorting algorithm-design data-structures

Answer key

1.21.3 Quick Sort: NIELIT Scientific Assistant A 2020 November: 83

Which of the following is correct recurrence for worst case of QuickSort?

A. $T(n) = T(n-4) + T(n-2) + O(1)$
B. $T(n) = T(n-1) + T(0) + O(n)$
C. $T(n) = 2T(n/2) + O(n)$
D. $T(n) = 4T(n/2) + O(n)$

nielit-sta-2020 algorithms quick-sort recurrence-relation

Answer key☟

1.21.4 Quick Sort: NIELIT Scientist B 2020 November: 59

What is the best case complexity of QuickSort?

A. $O(nlogn)$ B. $O(logn)$ C. $O(n)$ D. $O(n^2)$

nielit-scb-2020 sorting time-complexity quick-sort algorithm-design

Answer key☟

1.22 Radix Sort (1)

1.22.1 Radix Sort: NIELIT Scientific Assistant A 2020 November: 103

Consider an array of positive integers between 123456 to 876543, which sorting algorithm can be used to sort these number in linear time?

A. Impossible to sort in linear time
B. Radix Sort
C. Insertion Sort
D. Bubble Sort

nielit-sta-2020 algorithms sorting radix-sort

Answer key☟

1.23 Recurrence Relation (4)

1.23.1 Recurrence Relation: NIELIT 2016 DEC Scientist B (CS) - Section B: 27

What is the solution to the recurrence $T(n) = T\left(\frac{n}{2}\right) + n$?

A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$ D. None of these

nielit2016dec-scientistb-cs algorithms recurrence-relation time-complexity

Answer key☟

1.23.2 Recurrence Relation: NIELIT 2016 MAR Scientist B - Section C: 22

Time complexity of an algorithm $T(n)$, where n is the input size is given by

$$\begin{aligned} T(n) \quad &= T(n-1) + \tfrac{1}{n}, \text{ if } n > 1 \\ &= 1, \text{ otherwise} \end{aligned}$$

The order of this algorithm is

A. $\log n$ B. n C. n^2 D. n^n

nielit2016mar-scientistb algorithms recurrence-relation time-complexity

Answer key☟

1.23.3 Recurrence Relation: NIELIT 2017 July Scientist B (CS) - Section B: 55

Suppose $T(n) = 2T(n/2) + n$, $T(0) = T(1) = 1$ which one of the following is false?

A. $T(n) = O(n^2)$
B. $T(n) = \Theta(n \log n)$
C. $T(n) = \Omega(n^2)$
D. $T(n) = O(n \log n)$

nielit2017july-scientistb-cs algorithms recurrence-relation

Answer key☟

1.23.4 Recurrence Relation: NIELIT 2018-75

For the given recurrence equation

$$\begin{aligned}(n) &= 2T(n-1), && \text{if } n > 0\\ &= 1, && \text{otherwise}\end{aligned}$$

A. $O(n \log n)$
B. $O(n^2)$
C. $O(2^n)$
D. $O(n)$

nielit-2018 algorithms recurrence-relation

Answer key☟

1.24 Recursion (1)

1.24.1 Recursion: NIELIT Scientific Assistant A 2020 November: 101

What is the time complexity of the following recursive function?

```
int ComputFun(int n)
{
    if(n<=2)
      return 1;
    else
       return (ComputFun(floor(sqrt(n)))+n);
}
```

A. $\Theta(n)$
B. $\Theta(\log n)$
C. $\Theta(n \log n)$
D. $\Theta(\log \log n)$

nielit-sta-2020 algorithms recursion time-complexity

Answer key☟

1.25 Revision (1)

1.25.1 Revision: NIELIT 2021 Dec Scientist B - Section B: 7

Arrange the following in increasing orders of asymptotic complexity.

$f1(n) = 2^n, f2(n) = n^{\frac{3}{2}}, f3(n) = n \log n, f4(n) = n^{\log n}$

A. $f3, f2, f4, f1$
B. $f3, f2, f1, f4$
C. $f2, f3, f1, f4$
D. $f2, f3, f4, f1$

nielit-2021-it-dec-scientistb asymptotic-notations time-complexity algorithm-design revision

Answer key

1.26 Searching (3)

1.26.1 Searching: NIELIT 2017 DEC Scientist B - Section B: 8

Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A. The worst case number of probes performed by an optimal algorithm is

A. 2 B. 4 C. 3 D. 5

nielit2017dec-scientistb algorithms searching array

Answer key

1.26.2 Searching: NIELIT 2021 Dec Scientist B - Section B: 59

Which of the following type of search is easiest to implement?

A. Linear search
B. Non-linear search
C. Multidimensional search
D. Bidirectional search

nielit-2021-it-dec-scientistb data-structures searching algorithms

1.26.3 Searching: NIELIT Scientific Assistant A 2020 November: 65

Finding the location of the element with a given value is :

A. Traversal B. Search C. Sort D. None of the options

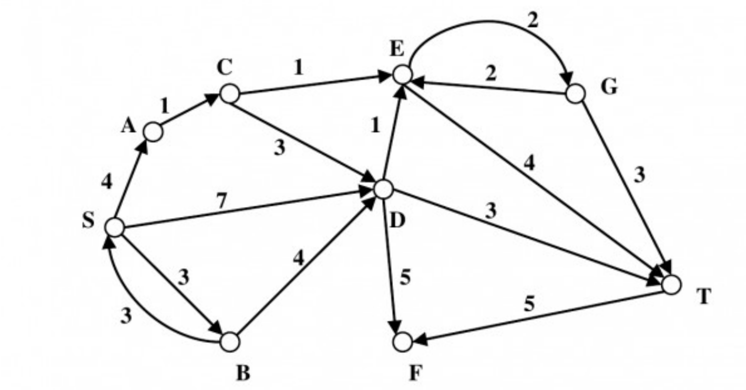
nielit-sta-2020 algorithms searching

Answer key

1.27 Shortest Path (1)

1.27.1 Shortest Path: NIELIT 2022 April Scientist B | Section B | Question: 51

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



A. SDT B. SBDT C. SACDT D. SACET

nielit2022apr-scientistb graph-algorithms shortest-path algorithm-design

1.28 Sorting (7)

1.28.1 Sorting: NIELIT 2016 DEC Scientist B (IT) - Section B: 12

Which of the following sorting procedures is the slowest?

A. Quick Sort B. Merge Sort C. Shell Sort D. Bubble Sort

nielit2016dec-scientistb-it algorithms sorting

Answer key☟

1.28.2 Sorting: NIELIT 2016 DEC Scientist B (IT) - Section B: 8

Selection sort, quick sort is a stable sorting method

A. True,True B. False,False C. True,False D. False,True

nielit2016dec-scientistb-it algorithms sorting

Answer key☟

1.28.3 Sorting: NIELIT 2016 MAR Scientist B - Section C: 12

Which of the following sorting algorithms does not have a worst case running time of $O(n^2\ )$?

A. Insertion sort. B. Merge sort. C. Quick sort. D. Bubble sort.

nielit2016mar-scientistb algorithms sorting time-complexity

Answer key☟

1.28.4 Sorting: NIELIT 2017 July Scientist B (CS) - Section B: 9

The worst case running times of Insertion sort, Merge sort and Quick sort, respectively, are

A. $\Theta(n \log n), \Theta(n \log n) \text{ and } \Theta(n^2)$

B. $\Theta(n^2), \Theta(n^2)$ and $\Theta(n \log n)$
C. $\Theta(n^2), \Theta(n \log n)$ and $\Theta(n \log n)$
D. $\Theta(n^2), \Theta(n \log n)$ and $\Theta(n^2)$

nielit2017july-scientistb-cs algorithms time-complexity sorting

Answer key☟

1.28.5 Sorting: NIELIT 2018-47

_______ sorting algorithms has the lowest worst-case complexity.

A. Selection Sort
B. Bubble Sort
C. Merge Sort
D. Quick Sort

nielit-2018 algorithms sorting

Answer key☟

1.28.6 Sorting: NIELIT 2021 Dec Scientist A - Section B: 86

Which of the following is not a stable sorting algorithm?

A. Insertion sort
B. Selection sort
C. Bubble sort
D. Merge sort

nielit2021dec-scientista sorting algorithms data-structures

Answer key☟

1.28.7 Sorting: NIELIT Scientific Assistant A 2020 November: 88

The given array is $\text{arr}=\{1,2,4,3\}$. Bubble sort is used to sort the array elements. How many passes will be done to sort the array?

A. 4
B. 2
C. 1
D. 3

nielit-sta-2020 algorithms sorting bubble-sort

Answer key☟

1.29 Time Complexity (7)

1.29.1 Time Complexity: NIELIT 2016 DEC Scientist B (CS) - Section B: 16

Two main measures for the efficiency of an algorithm are:

A. Processor and Memory
B. Complexity and Capacity
C. Time and Space
D. Data and Space

nielit2016dec-scientistb-cs algorithms time-complexity

Answer key☟

1.29.2 Time Complexity: NIELIT 2016 DEC Scientist B (CS) - Section B: 52

The concept of order Big O is important because

A. It can be used to decide the best algorithm that solves a given problem
B. It is the lower bound of the growth rate of algorithm
C. It determines the maximum size of a problem that can be solved in a given amount of time
D. Both (A) and (B)

nielit2016dec-scientistb-cs algorithms time-complexity

Answer key

1.29.3 Time Complexity: NIELIT 2016 MAR Scientist C - Section C: 51

The most efficient algorithm for finding the number of connected components in a n undirected graph on n vertices and m edges has time complexity

A. $\Theta(n)$
B. $\Theta(m)$
C. $\Theta(m+n)$
D. $\Theta(mn)$

nielit2016mar-scientistc algorithms time-complexity

Answer key

1.29.4 Time Complexity: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 30

An algorithm is made up of two modules $M1$ and $M2.$ If order of $M1$ is $f(n)$ and $M2$ is $g(n)$ then he order of algorithm is

A. $max(f(n), g(n))$
B. $min(f(n), g(n))$
C. $f(n) + g(n)$
D. $f(n) \times g(n)$

nielit2017oct-assistanta-cs algorithms time-complexity

Answer key

1.29.5 Time Complexity: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 8

Consider the following C code segment:

```
int Ls Prime(n)
{
  int i,n;
  for(i=2;i<=sqrt(n);i++)
  if(n%i ==0)
    {
        printf("NOT Prime.\n");
        return 0;
     }
   return 1;
```

```
}
```

Let $T(n)$ denote the number of times the for loop is executed by the program on input n. Which of the following is true?

A. $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(\sqrt{n})$
B. $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(1)$
C. $T(n) = O(n)$ and $T(n) = \Omega(\sqrt{n})$
D. None of these

nielit2017oct-assistanta-cs algorithms time-complexity

Answer key☟

1.29.6 Time Complexity: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 2

An algorithm is made up pf two modules $M1$ and $M2$. If order of $M1$ is $f(n)$ and $M2$ is $g(n)$ then the order of algorithm is

A. $max(f(n), g(n))$
B. $min(f(n), g(n))$
C. $f(n) + g(n)$
D. $f(n) \times g(n)$

nielit2017oct-assistanta-it algorithms time-complexity

Answer key☟

1.29.7 Time Complexity: NIELIT 2022 April Scientist B | Section B | Question: 44

What is the time complexity of the following function?

```
void myfun()
{
  int a,b;

  for(a=1; a<=n; a++)

    for(b=1; b<=log(a); b++)

        printf("My Function");
}
```

A. $\theta(n)$
B. $\theta(n^2)$
C. $\theta(n \log n)$
D. $\theta(n^2(\log n))$

nielit2022apr-scientistb algorithms time-complexity

Answer key☟

Answer Keys

1.0.1	D	1.1.1	C	1.1.2	C	1.1.3	B	1.1.4	B
1.1.5	D	1.1.6	C	1.1.7	A	1.1.8	D	1.1.9	D
1.1.10	A	1.1.11	A	1.1.12	C	1.1.13	C	1.2.1	A
1.2.2	A	1.3.1	B	1.3.2	B	1.4.1	C	1.4.2	D

1.5.1	D	1.5.2	C	1.6.1	C	1.6.2	A	1.7.1	B
1.8.1	A	1.9.1	B	1.9.2	B	1.10.1	A	1.10.2	A
1.10.3	D	1.11.1	D	1.11.2	B	1.11.3	A	1.11.4	D
1.11.5	C	1.11.6	A	1.12.1	C	1.12.2	D	1.12.3	C
1.13.1	C	1.13.2	D	1.13.3	A	1.13.4	A	1.13.5	A
1.14.1	C	1.15.1	A	1.15.2	C	1.16.1	D	1.17.1	C
1.17.2	C	1.17.3	B	1.18.1	C	1.18.2	B	1.19.1	D
1.19.2	C	1.19.3	D	1.19.4	B	1.20.1	B	1.20.2	C
1.21.1	D	1.21.2	B	1.21.3	B	1.21.4	A	1.22.1	B
1.23.1	B	1.23.2	A	1.23.3	C	1.23.4	C	1.24.1	D
1.25.1	A	1.26.1	D	1.26.2	A	1.26.3	B	1.27.1	D
1.28.1	D	1.28.2	B	1.28.3	B	1.28.4	D	1.28.5	C
1.28.6	B	1.28.7	D	1.29.1	C	1.29.2	A	1.29.3	C
1.29.4	A	1.29.5	B	1.29.6	A	1.29.7	C		

2 Artificial Intelligence (2)

2.1 Logical Reasoning (2)

2.1.1 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 121

The theory states that human mind will receive or accept only that information which it feels that it is relevant.

A. Perception theory
B. Selective Perception
C. Relevance Theory
D. None of the above

nielit2023feb-scientistd logical-reasoning verbal-aptitude

Answer key

2.1.2 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 127

Outgoing, Talkative, Social are part of ___________ Personality Trait as per big 5 personality trait.

A. Friendly
B. Openness to experience
C. Introversion
D. Extroversion

nielit2023feb-scientistd logical-reasoning analytical-aptitude

Answer Keys

2.1.1	B	2.1.2	D

3 CO & Architecture (61)

3.1 Addressing Modes (8)

3.1.1 Addressing Modes: NIELIT 2016 DEC Scientist B (CS) - Section B: 53

The addressing mode used in an instruction of the form $ADD\ X\ Y$, is

A. Direct B. Absolute C. Indirect D. Indexed

nielit2016dec-scientistb-cs co-and-architecture addressing-modes

Answer key

3.1.2 Addressing Modes: NIELIT 2016 DEC Scientist B (IT) - Section B: 55

The example of implied addressing is

A. Stack addressing
B. Indirect addressing
C. Immediate addressing
D. None of the above

nielit2016dec-scientistb-it co-and-architecture addressing-modes

Answer key

3.1.3 Addressing Modes: NIELIT 2016 MAR Scientist B - Section C: 9

A certain processor supports only the immediate and the direct addressing modes. Which of the following programming language features cannot be implemented on this processor?

A. Pointers. B. Arrays. C. Records. D. All of these.

nielit2016mar-scientistb co-and-architecture addressing-modes

Answer key

3.1.4 Addressing Modes: NIELIT 2017 DEC Scientist B - Section B: 47

INCA(Increase register A by 1) is an example of which of the following addressing mode?

A. Immediate addressing
B. Indirect addressing
C. Implied addressing
D. Relative addressing

nielit2017dec-scientistb co-and-architecture addressing-modes

Answer key

3.1.5 Addressing Modes: NIELIT 2017 DEC Scientist B - Section B: 59

A two-word instruction is stored in a location A. The operand part of instruction holds B. If the addressing mode is relative, the operand is available in location

A. $A+B+2$ B. $A+B+1$ C. $B+1$ D. $A+B$

nielit2017dec-scientistb co-and-architecture addressing-modes

Answer key

3.1.6 Addressing Modes: NIELIT 2018-82

Identify the true statement from the given statements. Program relocation at run time

1. requires transfer complete block to some memory locations
2. requires both base address and relative address
3. requires only absolute address

A. 1 B. 1 and 2 C. 1 , 2 and 3 D. 1 and 3

nielit-2018 co-and-architecture addressing-modes

Answer key

3.1.7 Addressing Modes: NIELIT 2021 Dec Scientist A - Section B: 48

The addressing mode/s, which uses the PC instead of a general-purpose register is :

A. Indexed with offset
B. Relative
C. Direct
D. Both Indexed with offset and direct

nielit2021dec-scientista co-and-architecture addressing-modes

Answer key

3.1.8 Addressing Modes: NIELIT Scientific Assistant A 2020 November: 117

In the following addressing mode, which of them performs better for accessing, array?

A. Register addressing mode
B. Direct addressing mode
C. Displacement addressing mode
D. Index addressing mode

nielit-sta-2020 co-and-architecture addressing-modes easy

Answer key

3.2 CO and Architecture (9)

3.2.1 CO and Architecture: NIELIT 2016 DEC Scientist B (CS) - Section B: 55

Process that periodically checks status of an I/O devices, is known as

A. Cold swapping B. I/O instructions C. Polling D. Dealing

nielit2016dec-scientistb-cs co-and-architecture

Answer key

3.2.2 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 19

When a subroutine is called, then address of the instruction following the CAL instruction is stored in/on the

A. Stack pointer B. Accumulator C. Program counter D. Stack

nielit2016mar-scientistc co-and-architecture

Answer key

3.2.3 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 27

The process of entering data into a storage location

A. causes variation in its address number
B. adds to the contents of the location
C. is called a readout operation
D. is destructive of previous contents

nielit2016mar-scientistc co-and-architecture

Answer key

3.2.4 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 28

Serial access memories are useful in applications where

A. data consists of numbers
B. short access time is required
C. each stored word is processed differently
D. data naturally needs to flow in and out in serial form

nielit2016mar-scientistc co-and-architecture

Answer key

3.2.5 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 47

If a processor does not have any stack pointer register, then

A. it cannot have subroutine call instruction
B. it can have subroutine call instruction, but no nested subroutine calls
C. nested subroutine calls are possible, but interrupts are not
D. all sequences of subroutine calls and also interrupts are possible

nielit2016mar-scientistc co-and-architecture

Answer key

3.2.6 CO and Architecture: NIELIT 2017 DEC Scientist B - Section B: 38

Which of the following is **false**?

A. The smallest and fastest computer imitating brain working is called quantum computer.
B. A computer with a speed of around 100 million instructions per second with the word length of around 64 bits is known as super computer.
C. The term Exa-byte $= 1024$ Tera Bytes.
D. None of the options.

nielit2017dec-scientistb co-and-architecture

Answer key

3.2.7 CO and Architecture: NIELIT 2017 July Scientist B (CS) - Section B: 56

The part of machine level instruction, which tells the central processor what has to be done, is

A. Operation code B. Address C. Locator D. Flip-Flop

nielit2017july-scientistb-cs co-and-architecture

Answer key

3.2.8 CO and Architecture: NIELIT 2018-60

If the period of a signal is 100 ms. Then its frequency in Hertz is ______

A. 10 B. 100 C. 1000 D. 10000

nielit-2018 co-and-architecture

Answer key

3.2.9 CO and Architecture: NIELIT 2018-61

The ALU uses _____ to store intermediate result

A. Cache B. Registers C. Accumulators D. Stack

nielit-2018 co-and-architecture

Answer key

3.3 Cache Memory (5)

3.3.1 Cache Memory: NIELIT 2017 DEC Scientist B - Section B: 4

In a cache memory if total number of sets are 's', then the set offset is:

A. 2^8 B. $\log_2 s$ C. s^2 D. s

nielit2017dec-scientistb co-and-architecture cache-memory

Answer key

3.3.2 Cache Memory: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 22

In time division switches if each memory access takes $100\ ns$ and one frame period is $125\ \mu s$, then the maximum number of lines that can be supported is

A. 625 lines B. 1250 lines C. 2300 lines D. 318 lines

nielit2017oct-assistanta-it co-and-architecture cache-memory

Answer key

3.3.3 Cache Memory: NIELIT 2021 Dec Scientist A - Section B: 45

More than one word is put in one cache block to:

A. exploit the temporal locality of reference in a program
B. exploit the spatial locality of reference in a program
C. reduce the miss penalty
D. none of the option

nielit2021dec-scientista co-and-architecture cache-memory

Answer key☟

3.3.4 Cache Memory: NIELIT 2021 Dec Scientist B - Section B: 15

The size of the physical address space of a processor is 2^{P} bytes. The word length is 2^{W} bytes. The capacity of cache memory is 2^{N} bytes. The size of each cache block 2^{M} words. For a $\mathrm{K}\text{-Way}$ set associative cache memory, the length (in number of bits) of the tag fields is:

A. $\mathrm{P}\text{- }\mathrm{N}\text{- }\mathrm{M}\text{- }\mathrm{W}+\log \mathrm{K}$
B. $\mathrm{P}\text{- }\mathrm{N}\text{- }\mathrm{M}\text{- }\mathrm{W}\ -\log_2 \mathrm{K}$
C. $\mathrm{P}\text{- }\mathrm{N}+\log_2 \mathrm{K}$
D. $\mathrm{P}\text{- }\mathrm{N}-\log_2 \mathrm{K}$

nielit-2021-it-dec-scientistb co-and-architecture cache-memory nielit2021dec-scientistb

Answer key☟

3.3.5 Cache Memory: NIELIT Scientist B 2020 November: 69

A direct mapped cache is of size $32\ \mathrm{KB}$ and has block size 32 Bytes. CPU also generates 32 bit address. Number of bits needed for indexing the cache:

A. 14
B. 15
C. 10
D. 17

nielit-scb-2020 cache-memory co-and-architecture

Answer key☟

3.4 Data Dependency (2)

3.4.1 Data Dependency: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 5

Which of the following is a desirable property of module?

A. Independency
B. Low cohesiveness
C. High Coupling
D. Multifunctional

nielit2017oct-assistanta-cs co-and-architecture data-dependency

Answer key☟

3.4.2 Data Dependency: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 10

Which of the following is a desirable property of module?

A. Independency
B. Low cohesiveness
C. High Coupling
D. Multifunctional

nielit2017oct-assistanta-it co-and-architecture data-dependency

Answer key

3.5 Disk (5)

3.5.1 Disk: NIELIT 2016 DEC Scientist B (CS) - Section B: 50

Where does the swap reside?

A. RAM
B. ROM
C. DISK
D. On-chip cache

nielit2016dec-scientistb-cs co-and-architecture disk

Answer key

3.5.2 Disk: NIELIT 2017 DEC Scientific Assistant A - Section B: 23

A 3.5 inch micro floppy high density disk contains the data _________ .

A. $720\,MB$
B. $1.44\,MB$
C. $720\,KB$
D. $1.44\,KB$

nielit2017dec-assistanta co-and-architecture disk

Answer key

3.5.3 Disk: NIELIT 2017 July Scientist B (CS) - Section B: 22

What is the average Access Time for a drum rotating at 4000 revolutions per minute?

A. 2.5 milliseconds
B. 5.0 milliseconds
C. 7.5 milliseconds
D. 4.0 milliseconds

nielit2017july-scientistb-cs co-and-architecture disk

Answer key

3.5.4 Disk: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 1

Which level of RAID refers to disk mirroring with block striping?

A. RAID level 1
B. RAID level 2
C. RAID level 0
D. RAID level 3

nielit2017oct-assistanta-cs co-and-architecture disk

Answer key

3.5.5 Disk: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 26

Which level of RAID refers to disk mirroring with block striping?

A. RAID level 1 B. RAID level 2 C. RAID level 0 D. RAID level 3

nielit2017oct-assistanta-it co-and-architecture disk

Answer key☟

3.6 Instruction Format (6)

3.6.1 Instruction Format: NIELIT 2017 DEC Scientist B - Section B: 40

Which of the following is/are **not** features of RISC processor?

i. Large number of addressing modes.
ii. Uniform instruction set.

A. (i) Only B. (ii) Only C. Both (i) and (ii) D. None of the options

nielit2017dec-scientistb co-and-architecture addressing-modes instruction-format

Answer key☟

3.6.2 Instruction Format: NIELIT 2017 DEC Scientist B - Section B: 49

Which of the following is **false**?

A. Interrupts which are initiated by an instruction are software interrupts
B. When a subroutine is called, the address of the instruction following the CALL instruction is stored in the stack pointer
C. A micro program which is written as 0's and 1's is a binary micro program
D. None of the options

nielit2017dec-scientistb co-and-architecture interrupts instruction-format

Answer key☟

3.6.3 Instruction Format: NIELIT 2017 DEC Scientist B - Section B: 6

A stack organized computer has which of the following instructions?

A. zero-address B. one-address C. two-address D. three-address

nielit2017dec-scientistb co-and-architecture instruction-format

Answer key☟

3.6.4 Instruction Format: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 3

Match list I with List II and select the correct answer using the codes given below the lists.

	List I		List II
$A.$	0-address instruction	$i.$	$T = TOP(T-1)$
$B.$	1-address instruction	$ii.$	$Y = Y + X$
$C.$	2-address instruction	$iii.$	$Y = A - B$
$D.$	3-address instruction	$iv.$	$ACC = ACC - X$

Codes:

A. A-i, B-ii, C-iii, D-iv
B. A-iii, B-ii, C-iv, D-i
C. A-ii, B-iii, C-i, D-iv
D. A-i, B-iv, C-ii, D-iii

nielit2017oct-assistanta-cs co-and-architecture instruction-format machine-instruction

Answer key☟

3.6.5 Instruction Format: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 35

In a 10-bit computer instruction format, the size of address field is 3-bits. The computer uses expanding OP code technique and has 4 two-address instructions and 16 one-address instructions. The number of zero address instructions it can support is

A. 256
B. 356
C. 640
D. 756

nielit2017oct-assistanta-it co-and-architecture machine-instruction instruction-format

Answer key☟

3.6.6 Instruction Format: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 8

Match list I with List II and select the correct answer using the codes given below the lists :

	List I		List II
$A.$	0-address instruction	$i.$	$T = TOP(T-1)$
$B.$	1-address instruction	$ii.$	$Y = Y + X$
$C.$	2-address instruction	$iii.$	$Y = A - B$
$D.$	3-address instruction	$iv.$	$ACC = ACC - X$

Codes:

A. A-i, B-ii, C-iii, D-iv
B. A-iii, B-ii, C-iv, D-i
C. A-ii, B-iii, C-i, D-iv
D. A-i, B-iv, C-ii, D-iii

nielit2017oct-assistanta-it co-and-architecture instruction-format machine-instruction

Answer key☟

3.7 Interrupts (1)

3.7.1 Interrupts: NIELIT 2016 DEC Scientist B (CS) - Section B: 39

External Interrupt may not arise because of

A. illegal or erroneous use of an instruction.
B. a timing device.
C. external source.
D. I/O devices.

nielit2016dec-scientistb-cs co-and-architecture interrupts

Answer key☟

3.8 Machine Instruction (2)

3.8.1 Machine Instruction: NIELIT 2016 DEC Scientist B (IT) - Section B: 39

MIMD stands for

A. Multiple Instruction Multiple Data
B. Multiple Instruction Memory Data
C. Memory Instruction Multiple data
D. Multiple Information Memory data

nielit2016dec-scientistb-it co-and-architecture machine-instruction

Answer key☟

3.8.2 Machine Instruction: NIELIT 2016 MAR Scientist C - Section C: 45

An instruction used to set the carry flag in a computer can be classified as

A. data transfer
B. process control
C. logical
D. program control

nielit2016mar-scientistc co-and-architecture machine-instruction

Answer key☟

3.9 Memory Interfacing (8)

3.9.1 Memory Interfacing: NIELIT 2016 DEC Scientist B (CS) - Section B: 58

CPU consists of _______.

A. ALU and Control Unit
B. ALU, Control Unit and Monitor
C. ALU, Control Unit and Hard disk
D. ALU, Control Unit and Register

nielit2016dec-scientistb-cs co-and-architecture memory-interfacing

Answer key☟

3.9.2 Memory Interfacing: NIELIT 2016 DEC Scientist B (IT) - Section B: 37

How many address lines are needed to address each memory location in a 2048×4 memory chip?

A. 10
B. 11
C. 8
D. 12

nielit2016dec-scientistb-it co-and-architecture memory-interfacing

Answer key☟

3.9.3 Memory Interfacing: NIELIT 2017 DEC Scientist B - Section B: 28

A RAM chip has 7 address lines, 8 data lines and 2 chips select lines. Then the number of memory locations is __________

A. 2^{12} B. 2^{10} C. 2^{19} D. 2^{13}

nielit2017dec-scientistb co-and-architecture memory-interfacing

Answer key

3.9.4 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 27

Which memory is difficult to interface with processor?

A. Static memory B. Dynamic memory C. ROM D. None of the option

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.5 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 28

For a memory system, the cycle time is

A. Same as the access time.
B. Longer than the access time.
C. Shorter than the access time.
D. Multiple of the access time.

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.6 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 29

In comparison with static RAM memory, the dynamic RAM memory has

A. Lower bit density and higher power consumption
B. Higher bit density and lower power consumption
C. Lower bit density and lower power consumption
D. None of the option

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.7 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 30

If each address space represents one byte of storage space, how many address lines are needed to access RAM chips arranged in a 4×6 array, where each chip is $8K \times 4$ bits?

A. 13 B. 14 C. 16 D. 17

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.8 Memory Interfacing: NIELIT Scientific Assistant A 2020 November: 55

A $26-$bit address bus has maximum accessible memory capacity of ________

A. 64 MB B. 16 MB C. 1 GB D. 4 GB

nielit-sta-2020 co-and-architecture memory-interfacing

Answer key☟

3.10 Memory Management (3)

3.10.1 Memory Management: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 2

Which of the following is not a form of main memory?

A. Instruction cache
B. Instruction register
C. Instruction opcode
D. Translation look-aside buffer

nielit2017oct-assistanta-cs co-and-architecture memory-management

Answer key☟

3.10.2 Memory Management: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 5

Which of the following is not a form of main memory?

A. Instruction cache
B. Instruction register
C. Instruction opcode
D. Translation look-aside buffer

nielit2017oct-assistanta-it co-and-architecture memory-management

Answer key☟

3.10.3 Memory Management: NIELIT 2021 Dec Scientist A - Section B: 110

Consider the following sequence of micro operations:

$$\text{MBR} \leftarrow \text{PC}$$

$$\text{MAR} \leftarrow \text{X}$$

$$\text{PC} \leftarrow \text{Y}$$

$$\text{MEMORY} \leftarrow \text{MBR}$$

Which one of the following is possible operation performed by this sequence ?

A. Instruction Fetch
B. Operand Fetch
C. Conditional Branch
D. Initiation of interrupt service

nielit2021dec-scientista co-and-architecture memory-management

3.11 Microprogramming (4)

3.11.1 Microprogramming: NIELIT 2016 MAR Scientist C - Section C: 46

Micro program is

A. the name of source program in micro computers
B. the set of instructions indicating the primitive operations in a system
C. primitive form of macros used in assembly language programming
D. program of very small size

nielit2016mar-scientistc co-and-architecture microprogramming

Answer key

3.11.2 Microprogramming: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 25

A micro programmed control unit

A. Is faster than a hardwired unit
B. Facilitates easy implementation of a new instruction
C. Is useful when small programs are to be run
D. All of the above

nielit2017oct-assistanta-cs co-and-architecture control-unit microprogramming

Answer key

3.11.3 Microprogramming: NIELIT 2021 Dec Scientist A - Section B: 50

A microprogrammed control unit :

A. is faster than hardwired control unit
B. allows easy implementation of new instructions
C. is useful when small programs are to be run
D. none of the options

nielit2021dec-scientista co-and-architecture microprogramming

Answer key

3.11.4 Microprogramming: NIELIT Scientist B 2020 November: 54

Consider a control unit generating the control signals. These control signals are divided into five mutually exclusive groups as shown below:

Groups	$G1$	$G1$	$G1$	$G1$	$G1$
Control Signals	3	7	10	12	2

How many bits are saved using the Vertical Micro-programmed instead of Horizontal Micro-programmed control unit?

A. 14
B. 34
C. 20
D. None

nielit-scb-2020 co-and-architecture microprogramming

Answer key

3.12 Pipelining (5)

3.12.1 Pipelining: NIELIT 2017 DEC Scientist B - Section B: 13

Consider a non-pipelined machine with 6 stages; the lengths of each stage are 20ns, 10ns, 30ns,25ns, 40 ns and 15ns respectively. Suppose for implementing the pipelining the machine adds 5 ns of overhead to each stage for clock skew and set up. What is the speed up factor of the pipelining system (ignoring any hazard impact)?

A. 7 B. 14 C. 3.11 D. 6.22

nielit2017dec-scientistb co-and-architecture pipelining

Answer key

3.12.2 Pipelining: NIELIT 2017 DEC Scientist B - Section B: 14

We have 10-stage pipeline, where the branch target conditions are resolved at stage 5. How many stalls are there for an incorrectly predicted branch?

A. 5 B. 6 C. 7 D. 4

nielit2017dec-scientistb co-and-architecture pipelining

Answer key

3.12.3 Pipelining: NIELIT 2017 July Scientist B (CS) - Section B: 23

Comparing the time $T1$ taken for a single instruction on a pipelined CPU, with time $T2$ taken on a non-pipelined but identical CPU, we can say that _______ ?

A. $T1 = T2$
B. $T1 > T2$
C. $T1 < T2$
D. $T1$ is $T2$ plus time taken for one instruction fetch cycle

nielit2017july-scientistb-cs co-and-architecture pipelining

Answer key

3.12.4 Pipelining: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 18

A pipeline is having speed up factor as 10 and operating with efficiency of 80%. What will be the number of stages in the pipeline?

A. 10 B. 8 C. 13 D. None

nielit2017oct-assistanta-it co-and-architecture pipelining

Answer key

3.12.5 Pipelining: NIELIT 2022 April Scientist B | Section B | Question: 99

A 5 stage pipelined CPU has the following sequence of stages:

- IF – instruction fetch from instruction memory
- RD – Instruction decode and register read
- EX – Execute: ALU operation for data and address computation
- MA – Data memory access – for write access, the register read at RD state is used.

- WB – Register write back

Consider the following sequence of instructions:

- $\mathrm{I1:L\ \ R0}, loc1; \mathrm{R0} <= \mathrm{M}[loc1]$
- $\mathrm{I2:A\ R0, R0; R0} <= \mathrm{R0} + \mathrm{R0}$
- $\mathrm{I3:S\ R2, R0; R2} <= \mathrm{R2} - \mathrm{R0}$

Let each stage take one clock cycle.

What is the number of clock cycles taken to complete the above sequence of instructions starting from the fetch of I_1?

A. 8 B. 10 C. 12 D. 15

nielit2022apr-scientistb co-and-architecture pipelining

3.13 Pushdown Automata (1)

3.13.1 Pushdown Automata: NIELIT 2021 Dec Scientist A - Section B: 111

In the case of, Zero-address instruction method the operands are stored in ______________ .

A. Registers B. Accumulators C. Push down stack D. Cache

nielit2021dec-scientista co-and-architecture pushdown-automata

Answer key☟

3.14 Speedup (1)

3.14.1 Speedup: NIELIT 2016 DEC Scientist B (CS) - Section B: 23

A nonpipeline system taken $50ns$ to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of $10ns.$ Determinant the speedup ration of the pipeline for 100 tasks. What is the maximum speedup that can be achieved?

A. $4.90, 5$ B. $4.76, 5$ C. $3.90, 5$ D. $4.30, 5$

nielit2016dec-scientistb-cs co-and-architecture pipelining speedup

Answer key☟

3.15 Virtual Memory (1)

3.15.1 Virtual Memory: NIELIT Scientist B 2020 November: 76

In ____________ VMs do not simulate the underlying hardware.

A. Para Virtualization B. Full Virtualization
C. Hardware-Assisted Virtualization D. Network Virtualization

nielit-scb-2020 operating-system virtual-memory

Answer key☟

Answer Keys

3.1.1	D	3.1.2	A	3.1.3	D	3.1.4	C	3.1.5	A
3.1.6	B	3.1.7	B	3.1.8	D	3.2.1	C	3.2.2	C
3.2.3	D	3.2.4	D	3.2.5	X	3.2.6	C	3.2.7	A
3.2.8	A	3.2.9	C	3.3.1	B	3.3.2	A	3.3.3	B
3.3.4	C	3.3.5	C	3.4.1	A	3.4.2	A	3.5.1	C
3.5.2	B	3.5.3	C	3.5.4	A	3.5.5	A	3.6.1	A
3.6.2	D	3.6.3	A	3.6.4	D	3.6.5	C	3.6.6	D
3.7.1	A	3.8.1	A	3.8.2	C	3.9.1	D	3.9.2	B
3.9.3	A	3.9.4	B	3.9.5	B	3.9.6	B	3.9.7	D
3.9.8	A	3.10.1	C	3.10.2	C	3.10.3	D	3.11.1	B
3.11.2	B	3.11.3	B	3.11.4	C	3.12.1	C	3.12.2	D
3.12.3	B	3.12.4	C	3.12.5	A	3.13.1	C	3.14.1	B
3.15.1	A								

4 Compiler Design (52)

4.0.1 NIELIT 2018-62

Among all given option, ____ must reside in the main memory.

A. Assembler B. Compiler C. Linker D. Loader

nielit-2018 compiler-design easy

Answer key☟

4.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 36

The graph that shows basic blocks and their successor relationship is called:

A. DAG B. Control graph C. Flow graph D. Hamiltonian graph

nielit2016dec-scientistb-cs compiler-design directed-acyclic-graph

Answer key☟

4.0.3 NIELIT 2016 DEC Scientist B (CS) - Section B: 22

The structure or format of data is called

A. Syntax B. Struct C. Semantic D. none of the above

nielit2016dec-scientistb-cs compiler-design easy

Answer key☟

4.0.4 NIELIT 2016 MAR Scientist C - Section C: 10

Relocation bits used by relocating loader are specified by

A. Relocating loader itself
B. Linker
C. Assembler
D. Macro processor

nielit2016mar-scientistc compiler-design

Answer key☟

4.0.5 NIELIT 2016 MAR Scientist C - Section C: 9

Which of the following can be accessed by transfer vector approach of linking?

A. External data segments
B. External subroutines
C. Data located in other procedure
D. All of these

nielit2016mar-scientistc compiler-design

Answer key☟

4.1 Assembler (1)

4.1.1 Assembler: NIELIT 2016 MAR Scientist B - Section C: 31

In a single pass assembler, most of the forward references can be avoided by putting the restriction

A. on the number of strings/lifereacts.
B. that the data segment must be defined after the code segment.
C. on unconditional rump.
D. that the data segment be defined before the code segment.

nielit2016mar-scientistb compiler-design assembler

Answer key

4.2 Assembly (1)

4.2.1 Assembly: NIELIT 2021 Dec Scientist B - Section B: 50

Assembly language ______________.

A. uses mnemonics or alphabetic codes in place of binary numbers used in machine language
B. is the easiest language to write programs
C. need not be translated into machine language
D. is high level language.

nielit-2021-it-dec-scientistb assembly programming-in-c co-and-architecture

Answer key

4.3 Code Optimization (9)

4.3.1 Code Optimization: NIELIT 2017 DEC Scientist B - Section B: 42

The optimization phase in a compiler generally

A. Reduces the space of the code
B. Optimizes the code to reduce execution time
C. Both (A) and (B)
D. Neither (A) nor (B)

nielit2017dec-scientistb compiler-design code-optimization

Answer key

4.3.2 Code Optimization: NIELIT 2017 DEC Scientist B - Section B: 5

Which of the following is machine independent optimization?

A. Loop optimization
B. Redundancy Elimination
C. Folding
D. All of the option

nielit2017dec-scientistb compiler-design code-optimization

Answer key

4.3.3 Code Optimization: NIELIT 2018-64

Which of the following code replacements is an example of operator strength reduction?

A. Replace P^2 by P^*P
B. Replace P^*16 by $P << 4$
C. Replace $\text{pow}(P,3)$ by P^*P^*P
D. Replace $(P << 5) - P$ by P^*3

nielit-2018 compiler-design code-optimization

Answer key

4.3.4 Code Optimization: NIELIT 2018-77

_____ merges the bodies of two loops

A. loop rolling
B. loop folding
C. loop merge
D. loop jamming

nielit-2018 compiler-design code-optimization easy

Answer key

4.3.5 Code Optimization: NIELIT 2022 April Scientist B | Section B | Question: 52

Consider the expression $(a-1)*(((b+c)/3))+d))$. Let X be the minimum number of registers required by an optimal code generation (without any register spill) algorithm for a load/store architecture, in which

i. only load and store instructions can have memory operands and
ii. arithmetic instructions can have only register or immediate operands.

The value of X is _____________ .

A. 2
B. 1
C. 4
D. 3

nielit2022apr-scientistb compiler-design code-optimization co-and-architecture

4.3.6 Code Optimization: NIELIT 2022 April Scientist B | Section B | Question: 56

Consider these two functions and two statements $\mathrm{S1}$ and $\mathrm{S2}$ about them.

`int work1(int *a, int i, int j)`
`{`
`int x = a[i+2];`
`a[j] = x+1;`
`return a[i+2] - 3;`
`}`
	`int work2(int *a, int i, int j)`
	`{`
	`int t1 = i+2;`
	`int t2 = a[t1];`
	`a[j] = t2+1;`
	`return t2 - 3;`
	`}`

$\mathrm{S1}$: The transformation form work1 to work2 is valid, i.e., for any program state and input arguments, work2 will compute the same output and have the same effect on program state as work1

$\mathrm{S2}$: All the transformations applied to work1 to get work2 will always improve the performance (i.e reduce CPU time) of work2 compared to work1

A. $S1$ is false and $S2$ is false
B. $S1$ is false and $S2$ is true
C. $S1$ is true and $S2$ is false
D. $S1$ is true and $S2$ is true

nielit2022apr-scientistb programming-in-c code-optimization data-structures

4.3.7 Code Optimization: NIELIT Scientific Assistant A 2020 November: 50

Some code optimizations are carried out on the intermediate code because:

A. they enhance the portability of the compiler to other target processors
B. program analysis is more accurate on intermediate code than on machine code
C. the information from data flow analysis cannot otherwise be used for optimization
D. the information from the front end cannot otherwise be used for optimization

nielit-sta-2020 compiler-design code-optimization normal

Answer key☟

4.3.8 Code Optimization: NIELIT Scientific Assistant A 2020 November: 53

Which of the following is not true for tree and graph?

A. A tree is a graph
B. A graph is a tree
C. Tree can have a cycle
D. Tree is a DAG

nielit-sta-2020 compiler-design code-optimization directed-acyclic-graph easy

Answer key☟

4.3.9 Code Optimization: NIELIT Scientific Assistant A 2020 November: 66

Peephole optimization is a :

A. Loop optimization
B. Local optimization
C. Constant folding
D. Data flow analysis

nielit-sta-2020 compiler-design code-optimization easy

Answer key☟

4.4 Compilation Phases (1)

4.4.1 Compilation Phases: NIELIT Scientific Assistant A 2020 November: 67

Which one of the following statements is FALSE?

A. Context-free grammar can be used to specify both lexical and syntax rules.
B. Type checking is done before parsing
C. High level language programs can be translated to different Intermediate Representations
D. Arguments to a function can be passed using the program stack

nielit-sta-2020 compiler-design easy compilation-phases

Answer key☟

4.5 Compilations (1)

4.5.1 Compilations: NIELIT 2017 DEC Scientist B - Section B: 26

The grammar $S \rightarrow aSb \mid bSa \mid SS \mid \varepsilon$ is:

A. Unambiguous CFG
B. Ambiguous CFG
C. Not a CFG
D. Deterministic CFG

nielit2017dec-scientistb compiler-design compilations context-free-grammar ambiguous

Answer key☟

4.6 Compiler tokenization (2)

4.6.1 Compiler tokenization: NIELIT 2018-56

Identify the total number of tokens in the given statement

```
printf("A%B=",&i);
```

A. 7
B. 8
C. 9
D. 13

nielit-2018 compiler-design compiler-tokenization easy

Answer key☟

4.6.2 Compiler tokenization: NIELIT Scientific Assistant A 2020 November: 100

The number of tokens in the following C/C++ statement is :

```
printf("i=%d, &i=%xx", i&i);
```

A. 9
B. 6
C. 10
D. 12

nielit-sta-2020 compiler-design lexical-analysis compiler-tokenization easy

Answer key☟

4.7 Context Free Grammar (2)

4.7.1 Context Free Grammar: NIELIT 2018-32

The context free grammar $S \rightarrow aSb \mid bSa \mid \epsilon$ generates:

A. Equal number of a’s and b’s
B. Unequal number of a’s and b’s
C. Any number of a’s followed by any number of b’s
D. Any number of b’s followed by any number of a’s

nielit-2018 compiler-design context-free-grammar

Answer key☟

4.7.2 Context Free Grammar: NIELIT 2021 Dec Scientist A - Section B: 92

Consider the grammar with nonterminals $\mathrm{N} = \{\mathrm{S}, \mathrm{C}, \mathrm{S}_1\}$ terminals $\mathrm{T} = \{\mathrm{a}, \mathrm{b}, \mathrm{i}, \mathrm{t}, \mathrm{e}\}$ With S as the start symbol, and the following set of rules :

$\mathrm{S} \rightarrow \mathrm{i\,Ct\,SS}_1 | \mathrm{a}$

$\mathrm{S}_1 \rightarrow \mathrm{eS} | \in$

$\mathrm{C} \rightarrow \mathrm{b}$

The grammar is not $\mathrm{LL}(1)$ because :

A. It is left recursive
B. It is right recursive
C. It is ambiguous
D. It is not context free

nielit2021dec-scientista grammar compiler-design parsing context-free-grammar

Answer key

4.8 Data Flow Analysis (1)

4.8.1 Data Flow Analysis: NIELIT 2021 Dec Scientist B - Section B: 117

A variable p is said to be line at point m if and only if :

A. p is assigned at point m
B. p is not assigned at point m
C. Value of p at m would be used along some path in the flow graph starting at point m
D. Value of p at m would be used along some path in the flow graph ending at point m

nielit2021dec-scientistb compiler-design data-flow-analysis

4.9 Dynamic Programming (1)

4.9.1 Dynamic Programming: NIELIT Scientific Assistant A 2020 November: 52

In case of the dynamic programming approach the value of an optimal solution is computed in :

A. Top down fashion
B. Bottom up fashion
C. Left to Right fashion
D. Right to Left fashion

nielit-sta-2020 compiler-design dynamic-programming

Answer key

4.10 First and Follow (1)

4.10.1 First and Follow: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 10

Consider an ε-tree CFG. If for every pair of productions $A \rightarrow u$ and $A \rightarrow v$

A. If $\mathrm{FIRST(u)} \cap \mathrm{FIRST(v)}$ is empty then the CFG has to be $LL(1)$.
B. If the CFG is $LL(1)$ then $\mathrm{FIRST(u)} \cap \mathrm{FIRST(v)}$ has to be empty.

C. Both (A) and (B)
D. None of the above

nielit2017oct-assistanta-cs compiler-design context-free-grammar first-and-follow

Answer key☟

4.11 Grammar (1)

4.11.1 Grammar: NIELIT 2017 July Scientist B (CS) - Section B: 47

What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon and unit production (i.e.,of type $A \to \epsilon$ and $A \to a$) to parse a string with n tokens?

A. $n/2$ B. $n-1$ C. $2n-1$ D. 2^n

nielit2017july-scientistb-cs compiler-design grammar

Answer key☟

4.12 Intermediate Code (2)

4.12.1 Intermediate Code: NIELIT 2018-76

Identify the correct nodes and edges in the given intermediate code

1. $i=1$
2. $t1=5*i$
3. $t2=4*t1$
4. $t3=t2$
5. $a[t3]=0$
6. $i=i+1$
7. if $i<15$ goto(2)

A. 33 B. 44 C. 43 D. 34

nielit-2018 compiler-design intermediate-code

Answer key☟

4.12.2 Intermediate Code: NIELIT 2021 Dec Scientist A - Section B: 54

One of the purposes of using intermediate code in compilers is to:

A. make parsing and semantic analysis simpler
B. improve error recovery and error reporting
C. increase the chances of reusing the machine – independent code optimizer in other compilers
D. improve the register allocation

nielit2021dec-scientista compiler-design intermediate-code

Answer key

4.13 Interpreter (1)

4.13.1 Interpreter: NIELIT 2017 DEC Scientist B - Section B: 54

Which of the following statements is/are true in the context of interpreters?

- $S1$: Interpreters process program according to the logical flow of control through the program.
- $S2$: Interpreter translates and executes the error-free first instruction before it goes to the second.
- $S3$: Interpreter processing time is less compared with compiler.
- $S4$: LISP and Prolog are interpreted languages.

A. Only $S1$
B. Only $S3$
C. Only $S1$, $S2$ and $S3$
D. Only $S1$, $S2$ and $S4$

nielit2017dec-scientistb compiler-design interpreter

Answer key

4.14 Is&software Engineering (1)

4.14.1 Is&software Engineering: NIELIT Scientist B 2020 November: 81

Debugger is a program that:

A. Allows to examine and modify the contents of registers
B. Allows to set breakpoints, execute a segment of program and display contents of register
C. Does not allow execution of a segment of program
D. All the options

nielit-scb-2020 operating-system is&software-engineering

Answer key

4.15 LR Parser (3)

4.15.1 LR Parser: NIELIT 2017 DEC Scientist B - Section B: 33

Which of the following statements is/are false?

$S1$: $LR(0)$ grammar and $SLR(1)$ grammar are equivalent

$S2$: $LR(1)$ grammar are subset of $LALR(1)$ grammars

A. $S1$ only
B. $S1$ and $S2$ both
C. $S2$ only
D. None of the options

nielit2017dec-scientistb compiler-design grammar lr-parser easy

Answer key

4.15.2 LR Parser: NIELIT 2022 April Scientist B | Section B | Question: 64

Which of the following statements about the parser is/are correct?

I. Canonical LR is more powerful than SLR.
II. SLR is more powerful than LALR.
III. SLR is more powerful than canonical LR.

A. (I) only
B. (II) only
C. (III) only
D. (II) and (III) only

nielit2022apr-scientistb parsing compiler-design lr-parser

Answer key☟

4.15.3 LR Parser: NIELIT Scientific Assistant A 2020 November: 94

Shift reduce parsing can also be called as:

A. Reverse of the Right Most Derivation
B. Right Most Derivation
C. Left Most Derivation
D. None of the options

nielit-sta-2020 compiler-design parsing lr-parser easy

Answer key☟

4.16 Lexical Analysis (3)

4.16.1 Lexical Analysis: NIELIT 2016 DEC Scientist B (CS) - Section B: 57

The output of lexical analyzer is:

A. A set of regular expressions
B. Strings of character
C. Syntax tree
D. Set of tokens

nielit2016dec-scientistb-cs compiler-design lexical-analysis easy

Answer key☟

4.16.2 Lexical Analysis: NIELIT 2017 July Scientist B (CS) - Section B: 49

In a compiler, keywords of a language are recognized during

A. parsing of the program
B. the code generation
C. the lexical analysis of the program
D. dataflow analysis

nielit2017july-scientistb-cs compiler-design lexical-analysis easy

Answer key☟

4.16.3 Lexical Analysis: NIELIT 2021 Dec Scientist B - Section B: 99

Two Important lexical categories are _____________ .

A. White Space
B. Comments

C. White Space & Comments

D. None of the options

nielit2021dec-scientistb compiler-design lexical-analysis

Answer key☟

4.17 Linker (2)

4.17.1 Linker: NIELIT 2016 MAR Scientist B - Section C: 32

A linker is given object module for a set of programs that were compiled separately. What information need not be included in an object module?

A. Object mode
B. Relocation bits
C. Names and locations of all external symbols defined in the object module.
D. Absolute addresses of internal symbols.

nielit2016mar-scientistb compiler-design linker

Answer key☟

4.17.2 Linker: NIELIT 2017 July Scientist B (CS) - Section B: 57

A system program that combines the separately complied modules of a program into a form suitable for execution

A. assembler
B. linking loader
C. cross compiler
D. load and go

nielit2017july-scientistb-cs compiler-design linker

Answer key☟

4.18 Parsing (8)

4.18.1 Parsing: NIELIT 2016 DEC Scientist B (CS) - Section B: 40

A top down parser generates

A. Left most derivation
B. Right most derivation
C. Left most derivation in reverse
D. Right most derivation in reverse

nielit2016dec-scientistb-cs compiler-design parsing easy

Answer key☟

4.18.2 Parsing: NIELIT 2016 MAR Scientist C - Section C: 11

The most powerful parser is

A. SLR
B. LALR
C. Canonical LR
D. Operator-precedence

nielit2016mar-scientistc compiler-design parsing easy

Answer key☟

4.18.3 Parsing: NIELIT 2016 MAR Scientist C - Section C: 12

YACC builds up

A. SLR parsing table
B. Canonical LR parsing table
C. LALR parsing table
D. None of these

nielit2016mar-scientistc compiler-design parsing easy

Answer key

4.18.4 Parsing: NIELIT 2016 MAR Scientist C - Section C: 13

Context-free grammar can be recognized by

A. finite state automation
B. 2- way linear bounded automata
C. push down automata
D. both (B) and (C)

nielit2016mar-scientistc compiler-design parsing

Answer key

4.18.5 Parsing: NIELIT 2021 Dec Scientist B - Section B: 70

A bottom-up parser generates _______________ .

A. Rightmost derivation
B. Rightmost derivation in reverse
C. Leftmost derivation
D. Leftmost derivation in reverse

nielit2021dec-scientistb compiler-design parsing

4.18.6 Parsing: NIELIT Scientific Assistant A 2020 November: 118

The $\mathrm{LL}(1)$ and $\mathrm{LR}(0)$ techniques are ___________

A. Both same in power
B. Both simulate reverse of right most derivation
C. Both simulate reverse of left most derivation
D. Incomparable

nielit-sta-2020 compiler-design parsing

Answer key

4.18.7 Parsing: NIELIT Scientific Assistant A 2020 November: 46

Assume that the SLR parser for a grammar G has n_1 states and the LALR parser for G has n_2 states. The relationship between n_1 and n_2 is :

A. n_1 is necessarily less than n_2
B. n_1 is necessarily equal to n_2
C. n_1 is necessarily greater than n_2
D. none of the options

nielit-sta-2020 compiler-design parsing

Answer key

4.18.8 Parsing: NIELIT Scientific Assistant A 2020 November: 69

Type of conflicts that can arise in $\mathrm{LR}(0)$ techniques are ________.

A. Shift-reduce conflict
B. Shift-Shift conflict
C. Both “Shift-reduce conflict” & “Shift-Shift conflict”
D. None of the options

nielit-sta-2020 compiler-design parsing easy

Answer key☟

4.19 Runtime Environment (1)

4.19.1 Runtime Environment: NIELIT 2016 DEC Scientist B (CS) - Section B: 14

The identification of common sub-expression and replacement of run time computations by compile-time computations is:

A. Local optimization
B. Constant folding
C. Loop Optimization
D. Data flow analysis

nielit2016dec-scientistb-cs compiler-design runtime-environment

Answer key☟

4.20 Stack (1)

4.20.1 Stack: NIELIT 2022 April Scientist B | Section B | Question: 103

Which of the following are true?

I. A programming language which does not permit global variables of any kind and has no nesting of procedures/functions, but permits recursion can be implemented with static storage allocation
II. Multi-level access link (or display) arrangement is needed to arrange activation records only if the programming language being implemented has nesting of procedures/functions
III. Recursion in programming languages cannot be implemented with dynamic storage allocation
IV. Nesting procedures/functions and recursion require a dynamic heap allocation scheme and cannot be implemented with a stack-based allocation scheme for activation records
V. Programming languages which permit a function to return a function as its result cannot be implemented with a stack-based storage allocation scheme for activation records

A. (II) and (V) only
B. (I), (III) and (IV) only
C. (I), (II) and (V) only
D. (II), (III) and (V) only

nielit2022apr-scientistb compiler-design operating-system stack memory-management programming-in-c

4.21 Syntax Directed Translation (3)

4.21.1 Syntax Directed Translation: NIELIT 2016 DEC Scientist B (CS) - Section B: 43

Syntax directed translation scheme is desirable because

A. It is based on the syntax
B. It is easy to modify
C. Its description is independent of any implementation
D. All of these

nielit2016dec-scientistb-cs compiler-design syntax-directed-translation

Answer key☟

4.21.2 Syntax Directed Translation: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 11

Synthesized attribute can easily be simulated by an

A. LL grammar
B. ambiguous grammar
C. LR grammar
D. none of the above

nielit2017oct-assistanta-cs compiler-design syntax-directed-translation

Answer key☟

4.21.3 Syntax Directed Translation: NIELIT 2022 April Scientist B | Section B | Question: 69

Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals $\{\text{S,A}\}$ and terminals $\{\text{a,b}\}$.

- $\text{S} \rightarrow \text{aA} \quad \{\text{print } 1\}$
- $\text{S} \rightarrow \text{a} \quad \{\text{print } 2\}$
- $\text{A} \rightarrow \text{Sb} \quad \{\text{print } 3\}$

Using the above SDTS, the output printed by a bottom-up parser, for the inputaab is:

A. $1\;3\;2$
B. $2\;2\;3$
C. $2\;3\;1$
D. Syntax Error

nielit2022apr-scientistb syntax-directed-translation parsing compiler-design

Answer key☟

4.22 Three Address Code (1)

4.22.1 Three Address Code: NIELIT 2021 Dec Scientist B - Section B: 105

A sequence of statement of the form $x = y \text{ op } z$ is called a :

A. Three address code
B. Syntax tree
C. Postfix notation
D. Operator

nielit2021dec-scientistb compiler-design three-address-code

Answer key☟

Answer Keys

4.0.1	D	4.0.2	C	4.0.3	A	4.0.4	B	4.0.5	B
4.1.1	D	4.2.1	A	4.3.1	C	4.3.2	D	4.3.3	B
4.3.4	D	4.3.5	2	4.3.6	A	4.3.7	A	4.3.8	C
4.3.9	B	4.4.1	B	4.5.1	B	4.6.1	A	4.6.2	A
4.7.1	A	4.7.2	C	4.8.1	C	4.9.1	B	4.10.1	C
4.11.1	B	4.12.1	A	4.12.2	C	4.13.1	D	4.14.1	B
4.15.1	B	4.15.2	A	4.15.3	A	4.16.1	D	4.16.2	C
4.16.3	C	4.17.1	D	4.17.2	B	4.18.1	A	4.18.2	C
4.18.3	C	4.18.4	D	4.18.5	B	4.18.6	A	4.18.7	B
4.18.8	A	4.19.1	B	4.20.1	A	4.21.1	D	4.21.2	C
4.21.3	C	4.22.1	A						

5 Computer Networks (111)

5.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 26

The process of converting the analog sample into discrete form is called

A. Modulation B. Multiplexing C. Quantization D. Sampling

nielit2017july-scientistb-cs computer-networks

Answer key☟

5.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 20

Bluetooth is an example of:

A. Personal area network
B. Virtual private network
C. Local area network
D. None of the above

nielit2016dec-scientistb-cs computer-networks

Answer key☟

5.0.3 NIELIT 2016 DEC Scientist B (CS) - Section B: 19

Communication between a computer and a keyboard involves ___________ transmission

A. Simplex B. Half-Duplex C. Automatic D. Full-Duplex

nielit2016dec-scientistb-cs computer-networks

Answer key☟

5.0.4 NIELIT 2016 DEC Scientist B (CS) - Section B: 9

The first Network:

A. ARPANET. B. NFSNET. C. CNNET. D. ASAPNET.

nielit2016dec-scientistb-cs computer-networks

Answer key☟

5.0.5 NIELIT 2016 DEC Scientist B (IT) - Section B: 54

Which of this is not a network edge device?

A. PC B. Server C. Smartphone D. Switch

nielit2016dec-scientistb-it computer-networks

Answer key☟

5.0.6 NIELIT 2016 DEC Scientist B (IT) - Section B: 22

Bluetooth is an example of:

A. Local area network.
B. Virtual private network.
C. Personal area network.
D. None of the mentioned above

nielit2016dec-scientistb-it computer-networks

Answer key☟

5.0.7 NIELIT 2016 DEC Scientist B (IT) - Section B: 5

Two alternative packages A and B are available for processing a database having $10k$ records. Package A requires $0.0001n2$ time units and package B requires $10n \log 10n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

A. 12
B. 10
C. 6
D. 5

nielit2016dec-scientistb-it computer-networks

Answer key☟

5.0.8 NIELIT 2016 DEC Scientist B (IT) - Section B: 2

Three or more devices share a link in ________ connection

A. Unipoint
B. Polarpoint
C. Point to point
D. Multipoint

nielit2016dec-scientistb-it computer-networks

Answer key☟

5.0.9 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 26

If the channel is band limited to $6\ kHz$ and signal to noise ratio is 16, what would be the capacity of channel?

A. 16.15 kbps
B. 23.24 kbps
C. 40.12 kbps
D. 24.74 kbps

nielit2017oct-assistanta-cs computer-networks

Answer key☟

5.0.10 NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 32

The maximum data rate of a channel of 3000-Hz bandwidth and SNR of 30 db is

A. 60000
B. 15000
C. 30000
D. 3000

nielit2017oct-assistanta-it computer-networks

Answer key☟

5.0.11 NIELIT 2016 MAR Scientist C - Section C: 33

In networking terminology UTP means

A. Unshielded Twisted pair
B. Ubiquitious Teflon port
C. Uniformly Terminating port
D. Unshielded T- connector port

nielit2016mar-scientistc computer-networks

Answer key☟

5.0.12 NIELIT 2016 MAR Scientist C - Section C: 22

What are the primary characteristics that distinguish a cell from a packet?

A. cells are generally smaller that packets
B. cells do not incorporate physical address
C. all cells have the same fixed length
D. packet cannot be switched

nielit2016mar-scientistc computer-networks

Answer key☟

5.0.13 NIELIT 2016 MAR Scientist C - Section C: 21

Repeaters function in

A. Physical layer
B. Data link layer
C. Network layer
D. Both (A) and (B)

nielit2016mar-scientistc computer-networks

Answer key☟

5.0.14 NIELIT 2016 MAR Scientist C - Section C: 20

Start and stop bits are used in serial communication for

A. error detection
B. error correction
C. synchronization
D. slowing down the communication

nielit2016mar-scientistc computer-networks

Answer key☟

5.0.15 NIELIT Scientific Assistant A 2020 November: 82

To guarantee correction of upto 5 errors in all cases, the minimum Hamming distance in a block code must be ______.

A. 11
B. 6
C. 5
D. 2

nielit-sta-2020 computer-networks hamming-code

Answer key☟

5.0.16 NIELIT Scientific Assistant A 2020 November: 43

In ICMP, in case of time exceeded error, when the datagram visits a router, the value of time to live field is ________

A. Remains constant
B. Decremented by 2
C. Incremented by 1
D. Decremented by 1

nielit-sta-2020 computer-networks icmp

Answer key☟

5.0.17 NIELIT Scientist B 2020 November: 88

In which modulation discrete values of carrier frequencies is used to transmit binary data?

A. Phase Shift Keying
B. Amplitude Shift Keying
C. Frequency Shift Keying
D. Disk Shift Keying

nielit-scb-2020 computer-networks

Answer key☟

5.0.18 NIELIT Scientist B 2020 November: 83

_____________ possible labels are allowed in the first level of generic domain.

A. 10
B. 12
C. 16
D. none of the options

nielit-scb-2020 computer-networks

Answer key☟

5.0.19 NIELIT 2021 Dec Scientist B - Section B: 93

The spectral efficiency of QPSK null to null having a width 4 Hz will be :

A. $1/2$
B. 1
C. $1/4$
D. 4

nielit2021dec-scientistb computer-networks

5.0.20 NIELIT 2021 Dec Scientist B - Section B: 59

A wireless network interface controller can work in _______________ .

A. infrastructure mode
B. ad–hoc mode
C. both infrastructure mode and ad-hoc mode
D. WDS mode

nielit2021dec-scientistb computer-networks

Answer key☟

5.1 Application Layer (1)

5.1.1 Application Layer: NIELIT 2016 DEC Scientist B (IT) - Section B: 49

Application layer protocol defines:

A. types of messages exchanged
B. rules for when and how processes send and respond to messages
C. message format, syntax and semantics
D. all of the above

nielit2016dec-scientistb-it computer-networks application-layer

Answer key☟

5.2 Application Layer Protocols (2)

5.2.1 Application Layer Protocols: NIELIT 2021 Dec Scientist A - Section B: 44

The File Transfer Protocol is built on ______________.

A. data centric architecture
B. service-oriented architecture
C. client server architecture
D. connection-oriented architecture

nielit2021dec-scientista computer-networks application-layer-protocols

Answer key☟

5.2.2 Application Layer Protocols: NIELIT 2021 Dec Scientist B - Section B: 28

Which TCP/IP protocol is used for file transfer with minimal capability and minimal overhead?

A. RARP
B. FTP
C. TFTP
D. TELNET

nielit-2021-it-dec-scientistb computer-networks transport-layer application-layer-protocols

Answer key☟

5.3 Baud Rate (1)

5.3.1 Baud Rate: NIELIT 2018-35

Baud rate measures the number of ____ transmitted per second

A. Symbols
B. Bits
C. Byte
D. None of these

nielit-2018 computer-networks baud-rate

Answer key☟

5.4 Bit Rate (1)

5.4.1 Bit Rate: NIELIT 2022 April Scientist B | Section B | Question: 60

For a bit-rate of 8 kbps, the best possible values of the transmitted frequencies in a coherent binary FSK system are :

A. 16 kHz and 20 kHz
B. 20 kHz and 32 kHz
C. 20 kHz and 40 kHz
D. 32 kHz and 40 kHz

nielit2022apr-scientistb computer-networks data-communication modulation bit-rate

5.5 Bit Stuffing (2)

5.5.1 Bit Stuffing: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 24

Bit stuffing refers to

A. Inserting a $'0'$ in user data stream to differentiate it with a flag
B. Inserting a $'0'$ in flag stream to avoid ambiguity
C. Appending a nibble to the flag sequence
D. Appending a nibble to the user data stream

nielit2017oct-assistanta-cs computer-networks bit-stuffing

Answer key

5.5.2 Bit Stuffing: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 4

Bit stuffing refers to

A. Inserting a $'0'$ in user data stream to differentiate it with a flag
B. Inserting a $'0'$ in flag stream to avoid ambiguity
C. Appending a nibble to the flag sequence
D. Appending a nibble to the user data stream

nielit2017oct-assistanta-it computer-networks bit-stuffing

Answer key

5.6 CO and Architecture (1)

5.6.1 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 68

What is the minimum no. of wires required for sending data over a serial communication link?

A. 1 B. 3 C. 2 D. 4

nielit-2021-it-dec-scientistb computer-networks co-and-architecture

5.7 CRC Polynomial (3)

5.7.1 CRC Polynomial: NIELIT 2018-79

Given message $M = 1010001101$. The CRC for this given message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is ______

A. 01011 B. 10101 C. 01110 D. 10110

nielit-2018 computer-networks crc-polynomial

Answer key

5.7.2 CRC Polynomial: NIELIT 2021 Dec Scientist A - Section B: 115

If the CRC has Polynomial of degree n, then what is the probability of detecting errors greater then n ?

A. $\frac{1}{2^{n-1}}$ B. $\frac{1}{2^{n+1}}$ C. $\frac{1}{2^{n}}$ D. $\frac{1}{2^{n+2}}$

nielit2021dec-scientista crc-polynomial probability error-detection computer-networks

5.7.3 CRC Polynomial: NIELIT Scientific Assistant A 2020 November: 120

In CRC calculation if divisor is 1011, and dataword is 1001 what will be the CRC?

A. 111 B. 101 C. 110 D. 100

nielit-sta-2020 computer-networks crc-polynomial

Answer key☟

5.8 CSMA CD (1)

5.8.1 CSMA CD: NIELIT 2021 Dec Scientist A - Section B: 73

In $\mathrm{CSMA/CD}$ after detecting the collision, station immediately stops transmission by sending the ______________.

A. Stop pattern
B. Preamble pattern
C. Jam signal
D. Block signal

nielit2021dec-scientista computer-networks csma-cd

Answer key☟

5.9 CSMA Ca (2)

5.9.1 CSMA Ca: NIELIT 2017 DEC Scientific Assistant A - Section B: 13

Which multiple access technique is used by $\mathrm{IEEE}\ 802.11$ standard for wireless LAN ?

A. CDMA B. $\mathrm{CSMA/CA}$ C. ALOHA D. None of the options

nielit2017dec-assistanta computer-networks csma-ca

Answer key☟

5.9.2 CSMA Ca: NIELIT 2021 Dec Scientist A - Section B: 112

Which multiple access technique is used by $\mathrm{IEEE}\ 802.11$ standard for wireless LAN ?

A. CDMA B. $\mathrm{CSMA/CA}$ C. ALOHA D. $\mathrm{CSMA/CD}$

nielit2021dec-scientista computer-networks csma-cd csma-ca lan-technologies

Answer key☟

5.10 Channel Capacity (1)

5.10.1 Channel Capacity: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 11

If the channel is band limited to $6\ kHz$ and signal to noise ratio is $16,$ what would be the capacity of channel?

A. 16.15 kbps
B. 23.24 kbps
C. 40.12 kbps
D. 24.74 kbps

nielit2017oct-assistanta-it computer-networks channel-capacity

Answer key☟

5.11 Communication (2)

5.11.1 Communication: NIELIT 2021 Dec Scientist A - Section B: 46

In DPSK technique, the technique used to encode bits is :

A. AMI
B. Differential code
C. Unipolar RZ format
D. Manchester formate

nielit2021dec-scientista communication

Answer key☟

5.11.2 Communication: NIELIT 2022 April Scientist B | Section B | Question: 119

In a binary data transmission DPSK is preferred to PSK because :

A. a coherent carrier is not required to be generated at the receiver
B. for a given energy per bit, the probability of error is less
C. the 1800 phase shifts of the carrier are unimportant
D. more protection is provided against impulse noise

nielit2022apr-scientistb computer-networks communication

Answer key☟

5.12 Congestion Control (4)

5.12.1 Congestion Control: NIELIT 2017 DEC Scientific Assistant A - Section B: 11

Closed - Loop control mechanism try to :

A. Remove congestion after it occurs
B. Remove congestion after sometime
C. Prevent congestion before it occurs
D. Prevent congestion before sending packets

nielit2017dec-assistanta computer-networks congestion-control

Answer key

5.12.2 Congestion Control: NIELIT 2017 DEC Scientific Assistant A - Section B: 43

When we use slow-start algorithm, the size of the congestion window increases _______ until it reaches a threshold.

A. Additively
B. Multiplicatively
C. Exponentially
D. None of the options

nielit2017dec-assistanta computer-networks congestion-control sliding-window

Answer key

5.12.3 Congestion Control: NIELIT Scientific Assistant A 2020 November: 87

Why does congestion occur?

A. Because the routers and switches have tables
B. Because the routers and switches have queues
C. Because the routers and switches have cross-points
D. None of the options

nielit-sta-2020 computer-networks congestion-control

Answer key

5.12.4 Congestion Control: NIELIT Scientist B 2020 November: 98

In the congestion avoidance algorithm, the size of the congestion window increases _________ until congestion is detected.

A. Exponentially
B. Additively
C. Multiplicatively
D. Suddenly

nielit-scb-2020 computer-networks congestion-control

Answer key

5.13 Cryptography (1)

5.13.1 Cryptography: NIELIT Scientist B 2020 November: 78

____________ uses pretty good privacy algorithm.

A. Electronic mails
B. File encryption
C. Both Electronic mails and File encryption
D. None of the options

nielit-scb-2020 network-security cryptography computer-networks

Answer key

5.14 Data Communication (2)

5.14.1 Data Communication: NIELIT 2021 Dec Scientist B - Section B: 86

The bit rate of digital communication system is $\mathrm{R}\ \mathrm{kbit/s}$. The modulation used is $32-\mathrm{QAM}$. The minimum bandwidth required for ISI free transmission is :

A. $\mathrm{R}/10 \mathrm{~Hz}$
B. $\mathrm{R}/10 \mathrm{~kHz}$
C. $\mathrm{R}/5 \mathrm{~Hz}$
D. $\mathrm{R}/5 \mathrm{~kHz}$

nielit2021dec-scientistb data-communication digital-logic

5.14.2 Data Communication: NIELIT Scientist B 2020 November: 114

A special PCM system uses 32 channels of data, one whose purpose is an identification (ID) and synchronization. The sampling rate is 4 kHz. The word length is 5 bits. Find the serial data rate.

A. $1280\ kHz$
B. $160\ kHz$
C. $320\ kHz$
D. $640\ kHz$

nielit-scb-2020 co-and-architecture data-communication

Answer key☟

5.15 Data Path (1)

5.15.1 Data Path: NIELIT 2016 DEC Scientist B (IT) - Section B: 47

The ________ is the physical path over which a message travels.

A. Path
B. Protocol
C. Route
D. Medium

nielit2016dec-scientistb-it computer-networks data-path

Answer key☟

5.16 Digital Signal Processing (1)

5.16.1 Digital Signal Processing: NIELIT 2021 Dec Scientist A - Section B: 65

Equalization process includes :

A. maximum likelihood sequence estimation and equalization with filters
B. maximum likelihood sequence estimation
C. equalization with filters
D. constant impulse response

nielit2021dec-scientista computer-networks digital-signal-processing

5.17 Digital Signature (1)

5.17.1 Digital Signature: NIELIT Scientific Assistant A 2020 November: 62

A digital signature is required:

A. for non-repudiation of communication by a sender
B. for all e-mail sending
C. for all DHCP server
D. for FTP transactions

nielit-sta-2020 computer-networks digital-signature

Answer key

5.18 Error Correction (1)

5.18.1 Error Correction: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 2

Which of the following is minimum error code?

A. Octal Code B. Binary Code C. Gray Code D. Excess-3 Code

nielit2017oct-assistanta-cs computer-networks error-correction

Answer key

5.19 Error Detection (2)

5.19.1 Error Detection: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 22

If the original size of data is 40 then after adding error detection redundancy bit the size of data length is

A. 26 B. 36 C. 46 D. 56

nielit2017oct-assistanta-cs computer-networks error-detection

Answer key

5.19.2 Error Detection: NIELIT Scientist B 2020 November: 60

To the detection of up to 5 errors in all cases, the minimum Hamming distance in a block code must be ______________.

A. 5 B. 6 C. 10 D. 8

nielit-scb-2020 error-detection digital-logic

Answer key

5.20 Fragmentation (3)

5.20.1 Fragmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 51

The demerits of the fragmentation are :

A. Complex routers
B. Open to DOS attack
C. No overlapping of fragments
D. (A) and (B) both

nielit2017dec-assistanta computer-networks fragmentation

Answer key

5.20.2 Fragmentation: NIELIT 2022 April Scientist B | Section B | Question: 75

In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are:

A. Last fragment, 2400 and 2789
B. First fragment, 2400 and 2759
C. Last fragment, 2400 and 2759
D. Middle fragment, 300 and 689

nielit2022apr-scientistb computer-networks ip-addressing fragmentation

5.20.3 Fragmentation: NIELIT Scientific Assistant A 2020 November: 75

The ___________ field in IPv4 datagram is not related to fragmentation.

A. Flag
B. Offset
C. TOS
D. Identifier

nielit-sta-2020 computer-networks fragmentation

Answer key☟

5.21 General Awareness (1)

5.21.1 General Awareness: NIELIT 2021 Dec Scientist B - Section B: 76

SPAM usually ___________.

A. Contains no valuable content.
B. Contains valuable content.
C. Contains content about website and is an operating system.
D. Is a module of operating system that handles scheduling of a process.

nielit-2021-it-dec-scientistb general-awareness analytical-aptitude

5.22 IP Addressing (7)

5.22.1 IP Addressing: NIELIT 2016 DEC Scientist B (IT) - Section B: 19

Which protocol finds the MAC address from IP address?

A. SMTP
B. ICMP
C. ARP
D. RARP

nielit2016dec-scientistb-it computer-networks ip-addressing

Answer key☟

5.22.2 IP Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 12

In classless addressing, there are no classes but addresses are still granted in :

A. Codes
B. Blocks
C. IPs
D. Sizes

nielit2017dec-assistanta computer-networks ip-addressing subnetting

Answer key☟

5.22.3 IP Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 24

A subnet mask in class C can have ______________ 1's with the remaining bits 0's.

A. 10 B. 24 C. 12 D. 7

nielit2017dec-assistanta computer-networks subnetting ip-addressing

Answer key

5.22.4 IP Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 45

In $IPv4$ Addresses, classfull addressing is replaced with :

A. Classless Addressing
B. Classfull Addressing
C. Subnet Advertising
D. None of the options

nielit2017dec-assistanta computer-networks ip-addressing ip-addressing

Answer key

5.22.5 IP Addressing: NIELIT 2018-68

_____ IP address can be used in WAN

A. $256.0.0.1$ B. $172.16.0.10$ C. $15.1.5.6$ D. $127.256.0.1$

nielit-2018 computer-networks ip-addressing

Answer key

5.22.6 IP Addressing: NIELIT 2022 April Scientist B | Section B | Question: 116

The routing table of a router is shown below:

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
default	Eth2	

On which interfaces will the router forward packets addressed to destinations $128.75.43.16$ and $192.12.17.10$ respectively?

A. Eth1 and Eth2 B. Eth0 and Eth2 C. Eth0 and Eth3 D. Eth1 and Eth3

nielit2022apr-scientistb computer-networks ip-addressing subnetting

Answer key

5.22.7 IP Addressing: NIELIT Scientist B 2020 November: 117

You are working with a network that is $172.16.0.0$ and would like to support 600 hosts per subnet. What subnet mask should you use?

A. $255.255.192.0$
B. $255.255.224.0$
C. $255.255.252.0$
D. None of the options

nielit-scb-2020 subnetting computer-networks ip-addressing

Answer key☟

5.23 IP Packet (1)

5.23.1 IP Packet: NIELIT 2016 MAR Scientist B - Section C: 55

A packet switching network

A. is free.
B. can reduce the cost of using an information utility.
C. allows communications channel to be shared among more than one user.
D. both (B) and (C).

nielit2016mar-scientistb computer-networks ip-packet

Answer key☟

5.24 Information Theory (1)

5.24.1 Information Theory: NIELIT 2021 Dec Scientist A - Section B: 90

To simulate a analog signal of frequency f, bandwidth requirement of channel is :

A. $2\mathrm{f}$
B. f
C. $\mathrm{f}/2$
D. $\mathrm{f}/4$

nielit2021dec-scientista information-theory computer-networks

Answer key☟

5.25 Logical Reasoning (2)

5.25.1 Logical Reasoning: NIELIT Scientist B 2020 November: 17

Directions: For the Assertion (A) and Reason (R) below, choose the correct alternative from the following:

I. Both (A) and (R) are true and (R) is the correct explanation of (A)
II. Both (A) and (R) are true and (R) is not the correct explanation of (A)
III. (A) is true but (R) is false
IV. (A) is false but (R) is true

Assertion (A):

Moon cannot be used as a satellite for communication

Reason (R)

Moon does not move in the equatorial plane of the Earth

A. I B. II C. III D. IV

nielit-scb-2020 computer-networks logical-reasoning

5.25.2 Logical Reasoning: NIELIT Scientist B 2020 November: 50

In __________, other nodes verify the validity of the block by checking that the hash of the data of the block is less than a present number.

A. Proof of Burn
B. Proof of STAKE
C. Proof of Work
D. All of the options

nielit-scb-2020 computer-networks logical-reasoning

Answer key☟

5.26 Network Addressing (6)

5.26.1 Network Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 27

Which of the following is a class B host address ?

A. $230.0.0.0$ B. $130.4.5.6$ C. $230.7.6.5$ D. $30.4.5.6$

nielit2017dec-assistanta computer-networks network-addressing ip-addressing

Answer key☟

5.26.2 Network Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 28

There is a need to create a network that has 5 subnets, each with at least 16 hosts. Which one is used as classful subnet mask ?

A. $255.255.255.192$
B. $255.255.255.248$
C. $255.255.255.240$
D. $255.255.255.224$

nielit2017dec-assistanta computer-networks network-addressing subnetting

Answer key☟

5.26.3 Network Addressing: NIELIT 2017 DEC Scientific Assistant A - Section B: 46

An Ethernet destination address $07-01-12-03-04-05$ is :

A. Unicast address
B. Multicast address
C. Broadcast address
D. All of the options

nielit2017dec-assistanta computer-networks network-addressing

Answer key☟

5.26.4 Network Addressing: NIELIT 2017 DEC Scientist B - Section B: 36

If the number of networks and number of hosts in class B are $2^m, (2^n - 2)$ respectively. Then the relation between m, n is

A. $3m = 2n$ B. $7m = 8n$ C. $8m = 7n$ D. $2m = 3n$

nielit2017dec-scientistb computer-networks network-addressing

Answer key☟

5.26.5 Network Addressing: NIELIT Scientist B 2020 November: 66

Which of the following can be used when creating a pool of global addresses instead of the netmask command?

A. /(slash notation) B. prefix-length C. no mask D. block-size

nielit-scb-2020 computer-networks network-addressing

Answer key☟

5.26.6 Network Addressing: NIELIT Scientist B 2020 November: 74

In classful addressing, a large part of the available addresses are __________.

A. Dispersed B. Blocked C. Wasted D. Reserved

nielit-scb-2020 computer-networks network-addressing

Answer key☟

5.27 Network Layer (5)

5.27.1 Network Layer: NIELIT 2016 MAR Scientist B - Section C: 54

Which of the following is not a transceiver function?

A. Transmission and receipt of data
B. Checking of line voltages
C. Addition and subtraction of headers
D. Collision detection

nielit2016mar-scientistb computer-networks network-layer

Answer key☟

5.27.2 Network Layer: NIELIT 2021 Dec Scientist B - Section B: 62

A device that allows two or more computers on different networks to communicate with each other is ___________.

A. Hub B. Bridge C. Repeater D. Gateway

nielit-2021-it-dec-scientistb computer-networks network-layer

5.27.3 Network Layer: NIELIT Scientist B 2020 November: 108

__________ tells a firewall about how to reassemble a data stream that has been divided into packets.

A. The source routing feature
B. The number in the header's identification field
C. The destination IP address
D. The header checksum field in the packet header

nielit-scb-2020 computer-networks network-layer

Answer key☟

5.27.4 Network Layer: NIELIT Scientist B 2020 November: 48

The __________ command will show you the translation table containing all the active NAT entries.

A. show ip nat translations
B. show ip nat tl
C. show ip nat states
D. none of the options

nielit-scb-2020 computer-networks network-layer

Answer key☟

5.27.5 Network Layer: NIELIT Scientist B 2020 November: 64

In an IPv6 header, the traffic class field is similar to the ________ field in the IPv4 header.

A. TOS field
B. Fragmentation field
C. Fast Switching
D. Option field

nielit-scb-2020 computer-networks ip-addressing network-layer

Answer key☟

5.28 Network Protocols (6)

5.28.1 Network Protocols: NIELIT 2016 DEC Scientist B (IT) - Section B: 31

A set of rules that governs data communication:

A. Rule
B. Medium
C. Link
D. Protocol

nielit2016dec-scientistb-it computer-networks network-protocols

Answer key☟

5.28.2 Network Protocols: NIELIT 2017 DEC Scientific Assistant A - Section B: 32

Which NetWare protocol provides link - state routing ?

A. $NLSP$
B. RIP
C. SAP
D. NCP

nielit2017dec-assistanta computer-networks routing network-protocols

Answer key

5.28.3 Network Protocols: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 23

Which of the following would not be specified in a communication protocol?

A. Header contents
B. Trailer contents
C. Error Checking
D. Data content of message

nielit2017oct-assistanta-cs computer-networks network-protocols

Answer key

5.28.4 Network Protocols: NIELIT 2018-53

Identify the true statement from the given statements.

1. HTTP may use different TCP connection for different objects of a webpage if non-persistent connections are used
2. FTP uses two TCP connections, one for data and another control
3. TELNET and FTP can only use TWO connections at a time

A. 1
B. 1 and 2
C. 2 and 3
D. 1, 2 and 3

nielit-2018 computer-networks network-protocols

Answer key

5.28.5 Network Protocols: NIELIT Scientific Assistant A 2020 November: 44

Which among the following types of Server filters Website Traffic?

A. POP Server
B. Database Server
C. Proxy Server
D. Mail Server

nielit-sta-2020 computer-networks network-protocols

Answer key

5.28.6 Network Protocols: NIELIT Scientific Assistant A 2020 November: 95

The router table contains addresses belonging to ___________ protocol(s).

A. a single
B. two
C. multiple
D. none of the options

nielit-sta-2020 computer-networks network-protocols

Answer key

5.29 Network Security (6)

5.29.1 Network Security: NIELIT 2016 MAR Scientist C - Section C: 38

How many bits are required to encode all twenty six letters, ten symbols, and ten numerals?

A. 5
B. 6
C. 7
D. 46

nielit2016mar-scientistc computer-networks network-security

Answer key

5.29.2 Network Security: NIELIT 2017 DEC Scientific Assistant A - Section B: 14

PGP encrypts data by using a block cipher called :

A. international data encryption algorithm
B. private data encryption algorithm
C. internet data encryption algorithm
D. none of the options

nielit2017dec-assistanta computer-networks network-security

Answer key

5.29.3 Network Security: NIELIT 2017 DEC Scientific Assistant A - Section B: 49

Why is one - time password safe ?

A. It is easy to generate
B. It cannot b shared
C. It is different for every access
D. It can be easily decrypted

nielit2017dec-assistanta computer-networks network-security

Answer key

5.29.4 Network Security: NIELIT 2021 Dec Scientist A - Section B: 101

The DoS attack, in which the attacker establishes a large number of half-open or fully open TCP connections at the target host is ________.

A. Vulnerability attack
B. Bandwidth flooding
C. Connection flooding
D. UDP flooding

nielit2021dec-scientista computer-networks tcp network-security

Answer key

5.29.5 Network Security: NIELIT 2021 Dec Scientist B - Section B: 75

What is WPA?

A. wi-fi protected access
B. wired protected access
C. wired process access
D. wi-fi process access

nielit2021dec-scientistb network-security computer-networks

Answer key

5.29.6 Network Security: NIELIT Scientist B 2020 November: 62

Data leakage threats are done by internal agents. Which of them is not an example of an internal data leakage threat?

A. Data leak from stolen credentials from desk
B. Data leak by partners
C. Data leak by 3rd party apps
D. All of the options

nielit-scb-2020 network-security logical-reasoning

Answer key☟

5.30 Osi Model (5)

5.30.1 Osi Model: NIELIT 2021 Dec Scientist A - Section B: 66

Which NetWare protocol works on layer 3-network layer of the OSI model ?

A. IPX B. NCP C. SPX D. NetBIOS

nielit2021dec-scientista computer-networks network-layer osi-model

Answer key☟

5.30.2 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 84

TCP/IP model does not have ______________ layer but OSI model have this layer.

A. session layer
B. transport layer
C. application layer
D. network layer

nielit2021dec-scientistb computer-networks osi-model tcp

Answer key☟

5.30.3 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 88

Each layer of the OSI model receives services or data from a _______________ layer.

A. below layer
B. above layer
C. both (A) and (B)
D. None of the above

nielit2021dec-scientistb osi-model computer-networks

Answer key☟

5.30.4 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 89

Who developed standards for the OSI reference model $?$

A. ANSI – American National Standards Institute
B. ISO – International Standards Organization
C. IEEE – Institute of Electrical and Electronics Engineering
D. ACM – Association for Computing Machinery

nielit2021dec-scientistb computer-networks osi-model

Answer key☟

5.30.5 Osi Model: NIELIT 2021 Dec Scientist B - Section B: 98

Which is the layer that converts packets to Frames and Frames to packets in the

OSI model ?

A. Physical Layer
B. Data Link Layer
C. Network Layer
D. Transport Layer

nielit2021dec-scientistb osi-model data-link-layer computer-networks

Answer key

5.31 Queuing Delay (1)

5.31.1 Queuing Delay: NIELIT 2021 Dec Scientist A - Section B: 75

In a network, If P is the only packet being transmitted and there was no earlier transmission, which of the following delays could be zero ?

A. Propagation delay
B. Queuing delay
C. Transmission delay
D. Processing delay

nielit2021dec-scientista computer-networks propagation-delay queuing-delay transmission-delay processing-delay

Answer key

5.32 Segmentation (1)

5.32.1 Segmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 25

What is the value of acknowledgement field in a segment

A. Number of previous bytes to receive
B. Total number of bytes to receive
C. Number of next bytes to receive
D. Sequence of zero's and one's

nielit2017dec-assistanta computer-networks segmentation

Answer key

5.33 Stop and Wait (1)

5.33.1 Stop and Wait: NIELIT 2022 April Scientist B | Section B | Question: 46

A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames.

Frames are of size 1000 bytes and the transmission rate at the sender is $80 \text{ Kbps} \; (1 \text{ Kbps} = 1000 \text{ bits/second}).$ Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is $8 \text{ Kbps}.$ The one-way propagation delay is $100 \text{ milliseconds}.$ Assuming no frame is lost, the sender throughput is __________ $\text{bytes/second}.$

A. 2500
B. 2000
C. 1500
D. 500

nielit2022apr-scientistb computer-networks stop-and-wait normal

5.34 Subnetting (3)

5.34.1 Subnetting: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 25

The address of a class B Host is to be split into subnets with a 3-bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet?

A. 8 subnets and 262142 hosts
B. 6 subnets and 262142 hosts
C. 6 subnets and 1022 hosts
D. 8 subnets and 1024 hosts

nielit2017oct-assistanta-it computer-networks subnetting

Answer key☟

5.34.2 Subnetting: NIELIT 2018-31

Identify the subnet mask for the given direct broadcast address of subnet is $201.15.16.31$

A. $255.255.192.192$
B. $255.255.255.198$
C. $255.255.255.240$
D. $255.255.257.240$

nielit-2018 computer-networks subnetting

Answer key☟

5.34.3 Subnetting: NIELIT Scientific Assistant A 2020 November: 102

You have a network ID of $192.168.10.0$ and require at least 25 host IDs for each subnet, with the largest amount of subnets available. Which subnet mask should you assign?

A. $255.255.255.192$
B. $255.255.255.224$
C. $255.255.255.240$
D. $255.255.255.248$

nielit-sta-2020 computer-networks subnetting

Answer key☟

5.35 TCP (3)

5.35.1 TCP: NIELIT 2016 MAR Scientist B - Section C: 53

In context of TCP/IP computer network models, which of the following is FALSE?

A. Besides span of geographical area, the other major difference between LAN and WAN is that the later uses switching element.
B. A repeater is used just to forward bits from one network to another one.
C. IP layer is connected oriented layer in TCP/IP.
D. A gateway is used to connect incompatible networks.

nielit2016mar-scientistb computer-networks tcp

Answer key☟

5.35.2 TCP: NIELIT 2021 Dec Scientist A - Section B: 105

Which of the following system calls results in the sending of SYN packets ?

A. Socket B. Bind C. Listen D. Connect

nielit2021dec-scientista operating-system tcp computer-networks

Answer key☟

5.35.3 TCP: NIELIT 2022 April Scientist B | Section B | Question: 71

Find out the maximum link speed at which a source can generate $1500-\text{byte TCP}$ payloads with packet lifetime of upto 120 msec before the $32-$bit sequence numbers wrap around? Take into account $\mathrm{TCP}, \mathrm{IPv4},$ and Ethernet header overheads while doing all calculations.

A. 137 Mbps B. 256 Mbps C. 299 Mbps D. 512 Mbps

nielit2022apr-scientistb computer-networks tcp

5.36 Transmission Delay (1)

5.36.1 Transmission Delay: NIELIT 2022 April Scientist B | Section B | Question: 101

Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is $10,000\text{ km}$ and the propagation speed along the link is $2\times 10^8\text{ m/sec}$. Host X send a file of $50,000\text{ bytes}$ as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q $\text{milliseconds},$ respectively. Then the value of p and q are

A. $\mathrm{p}=50$ and $\mathrm{q}=100$
B. $\mathrm{p}=50$ and $\mathrm{q}=400$
C. $\mathrm{p}=100$ and $\mathrm{q}=50$
D. $\mathrm{p}=400$ and $\mathrm{q}=50$

nielit2022apr-scientistb computer-networks propagation-delay transmission-delay analytical-aptitude

5.37 Transmission Media (1)

5.37.1 Transmission Media: NIELIT 2021 Dec Scientist B - Section B: 31

Which of the following is the efficient medium which can be used for higher bandwidth in broadband network?

A. Coaxial cable B. Optical fiber C. CAT-V D. STP

nielit-2021-it-dec-scientistb computer-networks transmission-media

5.38 Transport Layer (3)

5.38.1 Transport Layer: NIELIT 2017 DEC Scientific Assistant A - Section B: 33

Which layer connects the network support layers and user support layers ?

A. transport layer
B. network layer
C. data link layer
D. session layer

nielit2017dec-assistanta computer-networks network-layer transport-layer

Answer key☟

5.38.2 Transport Layer: NIELIT 2018-45

______ transport layer protocol is used to support electronic mail

A. SNMP
B. IP
C. SMTP
D. TCP

nielit-2018 computer-networks transport-layer

Answer key☟

5.38.3 Transport Layer: NIELIT 2021 Dec Scientist B - Section B: 92

Which address is used to identify a process on a host by the transport layer?

A. physical address
B. logical address
C. port address
D. specific address

nielit2021dec-scientistb computer-networks transport-layer

Answer key☟

5.39 Ttl (1)

5.39.1 Ttl: NIELIT 2018-44

Time to Live(TTL) field in the IP datagram is used ______

A. to optimize throughput
B. to prevent packet looping
C. to reduce delays
D. to prioritize packets

nielit-2018 computer-networks ttl

Answer key☟

5.40 UDP (2)

5.40.1 UDP: NIELIT 2017 DEC Scientific Assistant A - Section B: 1

Correct expression tor UDP user datagram length is

A. length of UDP = length of IP- length of IP header's
B. length of UDP= length of UDP-length of UDP header's
C. length of UDP = length of IP+length of IP header's
D. length of UDP = length of UDP+length of UDP header's

nielit2017dec-assistanta computer-networks udp

Answer key

5.40.2 UDP: NIELIT 2022 April Scientist B | Section B | Question: 63

Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet $\text{LAN}.$ Ethernet frames may carry data up to $1500 \text{ bytes (i.e. MTU} = 1500 \text{ bytes)}.$ Size of UDP header is 8 bytes and size of IP header is $20 \text{ bytes}.$ There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

A. 6 and 925 B. 6 and 7400 C. 7 and 1110 D. 7 and 8880

nielit2022apr-scientistb computer-networks udp ip-packet fragmentation ethernet

5.41 Wireless Lan (1)

5.41.1 Wireless Lan: NIELIT 2021 Dec Scientist B - Section B: 104

What is the access point (AP) in a wireless LAN ?

A. device that allows wireless devices to connect to a wired network
B. wireless devices itself
C. both device that allows wireless devices to connect to a wired network and wireless devices itself
D. all the nodes in the network

nielit2021dec-scientistb computer-networks wireless-lan network

Answer key

Answer Keys

5.0.1	C	5.0.2	A	5.0.3	A	5.0.4	A	5.0.5	D
5.0.6	C	5.0.7	C	5.0.8	D	5.0.9	D	5.0.10	B
5.0.11	A	5.0.12	C	5.0.13	A	5.0.14	C	5.0.15	A
5.0.16	D	5.0.17	C	5.0.18	D	5.0.19	B	5.0.20	C
5.1.1	D	5.2.1	C	5.2.2	C	5.3.1	A	5.4.1	D
5.5.1	A	5.5.2	A	5.6.1	C	5.7.1	C	5.7.2	C
5.7.3	C	5.8.1	C	5.9.1	B	5.9.2	D	5.10.1	D
5.11.1	B	5.11.2	A	5.12.1	A	5.12.2	C	5.12.3	D
5.12.4	B	5.13.1	C	5.14.1	B	5.14.2	D	5.15.1	D
5.16.1	A	5.17.1	A	5.18.1	C	5.19.1	C	5.19.2	B

5.20.1	D	5.20.2	C	5.20.3	C	5.21.1	A	5.22.1	C
5.22.2	B	5.22.3	B	5.22.4	A	5.22.5	C	5.22.6	A
5.22.7	C	5.23.1	D	5.24.1	A	5.25.1	A	5.25.2	C
5.26.1	B	5.26.2	D	5.26.3	B	5.26.4	C	5.26.5	B
5.26.6	C	5.27.1	C	5.27.2	D	5.27.3	A	5.27.4	A
5.27.5	A	5.28.1	D	5.28.2	A	5.28.3	D	5.28.4	B
5.28.5	C	5.28.6	A	5.29.1	B	5.29.2	A	5.29.3	C
5.29.4	C	5.29.5	A	5.29.6	C	5.30.1	A	5.30.2	A
5.30.3	C	5.30.4	B	5.30.5	B	5.31.1	B	5.32.1	C
5.33.1	A	5.34.1	X	5.34.2	D	5.34.3	B	5.35.1	C
5.35.2	D	5.35.3	C	5.36.1	D	5.37.1	B	5.38.1	A
5.38.2	D	5.38.3	C	5.39.1	B	5.40.1	A	5.40.2	C
5.41.1	A								

6 Databases (109)

6.0.1 NIELIT 2018-51

_____ is holding an entry for each terminal symbol and is acting as permanent database.

A. Variable Table
B. Terminal Table
C. Keyword Table
D. Identifier Table

nielit-2018 databases

Answer key☟

6.0.2 NIELIT Scientist B 2020 November: 110

An attribute(s) that is used to look up for records in a file is called a:

A. Function key
B. Catalog key
C. Access key
D. Search key

nielit-scb-2020 databases

Answer key☟

6.0.3 NIELIT Scientist B 2020 November: 107

Most NoSQL databases support automatic ___________ meaning that you get high availability and disaster recovery.

A. Processing
B. Scalability
C. Replication
D. All of the options

nielit-scb-2020 databases

Answer key☟

6.1 Aggregate Functions (1)

6.1.1 Aggregate Functions: NIELIT 2022 April Scientist B | Section B | Question: 115

The employee information in a company is stored in the relation

Employee (name, sex, salary, deptName)

Consider the following SQL query

```
Select deptName
  From Employee
  Where sex = 'M'
  Group by deptName
  Having avg(salary) >
    (select avg (salary) from Employee)
```

It returns the names of the department in which

A. the average salary is more than the average salary in the company
B. the average salary of male employees is more than the average salary of all male

employees in the company

C. the average salary of male employees is more than the average salary of employees in same the department

D. the average salary of male employees is more than the average salary in the company

nielit2022apr-scientistb sql databases query aggregate-functions

6.2 B Tree (3)

6.2.1 B Tree: NIELIT 2017 DEC Scientific Assistant A - Section B: 21

The $2-3-4$ tree is a self-balancing data structure, which is also called :

A. $2-4$ tree
B. $B+$ tree
C. $B-$ tree
D. None of the options

nielit2017dec-assistanta databases b-tree

Answer key☟

6.2.2 B Tree: NIELIT 2017 July Scientist B (CS) - Section B: 43

Which one of the following is a key factor for preferring B-trees to binary search trees for indexing database relations?

A. Database relations have a large number of records
B. Database relations are sorted on the primary key
C. B-trees require less memory than binary search trees
D. Data transfer from disks is in blocks

nielit2017july-scientistb-cs databases b-tree

Answer key☟

6.2.3 B Tree: NIELIT 2021 Dec Scientist A - Section B: 58

Non leaf nodes of $\mathrm{B}+$ tree structure from a :

A. Multilevel sparse indices
B. Multilevel dense indices
C. Sparse indices
D. Multilevel clustered indices

nielit2021dec-scientista data-structures b-tree databases

Answer key☟

6.3 Bcnf (2)

6.3.1 Bcnf: NIELIT 2017 DEC Scientist B - Section B: 24

Which of the following is TRUE?

A. Every relation in 3NF is also in BCNF
B. A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent

on every key of R

C. Every relation in BCNF is also in 3NF

D. No relation can be in both BCNF and 3NF.

nielit2017dec-scientistb database-normalization bcnf

Answer key☟

6.3.2 Bcnf: NIELIT 2017 DEC Scientist B - Section B: 55

Consider the relational schema $\mathrm{R}(\mathrm{A\ B\ C\ D})$ with following functional dependency set $F = \{A \rightarrow BC, C \rightarrow D\}$; The relation R is in

A. 2NF
B. BCNF
C. 3NF
D. 1NF

nielit2017dec-scientistb database-normalization bcnf

Answer key☟

6.4 Bcnf Decomposition (2)

6.4.1 Bcnf Decomposition: NIELIT 2016 DEC Scientist B (IT) - Section B: 44

Which one of the following statements about normal forms is FALSE?

A. BCNF is stricter than 3NF.

B. Lossless, dependency-preserving decomposition into BCNF is always possible.

C. Lossless, dependency-preserving decomposition into 3NF is always possible.

D. Any relation with two attributes is BCNF.

nielit2016dec-scientistb-it databases database-normalization bcnf-decomposition

Answer key☟

6.4.2 Bcnf Decomposition: NIELIT 2017 DEC Scientist B - Section B: 1

Which one of the following statements are not correct?

$S1$: 3NF decomposition is always lossless join and dependency preserving.

$S2$: 3NF decomposition is always lossless join but may or may not be dependency preserving.

$S3$: BCNF decomposition is always lossless join and dependency preserving.

$S4$: BCNF decomposition is always lossless join but may or may not be dependency preserving.

A. Only $S1$
B. Only $S4$
C. Both $S1$ and $S4$
D. Both $S2$ and $S3$

nielit2017dec-scientistb database-normalization bcnf-decomposition

Answer key☟

6.5 Btrees (1)

6.5.1 Btrees: NIELIT 2018-34

Identify the true statement from the given statements:

1. Number of child pointers in a B/B+ tree node is always equal to number of keys in it plus one
2. B/B+ tree is defined by a term minimum degree. And minimum degree depends on hard disk block size, key and address size

A. (1) B. (1) and (2) C. (2) D. None of these

nielit-2018 databases btrees

Answer key☟

6.6 Candidate Key (5)

6.6.1 Candidate Key: NIELIT 2016 DEC Scientist B (CS) - Section B: 54

If there is more than one key for relation schema in DBMS then each key in relation schema is classified as

A. prime key B. super key C. candidate key D. primary key

nielit2016dec-scientistb-cs databases candidate-key

Answer key☟

6.6.2 Candidate Key: NIELIT 2016 DEC Scientist B (IT) - Section B: 36

The primary key is selected from the:

A. Composite keys B. Determinants
C. Candidate keys D. Foreign keys

nielit2016dec-scientistb-it databases candidate-key

Answer key☟

6.6.3 Candidate Key: NIELIT 2016 MAR Scientist B - Section C: 43

L e t $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies:

$C \to F, E \to A, EC \to D, A \to B.$

Which of the following is a key for R?

A. CD B. EC C. AE D. AC

nielit2016mar-scientistb database-normalization candidate-key

Answer key☟

6.6.4 Candidate Key: NIELIT Scientific Assistant A 2020 November: 72

Let R=(A,B,C,D,E) having following FDs. $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$.

Which of the following is not a Candidate key?

A. A B. B C. E D. BC

nielit-sta-2020 databases easy database-normalization candidate-key

Answer key

6.6.5 Candidate Key: NIELIT Scientist B 2020 November: 96

An instance of relational schema $R(A, B, C)$ has distinct values of A including $NULL$ values. Which one of the following is true?

A. A is a candidate key
B. A is not a candidate key
C. A is a primary key
D. Both "A is a candidate key" and "A is a primary key"

nielit-scb-2020 relational-algebra databases candidate-key

Answer key

6.7 Data Dependency (4)

6.7.1 Data Dependency: NIELIT 2016 MAR Scientist C - Section C: 34

Given the following relation instance:

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Which of the following functional dependencies are satisfied by the instance?

A. $XY \rightarrow Z$ and $Z \rightarrow Y$
B. $YZ \rightarrow X$ and $Y \rightarrow Z$
C. $YZ \rightarrow X$ and $X \rightarrow Z$
D. $XZ \rightarrow Y$ and $Y \rightarrow X$

nielit2016mar-scientistc databases data-dependency

Answer key

6.7.2 Data Dependency: NIELIT 2016 MAR Scientist C - Section C: 35

Consider the schema $R = (\text{S T U V})$ and the dependencies $S \rightarrow T, T \rightarrow U, U \rightarrow V$, and $V \rightarrow S$. If $R = (R1 \text{ and } R2)$ be a decomposition such that $R1 \cap R2 = \phi$ then decomposition is

A. not in 2 NF
B. in 2 NF but not in 3 NF
C. in 3 NF but not in 2 NF
D. in both 2 NF and 3 NF

nielit2016mar-scientistc databases data-dependency

Answer key☟

6.7.3 Data Dependency: NIELIT 2016 MAR Scientist C - Section C: 4

A functional dependency of the form $x \rightarrow y$ is trivial if

A. $y \subseteq x$
B. $y \subset x$
C. $x \subseteq y$
D. $x \subset y$ and $y \subset x$

nielit2016mar-scientistc databases data-dependency

Answer key☟

6.7.4 Data Dependency: NIELIT 2018-58

Consider the following functional dependencies in a database:

$A \rightarrow B$

$B \rightarrow C$

$D \rightarrow E$

$E \rightarrow D$

$F \rightarrow G$

$F \rightarrow H$

$(E, F) \rightarrow I$

The relation (E, D, A, B) is

A. 2 NF
B. 3 NF
C. BCNF
D. None of the above

nielit-2018 databases data-dependency

Answer key☟

6.8 Database Design (2)

6.8.1 Database Design: NIELIT 2021 Dec Scientist A - Section B: 57

Consider a relation R with attributes $\{\mathrm{A, B, C}\}$ and functional dependency set $\mathrm{S}=\{\mathrm{A} \rightarrow \mathrm{B}, \mathrm{A} \rightarrow \mathrm{C}\}$. Then relation R can be decomposed into two relations :

A. $\mathrm{R1\{A, B\}\ AND\ R2\{A, C\}}$
B. $\mathrm{R1\{A, B\}\ AND\ R2\{B, C\}}$
C. $\mathrm{R1\{A, B, C\}\ AND\ R2\{A, C\}}$
D. None of the above

nielit2021dec-scientista databases database-design normal-forms functional-dependency relational-model

Answer key☟

6.8.2 Database Design: NIELIT 2021 Dec Scientist B - Section B: 65

Assume that a DBA issued the following create table command :

create table A (Aid, ...)

storage $\text{(initial 20480, next 20480, maxextents 8, minextents 3, pctincrease 0);}$

How many bytes of disk space will be allocated to this file when it is first created?

A. $163,840$ bytes
B. 20480 bytes
C. $61,440$ bytes
D. 8 bytes

nielit2021dec-scientistb databases disk database-design

Answer key

6.9 Database Normalization (11)

6.9.1 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 2

Process of analyzing relation schemas to achieve minimal redundancy and insertion or update anomalies is classified as:

A. normalization of data.
B. denomination of data.
C. isolation of data.
D. de-normalization of data.

nielit2016dec-scientistb-cs databases database-normalization

Answer key

6.9.2 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 31

Rule which states that addition of same attributes to right side and left side will result in other valid dependency is classified as

A. referential rule
B. inferential rule
C. augmentation rule
D. reflexive rule

nielit2016dec-scientistb-cs databases database-normalization

Answer key

6.9.3 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 47

If every functional dependency in set E is also in closure of F then this is classified as

A. FD is covered by E
B. E is covered by F
C. F is covered by E
D. F plus is covered by E

nielit2016dec-scientistb-cs databases database-normalization

Answer key

6.9.4 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 49

Considering relational database, functional dependency between two attributes A and B is denoted by

A. $A \rightarrow B$ B. $B \leftarrow A$ C. $AB \rightarrow R$ D. $R \leftarrow AB$

nielit2016dec-scientistb-cs databases database-normalization

Answer key☟

6.9.5 Database Normalization: NIELIT 2016 DEC Scientist B (IT) - Section B: 14

Normalization from which is based on transitive dependency is classified as:

A. First normal form.
B. Second normal form.
C. Fourth normal form.
D. Third normal form.

nielit2016dec-scientistb-it databases database-normalization

Answer key☟

6.9.6 Database Normalization: NIELIT 2016 DEC Scientist B (IT) - Section B: 40

In functional dependency between two sets of attribute A and B then set of attributes A of database is classified as:

A. top right side B. down left side C. left hand side D. right hand side

nielit2016dec-scientistb-it databases database-normalization

Answer key☟

6.9.7 Database Normalization: NIELIT 2016 DEC Scientist B (IT) - Section B: 57

If attribute of relation schema R is member of some candidate key then this type of attributes are classified as:

A. atomic attribute
B. candidate attribute
C. non-prime attribute
D. prime attribute

nielit2016dec-scientistb-it databases database-normalization

Answer key☟

6.9.8 Database Normalization: NIELIT 2016 MAR Scientist B - Section C: 44

For a database relation $R(a, b, c, d)$ where the domains of a, b, c, d include only atomic values, only the following functional dependencies and those that can be inferred from them hold:

$a \rightarrow c,$

$b \rightarrow d$

the relation is in

A. first normal form but not in second normal form.
B. second normal form but not in third normal form.
C. third normal form.
D. none of these.

nielit2016mar-scientistb databases database-normalization

Answer key☟

6.9.9 Database Normalization: NIELIT 2017 DEC Scientist B - Section B: 25

Consider the relational schema $R(ABCD)$ with following FD set $F = \{A \rightarrow CE, B \rightarrow D, AE \rightarrow D\}$. Identify the highest normal form satisfied by the relation R.

A. 2NF B. BCNF C. 3NF D. 1NF

nielit2017dec-scientistb databases database-normalization

Answer key☟

6.9.10 Database Normalization: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 16

For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values, only the following functional dependencies and those that can be inferred from them hold.

- $a \rightarrow c$
- $b \rightarrow d$

The relation is in

A. First normal form but not in second normal form
B. Second normal form but not in third normal form
C. Third normal form
D. None of the above

nielit2017oct-assistanta-it databases database-normalization

Answer key☟

6.9.11 Database Normalization: NIELIT Scientific Assistant A 2020 November: 73

Anomalies are avoided by splitting the offending relation into multiple relations, is also known as __________.

A. Accupressure
B. Decomposition
C. Precomposition
D. Both decomposition & precomposition

nielit-sta-2020 databases database-normalization

Answer key☟

6.10 Deadlock Prevention Avoidance Detection (2)

6.10.1 Deadlock Prevention Avoidance Detection: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 34

Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R. It will

A. result in deadlock situation
B. immediately be granted

C. immediately be rejected

D. be granted as soon as it is released by A

nielit2017oct-assistanta-cs databases transaction-and-concurrency deadlock-prevention-avoidance-detection

Answer key

6.10.2 Deadlock Prevention Avoidance Detection: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 5

Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R. It will

A. result in deadlock situation
B. immediately be granted
C. immediately be rejected
D. be granted as soon as it is released by A

nielit2017oct-assistanta-it databases transaction-and-concurrency deadlock-prevention-avoidance-detection

Answer key

6.11 Degree of Graph (1)

6.11.1 Degree of Graph: NIELIT 2021 Dec Scientist B - Section B: 118

Number of entity types participating in $\mathrm{E-R}$ diagrams is represented by :

A. Degree of relationship
B. Structure of relationship
C. Instance of relationship
D. Role of relationship

nielit2021dec-scientistb er-diagram databases degree-of-graph

Answer key

6.12 Dependency Preserving (2)

6.12.1 Dependency Preserving: NIELIT 2016 MAR Scientist C - Section C: 3

Every Boyce-Codd Normal Form (BCNF) decomposition is

A. dependency preserving
B. not dependency preserving
C. need be dependency preserving
D. none of these

nielit2016mar-scientistc databases dependency-preserving

Answer key

6.12.2 Dependency Preserving: NIELIT 2018-63

Identify the true statement from the given statements.

1. Lossless, dependency-preserving Decompositoin into 3 NF is always possible
2. Any relation with two attributes is BCNF

A. 1
B. 2
C. 1 and 2
D. None of these

nielit-2018 databases dependency-preserving

Answer key

6.13 ER Diagram (5)

6.13.1 ER Diagram: NIELIT 2021 Dec Scientist A - Section B: 49

In an ER Diagram, a double ellipse is used to represent :

A. Simple Attribute
B. Composite Attribute
C. Descriptive Attribute
D. Multi-valued Attribute

nielit2021dec-scientista er-diagram databases

Answer key☟

6.13.2 ER Diagram: NIELIT 2021 Dec Scientist B - Section B: 11

In $\mathrm{E\text{-}R}$ diagrams roles are indicated by labelling the lines that connect which two shapes:

A. Diamond, Diamond
B. Rectangle, Circle
C. Rectangle, Rectangle
D. Diamond, Rectangle

nielit-2021-it-dec-scientistb er-diagram databases

Answer key☟

6.13.3 ER Diagram: NIELIT 2021 Dec Scientist B - Section B: 30

Cardinality of relationship advisor to each entity sets instructor and student will be:

A. One to many
B. One to one
C. Many to many
D. Many to one

nielit-2021-it-dec-scientistb databases er-diagram relations

6.13.4 ER Diagram: NIELIT Scientist B 2020 November: 115

Considering binary relationships, possible cardinality ratios are:

A. one:one
B. $1 : N$
C. $M : N$
D. All the options

nielit-scb-2020 databases er-diagram relations

Answer key☟

6.13.5 ER Diagram: NIELIT Scientist B 2020 November: 43

Set of key attributes that identify weak entities related to some owner entity is classified as:

A. Structural key
B. String key
C. Partial key
D. Foreign key

nielit-scb-2020 databases er-diagram

Answer key☟

6.14 Hashing (1)

6.14.1 Hashing: NIELIT Scientist B 2020 November: 103

The physical location of a record is determined by a mathematical formula that transforms a file key into a record location is:

A. B-Tree File B. Hashed File C. Indexed File D. Sequential File

nielit-scb-2020 databases hashing data-structures

Answer key☟

6.15 Irrecoverable Error (1)

6.15.1 Irrecoverable Error: NIELIT 2022 April Scientist B | Section B | Question: 58

Which of the following scenarios may lead to an irrecoverable error in a database system?

A. A transaction writes a data item after it is read by an uncommitted transaction
B. A transaction reads a data item after it is read by an uncommitted transaction
C. A transaction reads a data item after it is written by a committed transaction
D. A transaction reads a data item after it is written by an uncommitted transaction

nielit2022apr-scientistb databases transaction-and-concurrency irrecoverable-error

6.16 Java (1)

6.16.1 Java: NIELIT 2021 Dec Scientist B - Section B: 66

Which one of the following contains date information?

A. java.sql.TimeStamp B. java.sql.Time
C. java.io.Time D. java.io.TimeStamp

nielit-2021-it-dec-scientistb nielit2021dec-scientistb java databases

6.17 Joins (1)

6.17.1 Joins: NIELIT 2017 July Scientist B (IT) - Section B: 55

A table joined with itself is called

A. Join B. Self Join C. Outer Join D. Equi Join

nielit2017july-scientistb-it databases joins

Answer key☟

6.18 Lock (1)

6.18.1 Lock: NIELIT 2016 MAR Scientist B - Section C: 34

Global locks

A. synchronize access to local resources.
B. synchronize access to global resources.
C. are used to avoid local locks.
D. prevent access to global resources.

nielit2016mar-scientistb databases transaction-and-concurrency lock

Answer key☟

6.19 Lossless Join (1)

6.19.1 Lossless Join: NIELIT 2022 April Scientist B | Section B | Question: 49

Let $\text{R}\ (\text{A}, \text{B}, \text{C}, \text{D})$ be a relational schema with the following functional dependencies:

$\text{A} \rightarrow \text{B},\ \ \text{B} \rightarrow \text{C}$

$\text{C} \rightarrow \text{D}$ and $\text{D} \rightarrow \text{B}$

The decomposition of R into

$(\text{A}, \text{B}), (\text{B}, \text{C}), (\text{B}, \text{D})$

A. gives a lossless join, and is dependency preserving
B. gives a lossless join, but is not dependency preserving
C. does not give a lossless join, but is dependency preserving
D. does not give a lossless join and is not dependency preserving

nielit2022apr-scientistb databases database-normalization lossless-join

Answer key☟

6.20 Normal Forms (2)

6.20.1 Normal Forms: NIELIT 2021 Dec Scientist B - Section B: 9

Specify, which of the following is preferred method for enforcing data integrity?

A. Constraints
B. Stored procedure
C. Triggers
D. Cursors

nielit-2021-it-dec-scientistb databases normal-forms database-normalization

Answer key☟

6.20.2 Normal Forms: NIELIT 2022 April Scientist B | Section B | Question: 54

The relation scheme $\text{Student Performance (name, courseNo, rollNo, grade)}$ has the following functional dependencies:

- name, courseNo, $\rightarrow$ grade
- rollNo, courseNo $\rightarrow$ grade
- name $\rightarrow$ rollNo
- rollNo $\rightarrow$ name

The highest normal form of this relation scheme is

A. $2NF$ B. $3NF$ C. $BCNF$ D. $4NF$

nielit2022apr-scientistb databases database-normalization normal-forms functional-dependency

Answer key☟

6.21 Operator Precedence (1)

6.21.1 Operator Precedence: NIELIT 2017 July Scientist B (IT) - Section B: 57

If an SQL query involves NOT,AND,OR with no parenthesis

A. NOT will be evaluated first; AND will be evaluated second; OR will be evaluated last.
B. NOT will be evaluated first; OR will be evaluated second; AND will be evaluated last.
C. AND will be evaluated first; OR will be evaluated second;NOT will be evaluated last.
D. The order of occurrence determines the order of evaluation.

nielit2017july-scientistb-it databases sql operator-precedence

Answer key☟

6.22 Optimization (1)

6.22.1 Optimization: NIELIT 2021 Dec Scientist B - Section B: 56

Which of the following is advantage of using JDBC connection pool?

A. Slow performance
B. Using more memory
C. Using less memory
D. Better performance

nielit-2021-it-dec-scientistb databases optimization

6.23 Primary Key (1)

6.23.1 Primary Key: NIELIT 2021 Dec Scientist A - Section B: 84

An instance of relational schema $\text{R}\,(\text{A}, \text{B}, \text{C})$ has distinct values of A including NULL values. Which one of the following is true ?

A. A is a candidate key
B. A is not a candidate key
C. A is a primary key
D. Both (A) and (C)

nielit2021dec-scientista databases relational-model candidate-key primary-key

Answer key☟

6.24 Query (1)

6.24.1 Query: NIELIT 2022 April Scientist B | Section B | Question: 111

Given the following schema:

employees (emp-id, first-name, last-name, hire-date, dept-id, salary

departments (dept-id, dept-name, manager-id, location-id)

You want to display the last names and hire dates of all latest hires in their respective departments in the location $\text{ID } 1700$. You issue the following query:

```
SQL>SELECT last-name, hire-date
   FROM employees
   WHERE (dept-id, hire-date) IN
   (SELECT dept-id, MAX(hire-date)
   FROM employees JOIN departments USING(dept-id)
   WHERE location-id =1700
   GROUP BY dept-id);
```

What is the outcome?

A. It executes but does not give the correct result
B. It executes and gives the correct result.
C. It generates an error because of pairwise comparison.
D. It generates an error because of the GROUP BY clause cannot be used with table joins in a sub-query.

nielit2022apr-scientistb sql databases query

6.25 Referential Integrity (3)

6.25.1 Referential Integrity: NIELIT 2018-49

The following table has two attributes X and Y where X is the primary key and Y is the foreign key referencing X with on-delete cascade.

X	Y
2	4
3	4
4	3
5	2
7	2
9	5
6	4

The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple $(3, 4)$ is deleted is

A. $(4,3)$ and $(6,4)$ B. $(2,4)$ and $(7,2)$ C. $(3,2)$ and $(9,5)$ D. $(3,4),(4,5)$ and $(6,4)$

nielit-2018 databases referential-integrity

Answer key☟

6.25.2 Referential Integrity: NIELIT 2021 Dec Scientist B - Section B: 10

Mention which of the following is a special type of integrity constraint that relates two relations & maintains consistency across the relations.

A. Entity Integrity Constraints.
B. Referential Integrity Constraints
C. Domain Integrity Constraints
D. Domain Constraints

nielit-2021-it-dec-scientistb relational-model referential-integrity databases

Answer key☟

6.25.3 Referential Integrity: NIELIT Scientific Assistant A 2020 November: 97

Domain constraints, functional dependency and referential integrity are special forms of _______

A. Foreign key
B. Primary key
C. Assertion
D. Referential constraint

nielit-sta-2020 databases database-normalization referential-integrity

Answer key☟

6.26 Relational Algebra (10)

6.26.1 Relational Algebra: NIELIT 2016 DEC Scientist B (IT) - Section B: 18

Which of the following is a fundamental operation in relational algebra?

A. Set intersection
B. Assignment
C. Natural Join
D. None of the above

nielit2016dec-scientistb-it databases relational-algebra

Answer key☟

6.26.2 Relational Algebra: NIELIT 2016 DEC Scientist B (IT) - Section B: 43

Which of the following is a fundamental operation in relational algebra?

A. Set intersection
B. Natural join
C. Assignment
D. None of the above

nielit2016dec-scientistb-it databases relational-algebra

Answer key☟

6.26.3 Relational Algebra: NIELIT 2017 July Scientist B (IT) - Section B: 54

Cross Product is a

A. Unary Operator
B. Ternary Operator
C. Binary Operator
D. Not an operator

nielit2017july-scientistb-it databases relational-algebra

Answer key☟

6.26.4 Relational Algebra: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 27

Which of the following desired features are beyond the capability of relational algebra?

A. Aggregate Computation
B. Multiplication
C. Finding transitive closure
D. All of the above

nielit2017oct-assistanta-it databases relational-algebra

Answer key☟

6.26.5 Relational Algebra: NIELIT 2021 Dec Scientist A - Section B: 72

Consider the relations :

$\mathrm{R}_1\{\mathrm{Roll\_no}, \mathrm{Name}, \mathrm{Grades}\}$ and $\mathrm{R}_2\{\mathrm{Roll\_no}, \mathrm{Subject\_ID}, \mathrm{Grades}\}$

Which of the following operations cannot be performed using the above relations?

A. Union
B. Select
C. Join
D. Project

nielit2021dec-scientista relational-algebra databases

Answer key☟

6.26.6 Relational Algebra: NIELIT 2021 Dec Scientist B - Section B: 108

What will be the result of following relational algebra expression ?

$\pi_{\mathrm{A,B}}(\sigma_{\mathrm{C}=10}(\mathrm{R}))$

A. Print columns A & B from relation R when $\mathrm{C}=10$
B. Print $\mathrm{C}=10$ from relation R
C. Print all data of relation R when $\mathrm{C}=10$
D. Print $\mathrm{A}, \mathrm{B}, \mathrm{C}$ from relation B when $\mathrm{C}=10$

nielit2021dec-scientistb relational-algebra databases

Answer key☟

6.26.7 Relational Algebra: NIELIT 2021 Dec Scientist B - Section B: 82

The equivalent relational algebra expression for the query “Find the names of suppliers who supplied all the items to all the customers”.

A. $\neg\, t/t\ \in$ supplier $(\mathrm{Q(t)})$
B. $\forall\, t\,[\mathrm{Sname}]\ /\ t\ \in$ supplier $(\mathrm{Q(t)})$
C. $t/\neg\, t\ \in$ supplier $(\mathrm{Q(t)})$
D. $\forall\, t/(\sim \mathrm{Q(t)})$

nielit2021dec-scientistb relational-algebra databases

6.26.8 Relational Algebra: NIELIT 2022 April Scientist B | Section B | Question: 66

Given the relations

employee (name, salary, deptno) and

department (deptno, deptname, address)

Which of the following queries cannot be expressed using the basic relational algebra operations $(\mathrm{U}, -, \mathrm{x}, \ , \ , \mathrm{p})$?

A. Department address of every employee
B. Employees whose name is the same as their department name
C. The sum of all employees' salaries
D. All employees of a given department

nielit2022apr-scientistb relational-algebra databases

6.26.9 Relational Algebra: NIELIT Scientific Assistant A 2020 November: 119

Let R and S be two relations with the following schema

$R(\underline{P, Q,}R1, R2, R3)$

$S(\underline{P, Q,}S1, S2)$

Where [P,Q] is the key for both schemas. Which of the following queries are equivalent?

(I) $II_P(R \bowtie S)$

(II) $II_P(R) \bowtie II_P(S)$

(III) $II_P(II_{P,Q}(R) - (II_{P,Q}(R) - II_{P,Q}(S)))$

A. Only (I) and (II)
B. Only (I) and (III)
C. Only (I), (II) and (III)
D. Only (I), (III) and (IV)

nielit-sta-2020 databases relational-algebra

Answer key☟

6.26.10 Relational Algebra: NIELIT Scientific Assistant A 2020 November: 48

What is meant by the following relational algebra statement: $\mathrm{STUDENT} \times \mathrm{COURSE}$?

A. Compute the natural join between the $\mathrm{STUDENT}$ and COURSE relations
B. Compute the left outer join between the $\mathrm{STUDENT}$ and COURSE relations
C. Compute the cartesian product between the $\mathrm{STUDENT}$ and COURSE relations
D. Compute the outer join between the $\mathrm{STUDENT}$ and COURSE relations

nielit-sta-2020 databases relational-algebra

Answer key☟

6.27 Relational Calculus (4)

6.27.1 Relational Calculus: NIELIT 2016 MAR Scientist B - Section C: 42

The relational algebra expression equivalent to the tuple calculus expression

$\{t \mid t \in r \wedge (t[A] = 10 \wedge t[B] = 20)\}$ is

A. $\sigma_{(A=10 \vee B=20)}(r)$
B. $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$
C. $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$
D. $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

nielit2016mar-scientistb databases relational-calculus

Answer key☟

6.27.2 Relational Calculus: NIELIT 2017 July Scientist B (IT) - Section B: 19

If R is a relation in Relational Data Model and $A_1, A_2, \ldots A_n$ are the attributes of relation R, what is the cardinality of R expressed in terms of domain of attributes?

A. $|R| \leq |\text{ dom }(A_1) \times \text{ dom }(A_2) \ldots \text{ dom }(A_n)|$
B. $|R| \geq |\text{ dom } \quad (A_1) \times \text{ dom } \quad (A_2) \ldots \text{ dom } \quad (A_n)|$
C. $|R| = \text{ max } (|\text{ dom}(A_1)|, |\text{ dom }(A_2)|, \cdots |\text{ dom }(A_n)|)$
D. $|R| = \text{ min } (|\text{ dom}(A_1)|, |\text{ dom }(A_2)|, \cdots |\text{ dom }(A_n)|)$

nielit2017july-scientistb-it databases relational-model relational-calculus

Answer key☟

6.27.3 Relational Calculus: NIELIT 2022 April Scientist B | Section B | Question: 83

Which of the following relational calculus expressions is not safe ?

i. $\{t \mid \exists u \in R_1 (t[A] = u[A]) \wedge \neg \exists s \in R_2 (t[A] = s[A])\}$
ii. $\{t \mid \forall u \in R_1 (u[A] = " x " \Rightarrow \exists s \in R_2 (t[A] = s[A] \wedge s[A] = u[A]))\}$
iii. $\{t \mid \neg(t \in R_1)\}$
iv. $\{t \mid \exists u \in R_1 (t[A] = u[A]) \wedge \exists s \in R_2 (t[A] = s[A])\}$

A. (i)
B. (ii)
C. (iii)
D. (iv)

nielit2022apr-scientistb relational-calculus databases

6.27.4 Relational Calculus: NIELIT Scientist B 2020 November: 111

An expression in the domain relational calculus is of the form:

A. $\{P(x_1, x_2, \ldots, x_n) \mid < x_1, x_2, \ldots, x_n >\}$
B. $\{x_1, x_2, \ldots, x_n \mid < x_1, x_2, \ldots, x_n >\}$
C. $\{x_1, x_2, \ldots, x_n \mid x_1, x_2, \ldots, x_n\}$
D. $\{< x_1, x_2, \ldots, x_n > \mid P(x_1, x_2, \ldots, x_n)\}$

nielit-scb-2020 databases relational-calculus

Answer key☟

6.28 Relational Model (11)

6.28.1 Relational Model: NIELIT 2016 MAR Scientist B - Section C: 41

The employee salary should not be greater than Rs.2000. This is

A. integrity constraint
B. referential constraint
C. over-defined constraint
D. feasible constraint

nielit2016mar-scientistb databases relational-model

Answer key

6.28.2 Relational Model: NIELIT 2016 MAR Scientist C - Section C: 5

A primary key, if combined with a foreign key creates

A. parent child relationship between the tables that connect them
B. many-to-many relationship between the tables that connect them
C. network model between the tables that connect them
D. none of these

nielit2016mar-scientistc databases relational-model

Answer key

6.28.3 Relational Model: NIELIT 2017 DEC Scientist B - Section B: 34

The condition for total participation of entity in a relationship is _______.

A. Maximum cardinality should be one
B. Minimum cardinality should be one
C. Minimum cardinality should be zero
D. None of the options

nielit2017dec-scientistb databases relational-model

Answer key

6.28.4 Relational Model: NIELIT 2017 July Scientist B (IT) - Section B: 53

Related fields in a database are grouped to form a

A. data file
B. data record
C. menu
D. bank

nielit2017july-scientistb-it databases relational-model

Answer key

6.28.5 Relational Model: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 14

E-R model uses this symbol to represent weak entity set?

A. Dotted rectangle
B. Diamond
C. Doubly outlined rectangle
D. None of these

nielit2017oct-assistanta-cs databases relational-model

Answer key

6.28.6 Relational Model: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 21

What is the modality of relationship, if there is no explicit need for relationship to occur?

A. Zero B. Two C. Three D. One

nielit2017oct-assistanta-cs databases relational-model

Answer key☟

6.28.7 Relational Model: NIELIT 2018-42

____ symbol is used to denote derived attributes in ER model

A. Dashed ellipse
B. Squared ellipse
C. Ellipse with attribute name underlined
D. Rectangular Box

nielit-2018 databases relational-model

Answer key☟

6.28.8 Relational Model: NIELIT 2021 Dec Scientist B - Section B: 51

The ____________ of a relationship is 0 if there is no explicit need for the relationship to occur or the relationship is optional.

A. modality B. cardinality C. entity D. structured analysis

nielit2021dec-scientistb databases er-diagram relational-model

Answer key☟

6.28.9 Relational Model: NIELIT 2021 Dec Scientist B - Section B: 58

Indicate which representation is the logical design of the database, and which is a snapshot of the data in the database at a given instant in time.

A. Instance, Range
B. Relation, Schema
C. Relation, Domain
D. Schema, Instance

nielit-2021-it-dec-scientistb databases relational-model

Answer key☟

6.28.10 Relational Model: NIELIT 2021 Dec Scientist B - Section B: 72

Consider a statement

```
Course (course_id, sec_id, semester)
```

Here the $\text{course\_id, sec\_id, and semester}$ will be termed as ________ and Course is a ________.

A. Relations, Attribute
B. Attributes, Relation
C. Tuple, Relation
D. Tuple, Attribute

nielit-2021-it-dec-scientistb relational-model databases

Answer key☟

6.28.11 Relational Model: NIELIT Scientist B 2020 November: 87

Point out the wrong statement :

A. Non-Relational databases require that schemas be defined before you can add data
B. No SQL databases are built to allow the insertion of data without a predefined schema
C. New SQL databases are built to allow the insertion of data without a predefined schema
D. All of the options

nielit-scb-2020 databases relational-model

Answer key☟

6.29 SQL (9)

6.29.1 SQL: NIELIT 2016 DEC Scientist B (IT) - Section B: 41

Which type of Statement can execute parameterized queries?

A. PreparedStatement
B. ParameterizedStatement
C. ParameterizedStatement and CallableStatement
D. All kinds of Statements

nielit2016dec-scientistb-it databases sql

Answer key☟

6.29.2 SQL: NIELIT 2017 July Scientist B (IT) - Section B: 52

'AS' clause is used in SQL for

A. Selection operation
B. Rename operation
C. Join Operation
D. Projection Operation

nielit2017july-scientistb-it databases sql

Answer key☟

6.29.3 SQL: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 13

Given relations $R(w, x)$ and $S(y, z)$, the result of
SELECT DISTINCT w, x from R, S

A. R has no duplicates and S is non-empty
B. R and S have no duplicates
C. S has no duplicates and R is non-empty
D. R and S has the same number of tuples

nielit2017oct-assistanta-cs databases sql

Answer key☟

6.29.4 SQL: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 36

Table employees has 10 records. It has a non-NULL SALARY column which is also UNIQUE. The SQL statement

SELECT COUNT(*)

FROM EMPLOYEE

WHERE SALARY > ALL (SELECT SALARY FROM EMPLOYEE);

A. 10 B. 9 C. 5 D. 0

nielit2017oct-assistanta-cs databases sql

Answer key☟

6.29.5 SQL: NIELIT 2021 Dec Scientist B - Section B: 74

DELETE [FROM] table $[\text{ WHERE condition}]$; from the syntax if you omit the WHERE clause.

A. All rows in the table are deleted.
B. It will give you an error.
C. No rows will be deleted.
D. Only one row will be deleted.

nielit2021dec-scientistb sql databases

Answer key☟

6.29.6 SQL: NIELIT Scientific Assistant A 2020 November: 45

$(< \text{ALL})$ comparison operator means:

A. more than the maximum value in the subquery
B. less than the minimum value in the subquery
C. is equivalent to IN
D. none of the options

nielit-sta-2020 databases sql

Answer key☟

6.29.7 SQL: NIELIT Scientific Assistant A 2020 November: 91

With the following syntax

```
INSERT INTO table[(column[,column...])]
VALUES (value[, value...]);
```

you can:

A. Insert one row at a time
B. Insert multiple rows at a time
C. Insert one column at a time
D. Insert multiple columns at a time

nielit-sta-2020 databases sql

Answer key☟

6.29.8 SQL: NIELIT Scientific Assistant A 2020 November: 98

When retrieving data in particular table in PostgreSQL, we use the ______ statement.

A. \dt B. ORDER BY C. SELECT FROM D. \i

nielit-sta-2020 databases sql

Answer key☟

6.29.9 SQL: NIELIT Scientist B 2020 November: 99

Limitations of the XML Data Type are:

A. It cannot be compared or sorted. This means an XML data type cannot be used in a GROUP BY statement
B. It cannot be used as a key column in an index
C. The value() method of the XML data type returns a scalar value, so it can be specified anywhere where scalar values are allowed
D. All of the options

nielit-scb-2020 databases sql database-normalization

Answer key☟

6.30 Serializability (3)

6.30.1 Serializability: NIELIT 2017 July Scientist B (CS) - Section B: 46

Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item x, denoted by $r(x)$ and $w(x)$respectively. Which one of them is conflict serializable?

1. $r_1(x); r_2(x); w_1(x); r_3(x); w_2(x)$
2. $r_2(x); r_1(x); w_2(x); r_3(x); w_1(x)$
3. $r_3(x); r_2(x); r_1(x); w_2(x); w_1(x)$
4. $r_2(x); w_2(x); r_3(x); r_1(x); w_1(x)$

A. 1 B. 2 C. 3 D. 4

nielit2017july-scientistb-cs databases serializability conflict-serializable

6.30.2 Serializability: NIELIT 2021 Dec Scientist B - Section B: 107

If all transactions obey the ______________ then all possible interleaved schedules (non-serial schedules) are serializable.

A. Lock based protocol B. Two phase Locking protocal
C. Read-write lock based protocol D. Time stamp based protocol

nielit2021dec-scientistb transaction-and-concurrency serializability databases

6.30.3 Serializability: NIELIT 2022 April Scientist B | Section B | Question: 57

Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{\text{O1}, \ldots, \text{Ok}\}$. This is done in the following manner:

- $\mathrm{Step}\ 1.\ \mathrm{T}$ acquires exclusive locks to $\mathrm{O1}, \ldots, \mathrm{Ok}$ in increasing order of their addresses.
- $\mathrm{Step}\ 2.$ The required operations are performed.
- $\mathrm{Step}\ 3.$ All locks are released

This protocol will

A. guarantee serializability and deadlock-freedom
B. guarantee neither serializability nor deadlock-freedom
C. guarantee serializability but not deadlock-freedom
D. guarantee deadlock-freedom but not serializability.

nielit2022apr-scientistb transaction-and-concurrency deadlock-prevention-avoidance-detection serializability databases

Answer key☟

6.31 Super Key (1)

6.31.1 Super Key: NIELIT 2018-55

Maximum number of superkeys for the relation schema $R(X, Y, Z, W)$ with X as the key is

A. 6 B. 8 C. 9 D. 12

nielit-2018 databases super-key

Answer key☟

6.32 Transaction and Concurrency (5)

6.32.1 Transaction and Concurrency: NIELIT 2017 DEC Scientist B - Section B: 18

In conservative two phase locking protocol, a transaction

A. Should release all the locks only at the beginning of transaction
B. Should release exclusive locks only after the commit operation
C. Should acquire all the exclusive locks at the beginning of transaction
D. Should acquire all the locks at the beginning of transaction

nielit2017dec-scientistb databases transaction-and-concurrency two-phase-locking-protocol

Answer key☟

6.32.2 Transaction and Concurrency: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 10

When transaction Ti requests a data item currently held by Tj, Ti is allowed to wait only if it has a timestamp smaller than that of Tj (that is Ti is order than Tj). Otherwise, Ti is rolled back (dies). This is

A. Wait-die B. Wait-wound C. Wound-wait D. Wait

nielit2017oct-assistanta-cs databases transaction-and-concurrency

Answer key☟

6.32.3 Transaction and Concurrency: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 31

When transaction Ti requests a data item currently held by Tj, Ti is allowed to wait only if it has a time stamp smaller than that of Tj (that is Ti is order than Tj). Otherwise, Ti is rolled back (dies). This is

A. Wait-die B. Wait-wound C. Wound-wait D. Wait

nielit2017oct-assistanta-it databases transaction-and-concurrency

Answer key☟

6.32.4 Transaction and Concurrency: NIELIT 2021 Dec Scientist B - Section B: 78

Which of the following is not a JDBC connection isolation levels?

A. TRANSACTION_NONE
B. $\text{TRANSACTION\_READ\_COMMITTED}$
C. $\text{TRANSACTION\_REPEATABLE\_READ}$
D. $\text{TRANSACTION\_NONREPEATABLE\_RE}$

nielit-2021-it-dec-scientistb databases transaction-and-concurrency sql

6.32.5 Transaction and Concurrency: NIELIT Scientific Assistant A 2020 November: 78

Which of the following scenarios may lead to an irrecoverable error in a database system?

A. A transaction writes a data item after it is read by an uncommitted transaction
B. A transaction reads a data item after it is read by an uncommitted transaction
C. A transaction reads a data item after it is written by a committed transaction
D. A transaction reads a data item after it is written by an uncommitted transaction

nielit-sta-2020 databases transaction-and-concurrency

Answer key☟

6.33 Transactions and Concurrency Control (1)

6.33.1 Transactions and Concurrency Control: NIELIT 2018-36

______ is NOT a part of the ACID properties

A. Inconsistency B. Consistency C. Atomicity D. Isolation

nielit-2018 databases transactions-and-concurrency-control

Answer key☟

6.34 Tuple Relational Calculus (3)

6.34.1 Tuple Relational Calculus: NIELIT 2016 MAR Scientist B - Section C: 40

In a relational Schema, each tuple is divided into fields called

A. relations B. domains C. queries D. none of these

nielit2016mar-scientistb databases relational-model tuple-relational-calculus

Answer key

6.34.2 Tuple Relational Calculus: NIELIT 2017 July Scientist B (IT) - Section B: 56

Consider the join of a relation R with relation S. If R has m tuples and S has n tuples, then the maximum size of join is

A. mn B. $m+n$ C. $(m+n)/2$ D. $2(m+n)$

nielit2017july-scientistb-it databases joins tuple-relational-calculus

Answer key

6.34.3 Tuple Relational Calculus: NIELIT 2017 July Scientist B (IT) - Section B: 58

In tuple relational calculus $P1 \rightarrow P2$ is equivalent to

A. $\neg P1 \vee P2$ B. $P1 \vee P2$
C. $P1 \wedge P2$ D. $P1 \wedge \neg P2$

nielit2017july-scientistb-it databases tuple-relational-calculus

Answer key

6.35 Undo Redo (1)

6.35.1 Undo Redo: NIELIT Scientist B 2020 November: 80

Consider a Simple Checkpointing Protocol and the following set of operations in the log.

$(start, T_4); (write, T_4, y, 2, 3); (start, T_1);$

$(commit, T_4); (write, T_1, z, 5, 7);$

$(Checkpoint);$

$(start, T_2); (write, T_2, x, 1, 9);$

$(commit, T_2); (start, T_3); (write, T_3, z, 7, 2);$

if a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list?

A. $Undo : T_3, T_1; Redo\ T_2$ B. $Undo : T_3, T_1; Redo\ T_2, T_4$
C. $Undo : none;\ Redo\ T_2, T_4, T_3, T_1$ D. $Undo : T_3, T_1, T_4;\ Redo\ T_2$

nielit-scb-2020 transaction-and-concurrency databases undo-redo checkpointing

Answer key

6.36 Unique Key (1)

6.36.1 Unique Key: NIELIT 2021 Dec Scientist A - Section B: 83

Consider the following statements :

I. The primary key of a relation cannot contain null values.
II. Unique Key can have null values.

Which among the following is true ?

A. Both I and II are true
B. Both I and II are false
C. Only I is true
D. Only II is true

nielit2021dec-scientista databases relational-model primary-key unique-key

Answer key

6.37 Web Technologies (1)

6.37.1 Web Technologies: NIELIT Scientist B 2020 November: 68

How to specify the comment in the XML document?

A. <?-- -->
B. <!-- --!>
C. <!-- -->
D. </-- -->

nielit-scb-2020 web-technologies

Answer key

Answer Keys

6.0.1	B	6.0.2	D	6.0.3	C	6.1.1	D	6.2.1	A
6.2.2	D	6.2.3	A	6.3.1	C	6.3.2	A	6.4.1	B
6.4.2	D	6.5.1	B	6.6.1	C	6.6.2	C	6.6.3	B
6.6.4	B	6.6.5	B	6.7.1	B	6.7.2	D	6.7.3	A
6.7.4	D	6.8.1	A	6.8.2	C	6.9.1	A	6.9.2	C
6.9.3	B	6.9.4	A	6.9.5	D	6.9.6	C	6.9.7	D
6.9.8	A	6.9.9	D	6.9.10	A	6.9.11	B	6.10.1	B
6.10.2	B	6.11.1	A	6.12.1	D	6.12.2	C	6.13.1	D
6.13.2	D	6.13.3	A	6.13.4	D	6.13.5	C	6.14.1	B
6.15.1	D	6.16.1	A	6.17.1	B	6.18.1	B	6.19.1	A
6.20.1	A	6.20.2	B	6.21.1	A	6.22.1	D	6.23.1	A

6.24.1	B
6.26.2	D
6.26.7	B
6.27.2	A
6.28.3	B
6.28.8	A
6.29.2	B
6.29.7	A
6.30.3	A
6.32.4	D
6.34.3	A

6.25.1	X
6.26.3	C
6.26.8	C
6.27.3	C
6.28.4	B
6.28.9	D
6.29.3	A
6.29.8	C
6.31.1	B
6.32.5	D
6.35.1	A

6.25.2	B
6.26.4	D
6.26.9	D
6.27.4	D
6.28.5	C
6.28.10	B
6.29.4	D
6.29.9	D
6.32.1	D
6.33.1	A
6.36.1	A

6.25.3	C
6.26.5	A
6.26.10	C
6.28.1	A
6.28.6	A
6.28.11	A
6.29.5	A
6.30.1	D
6.32.2	A
6.34.1	B
6.37.1	C

6.26.1	D
6.26.6	A
6.27.1	C
6.28.2	A
6.28.7	A
6.29.1	A
6.29.6	B
6.30.2	B
6.32.3	A
6.34.2	A

7 Digital Logic (88)

7.0.1 NIELIT 2016 MAR Scientist C - Section C: 62

The excess 3 code is also called

A. cyclic redundancy code
B. weighted code
C. self complimenting code
D. algebraic code

nielit2016mar-scientistc digital-logic

Answer key☟

7.0.2 NIELIT 2016 MAR Scientist C - Section C: 36

Odd parity of word can be conveniently tested by

A. OR gate
B. AND gate
C. NOR gate
D. XOR gate

nielit2016mar-scientistc digital-logic

Answer key☟

7.1 Adder (2)

7.1.1 Adder: NIELIT 2016 DEC Scientist B (IT) - Section B: 33

How many inputs are required in Full Adder Circuit?

A. 2
B. 3
C. More than two inputs
D. None of the above

nielit2016dec-scientistb-it digital-logic combinational-circuit adder

Answer key☟

7.1.2 Adder: NIELIT 2021 Dec Scientist A - Section B: 94

The number of full and half-adders required to add 16-bit numbers is :

A. 8 half-adders, 8 full-adders
B. 1 half-adders, 15 full-adders
C. 16 half-adders, 0 full-adders
D. 4 half-adders, 12 full-adders

nielit2021dec-scientista digital-circuits adder

Answer key☟

7.2 Binary Arithmetic (1)

7.2.1 Binary Arithmetic: NIELIT 2021 Dec Scientist B - Section B: 81

The two numbers represented in Signed $2's$ complement form are $\text{A} = 11101101$ and $\text{B} = 11100110.$ If B is subtracted from $\text{A},$ the value obtained in signed $2's$ complement is :

A. 111000101 B. 00000111 C. 11111000 D. 10000011

nielit2021dec-scientistb number-representation twos-complement binary-arithmetic data-structures

Answer key

7.3 Bit Representation (1)

7.3.1 Bit Representation: NIELIT 2021 Dec Scientist B - Section B: 4

Maximum number of bits required to represent any character from ASCII code set is:

A. 10 B. 8 C. 7 D. 3

nielit-2021-it-dec-scientistb number-representation ascii bit-representation

Answer key

7.4 Bitwise Operations (1)

7.4.1 Bitwise Operations: NIELIT 2021 Dec Scientist A - Section B: 113

What happens when a bit string is XORed with itself n times as shown below ?

$[\text{B} \oplus (\text{B} \oplus (\text{B} \oplus (\text{B} \ldots \ldots \text{n} \;\; \text{times}]$

A. Complements when n is even
B. Complements when n is odd
C. Divides by 2^n always
D. Remains unchanged when n is even

nielit2021dec-scientista bitwise-operations logical-reasoning

Answer key

7.5 Boolean Algebra (16)

7.5.1 Boolean Algebra: NIELIT 2016 MAR Scientist B - Section C: 1

Which of the following logic expression is incorrect?

A. $1 \oplus 0 = 1$
B. $1 \oplus 1 \oplus 0 = 1$
C. $1 \oplus 1 \oplus 1 = 1$
D. $1 \oplus 1 = 0$

nielit2016mar-scientistb digital-logic boolean-algebra

Answer key

7.5.2 Boolean Algebra: NIELIT 2017 DEC Scientific Assistant A - Section B: 55

In Boolean algebra $1 + 1 + 1 + 1 \ldots \ldots 800$ times ones $=$ ______ .

A. 1 B. 0 C. 11 D. 800

nielit2017dec-assistanta digital-logic boolean-algebra

Answer key

7.5.3 Boolean Algebra: NIELIT 2017 July Scientist B (CS) - Section B: 18

If $A \oplus B = C$, then which one of the following is true?

A. $A \oplus C = B$
B. $B \oplus C = A$
C. $A \oplus B \oplus C = 0$
D. Both (A) and (B)

nielit2017july-scientistb-cs digital-logic boolean-algebra

Answer key☟

7.5.4 Boolean Algebra: NIELIT 2017 July Scientist B (IT) - Section B: 51

In digital logic, if $A \oplus B = C$, then which one of the following is true?

A. $A \oplus C = B$
B. $B \oplus C = A$
C. $A \oplus B \oplus C = 0$
D. Both (A) and (B)

nielit2017july-scientistb-it digital-logic boolean-algebra

Answer key☟

7.5.5 Boolean Algebra: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 20

The possible number of Boolean function of 3 variables X, Y and Z such that $f(X, Y, Z) = f(X', Y', Z')$

A. 8
B. 16
C. 64
D. 32

nielit2017oct-assistanta-cs digital-logic boolean-algebra

Answer key☟

7.5.6 Boolean Algebra: NIELIT 2018-59

$(A + C')(B' + C')$ simplifies to

A. $AC' + B'$
B. $C(A' + B')$
C. $BC' + A$
D. $AB' + C'$

nielit-2018 digital-logic boolean-algebra

Answer key☟

7.5.7 Boolean Algebra: NIELIT 2018-66

____ number of gates are required to implement the Boolean function $(AB + C)$ with using only 2-input NOR gates.

A. 2
B. 3
C. 4
D. 5

nielit-2018 digital-logic boolean-algebra

Answer key☟

7.5.8 Boolean Algebra: NIELIT 2018-71

In the given truth table, $f(x, y)$ represent the Boolean function.

x	y	$f(x,y)$
0	0	1
0	1	0
1	0	0
1	1	1

A. $x \leftrightarrow y$ B. $x \wedge y$ C. $x \vee y$ D. $x \rightarrow y$

nielit-2018 boolean-algebra

Answer key☟

7.5.9 Boolean Algebra: NIELIT 2021 Dec Scientist A - Section B: 106

Let $\mathrm{f(A, B)} = \overline{\mathrm{A}} + \mathrm{B}$, Simplified expression for function $\mathrm{f(f(x{+}y, y), z)}$ is :

A. $\overline{\mathrm{x}} + \mathrm{z}$
B. xyz
C. $\mathrm{x\overline{y}} + \mathrm{z}$
D. None of the options

nielit2021dec-scientista boolean-algebra digital-logic

Answer key☟

7.5.10 Boolean Algebra: NIELIT 2021 Dec Scientist A - Section B: 52

Which of the following Boolean algebra rules is correct ?

A. $A.\overline{\mathrm{A}} = 1$
B. $A + AB = A + B$
C. $A(A + B) = B$
D. $A + \overline{\mathrm{A}}B = A + B$

nielit2021dec-scientista boolean-algebra logical-reasoning digital-logic

Answer key☟

7.5.11 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 101

Which of the following Boolean expressions is not logically equivalent to all of the rest ?

A. $\mathrm{ab + (cd)' + cd + bd'}$
B. $\mathrm{a\ (b{+}c) + cd}$
C. $\mathrm{ab + ac + (cd)'}$
D. $\mathrm{bd' + c'\ d' + ab + cd}$

nielit2021dec-scientistb boolean-algebra logical-reasoning digital-logic

7.5.12 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 114

If data stored in $\mathrm{AC = 5F\ h}$ and $\mathrm{DR = C2\ h}$ what is value of AC after $\mathrm{AC{\wedge}DR}$ operation ?

A. $\mathrm{9D}$ B. $\mathrm{42}$ C. DF D. DE

nielit2021dec-scientistb digital-logic boolean-algebra

Answer key☟

7.5.13 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 50

When factorizing the Boolean equation $\mathrm{Y}=\mathrm{A}\overline{\mathrm{B}}+\mathrm{AB}$, the result will be :

A. $\mathrm{A}\overline{\mathrm{B}}$ B. AB C. A D. B

nielit2021dec-scientistb boolean-algebra digital-logic

Answer key☟

7.5.14 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 57

The complement of the expression $Y=\mathrm{ABC}+\mathrm{AB}\overline{\mathrm{C}}+\overline{\mathrm{A}}\,\overline{\mathrm{B}}\mathrm{C}+\overline{\mathrm{A}}\mathrm{BC}$ is :

A. $(\overline{\mathrm{A}}+\overline{\mathrm{B}})(\mathrm{A}+\overline{\mathrm{C}})$

B. $(\overline{\mathrm{A}}+\mathrm{B})(\mathrm{A}+\mathrm{C})$

C. $(\mathrm{A}+\overline{\mathrm{B}})(\overline{\mathrm{A}}+\mathrm{C})$

D. $(\mathrm{A}+\overline{\mathrm{B}})(\mathrm{A}+\overline{\mathrm{C}})$

nielit-2021-it-dec-scientistb digital-logic boolean-algebra

7.5.15 Boolean Algebra: NIELIT 2021 Dec Scientist B - Section B: 94

The complement of the expression $\mathrm{Y}=\mathrm{ABC}+\mathrm{AB}\overline{\mathrm{C}}+\overline{\mathrm{A}}\,\overline{\mathrm{B}}\mathrm{C}+\overline{\mathrm{A}}\mathrm{BC}$ is :

A. $(\overline{\mathrm{A}}+\overline{\mathrm{B}})(\mathrm{A}+\overline{\mathrm{C}})$

B. $(\overline{\mathrm{A}}+\mathrm{B})(\mathrm{A}+\mathrm{C})$

C. $(\mathrm{A}+\overline{\mathrm{B}})(\overline{\mathrm{A}}+\mathrm{C})$

D. $(\mathrm{A}+\overline{\mathrm{B}})(\mathrm{A}+\overline{\mathrm{C}})$

nielit2021dec-scientistb digital-logic boolean-algebra

Answer key☟

7.5.16 Boolean Algebra: NIELIT Scientist B 2020 November: 97

If x, y, z are Boolean variable then $(x+\bar{y})(x\cdot\bar{y}+x\cdot z)(\bar{x}\cdot\bar{z}+\bar{y})$ is equal to :

A. $x\cdot\bar{y}$ B. $x\cdot\bar{y}+z$

C. $x\cdot\bar{z}$ D. none of the options

nielit-scb-2020 digital-logic boolean-algebra

Answer key☟

7.6 Booths Algorithm (1)

7.6.1 Booths Algorithm: NIELIT 2016 DEC Scientist B (IT) - Section B: 58

How many Addition and Subtraction are required if you perform multiplication of 5 (Multiplicand) and -30(Multiplier) using Booth Algorithm?

A. $2,1$ B. $1,2$ C. $1,1$ D. $2,2$

nielit2016dec-scientistb-it digital-logic booths-algorithm

Answer key☟

7.7 CO and Architecture (1)

7.7.1 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 5

ALU does not perform the operation of:

A. Data transfer
B. Logical operation
C. Arithmetic operation
D. Comparison operation

nielit-2021-it-dec-scientistb co-and-architecture

Answer key☟

7.8 Carry Generator (1)

7.8.1 Carry Generator: NIELIT 2016 MAR Scientist B - Section C: 2

In which of the following adder circuits, the carry look ripple delay is eliminated?

A. Half adder
B. Full adder
C. Parallel adder
D. Carry-look ahead adder

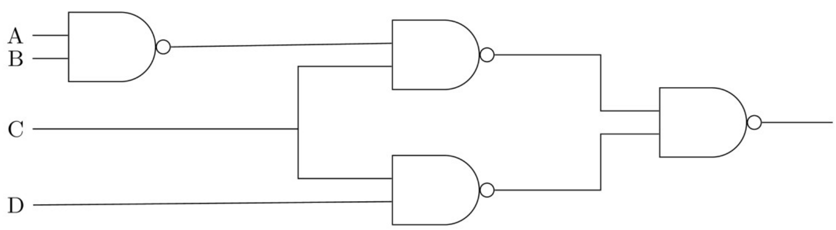
nielit2016mar-scientistb digital-logic combinational-circuit adder carry-generator

Answer key☟

7.9 Circuit Output (1)

7.9.1 Circuit Output: NIELIT 2021 Dec Scientist B - Section B: 73

The logic expression for the output of following circuit is:



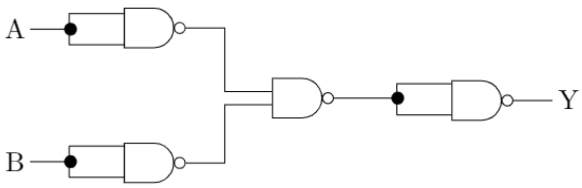
A. $\overline{A}C + \overline{B}C + CD$
B. $A\overline{C} + B\overline{C} + \overline{C}D$
C. $ABC + \overline{CD}$
D. $\overline{AB} + \overline{BC} + \overline{CD}$

nielit-2021-it-dec-scientistb digital-circuits circuit-output nielit2021dec-scientistb

Answer key☟

7.10 Combinational Circuit (11)

7.10.1 Combinational Circuit: NIELIT 2016 DEC Scientist B (CS) - Section B: 4



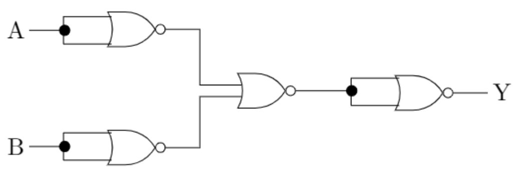
The Circuit is equivalent to:

A. OR Gate B. NOR Gate C. AND Gate D. $EX-OR$ Gate

nielit2016dec-scientistb-cs digital-logic combinational-circuit

Answer key☟

7.10.2 Combinational Circuit: NIELIT 2016 DEC Scientist B (IT) - Section B: 4



The Circuit is equivalent to:

A. EX-OR Gate B. NAND Gate C. OR Gate D. AND Gate

nielit2016dec-scientistb-it digital-logic combinational-circuit

Answer key☟

7.10.3 Combinational Circuit: NIELIT 2016 MAR Scientist B - Section C: 10

Disadvantage of dynamic RAM over static RAM is

A. higher power consumption.
B. variable speed.
C. need to refresh the capacitor charge every once in two milliseconds.
D. higher bit density.

nielit2016mar-scientistb digital-logic combinational-circuit

Answer key☟

7.10.4 Combinational Circuit: NIELIT 2016 MAR Scientist B - Section C: 7

How many RAM chips of size $(256K \times 1 \text{ bit})$ are required to build 1M Byte memory?

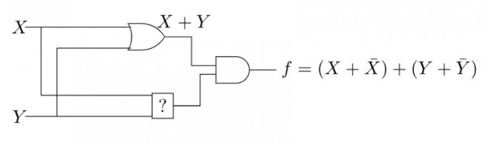
A. 8 B. 10 C. 24 D. 32

nielit2016mar-scientistb digital-logic combinational-circuit

Answer key

7.10.5 Combinational Circuit: NIELIT 2017 July Scientist B (CS) - Section B: 19

To make the following circuit a tautology '?' marked box should be



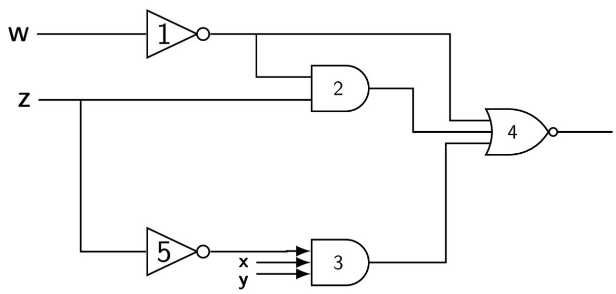
A. OR gate
B. AND gate
C. NAND gate
D. EX-OR gate

nielit2017july-scientistb-cs digital-logic combinational-circuit

Answer key

7.10.6 Combinational Circuit: NIELIT 2017 July Scientist B (CS) - Section B: 20

In the following gate network which gate is redundant?



A. Gate no.1
B. Gate no.2
C. Gate no.3
D. Gate no.4

nielit2017july-scientistb-cs digital-logic combinational-circuit

Answer key

7.10.7 Combinational Circuit: NIELIT 2017 July Scientist B (CS) - Section B: 21

The combinational circuit given below is implemented with two NAND gates. To which of the following individual gates is its equivalent?

A. NOT
B. OR
C. AND
D. XOR

nielit2017july-scientistb-cs digital-logic combinational-circuit easy

Answer key

7.10.8 Combinational Circuit: NIELIT 2017 July Scientist B (IT) - Section B: 49

Which one of the following is true?

A. NAND gate and AND gate both are universal gates
B. NOR gate and OR gate both are universal gates

C. NAND gate and OR gate both are universal gates
D. NAND gate and NOR gate both are universal gates

nielit2017july-scientistb-it digital-logic combinational-circuit easy

Answer key

7.10.9 Combinational Circuit: NIELIT 2018-73

Minimum ____ full adders and ____ half adders are required by the BCD adder to add two decimal digits.

A. $3, 2$ B. $9, 4$ C. $6, 5$ D. $5, 2$

nielit-2018 digital-logic combinational-circuit

Answer key

7.10.10 Combinational Circuit: NIELIT Scientific Assistant A 2020 November: 111

Encodes are made by three ___________ gates.

A. AND B. OR C. NAND D. XOR

nielit-sta-2020 digital-logic combinational-circuit

Answer key

7.10.11 Combinational Circuit: NIELIT Scientific Assistant A 2020 November: 54

How many AND, OR and XOR gates are required for implementation of full adder?

A. $1, 2, 2$ B. $2, 2, 1$ C. $3, 2, 2$ D. $3, 0, 1$

nielit-sta-2020 digital-logic combinational-circuit

Answer key

7.11 Communication (1)

7.11.1 Communication: NIELIT 2022 April Scientist B | Section B | Question: 74

In context of analog & digital communication, as the length of a long-wire antenna is increased, _____________

A. the number of lobes also increase
B. the number of lobes decrease
C. efficiency increase
D. the number of nodes decreases

nielit2022apr-scientistb communication

Answer key

7.12 Decoder (1)

7.12.1 Decoder: NIELIT 2018-39

A RAM chip has a capacity of 1024 words of 8 bits each $(1K \times 8)$. The number of 2×4 decoders with enable line needed to construct a $32K \times 8$ RAM from $1K \times 8$ RAM is

A. 4 B. 5 C. 6 D. 7

nielit-2018 digital-logic combinational-circuit decoder

Answer key

7.13 Digital Circuits (4)

7.13.1 Digital Circuits: NIELIT 2021 Dec Scientist A - Section B: 118

In VCO the output frequency is a linear function of its input :

A. Frequency B. Voltage C. Time period D. None of the option

nielit2021dec-scientista digital-circuits

7.13.2 Digital Circuits: NIELIT 2021 Dec Scientist A - Section B: 56

A demultiplexer is used to :

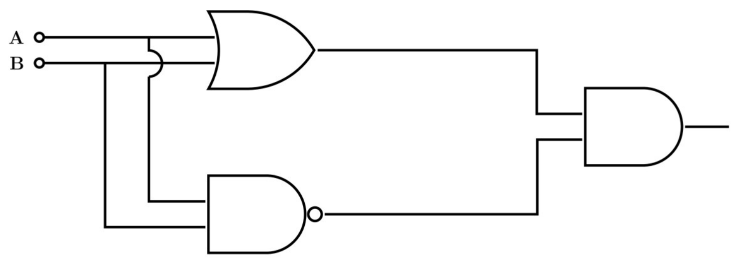
A. Route the data from single input to one of many outputs
B. Perform serial to parallel conversion
C. Both (A) and (B)
D. Select data from several inputs and route it to single output

nielit2021dec-scientista digital-circuits combinational-circuit

Answer key

7.13.3 Digital Circuits: NIELIT 2021 Dec Scientist B - Section B: 47

The following circuit depicts the implementation of __________



A. XOR Gate B. AND Gate C. OR Gate D. NAND Gate

nielit2021dec-scientistb digital-logic digital-circuits combinational-circuit easy

Answer key

7.13.4 Digital Circuits: NIELIT Scientist B 2020 November: 75

What is the basis of KVL?

A. Conservation of charge
B. Conservation of energy
C. Conservation of power
D. All of the options

nielit-scb-2020 digital-circuits

Answer key☟

7.14 Digital Counter (3)

7.14.1 Digital Counter: NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 10

A 4 bit ripple counter and a 4 bit synchronous counter are made using flip-flops having a propagation delay of 10 ns each. If the worst case delay in the ripple counter and the synchronous counter be R and S respectively, then

A. $R = 10$ ns, $S = 40$ ns
B. $R = 40$ ns, $S = 10$ ns
C. $R = 10$ ns, $S = 30$ ns
D. $R = 30$ ns, $S = 10$ ns

nielit2017oct-assistanta-cs digital-logic sequential-circuit flip-flop digital-counter

Answer key☟

7.14.2 Digital Counter: NIELIT 2017 OCT Scientific Assistant A (IT) - Section D: 10

A 4 bit ripple counter and a 4 bit synchronous counter are made using flip-flops having a propagation delay of 10 ns each. If the worst case delay in the ripple counter and the synchronous counter be R and S respectively, then

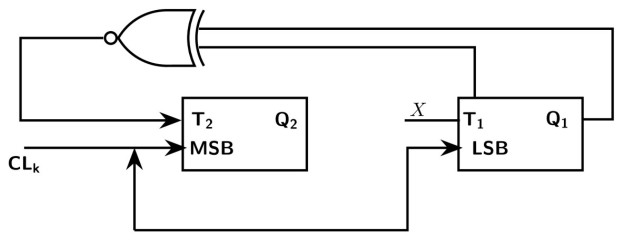
A. $R = 10$ ns, $S = 40$ ns
B. $R = 40$ ns, $S = 10$ ns
C. $R = 10$ ns, $S = 30$ ns
D. $R = 30$ ns, $S = 10$ ns

nielit2017oct-assistanta-it digital-logic sequential-circuit flip-flop digital-counter

Answer key☟

7.14.3 Digital Counter: NIELIT 2021 Dec Scientist B - Section B: 2

Consider the following 2 bit counter using T flip-flops following $0-2-3-1-0$ sequence. What should be the value of x?



A. Q_2
B. $\mathrm{Q}_1 + \mathrm{Q}_2$
C. $\mathrm{Q}_1 \oplus \mathrm{Q}_2$
D. $\mathrm{Q}_1 \oplus \overline{\mathrm{Q}}_2$

nielit-2021-it-dec-scientistb digital-logic digital-counter

7.15 Error Detection (1)

7.15.1 Error Detection: NIELIT 2021 Dec Scientist A - Section B: 77

A binary sequence $\mathrm{b[n]}$ is given as shown below

$$\mathrm{b[n]} = \{0,1,1,0,0,0,0,1,0,1,0,0,0,0,0,0,0,0,0,0,1,0,1,1,0,0,0,0,1,1,0,1\}$$

Consider the following statements regarding the above coded sequence :

i. It has a DC null in the PSD.
ii. It possesses error detecting capability.
iii. It possesses error correcting capability.
iv. It facilitates clock recovery at the receiver.

Which of the above statements are true?

A. $\mathrm{i, iii, and\ iv}$ B. $\mathrm{i, ii, and\ iv}$ C. $\mathrm{i, ii, and\ iii}$ D. $\mathrm{ii, iii, and\ iv}$

nielit2021dec-scientista digital-logic computer-networks error-detection

7.16 Flip Flop (12)

7.16.1 Flip Flop: NIELIT 2016 DEC Scientist B (IT) - Section B: 35

What will be the final output of D flip-Flop if the input string is 0010011100?

A. 1 B. 0 C. Don't Care D. None of the above

nielit2016dec-scientistb-it digital-logic sequential-circuit flip-flop

Answer key☟

7.16.2 Flip Flop: NIELIT 2016 MAR Scientist B - Section C: 4

In a ripple counter using edge triggered JK flip-flops, the pulse input is applied to the

A. clock input of all flip-flops
B. clock input of one flip-flop
C. J and K inputs of all flip-flops
D. J and K inputs of one flip flop

nielit2016mar-scientistb digital-logic sequential-circuit flip-flop

Answer key☟

7.16.3 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 40

If the input J is connected through K input of J-K, then flip-flop will behave as a

A. D type flip-flop
B. T type flip-flop
C. S-R flip-flop
D. Toggle switch

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.4 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 41

To build a mod-19 counter the number of flip-flop required is

A. 3 B. 5 C. 7 D. 8

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.5 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 44

Which of the following conditions must be met to avoid race around problem?

A. $\Delta t < t_p < T$
B. $T > \Delta t > t_p$
C. $2t_p < \Delta t < T$
D. none of these

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.6 Flip Flop: NIELIT 2016 MAR Scientist C - Section C: 64

How many flip-flop are needed to divide the input frequency by 64?

A. 4 B. 5 C. 6 D. 8

nielit2016mar-scientistc digital-logic flip-flop

Answer key

7.16.7 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 8

In a ripple counter using edge-triggered JK flip-flops, the pulse input is applied to

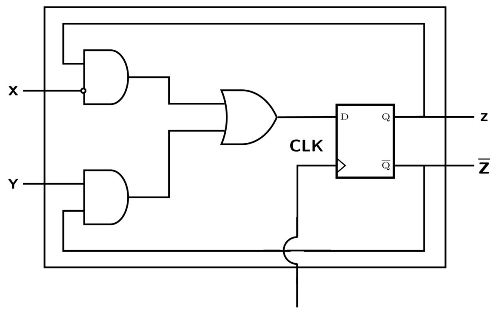
A. Clock input of all flip-flops
B. J and K input of one flip-flop
C. J and K input of all flip flops
D. Clock input of one flip-flop

nielit2017oct-assistanta-cs digital-logic sequential-circuit flip-flop

Answer key

7.16.8 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 9

A sequential circuit using D flip-flop and logic gates is shown in Figure, where X and Y are the inputs and Z is the output. The circuit is



A. S-R Flip-flop with inputs $X = R$ and $Y = S$
B. S-R Flip-flop with inputs $X = S$ and $Y = R$
C. J-K Flip-flop with inputs $X = J$ and $Y = K$
D. J-K Flip-flop with inputs $X = K$ and $Y = J$

nielit2017oct-assistanta-cs digital-logic sequential-circuit flip-flop

Answer key☟

7.16.9 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 19

In a ripple counter using edge-triggered JK flip-flops, the pulse input is applied to

A. Clock input of all flip-flops
B. J and K input of one flip-flop
C. J and K input of all flip-flops
D. Clock input of one flip-flop

nielit2017oct-assistanta-it digital-logic sequential-circuit flip-flop

7.16.10 Flip Flop: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 33

The number of columns in a state table for a sequential circuit with $'m'$ flip flops and $'n'$ input is

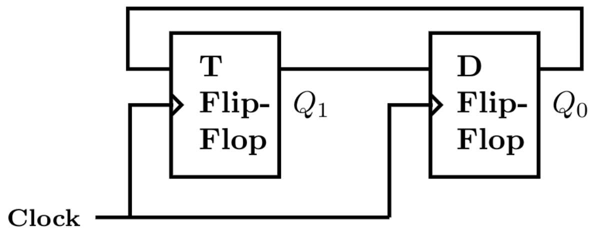
A. $m + n$
B. $m + 2n$
C. $2m + n$
D. $2m + 2n$

nielit2017oct-assistanta-it digital-logic sequential-circuit flip-flop

Answer key☟

7.16.11 Flip Flop: NIELIT 2022 April Scientist B | Section B | Question: 94

Consider a combination of T and D flip-flops connected as shown below. The output of the D flipflop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the $1\mathrm{st}$ clock cycle). The outputs ________

A. $\mathrm{Q}_1\mathrm{Q}_0$ after the 3^{rd} cycle are 11 and after the 4^{th} cycle are 00 respectively
B. $\mathrm{Q}_1\mathrm{Q}_0$ after the 3^{rd} cycle are 11 and after the 4^{th} cycle are 01 respectively
C. $\mathrm{Q}_1\mathrm{Q}_0$ after the 3^{rd} cycle are 00 and after the 4^{th} cycle are 11 respectively
D. $\mathrm{Q}_1\mathrm{Q}_0$ after the 3^{rd} cycle are 01 and after the 4^{th} cycle are 01 respectively

nielit2022apr-scientistb digital-circuits flip-flop sequential-circuit

7.16.12 Flip Flop: NIELIT Scientific Assistant A 2020 November: 85

Which flip-flop is used to make all types of shift registers?

A. JK flip-flop B. D flip-flop C. T flip-flop D. All the options

nielit-sta-2020 digital-logic sequential-circuit flip-flop

Answer key☟

7.17 Information Theory (1)

7.17.1 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 103

For a hamming code of parity bit $m = 5$, what is the total bits and data bits?

A. $(255, 247)$ B. $(127, 119)$ C. $(31, 26)$ D. $(0, 8)$

nielit2021dec-scientistb error-detection information-theory

Answer key☟

7.18 K Map (1)

7.18.1 K Map: NIELIT 2016 DEC Scientist B (IT) - Section B: 45

AB \ CD	00	01	11	10
00	X	1		1
01				
11				
10	X	1	1	X

Which will be the equation of simplification of the given K-map?

A. $AB' + B'CD' + A'B'C'$
B. $AB' + A'B'D' + A'B'C'$
C. $B'D' + AB' + B'C'$
D. $B'D' + A'B'C' + AB'$

nielit2016dec-scientistb-it digital-logic boolean-algebra k-map

Answer key☟

7.19 Minimization (1)

7.19.1 Minimization: NIELIT 2016 DEC Scientist B (CS) - Section B: 45

AB \ CD	00	01	11	10
00	X	X		1
01		1		
11		X		
10	X	1		X

What will be the equation of the given K-map?

A. $A'B'D' + C'D + AB'C'$
B. $B'CD' + AB'C' + A'C'$
C. $B'D' + C'D$
D. $C'D + B'CD'$

nielit2016dec-scientistb-cs digital-logic k-map minimization

Answer key

7.20 Multiplexer (3)

7.20.1 Multiplexer: NIELIT 2017 July Scientist B (CS) - Section B: 17

The digital multiplexer is basically a combination logic circuit to perform the operation

A. AND-AND B. OR-OR C. AND-OR D. OR-AND

nielit2017july-scientistb-cs digital-logic combinational-circuit multiplexer

Answer key

7.20.2 Multiplexer: NIELIT 2017 July Scientist B (IT) - Section B: 50

Which one of the following is the function of a multiplexer?

A. To decode information
B. To select 1 out of N input data sources and to transmit it to single channel
C. To transmit data on N lines
D. To perform serial to parallel conversion

nielit2017july-scientistb-it digital-logic multiplexer

Answer key

7.20.3 Multiplexer: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 30

How many 2-input multiplexers are required to construct a 2^{10} input multiplexer?

A. 1023 B. 31 C. 10 D. 127

nielit2017oct-assistanta-it digital-logic combinational-circuit multiplexer

Answer key

7.21 Number Representation (3)

7.21.1 Number Representation: NIELIT 2018-46

The hexadecimal representation of $(632)_8$ is

A. $19A$ B. 198 C. $29A$ D. 291

nielit-2018 digital-logic number-representation easy

Answer key

7.21.2 Number Representation: NIELIT 2021 Dec Scientist B - Section B: 97

If integer means to 2 bytes of storage then maximum value of a signed integer is :

A. $2^{16}-1$ B. $2^{15}-1$ C. 2^{16} D. 2^{15}

nielit2021dec-scientistb number-representation programming-in-c

Answer key☟

7.21.3 Number Representation: NIELIT Scientist B 2020 November: 120

One-megabyte memory storage in form of bytes is equal to ___________

A. 1024 bytes B. 1024^2 bytes C. 1024^3 bytes D. 1024^4 bytes

nielit-scb-2020 number-representation co-and-architecture memory-management

Answer key☟

7.22 Number System (12)

7.22.1 Number System: NIELIT 2016 DEC Scientist B (CS) - Section B: 33

What will be the Excess-3 code for 1001?

A. 1001 B. 1010 C. 1011 D. 1100

nielit2016dec-scientistb-cs digital-logic number-system

Answer key☟

7.22.2 Number System: NIELIT 2016 DEC Scientist B (CS) - Section B: 59

The Decimal equivalent of the Hexadecimal number $(A09D)_{16}$ is

A. 31845 B. 41117 C. 41052 D. 32546

nielit2016dec-scientistb-cs digital-logic number-system

Answer key☟

7.22.3 Number System: NIELIT 2016 DEC Scientist B (IT) - Section B: 53

What is 2's complement of $(101)_3$?

A. $(010)_3$ B. $(011)_3$ C. $(121)_3$ D. $(121)_2$

nielit2016dec-scientistb-it digital-logic number-system

Answer key☟

7.22.4 Number System: NIELIT 2016 DEC Scientist B (IT) - Section B: 59

The Decimal equivalent of the Hexadecimal number $(AC7B)_{16}$ is:

A. 32564 B. 44155 C. 50215 D. 43562

nielit2016dec-scientistb-it digital-logic number-system

Answer key☟

7.22.5 Number System: NIELIT 2016 MAR Scientist B - Section C: 5

A decimal number has 30 digits. Approximately, how many digits would the binary representation have?

A. 30 B. 60 C. 90 D. 120

nielit2016mar-scientistb digital-logic number-system

Answer key☟

7.22.6 Number System: NIELIT 2016 MAR Scientist B - Section C: 6

The result of the subtraction $FD_{16} - 88_{16}$ is

A. 75_{16} B. 65_{16} C. $5E_{16}$ D. 10_{16}

nielit2016mar-scientistb digital-logic number-system

Answer key☟

7.22.7 Number System: NIELIT 2017 DEC Scientific Assistant A - Section B: 16

The smallest integer that can be represented by an 8-bit number in 2's complement form is :

A. -256 B. -128 C. -127 D. 0

nielit2017dec-assistanta digital-logic number-system

Answer key☟

7.22.8 Number System: NIELIT 2017 DEC Scientific Assistant A - Section B: 26

The number $(25)_6$, in base 6 is equivalent to _________ in binary number system.

A. 11001 B. 10001 C. 11000 D. 10000

nielit2017dec-assistanta digital-logic number-system

Answer key☟

7.22.9 Number System: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 1

A decimal has 25 digits. The number of bits needed for its equivalent binary representation is approximately,

A. 50 B. 74 C. 40 D. 60

nielit2017oct-assistanta-cs digital-logic number-system

Answer key☟

7.22.10 Number System: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 7

A decimal has 25 digits. The number of bits needed for its equivalent binary representation is approximately

A. 50 B. 74 C. 40 D. 60

nielit2017oct-assistanta-it digital-logic number-system

Answer key☟

7.22.11 Number System: NIELIT 2021 Dec Scientist B - Section B: 87

Convert $(503201)_6$ into $(?)_4$

A. 12122231 B. 11000011 C. 21222301 D. 22323301

nielit2021dec-scientistb number-system

Answer key☟

7.22.12 Number System: NIELIT 2022 April Scientist B | Section B | Question: 73

Given $\sqrt{(224)_r} = (13)_r$. The value of the radix r is:

A. 10 B. 8 C. 5 D. 6

nielit2022apr-scientistb number-system quantitative-aptitude

Answer key☟

7.23 Output (1)

7.23.1 Output: NIELIT 2022 April Scientist B | Section B | Question: 120

Given the function $\text{F} = \text{P'} + \text{QR},$ where F is a function in three Boolean variables P, Q and R and $\text{P'} = !\text{P}$, consider the following statements.

$\text{S1: F} = \sum(4, 5, 6)$

$\text{S2: F} = \sum(0, 1, 2, 3, 7)$

$\text{S3: F} = \Pi(4, 5, 6)$

$\text{S4: F} = \Pi(0, 1, 2, 3, 7)$

Which of the following is true?

A. (S1)-False, (S2)-True, (S3)-True, (S4)-False
B. (S1)-True, (S2)-False, (S3)-False, (S4)-True
C. (S1)-False, (S2)-False, (S3)-True, (S4)-True
D. (S1)-True, (S2)-True, (S3)-False, (S4)-False

nielit2022apr-scientistb digital-logic boolean-algebra output

7.24 Register (1)

7.24.1 Register: NIELIT 2021 Dec Scientist B - Section B: 111

If the value of Register $\mathrm{A} = 9\mathrm{B}\ \mathrm{h}\ \&\ \mathrm{Carry} = 1.$ What will be the value of Register A after executing the RORC instruction 1 time ?

A. $\mathrm{AB}\ \mathrm{h}$ B. $\mathrm{CD}\ \mathrm{h}$ C. $\mathrm{EF}\ \mathrm{h}$ D. $\mathrm{AC}\ \mathrm{h}$

nielit2021dec-scientistb digital-logic co-and-architecture register programming-in-c

Answer key

7.25 Register Allocation (1)

7.25.1 Register Allocation: NIELIT 2016 MAR Scientist C - Section C: 63

The range of the numbers which can be stored in an eight bit register is

A. -128 to $+127$
B. -128 to $+128$
C. $-999999 + \ +999999$
D. none of these

nielit2016mar-scientistc digital-logic register-allocation

Answer key

7.26 Sequential Circuit (4)

7.26.1 Sequential Circuit: NIELIT 2016 DEC Scientist B (CS) - Section B: 35

What will be the final output of D Flip-Flop, if the input string is 11010011?

A. 1 B. 0 C. Don't Care D. None of the above

nielit2016dec-scientistb-cs digital-logic sequential-circuit

Answer key

7.26.2 Sequential Circuit: NIELIT 2016 MAR Scientist B - Section C: 3

The output of a sequential circuit depends on

A. present inputs only
B. past inputs only
C. both present and past inputs
D. present outputs only

nielit2016mar-scientistb digital-logic sequential-circuit

Answer key

7.26.3 Sequential Circuit: NIELIT 2016 MAR Scientist C - Section C: 37

A sequential circuit outputs a ONE when an even number(> 0) of one's are input; otherwise the output is $\mathrm{ZERO}.$ The minimum number of states required is

A. 0 B. 1 C. 2 D. 3

nielit2016mar-scientistc digital-logic sequential-circuit

Answer key

7.26.4 Sequential Circuit: NIELIT 2016 MAR Scientist C - Section C: 39

If a clock with time period "T" is used with n stage shift register, then output of final stage will be delayed by

A. nT sec B. $(n-1)T$ sec C. n/T sec D. $(2n-1)T$ sec

nielit2016mar-scientistc digital-logic sequential-circuit

Answer key☟

Answer Keys

7.0.1	C	7.0.2	D	7.1.1	B	7.1.2	B	7.2.1	B
7.3.1	B	7.4.1	D	7.5.1	B	7.5.2	A	7.5.3	A;B;C;D
7.5.4	A;B;C;D	7.5.5	B	7.5.6	D	7.5.7	B	7.5.8	A
7.5.9	C	7.5.10	D	7.5.11	A	7.5.12	B	7.5.13	C
7.5.14	A	7.5.15	A	7.5.16	A	7.6.1	B	7.7.1	A
7.8.1	D	7.9.1	A	7.10.1	B	7.10.2	B	7.10.3	B
7.10.4	D	7.10.5	X	7.10.6	B	7.10.7	C	7.10.8	D
7.10.9	D	7.10.10	B	7.10.11	C	7.11.1	A	7.12.1	B
7.13.1	B	7.13.2	C	7.13.3	A	7.13.4	B	7.14.1	B
7.14.2	B	7.14.3	D	7.15.1	TBA	7.16.1	B	7.16.2	B
7.16.3	B	7.16.4	B	7.16.5	B	7.16.6	C	7.16.7	C
7.16.8	D	7.16.9	D	7.16.10	C	7.16.11	B	7.16.12	A
7.17.1	C	7.18.1	C	7.19.1	C	7.20.1	C	7.20.2	B
7.20.3	A	7.21.1	A	7.21.2	B	7.21.3	B	7.22.1	D
7.22.2	B	7.22.3	C	7.22.4	B	7.22.5	C	7.22.6	A
7.22.7	B	7.22.8	B	7.22.9	X	7.22.10	X	7.22.11	C
7.22.12	C	7.23.1	A	7.24.1	B	7.25.1	A	7.26.1	A
7.26.2	C	7.26.3	C	7.26.4	B				

8 Discrete Mathematics: Combinatory (12)

8.1 Circular Permutation (1)

8.1.1 Circular Permutation: NIELIT 2017 DEC Scientist B - Section B: 15

In how many ways 8 girls and 8 boys can sit around a circular table so that no two boys sit together?

A. $(7!)^2$ B. $(8!)^2$ C. $7!8!$ D. $15!$

nielit2017dec-scientistb discrete-mathematics circular-permutation

Answer key☟

8.2 Combinatory (3)

8.2.1 Combinatory: NIELIT 2016 DEC Scientist B (IT) - Section B: 34

Mala has a colouring book in which each English letter is drawn two times. She wants to paint each of these 52 prints with one of k colours, such that the colour-pairs used to colour ay two letters are different. Both prints of a letter can also be coloured with the same colour. What is the minimum value of k that satisfies this requirement?

A. 9 B. 8 C. 7 D. 6

nielit2016dec-scientistb-it discrete-mathematics combinatory

Answer key☟

8.2.2 Combinatory: NIELIT 2016 MAR Scientist B - Section B: 2

The number of ways in which a team of eleven players can be selected from 22 players including 2 of them and excluding 4 of them is

A. ${}^{16}C_{11}$ B. ${}^{16}C_{5}$ C. ${}^{16}C_{9}$ D. ${}^{20}C_{9}$

nielit2016mar-scientistb discrete-mathematics combinatory

Answer key☟

8.2.3 Combinatory: NIELIT 2017 July Scientist B (IT) - Section B: 14

There are four bus lines between A and B; and three bus lines between B and C. The number of way a person roundtrip by bus from A to C by way of B will be

A. 12 B. 7 C. 144 D. 264

nielit2017july-scientistb-it discrete-mathematics combinatory

Answer key☟

8.3 Counting (1)

8.3.1 Counting: NIELIT Scientist B 2020 November: 21

There are 6 boxes numbered $1, 2, \ldots\ldots, 6$. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is :

A. 18 B. 19 C. 20 D. 21

nielit-scb-2020 counting combinatory

Answer key☟

8.4 Permutation and Combination (2)

8.4.1 Permutation and Combination: NIELIT 2021 Dec Scientist A - Section A: 37

How many eight letter words can be formed from the letters of the word "COURTESY" beginning with C and ending with Y ?

A. 120 B. 256 C. 720 D. 750

nielit2021dec-scientista counting permutation-and-combination

Answer key☟

8.4.2 Permutation and Combination: NIELIT Scientist B 2020 November: 118

The number of 4 digit numbers which contain not more than two different digits is:

A. 576 B. 567 C. 513 D. 504

nielit-scb-2020 combinatory counting permutation-and-combination

8.5 Recurrence Relation (4)

8.5.1 Recurrence Relation: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 6

The solution of the recurrence relation

$a_r = a_{r-1} + 2a_{r-2}$ with $a_0 = 2, a_1 = 7$ is

A. $a_r = (3)^r + (1)^r$
B. $2a_r = (2)^r/3 – (1)^r$
C. $a_r = 3^{r+1} – (-1)^r$
D. $a_r = 3(2)^r – (-1)^r$

nielit2017oct-assistanta-cs discrete-mathematics combinatory recurrence-relation

Answer key☟

8.5.2 Recurrence Relation: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 34

The solution of the recurrence relation

$a_r = a_{r-1} + 2a_{r-2}$ with $a_0 = 2, a_1 = 7$ is

A. $a_r = (3)^r + (1)^r$
B. $2a_r = (2)^r/3 – (1)^r$

C. $a_r = 3^{r+1} - (-1)^r$

D. $a_r = 3(2)^r - (-1)^r$

nielit2017oct-assistanta-it combinatory recurrence-relation

Answer key☟

8.5.3 Recurrence Relation: NIELIT 2022 April Scientist B | Section B | Question: 48

The particular solution of the recurrence relation $a_{r+2} - 4a_{r+1} + 4a_r = 2^r$ is:

A. $r.2^r$

B. $r(r-1)2^{r-1}$

C. $r(r-1)2^{r-2}$

D. $r(r-1)2^{r-3}$

nielit2022apr-scientistb combinatory recurrence-relation

Answer key☟

8.5.4 Recurrence Relation: NIELIT Scientific Assistant A 2020 November: 51

Given $r_{12} = 0.6, r_{13} = 0.5$ and $r_{23} = 0.8,$ the value of $r_{12.3}$ is :

A. 0.47

B. 0.40

C. 0.74

D. 0.64

nielit-sta-2020 combinatory recurrence-relation

8.6 Summation (1)

8.6.1 Summation: NIELIT 2021 Dec Scientist A - Section B: 59

Let $\mathrm{C}(n,\ r) = \binom{n}{r}$. The value of $\sum\limits_{k=0}^{20}(2k+1)\mathrm{C}(41, 2k+1)$, is :

A. $40(2)^{40}$

B. $40(2)^{39}$

C. $41(2)^{40}$

D. $41(2)^{39}$

nielit2021dec-scientista combinatory permutation-and-combination summation

Answer key☟

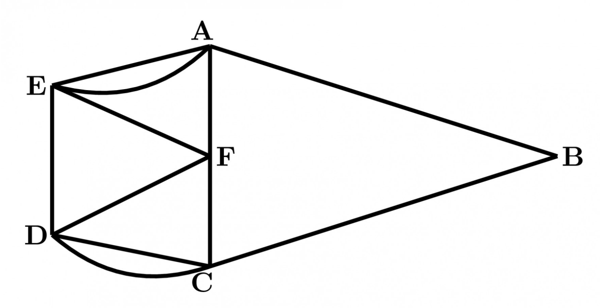
Answer Keys

8.1.1	C	8.2.1	C	8.2.2	C	8.2.3	C	8.3.1	D
8.4.1	C	8.4.2	A	8.5.1	D	8.5.2	D	8.5.3	D
8.5.4	B	8.6.1	D						

9 Discrete Mathematics: Graph Theory (34)

9.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 13

For the graph shown, which of the following paths is a Hamilton circuit?



A. $ABCDCFDEFAEA$
B. $AEDCBAF$
C. $AEFDCBA$
D. $AFCDEBA$

nielit2017july-scientistb-cs discrete-mathematics graph-theory

Answer key☟

9.0.2 NIELIT 2017 July Scientist B (IT) - Section B: 15

The number of diagonals that can be drawn by joining the vertices of an octagon is

A. 28 B. 48 C. 20 D. None of the option

nielit2017july-scientistb-it discrete-mathematics graph-theory

Answer key☟

9.0.3 NIELIT 2017 July Scientist B (IT) - Section B: 3

A path in graph G, which contains every vertex of G and only once?

A. Euler circuit
B. Hamiltonian path
C. Euler Path
D. Hamiltonian Circuit

nielit2017july-scientistb-it discrete-mathematics graph-theory

Answer key☟

9.0.4 NIELIT 2017 July Scientist B (IT) - Section B: 2

Which of the following is an advantage of adjacency list representation over adjacency matrix representation of a graph?

A. In adjacency list representation, space is saved for sparse graphs.
B. Deleting a vertex in adjacency list representation is easier than adjacency matrix

representation.
C. Adding a vertex in adjacency list representation is easier than adjacency matrix representation.
D. All of the option.

nielit2017july-scientistb-it discrete-mathematics graph-theory

Answer key

9.0.5 NIELIT 2016 MAR Scientist B - Section B: 1

The number of ways to cut a six sided convex polygon whose vertices are labeled into four triangles using diagonal lines that do not cross is

A. 13 B. 14 C. 12 D. 11

nielit2016mar-scientistb discrete-mathematics graph-theory

Answer key

9.0.6 NIELIT 2017 DEC Scientific Assistant A - Section B: 20

Total number of simple graphs that can be drawn using six vertices are:

A. 2^{15} B. 2^{14} C. 2^{13} D. 2^{12}

nielit2017dec-assistanta discrete-mathematics graph-theory

Answer key

9.0.7 NIELIT 2016 MAR Scientist C - Section B: 20

Degree of each vertex in K_n is

A. n B. $n-1$ C. $n-2$ D. $2n-1$

nielit2016mar-scientistc discrete-mathematics graph-theory

Answer key

9.0.8 NIELIT 2021 Dec Scientist A - Section B: 60

If R and D are the radius and diameter of the graph $\mathrm{K}_{4,7}$, then the ordered pair (R, D) is equal to :

A. $(2,2)$ B. $(1,2)$ C. $(2,4)$ D. $(1,3)$

nielit2021dec-scientista graph-theory

Answer key

9.0.9 NIELIT 2021 Dec Scientist A - Section B: 53

The largest number of faces in a simple connected maximal planar graph with 100 vertices is :

A. 200 B. 198 C. 196 D. 96

nielit2021dec-scientista graph-theory

Answer key☟

9.0.10 NIELIT 2022 April Scientist B | Section B | Question: 104

A connected planar graph having 6 vertices, 7 edges contains _____________ regions.

A. 15 B. 11 C. 3 D. 1

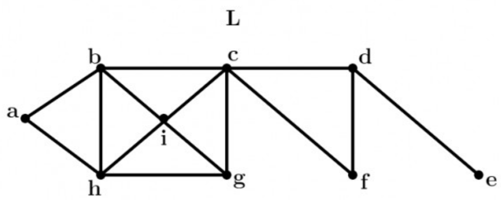
nielit2022apr-scientistb graph-theory

Answer key☟

9.1 Bridges (2)

9.1.1 Bridges: NIELIT 2017 July Scientist B (CS) - Section B: 11

Consider the following graph L and find the bridges,if any.



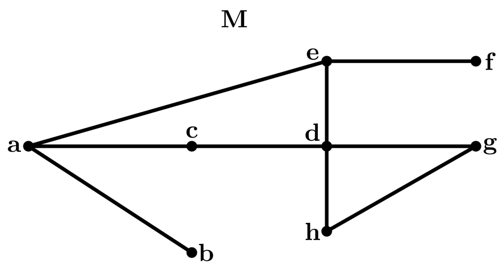
A. No bridge B. $\{d, e\}$ C. $\{c, d\}$ D. $\{c, d\}$ and $\{c, f\}$

nielit2017july-scientistb-cs discrete-mathematics graph-theory bridges

Answer key☟

9.1.2 Bridges: NIELIT 2017 July Scientist B (IT) - Section B: 6

Considering the following graph, which one of the following set of edges represents all the bridges of the given graph?



A. $(a, b), (e, f)$ B. $(a, b), (a, c)$ C. $(c, d), (d, h)$ D. (a, b)

nielit2017july-scientistb-it discrete-mathematics graph-theory bridges

Answer key☟

9.2 Counting (1)

9.2.1 Counting: NIELIT 2021 Dec Scientist B - Section B: 1

The number of un-labeled non-isomorphic graphs with four vertices is:

A. 12 B. 11 C. 10 D. 9

nielit-2021-it-dec-scientistb graph-theory counting

Answer key☟

9.3 Cycle (1)

9.3.1 Cycle: NIELIT 2017 DEC Scientist B - Section B: 23

Let G be a complete undirected graph on 8 vertices. If vertices of G are labelled, then the number of distinct cycles of length 5 in G is equal to:

A. 15 B. 30 C. 56 D. 60

nielit2017dec-scientistb discrete-mathematics graph-theory cycle

Answer key☟

9.4 Degree of Graph (7)

9.4.1 Degree of Graph: NIELIT 2016 MAR Scientist B - Section B: 3

Maximum degree of any node in a simple graph with n vertices is

A. $n-1$ B. n C. $n/2$ D. $n-2$

nielit2016mar-scientistb discrete-mathematics graph-theory degree-of-graph

Answer key☟

9.4.2 Degree of Graph: NIELIT 2017 July Scientist B (CS) - Section B: 2

Which of the following statements is/are TRUE for an undirected graph?

P. Number of odd degree vertices is even
Q. Sum of degrees of all vertices is even

A. P only
B. Q only
C. Both P and Q
D. Neither P nor Q

nielit2017july-scientistb-cs discrete-mathematics graph-theory degree-of-graph

9.4.3 Degree of Graph: NIELIT 2017 July Scientist B (IT) - Section B: 1

Given an undirected graph G with V vertices and E edges, the sum of the degrees of all vertices is

A. E B. $2E$ C. V D. $2V$

nielit2017july-scientistb-it discrete-mathematics graph-theory degree-of-graph

Answer key☟

9.4.4 Degree of Graph: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 7

The number of the edges in a regular graph of degree $'d'$ and $'n'$ vertices is

A. Maximum of n, d B. $n + d$ C. nd D. $nd/2$

nielit2017oct-assistanta-cs discrete-mathematics graph-theory degree-of-graph

Answer key☟

9.4.5 Degree of Graph: NIELIT 2021 Dec Scientist B - Section B: 13

For which one of the following sequences CAN NOT be a degree sequence of a graph of order 5?

A. $3, 3, 2, 2, 2$ B. $3, 3, 3, 3, 2$
C. $3, 3, 3, 2, 2$ D. $4, 3, 3, 2, 2$

nielit-2021-it-dec-scientistb graph-theory degree-of-graph

Answer key☟

9.4.6 Degree of Graph: NIELIT 2021 Dec Scientist B - Section B: 96

For which one of the following sequences CAN NOT be a degree sequence if a graph of order 5 ?

A. $3, 3, 2, 2, 2$ B. $3, 3, 3, 3, 2$ C. $3, 3, 3, 2, 2$ D. $4, 3, 3, 2, 2$

nielit2021dec-scientistb graph-theory degree-of-graph graph-algorithms discrete-mathematics

9.4.7 Degree of Graph: NIELIT Scientific Assistant A 2020 November: 96

In an undirected graph, if we add the degrees of all vertices, it is:

A. odd B. even
C. cannot be determined D. always $n + 1$, where n is number of nodes

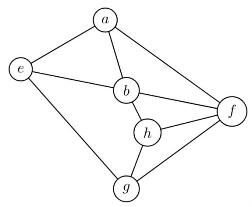
nielit-sta-2020 graph-theory easy degree-of-graph

Answer key☟

9.5 Depth First Search (1)

9.5.1 Depth First Search: NIELIT 2017 July Scientist B (IT) - Section B: 5

In a given following graph among the following sequences:



I. abeghf
II. abfehg
III. abfhge
IV. afghbe

Which are depth first traversals of the above graph?

A. I,II and IV only B. I and IV only C. II,III and IV only D. I,III and IV only

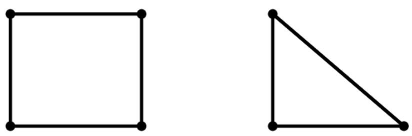
nielit2017july-scientistb-it discrete-mathematics graph-theory depth-first-search

Answer key☟

9.6 Euler Graph (2)

9.6.1 Euler Graph: NIELIT 2017 July Scientist B (CS) - Section B: 12

The following graph has no Euler circuit because



A. It has 7 vertices.
B. It is even-valent (all vertices have even valence).
C. It is not connected.
D. It does not have a Euler circuit.

nielit2017july-scientistb-cs discrete-mathematics graph-theory euler-graph

Answer key☟

9.6.2 Euler Graph: NIELIT 2017 July Scientist B (IT) - Section B: 7

Which of the following statements is/are TRUE?

$S1$:The existence of an Euler circuit implies that an Euler path exists.

$S2$:The existence of an Euler path implies that an Euler circuit exists.

A. $S1$ is true.
B. $S2$ is true.
C. $S1$ and $S2$ both are true.
D. $S1$ and $S2$ both are false.

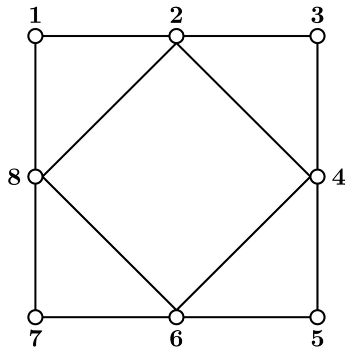
nielit2017july-scientistb-it discrete-mathematics graph-theory euler-graph

Answer key☟

9.7 Graph Algorithms (1)

9.7.1 Graph Algorithms: NIELIT 2021 Dec Scientist B - Section B: 73

What will be the Eulerian tour of the following graph G ?



G

A. $1\text{–}2\text{–}3\text{–}4\text{–}5\text{–}6\text{–}7\text{–}8\text{–}6\text{–}4\text{–}2\text{–}8\text{–}1$
B. $1\text{–}2\text{–}3\text{–}4\text{–}5\text{–}6\text{–}7\text{–}8$
C. $1\text{–}2\text{–}3\text{–}4\text{–}5\text{–}6\text{–}7\text{–}8\text{–}1$
D. $8\text{–}7\text{–}6\text{–}5\text{–}4\text{–}3\text{–}2\text{–}1$

nielit2021dec-scientistb graph-theory graph-algorithms

Answer key☟

9.8 Graph Coloring (1)

9.8.1 Graph Coloring: NIELIT 2017 DEC Scientist B - Section B: 52

Let G be a simple undirected graph on $n = 3x$ vertices $(x \geq 1)$ with chromatic number 3, then maximum number of edges in G is

A. $n(n-1)/2$ B. n^{n-2} C. nx D. n

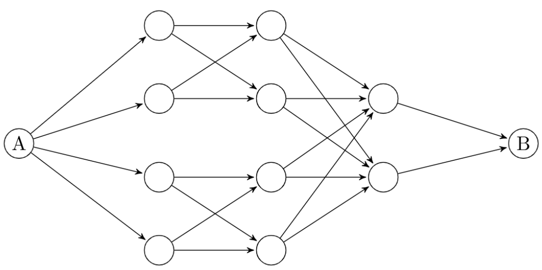
nielit2017dec-scientistb discrete-mathematics graph-theory graph-coloring

Answer key☟

9.9 Graph Connectivity (1)

9.9.1 Graph Connectivity: NIELIT Scientific Assistant A 2020 November: 19

What is the total number of ways to reach from A to B in the network given?



A. 12 B. 16 C. 20 D. 22

nielit-sta-2020 graph-theory graph-connectivity

Answer key☟

9.10 Graph Isomorphism (1)

9.10.1 Graph Isomorphism: NIELIT 2021 Dec Scientist B - Section B: 83

The number of un-labeled non-isomorphic graphs with four vertices is :

A. 12 B. 11 C. 10 D. 9

nielit2021dec-scientistb graph-theory combinatory graph-isomorphism

9.11 Graph Planarity (6)

9.11.1 Graph Planarity: NIELIT 2016 DEC Scientist B (CS) - Section B: 5

Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of bounded faces in any embedding of G on the plane is equal to:

A. 3 B. 4 C. 5 D. 6

nielit2016dec-scientistb-cs discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.2 Graph Planarity: NIELIT 2017 DEC Scientific Assistant A - Section B: 38

If a planner graph, having 25 vertices divides the plane into 17 different regions. Then how many edges are used to connect the vertices in this graph.

A. 20 B. 30 C. 40 D. 50

nielit2017dec-assistanta discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.3 Graph Planarity: NIELIT 2017 July Scientist B (CS) - Section B: 14

If G is an undirected planar graph on n vertices with e edges then

A. $e \leq n$ B. $e \leq 2n$ C. $e \leq 3n$ D. None of the option

nielit2017july-scientistb-cs discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.4 Graph Planarity: NIELIT 2017 July Scientist B (CS) - Section B: 15

Choose the most appropriate definition of plane graph.

A. A simple graph which is isomorphic to hamiltonian graph.

B. A graph drawn in a plane in such a way that if the vertex set of graph can be partitioned into two non-empty disjoint subset X and Y in such a way that each edge of G has one end in X and one end in Y.
C. A graph drawn in a plane in such a way that any pair of edges meet only at their end vertices.
D. None of the option.

nielit2017july-scientistb-cs discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.5 Graph Planarity: NIELIT 2017 July Scientist B (IT) - Section B: 12

Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is

A. 6 B. 8 C. 9 D. 13

nielit2017july-scientistb-it discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.6 Graph Planarity: NIELIT 2017 July Scientist B (IT) - Section B: 8

A connected planar graph divides the plane into a number of regions. If the graph has eight vertices and these are linked by 13 edges, then the number of regions is:

A. 5 B. 6 C. 7 D. 8

nielit2017july-scientistb-it discrete-mathematics graph-theory graph-planarity

Answer key☟

Answer Keys

9.0.1	C	9.0.2	C	9.0.3	B	9.0.4	D	9.0.5	B
9.0.6	A	9.0.7	B	9.0.8	A	9.0.9	C	9.0.10	C
9.1.1	B	9.1.2	A	9.2.1	B	9.3.1	C	9.4.1	A
9.4.2	C	9.4.3	B	9.4.4	D	9.4.5	C	9.4.6	C
9.4.7	B	9.5.1	D	9.6.1	C	9.6.2	A	9.7.1	A
9.8.1	C	9.9.1	B	9.10.1	B	9.11.1	D	9.11.2	C
9.11.3	B	9.11.4	C	9.11.5	B	9.11.6	C		

10 Discrete Mathematics: Mathematical Logic (8)

10.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 16

Which of the following propositions is tautology?

A. $(p \vee q) \rightarrow q$
B. $p \vee (q \rightarrow p)$
C. $p \vee (p \rightarrow q)$
D. Both (B) and (C)

nielit2017july-scientistb-cs   mathematical-logic

Answer key☟

10.0.2 NIELIT 2016 MAR Scientist C - Section C: 65

In propositional logic, which of the following is equivalent to $p \rightarrow q$?

A. $\sim p \rightarrow q$
B. $\sim p \vee q$
C. $\sim p \vee \sim q$
D. $p \rightarrow \sim q$

nielit2016mar-scientistc   discrete-mathematics   mathematical-logic

Answer key☟

10.0.3 NIELIT 2016 MAR Scientist C - Section C: 25

Which of the following is FALSE?

$Read\ \wedge as\ AND, \vee\ as\ OR, \sim as\ NOT,$ $\rightarrow$ as one way implication and $\leftrightarrow$ as two way implication?

A. $((x \rightarrow y) \wedge x) \rightarrow y$
B. $((\sim x \rightarrow y) \wedge (\sim x \wedge \sim y)) \rightarrow x$
C. $(x \rightarrow (x \vee y))$
D. $((x \vee y) \leftrightarrow (\sim x \vee \sim y))$

nielit2016mar-scientistc   discrete-mathematics   mathematical-logic

Answer key☟

10.1 First Order Logic (3)

10.1.1 First Order Logic: NIELIT 2017 DEC Scientist B - Section B: 43

Which one is the correct translation of the following statement into mathematical logic?

“None of my friends are perfect.”

A. $\neg \exists x (p(x) \wedge q(x))$
B. $\exists x (\neg p(x) \wedge q(x))$
C. $\exists x (\neg p(x) \wedge \neg q(x))$
D. $\exists x (p(x) \wedge \neg q(x))$

nielit2017dec-scientistb   discrete-mathematics   mathematical-logic   first-order-logic

Answer key☟

10.1.2 First Order Logic: NIELIT 2021 Dec Scientist B - Section B: 25

Let $\mathrm{P}(x)$ be "x is perfect", $\mathrm{F}(x)$ be "x is your friend" and the domain be all people. The statement, "At least one of your friends is perfect" is :

A. $\forall x\ (\mathrm{F}(x) \rightarrow \mathrm{P}(x))$
B. $\forall x\ (\mathrm{F}(x) \wedge \mathrm{P}(x))$
C. $\exists x\ (\mathrm{F}(x) \wedge \mathrm{P}(x))$
D. $\exists x\ (\mathrm{F}(x) \rightarrow \mathrm{P}(x))$

nielit-2021-it-dec-scientistb first-order-logic logical-reasoning

Answer key☟

10.1.3 First Order Logic: NIELIT 2022 April Scientist B | Section B | Question: 67

The first order logic statement $((\mathrm{R} \vee \mathrm{Q}) \wedge (\mathrm{P} \vee \sim \mathrm{Q}))$ is equivalent to which of the following?

A. $((\mathrm{R} \vee \sim \mathrm{Q}) \wedge (\mathrm{P} \vee \sim \mathrm{Q}) \wedge (\mathrm{R} \vee \mathrm{P}))$
B. $((\mathrm{R} \vee \mathrm{Q}) \wedge (\mathrm{P} \vee \sim \mathrm{Q}) \wedge (\mathrm{R} \vee \mathrm{P}))$
C. $((\mathrm{R} \vee \mathrm{Q}) \wedge (\mathrm{P} \vee \sim \mathrm{Q}) \wedge (\mathrm{R} \vee \sim \mathrm{P}))$
D. $((\mathrm{R} \vee \mathrm{Q}) \wedge (\mathrm{P} \vee \sim \mathrm{Q}) \wedge (\sim \mathrm{R} \vee \mathrm{P}))$

nielit2022apr-scientistb first-order-logic propositional-logic logical-reasoning

Answer key☟

10.2 Propositional Logic (2)

10.2.1 Propositional Logic: NIELIT 2017 July Scientist B (IT) - Section B: 13

Which of the following statements is false?

A. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $\sim Q \wedge \sim P$
B. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $Q \vee P$
C. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $Q \vee (P \wedge \sim Q)$
D. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $P \vee (Q \wedge \sim P)$

nielit2017july-scientistb-it mathematical-logic propositional-logic

Answer key☟

10.2.2 Propositional Logic: NIELIT Scientist B 2020 November: 84

Given the truth table of a Binary Operation \$ as follows:

X	Y	X$Y
1	0	1
1	1	1
0	1	0
0	0	1

Identify the matching Boolean Expression.

A. $X\$\neg Y$
B. $\neg X\$Y$
C. $\neg X\$\neg Y$
D. none of the options

nielit-scb-2020 mathematical-logic propositional-logic discrete-mathematics

Answer key☟

Answer Keys

10.0.1	C	10.0.2	B	10.0.3	D	10.1.1	A	10.1.2	C
10.1.3	C	10.2.1	A	10.2.2	D				

11 Discrete Mathematics: Set Theory & Algebra (20)

11.1 Abelian Group (2)

11.1.1 Abelian Group: NIELIT 2016 DEC Scientist B (CS) - Section B: 28

Which one of the following is NOT necessarily a property of a Group?

A. Commutativity
B. Associativity
C. Existence of inverse for every element
D. Existence of identity

nielit2016dec-scientistb-cs discrete-mathematics group-theory abelian-group

Answer key☟

11.1.2 Abelian Group: NIELIT 2017 DEC Scientist B - Section B: 48

On a set $A = \{a, b, c, d\}$ a binary operation $*$ is defined as given in the following table.

*	a	b	c	d
a	a	c	b	d
b	c	b	d	a
c	b	d	a	c
d	d	a	c	b

The relation is

A. Commutative but not associative
B. Neither commutative nor associative
C. Both commutative and associative
D. Associative but not commutative

nielit2017dec-scientistb discrete-mathematics group-theory abelian-group

Answer key☟

11.2 Boolean Algebra (1)

11.2.1 Boolean Algebra: NIELIT 2017 July Scientist B (IT) - Section B: 18

If B is a Boolean algebra, then which of the following is true?

A. B is a finite but not complemented lattice
B. B is a finite, complemented and distributive lattice
C. B is a finite,distributive but not complemented lattice
D. B is not distributive lattice

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra boolean-algebra

Answer key☟

11.3 Cartesian Product (1)

11.3.1 Cartesian Product: NIELIT 2017 July Scientist B (IT) - Section B: 10

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b\}$?

A. $\{(1, a), (1, b), (2, a), (b, b)\}$
B. $\{(1, 1), (2, 2), (a, a), (b, b)\}$
C. $\{(1, a), (2, a), (1, b), (2, b)\}$
D. $\{(1, 1), (a, a), (2, a), (1, b)\}$

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra cartesian-product

Answer key☟

11.4 Functions (3)

11.4.1 Functions: NIELIT 2016 DEC Scientist B (CS) - Section B: 34

How many onto (or surjective) functions are there from an n-element $(n >= 2)$ set to a 2-element set?

A. 2^n
B. $2^n - 1$
C. $2^n - 2$
D. $2(2^n - 2)$

nielit2016dec-scientistb-cs discrete-mathematics set-theory&algebra functions

Answer key☟

11.4.2 Functions: NIELIT 2016 MAR Scientist B - Section B: 8

If $\Delta f(x) = f(x + h) - f(x)$, then a constant k, Δk

A. 1
B. 0
C. $f(k) - f(0)$
D. $f(x + k) - f(x)$

nielit2016mar-scientistb discrete-mathematics set-theory&algebra functions

11.4.3 Functions: NIELIT 2016 MAR Scientist C - Section C: 16

If $f : \{a, b\}^* \to \{a, b\}^*$ be given by $f(n) = ax$ for every value of $n \in \{a, b\}$, then f is

A. one to one not onto
B. one to one and onto
C. not one to one and not onto
D. not one to one and onto

nielit2016mar-scientistc discrete-mathematics set-theory&algebra functions

Answer key☟

11.5 Group Theory (1)

11.5.1 Group Theory: NIELIT 2016 DEC Scientist B (IT) - Section B: 21

Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $(S, *)$ forms:

A. A group
B. A ring
C. An integral domain
D. A field

nielit2016dec-scientistb-it discrete-mathematics set-theory&algebra group-theory

Answer key

11.6 Inclusion Exclusion Principle (1)

11.6.1 Inclusion Exclusion Principle: NIELIT 2017 DEC Scientist B - Section B: 46

The number of integers between 1 and 500(both inclusive) that are divisible by 3 or 5 or 7 is _________.

A. 269 B. 270 C. 271 D. 272

nielit2017dec-scientistb discrete-mathematics set-theory&algebra inclusion-exclusion-principle

Answer key

11.7 Lattice (1)

11.7.1 Lattice: NIELIT 2017 July Scientist B (IT) - Section B: 22

Let L be a lattice. Then for every a and b in L which one of the following is correct?

A. $a \vee b = a \wedge b$
B. $a \vee (b \vee c) = (a \vee b) \vee c$
C. $a \vee (b \wedge c) = a$
D. $a \vee (b \vee c) = b$

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra lattice

Answer key

11.8 Partial Order (2)

11.8.1 Partial Order: NIELIT 2017 July Scientist B (IT) - Section B: 16

A partial ordered relation is transitive, reflexive and

A. antisymmetric B. bisymmetric C. antireflexive D. asymmetric

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra partial-order

Answer key

11.8.2 Partial Order: NIELIT 2017 July Scientist B (IT) - Section B: 17

Let $N = \{1, 2, 3, \ldots\}$ be ordered by divisibility, which of the following subset is totally ordered?

A. $(2, 6, 24)$ B. $(3, 5, 15)$ C. $(2, 9, 16)$ D. $(4, 15, 30)$

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra partial-order

Answer key

11.9 Polynomials (1)

11.9.1 Polynomials: NIELIT 2016 MAR Scientist B - Section C: 21

A polynomial $p(x)$ is such that $p(0) = 5,\ p(1) = 4,\ p(2) = 9$ and $p(3) = 20$. The minimum degree it can have is

A. 1 B. 2 C. 3 D. 4

nielit2016mar-scientistb engineering-mathematics functions polynomials

Answer key☟

11.10 Relations (3)

11.10.1 Relations: NIELIT 2016 DEC Scientist B (IT) - Section B: 28

What is the possible number of reflexive relation on a set of 5 elements?

A. 2^{10} B. 2^{15} C. 2^{20} D. 2^{25}

nielit2016dec-scientistb-it discrete-mathematics set-theory&algebra relations

Answer key☟

11.10.2 Relations: NIELIT 2017 July Scientist B (IT) - Section B: 21

The relation $\{(1,2),(1,3)(3,1),(1,1),(3,3),(3,2),(1,4),(4,2),(3,4)\}$ is

A. Reflexive B. Transitive C. Symmetric D. Asymmetric

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra relations

Answer key☟

11.10.3 Relations: NIELIT 2021 Dec Scientist B - Section B: 115

If A and B are two sets. A binary relation from set A and set B is any subset of the _____________ .

A. Cartesian Product $\text{A} \times \text{B}$
B. Union $\text{A} \cup \text{B}$
C. Intersection $\text{A} \cap \text{B}$
D. Addition $\text{A} + \text{B}$

nielit2021dec-scientistb set-theory relations

Answer key☟

11.11 Set Theory (4)

11.11.1 Set Theory: NIELIT 2016 MAR Scientist B - Section C: 23

I f S be an infinite set and $S_1 \ldots\ldots, S_n$ be sets such that $S_1 \cup S_2 \cup \cdots \cup S_n = S$, then

A. at least one of the set S_i is a finite set.
B. not more than one of the sets S_i can be finite.
C. at least one of the sets S_i is an infinite set.

D. not more than one of the sets S_i can be infinite.

nielit2016mar-scientistb discrete-mathematics set-theory&algebra set-theory

Answer key

11.11.2 Set Theory: NIELIT 2017 July Scientist B (IT) - Section B: 11

What is the Cardinality of the Power set of the set $\{0, 1, 2\}$?

A. 8 B. 6 C. 7 D. 9

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra set-theory

Answer key

11.11.3 Set Theory: NIELIT 2017 July Scientist B (IT) - Section B: 20

If A and B are two sets and $A \cup B = A \cap B$ then

A. $A = \phi$ B. $B = \phi$ C. $A \neq B$ D. $A = B$

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra set-theory

Answer key

11.11.4 Set Theory: NIELIT 2017 July Scientist B (IT) - Section B: 9

Power set of empty set has exactly ________ subset

A. One B. Two C. Zero D. Three

nielit2017july-scientistb-it discrete-mathematics set-theory&algebra set-theory

Answer key

Answer Keys

11.1.1	A	11.1.2	A	11.2.1	B	11.3.1	C	11.4.1	C
11.4.2	B	11.4.3	A	11.5.1	A	11.6.1	C	11.7.1	B
11.8.1	A	11.8.2	A	11.9.1	B	11.10.1	C	11.10.2	B
11.10.3	A	11.11.1	C	11.11.2	A	11.11.3	D	11.11.4	B

12 Engineering Mathematics: Calculus (30)

12.0.1 NIELIT 2016 MAR Scientist C - Section B: 17

A ladder 13 feet long rests against the side of a house. The bottom of the ladder slides away from the house at a rate of 0.5 ft/s. How fast is the top of the ladder sliding down the wall when the bottom of the ladder is 5 feet from the house?

A. $\frac{5}{24}\ \mathrm{ft/s}$ B. $\frac{5}{12}\ \mathrm{ft/s}$ C. $-\frac{5}{24}\mathrm{ft/s}$ D. $-\frac{5}{12}\ \mathrm{ft/s}$

nielit2016mar-scientistc engineering-mathematics calculus

Answer key☟

12.0.2 NIELIT 2016 MAR Scientist C - Section B: 13

$$\lim_{x\to 0}\frac{1}{x^6}\int_0^{x^2}\frac{t^2dt}{t^6+1}=?$$

A. $1/4$ B. $1/3$ C. $1/2$ D. 1

nielit2016mar-scientistc calculus

Answer key☟

12.0.3 NIELIT 2016 MAR Scientist C - Section B: 12

$$\lim_{x\to a}\frac{1}{x^2-a^2}\int_a^x \sin(t^2)dt=?$$

A. $2a\sin(a^2)$ B. $2a$ C. $\sin(a^2)$ D. None of the above

nielit2016mar-scientistc engineering-mathematics calculus

Answer key☟

12.0.4 NIELIT 2023 Feb Scientist D | Section D | Question: 98

Given $\mathrm{A}=\sin^2\theta+\cos^4\theta$ then for all real values of θ :

A. $1\le \mathrm{A}\le 2$
B. $\frac{3}{4}\le \mathrm{A}\le 1$
C. $\frac{13}{16}\le \mathrm{A}\le 1$
D. $\frac{3}{4}\le \mathrm{A}\le \frac{13}{16}$

nielit2023feb-scientistd calculus

12.1 Continuity (2)

12.1.1 Continuity: NIELIT 2017 DEC Scientific Assistant A - Section B: 10

The function $f\left(x\right)=\dfrac{x^2-1}{x-1}$ at $x=1$ is :

A. Continuous and differentiable
B. Continuous but not differentiable
C. Differentiable but not continuous
D. Neither continuous nor differentiable

nielit2017dec-assistanta engineering-mathematics calculus continuity

Answer key☟

12.1.2 Continuity: NIELIT2017 STA-set-c-119

The function $f(x)=\frac{x^2-1}{x-1}$ at $x=1$ is:

A. Continuous and Differentiable
B. Continuous but not Differentiable
C. Differentiable but not Continuous
D. Neither Continuous nor Differentiable

nielit-july-2017 engineering-mathematics calculus continuity

Answer key☟

12.2 Coordinate Transformation (2)

12.2.1 Coordinate Transformation: NIELIT 2021 Dec Scientist B - Section B: 102

On changing to spherical co-ordinates, the integral $\iiint_V \, dy \, dx \, dz$, where V is the volume of the hemisphere $x^2+y^2+z^2=a^2$, is equivalent to the integral ______________.

A. $\displaystyle\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{a} r^2 \ \sin\theta \, dr \, d\theta \, d\phi$
B. $\displaystyle\int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2 \ \sin\theta \, dr \, d\theta \, d\phi$
C. $\displaystyle\int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\int_0^{a} r^2 \ \sin\theta \, dr \, d\theta \, d\phi$
D. $\displaystyle\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{a} r^2 \ \cos\theta \, dr \, d\theta \, d\phi$

nielit2021dec-scientistb calculus coordinate-transformation integrals

12.2.2 Coordinate Transformation: NIELIT 2021 Dec Scientist B - Section B: 35

On changing to spherical co-ordinates, the integral $\iiint_V \, dy \, dx \, dz$, where V is the volume of the hemisphere $x^2+y^2+z^2=a^2$, is equivalent to the integral ______________.

A. $\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^{a} r^2 \sin\theta \, dr \, d\theta \, d\phi$

B. $\int_0^{2\pi} \int_0^{\pi} \int_0^{a} r^2 \sin\theta \, dr \, d\theta \, d\phi$

C. $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^{a} r^2 \sin\theta \, dr \, d\theta \, d\phi$

D. $\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^{a} r^2 \cos\theta \, dr \, d\theta \, d\phi$

nielit-2021-it-dec-scientistb calculus integral coordinate-transformation

12.3 Definite Integral (4)

12.3.1 Definite Integral: NIELIT 2016 MAR Scientist B - Section B: 11

What is the derivative w.r.t x of the function given by

$$\Phi(x) = \int_0^{x^2} \sqrt{t} \, dt,$$

A. $2x^2$ B. $\sqrt{x}$ C. 0 D. 1

nielit2016mar-scientistb engineering-mathematics calculus integration definite-integral

Answer key☟

12.3.2 Definite Integral: NIELIT 2016 MAR Scientist B - Section B: 9

The value of improper integral

$$\int_0^1 x \ln x = ?$$

A. $1/4$ B. 0 C. $-1/4$ D. 1

nielit2016mar-scientistb engineering-mathematics calculus integration definite-integral

Answer key☟

12.3.3 Definite Integral: NIELIT 2016 MAR Scientist C - Section B: 11

$$\int_0^{\frac{\pi}{2}} \sin^7 \theta \cos^4 \theta d\theta = ?$$

A. $16/1155$ B. $16/385$ C. $16\pi/385$ D. $8\pi/385$

nielit2016mar-scientistc engineering-mathematics calculus definite-integral

Answer key☟

12.3.4 Definite Integral: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 19

The value of the Integral $I = \int_0^{\pi/2} x^2 \sin x dx$ is

A. $(x+2)/2$ B. $2/(\pi-2)$ C. $\pi-2$ D. $\pi+2$

nielit2017oct-assistanta-cs engineering-mathematics calculus definite-integral

Answer key☟

12.4 Differential Equation (1)

12.4.1 Differential Equation: NIELIT 2016 MAR Scientist B - Section B: 15

Differential equation, $\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$ will have a solution of the form

A. $(C_1 + C_2 t)e^{-5t}$
B. $C_1 e^{-2t}$
C. $C_1 e^{-5t} + C_2 e^{5t}$
D. $C_1 e^{-5t} + C_2 e^{2t}$

where C_1 and C_2 are constants.

nielit2016mar-scientistb non-gatecse differential-equation

12.5 Functions (2)

12.5.1 Functions: NIELIT 2022 April Scientist B | Section B | Question: 62

The function $f(x) = x(x+3)e^{-\frac{x}{2}}$ satisfies all the conditions of Rolle's theorem in $[-3, 0]$. The value of c is:

A. -3 B. -2 C. 3 D. 0

nielit2022apr-scientistb calculus functions

Answer key☟

12.5.2 Functions: NIELIT 2023 Feb Scientist C | Section D | Question: 105

The value of $f(x,y) = \left(\frac{\sqrt[4]{x^3y} - \sqrt[4]{xy^3}}{\sqrt{y} - \sqrt{x}} + \frac{1 + \sqrt[2]{xy}}{\sqrt[4]{xy}}\right)^{-2} \cdot \left(1 + 2\sqrt{\frac{y}{x}} + \frac{y}{x}\right)^{\frac{1}{2}}$ when $x = 9, y = 0.04$:

A. 0.64 B. 0.46 C. 0.32 D. 0.11

nielit2023feb-scientistc functions calculus

12.6 Integral (1)

12.6.1 Integral: NIELIT 2021 Dec Scientist A - Section B: 116

$\iint \frac{xy}{\sqrt{1-y^2}} dxdy.$ Over the positive quadrant of the circle $x^2+y^2=1$ is _______________.

A. $\frac{1}{6}$ B. $\frac{2}{3}$ C. $\frac{5}{6}$ D. $\frac{5}{3}$

nielit2021dec-scientista calculus integral geometry

12.7 Limits (2)

12.7.1 Limits: NIELIT 2016 MAR Scientist B - Section B: 13

$\lim\limits_{x \to 0} \frac{(1-\cos x)}{2}$ is equal to

A. 0 B. 1 C. $1/3$ D. $1/2$

nielit2016mar-scientistb engineering-mathematics calculus limits

Answer key☟

12.7.2 Limits: NIELIT 2016 MAR Scientist C - Section B: 10

$\lim\limits_{x \to 0} \frac{x^3+x^2-5x-2}{2x^3-7x^2+4x+4} = ?$

A. -0.5 B. (0.5) C. ∞ D. None of the above

nielit2016mar-scientistc engineering-mathematics calculus limits

Answer key☟

12.8 Maxima Minima (8)

12.8.1 Maxima Minima: NIELIT 2016 DEC Scientist B (CS) - Section B: 26

Consider the function $f(x) = \sin(x)$ in the interval $\left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$. The number and location(s) of the minima of this function are:

A. One, at $\frac{\pi}{2}$

B. One, at $\frac{3\pi}{2}$

C. Two, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$

D. Two, at $\frac{\pi}{4}$ and $\frac{3\pi}{2}$

nielit2016dec-scientistb-cs engineering-mathematics calculus maxima-minima

Answer key☟

12.8.2 Maxima Minima: NIELIT 2016 MAR Scientist B - Section B: 10

Maxima and minimum of the function

$f(x) = 2x^3 - 15x^2 + 36x + 10$ occur; respectively at

A. $x = 3$ and $x = 2$ B. $x = 1$ and $x = 3$ C. $x = 2$ and $x = 3$ D. $x = 3$ and $x = 4$

nielit2016mar-scientistb engineering-mathematics calculus maxima-minima

Answer key☟

12.8.3 Maxima Minima: NIELIT 2016 MAR Scientist B - Section B: 14

The minimum value of $|x^2 - 5x + 2|$ is

A. -5 B. 0 C. -1 D. -2

nielit2016mar-scientistb engineering-mathematics calculus maxima-minima

Answer key☟

12.8.4 Maxima Minima: NIELIT 2016 MAR Scientist B - Section B: 5

The greatest and the least value of

$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 1$ in $[0, 2]$ are

A. $0, 8$ B. $0, -8$ C. $1, 8$ D. $1, -8$

nielit2016mar-scientistb engineering-mathematics calculus maxima-minima

Answer key☟

12.8.5 Maxima Minima: NIELIT 2016 MAR Scientist C - Section B: 2

The function $f(x) = x^5 - 5x^4 + 5x^3 - 1$ has

A. one minima and two maxima
B. two minima and one maxima
C. two minima and two maxima
D. one minima and one maxima

nielit2016mar-scientistc calculus maxima-minima

Answer key☟

12.8.6 Maxima Minima: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 18

What is the maximum value of the function $f(x) = 2x^2 - 2x + 6$ in the interval $[0, 2]$?

A. 6 B. 10 C. 12 D. $5, 5$

nielit2017oct-assistanta-cs engineering-mathematics calculus maxima-minima

Answer key☟

12.8.7 Maxima Minima: NIELIT 2021 Dec Scientist A - Section B: 70

With usual notations, the properties of maxima and minima under various conditions are ____________ .

	I		II
(P)	Maxima	(i)	$rt-s^2=0$
(Q)	Minima	(ii)	$rt-s^2<0$
(R)	Saddle Point	(iii)	$rt-s^2>0, r>0$
(S)	Case of failure	(iv)	$rt-s^2>0, r<0$

A. P – i, Q – iii, R – iv, S – ii
B. P – ii, Q – i, R – iii, S – iv
C. P – iii, Q – iv, R – ii, S – i
D. P – iv, Q – iii, R – ii, S – i

nielit2021dec-scientista calculus maxima-minima

12.8.8 Maxima Minima: NIELIT 2022 April Scientist B | Section B | Question: 112

The function $f(x,y)=x^2+y^2+6x+12$ has :

A. minimum value, -3 and maximum value, 0.
B. only minimum value, 3.
C. only maximum value, 12.
D. neither maxima nor minima.

nielit2022apr-scientistb calculus maxima-minima functions

12.9 Numerical Methods (1)

12.9.1 Numerical Methods: NIELIT 2022 April Scientist B | Section B | Question: 88

The secant method formula for finding the square root of a real number R from the equation $x^2-\mathrm{R}=0$ is:

A. $x_{i+1}=\frac{x_i.x_{i-1}}{x_i+x_{i+1}}$
B. $x_{i+1}=\frac{x_i.x_{i-1}+\mathrm{R}}{x_i+x_{i-1}}$
C. $x_{i+1}=\frac{1}{2}\left(x_i+\frac{\mathrm{R}}{x_i}\right)$
D. $x_{i+1}=\frac{2x_i^2+x_i.x_{i-1}-\mathrm{R}}{x_i+x_{i-1}}$

nielit2022apr-scientistb calculus numerical-methods

Answer key☟

12.10 Partial Order (2)

12.10.1 Partial Order: NIELIT 2021 Dec Scientist B - Section B: 41

If $z=\cos\left(\frac{x}{y}\right)+\sin\left(\frac{x}{y}\right)$, then xz_x+yz_y is equal to _________.

A. z B. $2z$ C. $4z$ D. 0

nielit-2021-it-dec-scientistb calculus partial-order

12.10.2 Partial Order: NIELIT 2021 Dec Scientist B - Section B: 95

If $z = \cos\left(\frac{x}{y}\right) + \sin\left(\frac{x}{y}\right)$, then $xz_x + yz_y$ is equal to ______________ .

A. z B. $2z$ C. $4z$ D. 0

nielit2021dec-scientistb calculus partial-order functions

12.11 Quadratic Equations (1)

12.11.1 Quadratic Equations: NIELIT 2023 Feb Scientist C | Section D | Question: 95

Find the roots of equation $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x} : a + b \neq 0$.

A. a, b
B. $-\mathrm{a}, -\mathrm{b}$
C. $2\mathrm{a}, 3\ \mathrm{b}$
D. $3\mathrm{a}, 2\ \mathrm{b}$

nielit2023feb-scientistc quadratic-equations algebra

Answer Keys

12.0.1	C	12.0.2	B	12.0.3	D	12.0.4	B	12.1.1	D
12.1.2	D	12.2.1	A	12.2.2	A	12.3.1	A	12.3.2	C
12.3.3	A	12.3.4	C	12.4.1	A	12.5.1	B	12.5.2	A
12.6.1	A	12.7.1	A	12.7.2	A	12.8.1	D	12.8.2	C
12.8.3	B	12.8.4	D	12.8.5	A	12.8.6	B	12.8.7	D
12.8.8	B	12.9.1	C	12.10.1	D	12.10.2	D	12.11.1	B

13 Engineering Mathematics: Linear Algebra (21)

13.0.1 NIELIT 2018-27

The following vectors $(1,9,9,8),(2,0,0,8),(2,0,0,3)$ are

A. Linearly dependent
B. Linearly independent
C. Constant
D. None of these

nielit-2018 engineering-mathematics linear-algebra

Answer key☟

13.1 Determinant (8)

13.1.1 Determinant: NIELIT 2016 DEC Scientist B (CS) - Section B: 21

Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = 1$, then B^{-1} is:

A. $D^{-1}C^{-1}A^{-1}$
B. CDA
C. ADC
D. Does not necessarily exist.

nielit2016dec-scientistb-cs engineering-mathematics linear-algebra matrix determinant

Answer key☟

13.1.2 Determinant: NIELIT 2016 DEC Scientist B (IT) - Section B: 26

Two eigenvalues of a 3×3 real matrix P are $(2+\sqrt{-1})$ and 3. The determinant of P is ________.

A. 0
B. 1
C. 15
D. -1

nielit2016dec-scientistb-it engineering-mathematics linear-algebra determinant eigen-value

Answer key☟

13.1.3 Determinant: NIELIT 2016 MAR Scientist B - Section B: 12

If A and B are square matrices of size $n \times n$, then which of the following statements is not true?

A. $\det(AB) = \det(A)\det(B)$
B. $\det(kA) = k^n \det(A)$
C. $\det(A+B) = \det(A) + \det(B)$
D. $\det(A^T) = 1/\det(A^{-1})$

nielit2016mar-scientistb engineering-mathematics linear-algebra determinant

Answer key☟

13.1.4 Determinant: NIELIT 2016 MAR Scientist B - Section B: 4

What is the determinant of the matrix

$$\begin{bmatrix} 5 & 3 & 2 \\ 1 & 2 & 6 \\ 3 & 5 & 10 \end{bmatrix}$$

A. -76 B. -28 C. $+28$ D. $+72$

nielit2016mar-scientistb engineering-mathematics linear-algebra determinant

Answer key☟

13.1.5 Determinant: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 9

M is a square matrix of order $'n'$ and its determinant value is $5.$ If all the elements of M are multiple by $2,$ its determinant value becomes $40.$ The value of $'n'$ is

A. 2 B. 3 C. 5 D. 4

nielit2017oct-assistanta-cs engineering-mathematics linear-algebra matrix determinant

Answer key☟

13.1.6 Determinant: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 7

M is a square matrix of order $'n'$ and its determinant value is $5.$ If all the elements of M are multiplied by $2,$ its determinant value becomes $40.$ The value of $'n'$ is

A. 2 B. 3 C. 5 D. 4

nielit2017oct-assistanta-it linear-algebra matrix determinant

Answer key☟

13.1.7 Determinant: NIELIT 2021 Dec Scientist B - Section B: 18

The determinant of matrix $\begin{bmatrix} 0 & p-q & p-r \\ q-p & 0 & q-r \\ r-p & r-q & 0 \end{bmatrix}$ is ____________.

A. 0
B. $(p-q)(q-r)(r-p)$
C. pqr
D. $3pqr$

nielit-2021-it-dec-scientistb matrix determinant linear-algebra

Answer key☟

13.1.8 Determinant: NIELIT 2021 Dec Scientist B - Section B: 58

The determinant of matrix $\begin{bmatrix} 0 & p-q & p-r \\ q-p & 0 & q-r \\ r-p & r-q & 0 \end{bmatrix}$ is ____________ .

A. 0 B. $(p-q)(q-r)(r-p)$ C. pqr D. $3pqr$

nielit2021dec-scientistb matrix determinant

Answer key

13.2 Eigen Value (3)

13.2.1 Eigen Value: NIELIT 2016 MAR Scientist C - Section B: 8

The eigenvalues of the matrix $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ are

A. 5 and -5 B. 5 and -1 C. 1 and -5 D. 2 and 3

nielit2016mar-scientistc engineering-mathematics linear-algebra eigen-value

Answer key

13.2.2 Eigen Value: NIELIT 2018-21

The value of p such that the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is an eigen vector of the matrix $\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$ is

A. 15 B. 16 C. 17 D. 18

nielit-2018 engineering-mathematics linear-algebra matrix eigen-value

Answer key

13.2.3 Eigen Value: NIELIT 2022 April Scientist B | Section B | Question: 72

The Eigen values of a matrix

$\mathrm{A} = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ are $-3, -3$ and 5, then the trace of the matrix $A^3 - 3A^2$ is:

A. 200 B. 71 C. -58 D. -200

nielit2022apr-scientistb linear-algebra matrix eigen-value

Answer key

13.3 Eigen Vector (1)

13.3.1 Eigen Vector: NIELIT 2016 MAR Scientist C - Section B: 9

Consider three vectors $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, y = \begin{bmatrix} 4 \\ 8 \end{bmatrix}, z = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Which of the folowing statements is true

A. x and y are linearly independent
B. x and y are linearly dependent
C. x and z are linearly dependent
D. y and z are linearly dependent

nielit2016mar-scientistc engineering-mathematics linear-algebra eigen-vector

Answer key☟

13.4 Matrix (5)

13.4.1 Matrix: NIELIT 2016 MAR Scientist C - Section B: 19

If product of matrix $A = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$ is a null matrix, then θ and ϕ differ by an

A. odd multiple of π
B. even multiple of π
C. odd multiple of $\frac{\pi}{2}$
D. even multiple of $\frac{\pi}{2}$

nielit2016mar-scientistc linear-algebra matrix

13.4.2 Matrix: NIELIT 2016 MAR Scientist C - Section B: 3

The matrices $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & cos\theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under the multiplication

A. i f $a = b(\text{or})\theta = n\pi$, n is an integer
B. always
C. never
D. if $a\cos\theta \neq b\sin\theta$

nielit2016mar-scientistc linear-algebra matrix

Answer key☟

13.4.3 Matrix: NIELIT 2017 DEC Scientist B - Section B: 60

Consider two matrices M_1 and M_2 with $M_1^* M_2 = 0$ and M_1 is non singular. Then which of the following is true?

A. M_2 is non singular
B. M_2 is null matrix
C. M_2 is the identity matrix
D. M_2 is transpose of M_1

nielit2017dec-scientistb engineering-mathematics linear-algebra matrix

Answer key☟

13.4.4 Matrix: NIELIT 2018-30

If C is a non-singular matrix and $B = C \begin{bmatrix} 0 & x & y \\ 0 & 0 & x \\ 0 & 0 & 0 \end{bmatrix} C^{-1}$ then:

A. $B^2 = I$
B. $B^2 = \text{Null Matrix}$
C. $B^3 = I$
D. $B^3 = \text{Null Matrix}$

nielit-2018 engineering-mathematics linear-algebra matrix

Answer key☟

13.4.5 Matrix: NIELIT 2021 Dec Scientist A - Section B: 82

If for the matrix $\text{A}, \text{A}^3 = I$ then $\text{A}^{-1} =$ __________ .

A. A^2
B. A^3
C. A
D. None of these

nielit2021dec-scientista matrix linear-algebra

Answer key☟

13.5 System of Equations (3)

13.5.1 System of Equations: NIELIT 2016 MAR Scientist B - Section B: 6

The system of simultaneous equations

$x + 2y + z = 6$

$2x + y + 2z = 6$

$x + y + z = 5$

has

A. unique solution.
B. infinite number of solutions.
C. no solution.
D. exactly two solutions.

nielit2016mar-scientistb engineering-mathematics linear-algebra system-of-equations

Answer key☟

13.5.2 System of Equations: NIELIT 2022 April Scientist B | Section B | Question: 80

The upper triangular matrix U in the LU- decomposition of the matrix given below :

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -5 & 8 \\ 1 & 2 & -10 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \text{ is :}$$

A. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -4 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 11 \end{bmatrix}$

D. $\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 10 \end{bmatrix}$

nielit2022apr-scientistb matrix linear-algebra system-of-equations

Answer key☟

13.5.3 System of Equations: NIELIT Scientist B 2020 November: 27

What is the value of k for which the following system of equations has no solution: $\text{2x-8y=3 and kx+4y=10}$

A. -2 B. 1 C. -1 D. 2

nielit-scb-2020 system-of-equations

Answer key☟

Answer Keys

13.0.1	D	13.1.1	B	13.1.2	C	13.1.3	C	13.1.4	B
13.1.5	B	13.1.6	B	13.1.7	A	13.1.8	A	13.2.1	B
13.2.2	D	13.2.3	C	13.3.1	B	13.4.1	C	13.4.2	A
13.4.3	B	13.4.4	A	13.4.5	A	13.5.1	C	13.5.2	TBA
13.5.3	C								

14 Engineering Mathematics: Probability (21)

14.1 Conditional Probability (1)

14.1.1 Conditional Probability: NIELIT Scientific Assistant A 2020 November: 99

A bag contains 10 white balls and 5 blue balls. A ball is drawn from the bag and its color is noted. This ball is put back in the bag along with 3 more balls of the same color. A ball is drawn again from the bag at random. The probability that the first ball drawn is blue, given that the second ball drawn is blue, is:

A. $1/3$ B. $3/4$ C. $8/9$ D. $4/9$

nielit-sta-2020 probability conditional-probability

Answer key☟

14.2 Counting (2)

14.2.1 Counting: NIELIT 2021 Dec Scientist A - Section A: 25

Out of 13 applicants for a job there are 5 women and 8 men. Two persons are to be selected for the job. Find the probability that at least one of the selected persons will be a women.

A. $25/39$ B. $10/21$ C. $14/27$ D. $12/51$

nielit2021dec-scientista probability counting

Answer key☟

14.2.2 Counting: NIELIT Scientist B 2020 November: 38

In a group of 24 members, each member drinks either tea or coffee or both. If 15 of them drink tea and 18 drink coffee, find the probability that a person selected from the group drinks both tea and coffee.

A. $1/8$ B. $3/8$ C. $5/24$ D. None of the options

nielit-scb-2020 probability counting analytical-aptitude

Answer key☟

14.3 Expectation (3)

14.3.1 Expectation: NIELIT 2021 Dec Scientist B - Section B: 52

Let the random variable X has a mixed distributions with probability $\mathrm{P(X=0)}=\alpha$, and the density function.

$$f_x(x)=\begin{cases}\beta x^2(1-x), & 0<x<1\\ 0, & \text{otherwise}\end{cases}$$

If the expectation of X is α, then the value of $4\alpha+\beta$ is equal to :

A. $9/2$ B. 6 C. 9 D. $5/2$

nielit-2021-it-dec-scientistb probability expectation random-variable nielit2021dec-scientistb

14.3.2 Expectation: NIELIT 2021 Dec Scientist B - Section B: 71

Let the random variable X has a mixed distributions with probability $\mathrm{P}(\mathrm{X}=0)=\alpha$, and the density function.

$$f_x(x)=\begin{cases}\beta x^2(1-x), 0<x<1\\ 0, \text{otherwise}\end{cases}$$

If the expectation of X is α, then the value of $4\alpha+\beta$ is equal to :

A. $9/2$ B. 6 C. 9 D. $5/2$

nielit2021dec-scientistb probability random-variable expectation

14.3.3 Expectation: NIELIT Scientist B 2020 November: 29

A player rolls a die and receives the same number of rupees as the number of dots on the face that turns up. What should the player pay for each roll if he wants to make a profit of one rupee per throw of the die in the long run?

A. ₹ 2.50 B. ₹ 2 C. ₹ 3.50 D. ₹ 4

nielit-scb-2020 probability expectation quantitative-aptitude

Answer key☟

14.4 Independent Events (1)

14.4.1 Independent Events: NIELIT 2021 Dec Scientist A - Section B: 79

Let X be uniform random variable on $[0,4]$ and Y be uniform random variable on $[0,1]$. If X and Y are independent, then $\mathrm{P}\left(\max\{\mathrm{X},\mathrm{Y}\}>3\right)$ is equal to :

A. $1/4$ B. $1/2$ C. $1/8$ D. 1

nielit2021dec-scientista probability random-variable independent-events

Answer key☟

14.5 Normal Distribution (1)

14.5.1 Normal Distribution: NIELIT 2018-24

If $f(x)=k$ exp, $\{-(9x^2-12x+13)\}$, is a p,d,f of a normal distribution (k, being a constant), the mean and standard deviation of the distribution:

A. $\mu=\frac{2}{3},\sigma=\frac{1}{3\sqrt{2}}$ B. $\mu=2,\sigma=\frac{1}{\sqrt{2}}$

C. $\mu = \frac{1}{3}, \sigma = \frac{1}{3\sqrt{2}}$

D. $\mu = \frac{2}{3}, \sigma = \frac{1}{\sqrt{3}}$

nielit-2018 engineering-mathematics probability normal-distribution

Answer key

14.6 Probability (10)

14.6.1 Probability: NIELIT 2016 MAR Scientist B - Section B: 16

A box contains 10 screws, 3 of which are defective. Two screws are drawn at random with replacement. The probability that none of two screws is defective will be

A. 100%

B. 50%

C. 49%

D. None of these.

nielit2016mar-scientistb engineering-mathematics probability

Answer key

14.6.2 Probability: NIELIT 2016 MAR Scientist C - Section B: 4

If A and B are two related events, and $P(A \mid B)$ represents the conditional probability, Bayes' theorem states that

A. $P(A \mid B) = \dfrac{P(A)}{P(B)} P(B \mid A)$

B. $P(A \mid B) = P(A)P(B)P(B \mid A)$

C. $P(A \mid B) = \dfrac{P(A)}{P(B)}$

D. $P(A \mid B) = P(A) + P(B)$

nielit2016mar-scientistc engineering-mathematics probability

Answer key

14.6.3 Probability: NIELIT 2017 DEC Scientific Assistant A - Section B: 41

If a random coin is tossed 11 times, then what is the probability that for 7th toss head appears exactly 4 times?

A. $5/32$

B. $15/128$

C. $35/128$

D. None of the options

nielit2017dec-assistanta engineering-mathematics probability

Answer key

14.6.4 Probability: NIELIT 2017 DEC Scientific Assistant A - Section B: 44

If X, Y and Z are three exhaustive and mutually exclusive events related with any experiment and the $P(X) = 0.5P(Y)$ and $P(Z) = 0.3P(Y)$. Then $P(Y)$ = __________ .

A. 0.54

B. 0.66

C. 0.33

D. 0.44

nielit2017dec-assistanta engineering-mathematics probability

Answer key

14.6.5 Probability: NIELIT 2017 July Scientist B (IT) - Section B: 23

The probability that top and bottom cards of a randomly shuffled deck are both aces is:

A. $4/52 \times 4/52$
B. $4/52 \times 3/52$
C. $4/52 \times 3/51$
D. $4/52 \times 4/51$

nielit2017july-scientistb-it engineering-mathematics probability

Answer key☟

14.6.6 Probability: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 16

If P is risk probability, L is loss, then Risk Exposure (RE) is computed as.

A. $RE = P/L$
B. $RE = P + L$
C. $RE = P * L$
D. $RE = 2 * P * L$

nielit2017oct-assistanta-cs engineering-mathematics probability

Answer key☟

14.6.7 Probability: NIELIT 2018-26

If $P(A \cap B) = \frac{1}{2}$, $P(\overline{A} \cap \overline{B}) = \frac{1}{2}$ and $2P(A) = P(B) = p$, then the value of p is given by:

A. $\frac{1}{4}$
B. $\frac{1}{2}$
C. $\frac{1}{3}$
D. $\frac{2}{3}$

nielit-2018 engineering-mathematics probability

14.6.8 Probability: NIELIT 2021 Dec Scientist A - Section A: 34

First bag contains 5 white and 4 black balls. Second bag contains 7 white and 9 black balls. A ball is transferred from the first bag to the second bag and then a ball is drawn from the second bag. Find the probability that the ball drawn is white.

A. $7/18$
B. $5/9$
C. $4/9$
D. $11/18$

nielit2021dec-scientista probability

Answer key☟

14.6.9 Probability: NIELIT 2021 Dec Scientist A - Section B: 108

The covariance function of a band limited white noise is :

A. A Dirac delta function
B. An exponentially decreasing function
C. A sinc function
D. A $\operatorname{sinc}2$ function

nielit2021dec-scientista probability

Answer key☟

14.6.10 Probability: NIELIT 2022 April Scientist B | Section B | Question: 114

For the probability density function $f(x) = cx^2(1-x), 0 < x < 1$, the value of constant c and mean of the distribution are :

A. $6, 0.4$ B. $8, 0.5$ C. $12, 0.6$ D. $8, 0.6$

nielit2022apr-scientistb probability

Answer key☟

14.7 Random Variable (1)

14.7.1 Random Variable: NIELIT 2018-14

A continuous random variable x is distributed over the interval $[0, 2]$ with probability density function $f(x) = ax^2 + bx$, where a and b are constants. If the mean of the distribution is $\frac{1}{2}$. Find the values of the constants a and b.

A. $a = 2, b = -\frac{13}{6}$
B. $a = -\frac{15}{8}, b = 3$
C. $a = -\frac{29}{6}, b = 2$
D. $a = 3, b = -\frac{7}{2}$

nielit-2018 engineering-mathematics probability random-variable

Answer key☟

14.8 Statistics (1)

14.8.1 Statistics: NIELIT 2016 MAR Scientist B - Section B: 17

Following marks are obtained by the students in a test:
$81, 72, 90, 90, 86, 85, 92, 70, 71, 83, 89, 95, 85, 79, 62$. Range of the marks is

A. 9 B. 17 C. 27 D. 33

nielit2016mar-scientistb statistics

Answer key☟

14.9 Variance (1)

14.9.1 Variance: NIELIT Scientist B 2020 November: 112

Let $X_1, \dots X_{50}$ be independent random variables following $N(0, 1)$ distribution. Let $\Sigma_{i=1}^{50} X_i^2$, and $E(Y) = a$ and $Var(Y) = b$. Then, the ordered pair (a, b) is:

A. $(50, 100)$ B. $(50, 50)$ C. $(25, 50)$ D. $(25, 100)$

nielit-scb-2020 normal-distribution probability random-variable expectation variance

Answer Keys

14.1.1	D	14.2.1	A	14.2.2	B	14.3.1	C	14.3.2	C
14.3.3	A	14.4.1	A	14.5.1	A	14.6.1	C	14.6.2	A
14.6.3	A	14.6.4	A	14.6.5	C	14.6.6	A	14.6.7	D
14.6.8	C	14.6.9	C	14.6.10	C	14.7.1	B	14.8.1	D
14.9.1	A								

15 General Aptitude: Analytical Aptitude (183)

15.0.1 NIELIT Scientist B 2020 November: 23

If $09/12/2001(DD/MM/YYYY)$ happens to be Sunday, then $09/12/1971$ would have been a:

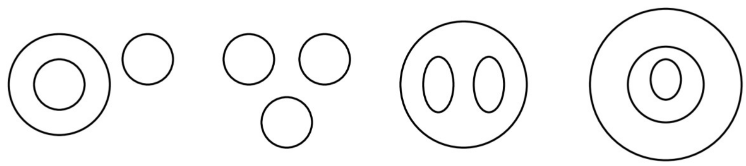
A. Wednesday B. Tuesday C. Saturday D. Thursday

nielit-scb-2020 analytical-aptitude quantitative-aptitude

Answer key☟

15.0.2 NIELIT 2021 Dec Scientist B - Section A: 1

Use following diagrams to answer question number 1 to 2 :



(I) (II) (III) (IV)

Cloud, River, Mountain :

A. II B. I C. IV D. III

nielit2021dec-scientistb analytical-aptitude spatial-aptitude

Answer key☟

15.0.3 NIELIT 2021 Dec Scientist A - Section A: 12

Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

A Training college has to conduct a refresher course for teachers of seven different subjects – Mechanics, Psychology, Philosophy, Sociology, Economics, Science and Engineering from November 22 to November 29.

i. Course should start with Psychology.
ii. November 23, being Sunday, should be a holiday.
iii. Science subject should be on the previous day of the Engineering subject.
iv. Course should end with Mechanics subject.
v. Philosophy should be immediately after holiday.
vi. There should be a gap of one day between Economics and Engineering.
vii. There should be a gap of two days between Sociology and Economics.

Which subject will be on Tuesday?

A. Psychology B. Mechanics C. Economics D. Sociology

nielit2021dec-scientista analytical-aptitude

Answer key☟

15.0.4 NIELIT 2021 Dec Scientist A - Section A: 10

Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

A Training college has to conduct a refresher course for teachers of seven different subjects – Mechanics, Psychology, Philosophy, Sociology, Economics, Science and Engineering from November 22 to November $29.$

i. Course should start with Psychology.
ii. November $23,$ being Sunday, should be a holiday.
iii. Science subject should be on the previous day of the Engineering subject.
iv. Course should end with Mechanics subject.
v. Philosophy should be immediately after holiday.
vi. There should be a gap of one day between Economics and Engineering.
vii. There should be a gap of two days between Sociology and Economics.

How many days' gap is there between Science and Philosophy?

A. 1 B. 2 C. 3 D. No gap

nielit2021dec-scientista analytical-aptitude

Answer key☟

15.0.5 NIELIT 2022 April Scientist B | Section A | Question: 30

Find the odd one out :

$4, 9, 256, 529, 573$

A. 256 B. 573 C. 529 D. 9

nielit2022apr-scientistb analytical-aptitude quantitative-aptitude

Answer key☟

15.0.6 NIELIT 2023 Feb Scientist D | Section D | Question: 113

I f $47.2506 = 4\text{A} + \frac{7}{\text{B}} + 2\text{C} + \frac{5}{\text{D}} + 6\text{E}$ then the value of $5\text{ A}+3\text{ B}+6\text{ C}+\text{D}+3\text{ E}$ is :

A. 53.6003 B. 53.6031 C. 153.6003 D. 213.0003

nielit2023feb-scientistd analytical-aptitude mathematical-logic

Answer key☟

15.0.7 NIELIT 2023 Feb Scientist C | Section A | Question: 8

$3265 : 4376 :: 4673 :?$

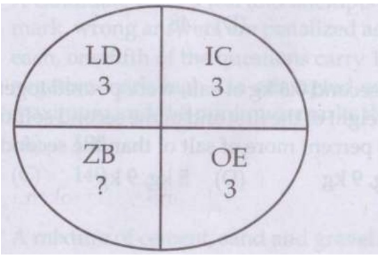
A. 2154 B. 3562 C. 5487 D. 5784

nielit2023feb-scientistc analytical-aptitude

Answer key☟

15.0.8 NIELIT 2023 Feb Scientist C | Section C | Question: 78

In the given below question, which character when placed at the sign of (?) shall complete the pattern?



A. 3 B. 16 C. 5 D. 13

nielit2023feb-scientistc analytical-aptitude

15.1 Alphabetic Series (3)

15.1.1 Alphabetic Series: NIELIT 2023 Feb Scientist D | Section A | Question: 1

If you pick up from following alphabet, the sixth and fourteenth letters from your right and then pick up the fifth and twentieth letters from your left and form a meaningful word, what is the first letter of that word?

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

A. M
B. E
C. No word can be formed
D. More than one word can be formed

nielit2023feb-scientistd logical-reasoning analytical-aptitude alphabetic-series

Answer key☟

15.1.2 Alphabetic Series: NIELIT Scientific Assistant A 2020 November: 13

In a certain code, 'CONSIDER' is written as RMNBSFEJ, how is 'MONOPOLY' written in that code?

A. LNMNZMPQ B. NMNLZMPQ C. POPNXKNO D. NMNLXKNO nielit-sta-2020 analytical-aptitude alphabetic-series

Answer key

15.1.3 Alphabetic Series: NIELIT Scientific Assistant A 2020 November: 8

If 'CONTEMPORARY' is coded as 'NOCTEMROPARY' then 'BODARDSITAND' is the code of which letter?

A. DOBARDTISAND B. BODDRASITDNA C. DOBDRATISDNA D. DOBARDSITAND

nielit-sta-2020 analytical-aptitude alphabetic-series

Answer key

15.2 Arithmetic Series (1)

15.2.1 Arithmetic Series: NIELIT 2023 Feb Scientist D | Section D | Question: 111

The value of $2-[3-\{6-(5\div\overline{4-3})\}]$ is equal to:

A. 0 B. 1 C. 3 D. -3

nielit2023feb-scientistd arithmetic-series analytical-aptitude

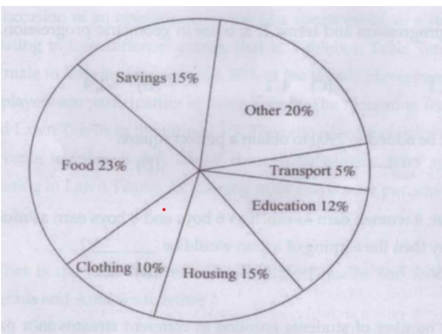
Answer key

15.3 Bar Graph (1)

15.3.1 Bar Graph: NIELIT 2023 Feb Scientist D | Section C | Question: 81

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



Graph shows that the maximum amount was spent on :

A. Food B. Housing C. Clothing D. Others nielit2023feb-scientistd analytical-aptitude data-interpretation bar-graph

15.4 Blood Relation (2)

15.4.1 Blood Relation: NIELIT Scientific Assistant A 2020 November: 3

A is the father of B and C is the son of D. E is the brother of A. B is the sister of C. How is D related to E?

A. Daughter B. Brother C. Brother in Law D. Sister in Law

nielit-sta-2020 analytical-aptitude blood-relation

Answer key☟

15.4.2 Blood Relation: NIELIT Scientific Assistant A 2020 November: 4

Pointing towards a picture, Ramesh said, "That picture is of sister of grandson of father of my maternal uncle." How is that lady in the picture related to Ramesh?

A. Mother's sister
B. Cousin (maternal brother)
C. Cousin (maternal sister)
D. Father's sister

nielit-sta-2020 analytical-aptitude blood-relation

Answer key☟

15.5 Counting (1)

15.5.1 Counting: NIELIT Scientist B 2020 November: 30

Five of India's leading models are posing for a photograph promoting "World Peace and Understanding".

But then, Sachin Malhotra the photographer is having a tough time getting them to stand in a straight line, because Natasha refuses to stand in a straight line, because Natasha refuses to stand next to Jessica since Jessica had said something about her in a leading gossip magazine. Rachel and Anna want to stand together because they are good friends. Ria on the other hand cannot get along well with Rachel, because there is some talk about Rachel scheming to get a contract already awarded to Ria. Anna believes her friendly astrologer who has asked her to stand at the extreme right for all group photographs. Finally, Sachin managed to pacify the girls and got a beautiful picture of five beautiful girls smiling beautifully in a straight line, promoting world peace.

If Anna's astrologer tells her to stand second from left and Natasha decides to stand second from right, then who is the girl standing at the extreme right?

A. Rachel B. Jessica C. Ria D. None of the options

nielit-scb-2020 analytical-aptitude counting

Answer key☟

15.6 Direction Sense (3)

15.6.1 Direction Sense: NIELIT 2021 Dec Scientist B - Section A: 6

Rakesh is standing at a point. He walks 20 m towards the East and further 10 m towards the South, then he walks 35 m towards the West and further 5 m towards the North, then he walks 15 m towards the East. What is the straight distance in metres between his starting point and the point where he reached last ?

A. 0 B. 5 C. 10 D. 15

nielit2021dec-scientistb analytical-aptitude direction-sense quantitative-aptitude

Answer key☟

15.6.2 Direction Sense: NIELIT 2023 Feb Scientist C | Section A | Question: 26

Sanjeev walks 10 metres towards the South. Turning to the left, he walks 20 metres and then moves to his right. After moving a distance of 20 metres, he turns to the right and walks 20 metres. Finally, he turns to the right and moves a distance of 10 metres. How far and in which direction is he from the starting point?

A. 10 metres North
B. 20 metres South
C. 20 metres North
D. 10 metres South

nielit2023feb-scientistc analytical-aptitude direction-sense

15.6.3 Direction Sense: NIELIT 2023 Feb Scientist D | Section C | Question: 88

A woman walks 5 km towards North and then turns to left. After walking 3 km she turns to right and walks 5 km. Now in which direction is she from the starting point?

A. North - West B. North C. South - West D. North - East

nielit2023feb-scientistd direction-sense analytical-aptitude

15.7 Distance (1)

15.7.1 Distance: NIELIT 2021 Dec Scientist A - Section A: 5

John's house is 100 m North of his uncle's office. His uncle's house is located 200 m West of his (uncle's) office. Kabir is the friend of John and he stays 100 m East of John's house. The office of Kabir is located 100 m South of his house. Then, how far is his uncle's house from Kabir's office ?

A. 200 m B. 300 m C. 400 m D. 500 m

nielit2021dec-scientista analytical-aptitude distance spatial-aptitude

Answer key☟

15.8 Family Relationship (23)

15.8.1 Family Relationship: NIELIT 2021 Dec Scientist A - Section A: 14

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

There are six women, Shalini, Divya, Ritu, Rashmi, Nisha and Renu in a family of 12 members. There are few married couples in the family and none of the grand children are married. Sunil is married into the family. Rohan, Mahesh and Jatin have a nephew Dipesh who is the son of Rashmi. Ravi is the paternal grandfather of Nisha. Ritu is the daughter-in-law of Shalini. Renu is the first cousin of Dipesh. Shalini has only three grand children. Mahesh has two brothers and only one sister Rashmi and a sister-in-law Divya. Dipesh's only unmarried maternal uncle Jatin is the brother-in-law of Sunil. Rohan is the paternal uncle of Nisha. Ritu has two daughters one of Whom is Nisha.

Rashmi is _____________ .

A. Mahesh's wife
B. Renu's aunt
C. Nisha's Mother
D. None of these

nielit2021dec-scientista logical-reasoning family-relationship

Answer key☟

15.8.2 Family Relationship: NIELIT 2021 Dec Scientist A - Section A: 15

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

There are six women, Shalini, Divya, Ritu, Rashmi, Nisha and Renu in a family of 12 members. There are few married couples in the family and none of the grand children are married. Sunil is married into the family. Rohan, Mahesh and Jatin have a nephew Dipesh who is the son of Rashmi. Ravi is the paternal grandfather of Nisha. Ritu is the daughter-in-law of Shalini. Renu is the first cousin of Dipesh. Shalini has only three grand children. Mahesh has two brothers and only one sister Rashmi and a sister-in-law Divya. Dipesh's only unmarried maternal uncle Jatin is the brother-in-law of Sunil. Rohan is the paternal uncle of Nisha. Ritu has two daughters one of Whom is Nisha.

Which one of the following is a married couple ?

A. Rohan and Ritu
B. Shalini and Mahesh
C. Renu and Sunil
D. Mahesh and Ritu

nielit2021dec-scientista logical-reasoning family-relationship

Answer key☟

15.8.3 Family Relationship: NIELIT 2021 Dec Scientist A - Section A: 16

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

There are six women, Shalini, Divya, Ritu, Rashmi, Nisha and Renu in a family of 12 members. There are few married couples in the family and none of the grand children are married. Sunil is married into the family. Rohan, Mahesh and Jatin have a nephew Dipesh who is the son of Rashmi. Ravi is the paternal grandfather of Nisha. Ritu is the daughter-in-law of Shalini. Renu is the first cousin of Dipesh. Shalini has only three grand children. Mahesh has two brothers and only one sister Rashmi and a sister-in-law Divya. Dipesh's only unmarried maternal uncle Jatin is the brother-in-law of Sunil. Rohan is the paternal uncle of Nisha. Ritu has two daughters one of Whom is Nisha.

Dipesh is __________________ .

A. Mahesh's son
B. Ravi's grand son
C. Rohan's son
D. Sunil's nephew

nielit2021dec-scientista logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.4 Family Relationship: NIELIT 2021 Dec Scientist A - Section A: 17

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

There are six women, Shalini, Divya, Ritu, Rashmi, Nisha and Renu in a family of 12 members. There are few married couples in the family and none of the grand children are married. Sunil is married into the family. Rohan, Mahesh and Jatin have a nephew Dipesh who is the son of Rashmi. Ravi is the paternal grandfather of Nisha. Ritu is the daughter-in-law of Shalini. Renu is the first cousin of Dipesh. Shalini has only three grand children. Mahesh has two brothers and only one sister Rashmi and a sister-in-law Divya. Dipesh's only unmarried maternal uncle Jatin is the brother-in-law of Sunil. Rohan is the paternal uncle of Nisha. Ritu has two daughters one of Whom is Nisha.

How many married couples are there in the second generation?

A. 1
B. 2
C. 3
D. 4

nielit2021dec-scientista logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.5 Family Relationship: NIELIT 2021 Dec Scientist A - Section A: 18

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

There are six women, Shalini, Divya, Ritu, Rashmi, Nisha and Renu in a family of 12 members. There are few married couples in the family and none of the grand children are married. Sunil is married into the family. Rohan, Mahesh and Jatin have a nephew Dipesh who is the son of Rashmi. Ravi is the paternal grandfather of Nisha. Ritu is the daughter-in-law of Shalini. Renu is the first cousin of Dipesh. Shalini has only three grand children. Mahesh has two brothers and only one sister Rashmi and a sister-in-law Divya. Dipesh's only unmarried maternal uncle Jatin is the brother-in-law of Sunil. Rohan is the paternal uncle of Nisha. Ritu has two daughters one of Whom is Nisha.

Which of the following is true?

A. Dipesh is Mahesh's son.
B. Ravi has only two married children.
C. Ravi is the paternal grandfather of Renu.
D. None of these.

nielit2021dec-scientista logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.6 Family Relationship: NIELIT 2021 Dec Scientist B - Section A: 10

Directions $(9-13)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

There are five persons P, Q, R, S and T. One is football player, one is chess player, one is hockey player. P and S are unmarried ladies and do not participate in any game. None of the ladies plays chess or football. There is a married couple in which T is the husband. Q is the brother of R and is neither a chess player nor a hockey player.

Who is the football player $?$

A. Q
B. R
C. S
D. T

nielit2021dec-scientistb logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.7 Family Relationship: NIELIT 2021 Dec Scientist B - Section A: 13

Directions $(9-13)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

There are five persons P, Q, R, S and T. One is football player, one is chess player, one is hockey player. P and S are unmarried ladies and do not participate in any game. None of the ladies plays chess or football. There is a married couple in which T is the husband. Q is the brother of R and is neither a chess player nor a hockey player.

Who is the chess player $?$

A. Q
B. R
C. S
D. T

nielit2021dec-scientistb logical-reasoning family-relationship

Answer key☟

15.8.8 Family Relationship: NIELIT 2022 April Scientist B | Section A | Question: 31

In a family of six persons A, B, C, D, E and F, there are two married couples. The family has equal number of male and female.

i. D is the grandmother of A and mother of B
ii. C is the wife of B and mother of F.
iii. F is the granddaughter of E.

Which of the following is true ?

A. A is the brother of E.
B. A is the sister of F.
C. D has two grandsons$
D. None of these

nielit2022apr-scientistb logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.9 Family Relationship: NIELIT 2022 April Scientist B | Section A | Question: 38

There are five persons sitting on a bench. Blue eyed lady sitting in the middle is my mother. At the extreme left, the man with grey hair is my maternal uncle. Lady sitting at the extreme right is having a pimple on her nose and is the wife of person who is sitting between blue eyed lady and grey haired man and has a pointed nose. A lady having marks on her face is the younger sister of the blue eyed lady and sitting on the remaining fifth place. The person having a pointed nose is the son of the blue eyed lady. What is the lady having marks on her face to the man sitting at extreme left ?

A. Wife B. Maternal Aunt C. Sister D. Sister-in-law

nielit2022apr-scientistb logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.10 Family Relationship: NIELIT 2022 April Scientist B | Section A | Question: 4

Consider the following relationships among members of a family of six persons $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}$ and F :

i. The number of males equals that of females.
ii. A and E are sons of F.
iii. D is the mother of two, one boy and one girl.
iv. B is the son of A.
v. There is only one married couple in the family at present.

Which one of the following inferences can be drawn from the above?

A. A, B and C are all females B. A is the husband of D
C. E and F are children of D D. D is the daughter of F

nielit2022apr-scientistb logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.11 Family Relationship: NIELIT 2023 Feb Scientist C | Section A | Question: 10

A woman Manpreet asks a man Harpreet, "You are the brother of my uncle's daughter. How are you related to me?". What was the man's answer?

A. Cousin B. Son C. Brother-in-law D. Nephew

nielit2023feb-scientistc logical-reasoning family-relationship

Answer key☟

15.8.12 Family Relationship: NIELIT 2023 Feb Scientist C | Section A | Question: 12

Rahul has two mothers. Shalini is the step-daughter of Seema and step-sister of Rahul. How is Seema related to Rahul?

A. Step-mother
B. Brother
C. Can't be determined
D. None of these

nielit2023feb-scientistc logical-reasoning family-relationship analytical-aptitude

15.8.13 Family Relationship: NIELIT 2023 Feb Scientist C | Section A | Question: 18

A woman said, "His wife is the only daughter of my father", after introducing a man. How is that man related to the woman?

A. Husband
B. Brother
C. Father-in-law
D. None of these

nielit2023feb-scientistc logical-reasoning family-relationship analytical-aptitude

15.8.14 Family Relationship: NIELIT 2023 Feb Scientist C | Section A | Question: 2

Chlorophyll: Plant: : Haemoglobin:?

A. Hemorrhage
B. Blood
C. Oxygen
D. Red

nielit2023feb-scientistc analytical-aptitude family-relationship

Answer key

15.8.15 Family Relationship: NIELIT 2023 Feb Scientist C | Section A | Question: 20

If

I. A is brother of B
II. B is brother of C
III. C is brother of D
IV. D is sister of E

Then which of the following statements is not necessarily true?

A. D is sister of C
B. B and C are brothers
C. $\text{A}, \text{B}, \text{C}$ are male and D is a female
D. D and E are sisters

nielit2023feb-scientistc analytical-aptitude verbal-reasoning family-relationship

15.8.16 Family Relationship: NIELIT Scientific Assistant A 2020 November: 26

Answer the question on the basis of the data given below:

- O is $X's$ father
- Y is $Z's$ mother

- P is $O's$ mother
- X is $Z's$ Sister

How is O related to Z?

A. Brother B. Cousin C. Father D. Uncle

nielit-sta-2020 analytical-aptitude family-relationship

Answer key

15.8.17 Family Relationship: NIELIT Scientific Assistant A 2020 November: 27

Answer the question on the basis of the data given below:

- O is $X's$ father
- Y is $Z's$ mother
- P is $O's$ mother
- X is $Z's$ Sister

How is P related to X?

A. Mother B. Grandmother C. Sister D. Daughter

nielit-sta-2020 analytical-aptitude family-relationship

Answer key

15.8.18 Family Relationship: NIELIT Scientific Assistant A 2020 November: 28

Answer the question on the basis of the data given below:

- O is $X's$ father
- Y is $Z's$ mother
- P is $O's$ mother
- X is $Z's$ Sister

How is Y related to O?

A. Wife B. Sister C. Mother D. Daughter

nielit-sta-2020 analytical-aptitude family-relationship

Answer key

15.8.19 Family Relationship: NIELIT Scientific Assistant A 2020 November: 29

Answer the question on the basis of the data given below:

- O is $X's$ father
- Y is $Z's$ mother
- P is $O's$ mother
- X is $Z's$ Sister

If P has a daughter Q, then how is Q related to Z?

A. Aunt B. Mother C. Sister D. Daughter

nielit-sta-2020 analytical-aptitude family-relationship

Answer key

15.8.20 Family Relationship: NIELIT Scientist B 2020 November: 18

Read the following information carefully and then answer the question given below it.

$\text{A, B, C, D, E and F}$ are six members of a family.

There are two married couples among them.

C is the mother of A and F.

E is the father of D.

A is the grandson of B.

The total number of female members in the family is three.

Which of the following pairs is one of the married couples?

A. $E-F$ B. $B-D$ C. $E-B$ D. $A-F$

nielit-scb-2020 logical-reasoning family-relationship analytical-aptitude

Answer key

15.8.21 Family Relationship: NIELIT Scientist B 2020 November: 19

Read the following information carefully and then answer the question given below it.

$\text{A, B, C, D, E and F}$ are six members of a family.

There are two married couples among them.

C is the mother of A and F.

E is the father of D.

A is the grandson of B.

The total number of female members in the family is three.

How is B related to F?

A. Sister B. Grandmother C. Wife D. Data inadequate

nielit-scb-2020 logical-reasoning family-relationship analytical-aptitude

Answer key

15.8.22 Family Relationship: NIELIT Scientist B 2020 November: 20

Read the following information carefully and then answer the question given below it.

$\mathrm{A, B, C, D, E}$ and F are six members of a family.

There are two married couples among them.

C is the mother of A and F.

E is the father of D.

A is the grandson of B.

The total number of female members in the family is three.

How is F related to A?

A. Brother B. Daughter C. Son D. None of the options

nielit-scb-2020 logical-reasoning family-relationship analytical-aptitude

Answer key☟

15.8.23 Family Relationship: NIELIT Scientist B 2020 November: 42

Consists of a question and two statements, labelled (1) and (2), in which certain data are given. You have to decide whether the data given in the statements are sufficient for answering the question. Using the data given in the statements, plus your knowledge of mathematics and everyday facts.

How is R related to S?

$\text{Statement}(1)$:

T, the wife of$R's$ only brother C does not have any siblings.

Statement (2):

S is $T's$ brother-in-law's wife.

A. Only Statement (1) is required for answering the question
B. Only Statement (2) is required for answering the question
C. Both statement together are required to answer the question
D. Answer cannot be ascertained with the given information

nielit-scb-2020 logical-reasoning family-relationship

Answer key☟

15.9 General (4)

15.9.1 General: NIELIT 2023 Feb Scientist C | Section E | Question: 142

This is a joint trait of a leader which is defined as his ability to possess honesty, responsibility and maturity in the working area.

A. Integrity
B. Personality
C. Intelligence
D. Flexibility

nielit2023feb-scientistc general logical-reasoning

15.9.2 General: NIELIT 2023 Feb Scientist C | Section E | Question: 149

What make you think is the most effective leadership style that can be used during emergency situations?

A. Democratic B. Laissez-fire C. Autocratic D. Supportive

nielit2023feb-scientistc general logical-reasoning

15.9.3 General: NIELIT Scientist B 2020 November: 15

Directions: For the Assertion (A) and Reason (R) below, choose the correct alternative from the following:

I. Both (A) and (R) are true and (R) is the correct explanation of (A)
II. Both (A) and (R) are true and (R) is not the correct explanation of (A)
III. (A) is true but (R) is false
IV. (A) is false but (R) is true

Assertion (A):

Salt is added to cook food at higher altitudes.

Reason (R) :

Temperature is lower at higher altitudes

A. I B. II C. III D. IV

nielit-scb-2020 general logical-reasoning

Answer key☟

15.9.4 General: NIELIT Scientist B 2020 November: 16

Directions: For the Assertion (A) and Reason (R) below, choose the correct alternative from the following:

I. Both (A) and (R) are true and (R) is the correct explanation of (A)
II. Both (A) and (R) are true and (R) is not the correct explanation of (A)
III. (A) is true but (R) is false
IV. (A) is false but (R) is true

Assertion (A):

Ventilators are provided near the roof.

Reason (R):

Conduction takes place better near the roof.

A. I B. II C. III D. IV

nielit-scb-2020 general logical-reasoning

Answer key☟

15.10 General Awareness (3)

15.10.1 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 133

LBDQ stands for:

A. Leader Behavior Description Questionnaire
B. Leader Behavior Defining Questionnaire
C. Leader Behavior Dominance Questionnaire
D. None of these

nielit2023feb-scientistc general-awareness

15.10.2 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 146

The two types of attitudes in a workplace can be termed as ___________.

A. Favorable and unfavorable
B. Optimistic and Pessimistic
C. Individual and group
D. Satisfied and dissatisfied

nielit2023feb-scientistc general-awareness

15.10.3 General Awareness: NIELIT 2023 Feb Scientist D | Section A | Question: 25

Directions: Answer questions **(24-25)** on the basis of following instructions :

Instructions: For the Assertions (A) and Reason (R) below, choose the correct alternative from the following :

(I) Both (A) and (R) are true and (R) is the correct explanation of (A).

(II) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(III) (A) is true but (R) is false.

(IV) (A) is false but (R) is true.

Assertion (A): Salt is added to cook food at higher altitudes.

Reason (R): Temperature is lower at higher altitudes.

A. (I) B. (II) C. (III) D. (IV)

nielit2023feb-scientistd general-awareness logical-reasoning analytical-aptitude

15.11 Inequality (3)

15.11.1 Inequality: NIELIT Scientific Assistant A 2020 November: 35

Relationship between different elements is provided in the statements. The statements are followed by conclusions. Study the conclusions based on the given statement and choose the correct answer.

$T>=U=V<=W<X; V>=Y$

Conclusions :

I. $Y<=T$
II. $U>=X$

A. if only conclusion (I) follows
B. if only conclusion (II) follows
C. if neither (I) nor (II) conclusion follows
D. if both (I) and (II) conclusions follow

nielit-sta-2020 analytical-aptitude inequality

Answer key☟

15.11.2 Inequality: NIELIT Scientific Assistant A 2020 November: 36

Relationship between different elements is provided in the statements. The statements are followed by conclusions. Study the conclusions based on the given statement and choose the correct answer.

$P<=Q<=R>S; T>=R; S>=U$

Conclusions:

I. $T>S$
II. $U<R$

A. if only conclusion (I) follows
B. if only conclusion (II) follows
C. if neither (I) nor (II) conclusion follows
D. if both (I) and (II) conclusions follow

nielit-sta-2020 analytical-aptitude inequality

Answer key☟

15.11.3 Inequality: NIELIT Scientific Assistant A 2020 November: 37

Relationship between different elements is provided in the statements. The statements are followed by conclusions. Study the conclusions based on the given statement and choose the correct answer.

$A<=B<C>=D;C<=E<=F$

Conclusions:

I. $F>=D$
II. $A>E$

A. if only conclusion (I) follows
B. if only conclusion (II) follows
C. if neither (I) nor (II) conclusion follows
D. both (I) and (II) conclusions follow

nielit-sta-2020 analytical-aptitude inequality

Answer key☟

15.12 Is&software Engineering (3)

15.12.1 Is&software Engineering: NIELIT 2022 April Scientist B | Section B | Question: 113

A company needs to develop a strategy for software product development for which it has a choice of two programming languages $\mathrm{L1}$ and $\mathrm{L2}$. The number lines of code (LOC) developed using $\mathrm{L2}$ is estimated to be twice of the LOC developed with $\mathrm{L1}$. The product will have to be maintained for five years. Various parameters for the company are given in the table below.

Parameter	**Language L1**	**Language L2**
Man years needed for development	LOC/10000	LOC/10000
Development Cost per man year	₹ 10, 00, 000	₹ 7, 50, 000
Maintenance Time	5 years	5 years
Cost of maintenance per year	₹ 1,00,000	₹ 50, 000

Total cost of the project includes cost of development and maintenance. What is the (LOC) for $\mathrm{L1}$ for which of the cost of the project using $\mathrm{L1}$ is equal to the cost of the project using $\mathrm{L2}$?

A. 4000
B. 5000
C. 4333
D. 4667

nielit2022apr-scientistb is&software-engineering quantitative-aptitude

15.12.2 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 147

A satisfied employee will be a :

A. Motivator to others
B. Manager
C. High performer
D. Team Leader

nielit2023feb-scientistc is&software-engineering

15.12.3 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 130

Participation, recognition and power are some of the example of ___________.

A. Extrinsic motivation
B. Intrinsic motivation

C. Non-financial motivation

D. Financial motivation

nielit2023feb-scientistd is&software-engineering

15.13 Logic Puzzle (5)

15.13.1 Logic Puzzle: NIELIT Scientific Assistant A 2020 November: 38

There are twelve persons named $\text{O, P, Q, R, S, T, U, V,W, X, Y and Z}$ who live in a multi-store apartment. The apartment has three floors and each floor has four rooms. These 12 persons who live in a set of 12 Rooms can be represented by a Matrix of 3 rows and 4 columns.

- Q lives immediate left below diagonally of a person who lives immediate left below diagonally of T.
- S lives immediate left above diagonally of a person who lives immediate left above diagonally of Z.
- X lives immediate right above diagonally of a person who lives immediate right below diagonally of O.
- P lives immediate right above diagonally of a person who lives immediate right above diagonally of Y.
- T lives immediate left above diagonally of a person who lives third to the right of V.
- Q lives immediate left of a person who lives two rooms below W in the same column.
- R lives to the immediate right of a person who lives immediate right above diagonally of Q. Z is living to the immediate left of U who receives ₹ 46000 as salary.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ $50000,$ ₹ $48000,$ ₹ 47000 and ₹ $46000.$
- The person who live on one of the floors (right to left) receive salary in the same order ₹ $45000,$ ₹ $38000,$ ₹ 35000 and ₹ $40000.$
- The person who live on one of the floors (left to right) receive salary in the same order ₹ $37000,$ ₹ $42000,$ ₹ 36000 and ₹ $43000.$

What is the salary received by a person who lives second to the right of S?

A. ₹ 35000

B. ₹ 45000

C. ₹ 37000

D. ₹ 38000

nielit-sta-2020 analytical-aptitude logical-reasoning logic-puzzle

Answer key☟

15.13.2 Logic Puzzle: NIELIT Scientific Assistant A 2020 November: 39

There are twelve persons named $\text{O, P, Q, R, S, T, U, V,W, X, Y and Z}$ who live in a multi-store apartment. The apartment has three floors and each floor has four rooms. These 12 persons who live in a set of 12 Rooms can be represented by a Matrix of 3 rows and 4 columns.

- Q lives immediate left below diagonally of a person who lives immediate left below diagonally of T.

- S lives immediate left above diagonally of a person who lives immediate left above diagonally of Z.
- X lives immediate right above diagonally of a person who lives immediate right below diagonally of O.
- P lives immediate right above diagonally of a person who lives immediate right above diagonally of Y.
- T lives immediate left above diagonally of a person who lives third to the right of V.
- Q lives immediate left of a person who lives two rooms below W in the same column.
- R lives to the immediate right of a person who lives immediate right above diagonally of Q. Z is living to the immediate left of U who receives ₹ 46000 as salary.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 50000, ₹ 48000, ₹ 47000 and ₹ 46000.
- The person who live on one of the floors (right to left) receive salary in the same order ₹ 45000, ₹ 38000, ₹ 35000 and ₹ 40000.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 37000, ₹ 42000, ₹ 36000 and ₹ 43000.

Who among the following lives third to the left of U?

A. O B. Q C. T D. S

nielit-sta-2020 analytical-aptitude logical-reasoning logic-puzzle

Answer key☟

15.13.3 Logic Puzzle: NIELIT Scientific Assistant A 2020 November: 40

There are twelve persons named O, P, Q, R, S, T, U, V,W, X, Y and Z who live in a multi-store apartment. The apartment has three floors and each floor has four rooms. These 12 persons who live in a set of 12 Rooms can be represented by a Matrix of 3 rows and 4 columns.

- Q lives immediate left below diagonally of a person who lives immediate left below diagonally of T.
- S lives immediate left above diagonally of a person who lives immediate left above diagonally of Z.
- X lives immediate right above diagonally of a person who lives immediate right below diagonally of O.
- P lives immediate right above diagonally of a person who lives immediate right above diagonally of Y.
- T lives immediate left above diagonally of a person who lives third to the right of V.
- Q lives immediate left of a person who lives two rooms below W in the same column.
- R lives to the immediate right of a person who lives immediate right above diagonally of Q. Z is living to the immediate left of U who receives ₹ 46000 as salary.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 50000, ₹ 48000, ₹ 47000 and ₹ 46000.
- The person who live on one of the floors (right to left) receive salary in the same order ₹ 45000, ₹ 38000, ₹ 35000 and ₹ 40000.
- The person who live on one of the floors (left to right) receive salary in the same order

₹ 37000, ₹ 42000, ₹ 36000 and ₹ 43000.

What is the sum of salaries of Y and P?

A. ₹ 90000 B. ₹ 99000 C. ₹ 93000 D. ₹ 89000

nielit-sta-2020 analytical-aptitude logical-reasoning logic-puzzle

Answer key

15.13.4 Logic Puzzle: NIELIT Scientific Assistant A 2020 November: 41

There are twelve persons named O, P, Q, R, S, T, U, V, W, X, Y and Z who live in a multi-store apartment. The apartment has three floors and each floor has four rooms. These 12 persons who live in a set of 12 Rooms can be represented by a Matrix of 3 rows and 4 columns.

- Q lives immediate left below diagonally of a person who lives immediate left below diagonally of T.
- S lives immediate left above diagonally of a person who lives immediate left above diagonally of Z.
- X lives immediate right above diagonally of a person who lives immediate right below diagonally of O.
- P lives immediate right above diagonally of a person who lives immediate right above diagonally of Y.
- T lives immediate left above diagonally of a person who lives third to the right of V.
- Q lives immediate left of a person who lives two rooms below W in the same column.
- R lives to the immediate right of a person who lives immediate right above diagonally of Q. Z is living to the immediate left of U who receives ₹ 46000 as salary.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 50000, ₹ 48000, ₹ 47000 and ₹ 46000.
- The person who live on one of the floors (right to left) receive salary in the same order ₹ 45000, ₹ 38000, ₹ 35000 and ₹ 40000.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 37000, ₹ 42000, ₹ 36000 and ₹ 43000.

What is the sum of the salaries received by the persons living on the top floor of the apartment?

A. ₹ 158000 B. ₹ 193000 C. ₹ 157000 D. ₹ 161000

nielit-sta-2020 analytical-aptitude logical-reasoning logic-puzzle

Answer key

15.13.5 Logic Puzzle: NIELIT Scientific Assistant A 2020 November: 42

There are twelve persons named O, P, Q, R, S, T, U, V, W, X, Y and Z who live in a multi-store apartment. The apartment has three floors and each floor has four rooms. These 12 persons who live in a set of 12 Rooms can be represented by a Matrix of 3 rows and 4 columns.

- Q lives immediate left below diagonally of a person who lives immediate left below diagonally of T.
- S lives immediate left above diagonally of a person who lives immediate left above diagonally of Z.
- X lives immediate right above diagonally of a person who lives immediate right below diagonally of O.
- P lives immediate right above diagonally of a person who lives immediate right above diagonally of Y.
- T lives immediate left above diagonally of a person who lives third to the right of V.
- Q lives immediate left of a person who lives two rooms below W in the same column.
- R lives to the immediate right of a person who lives immediate right above diagonally of Q. Z is living to the immediate left of U who receives ₹ 46000 as salary.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 50000, ₹ 48000, ₹ 47000 and ₹ 46000.
- The person who live on one of the floors (right to left) receive salary in the same order ₹ 45000, ₹ 38000, ₹ 35000 and ₹ 40000.
- The person who live on one of the floors (left to right) receive salary in the same order ₹ 37000, ₹ 42000, ₹ 36000 and ₹ 43000.

What is the aggregate salary of the people living at the right end of the apartment?

A. ₹ 137000 B. ₹ 134000 C. ₹ 125000 D. ₹ 131000

nielit-sta-2020 analytical-aptitude logical-reasoning logic-puzzle

Answer key☟

15.14 Logical Reasoning (62)

15.14.1 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 1

Read the following information carefully and answer question given below :

In an engineering college, four students Diksha, Shreya, Tanvi and Akriti exhibit a very strange mix of hobbies and subject interests. One of them studies Computer Science and plays Golf and Lawn Tennis. Diksha and Shreya study Mechanical engineering. Diksha plays Billiards. Both the Mechanical Engineering students play chess. Tanvi is a student of Physics. The physics student play Chess and Badminton. All the friends play two games each and study one subject each. One of the students also does weight training.

How many games are played and subjects studied by all the four students?

A. $2,1$ B. $3,2$ C. $6,3$ D. $5,4$

nielit2021dec-scientista logical-reasoning analytical-aptitude

Answer key☟

15.14.2 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 11

Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

A training college has to conduct a refresher course for teachers of seven different subjects – Mechanics, Psychology, Philosophy, Sociology, Economics, Science and Engineering from November 22 to November 29.

i. Course should start with Psychology.
ii. November 23, being Sunday, should be a holiday.
iii. Science subject should be on the previous day of the Engineering subject.
iv. Course should end with Mechanics subject.
v. Philosophy should be immediately after holiday.
vi. There should be a gap of one day between Economics and Engineering.
vii. There should be a gap of two days between Sociology and Economics.

The refresher course will start with which one of the following subjects?

A. Psychology B. Mechanics C. Economics D. Sociology

nielit2021dec-scientista analytical-aptitude logical-reasoning

Answer key☟

15.14.3 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 13

Read the information given below and on the basis of the information, select the correct alternative for each question (9–13) given after the information.

A training college has to conduct a refresher course for teachers of seven different subjects – Mechanics, Psychology, Philosophy, Sociology, Economics, Science and Engineering from November 22 to November 29.

i. Course should start with Psychology.
ii. November 23, being Sunday, should be a holiday.
iii. Science subject should be on the previous day of the Engineering subject.
iv. Course should end with Mechanics subject.
v. Philosophy should be immediately after holiday.
vi. There should be a gap of one day between Economics and Engineering.
vii. There should be a gap of two days between Sociology and Economics.

Which subject succeeds Science?

A. Psychology B. Mechanics C. Engineering D. Sociology

nielit2021dec-scientista logical-reasoning analytical-aptitude

Answer key☟

15.14.4 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 2

Read the following information carefully and answer question given below :

In an engineering college, four students Diksha, Shreya, Tanvi and Akriti exhibit a very strange mix of hobbies and subject interests. One of them studies Computer Science and plays Golf and Lawn Tennis. Diksha and Shreya study Mechanical engineering. Diksha plays Billiards. Both the Mechanical Engineering students play chess. Tanvi is a student of Physics. The physics student play Chess and Badminton. All the friends play two games each and study one subject each. One of the students also does weight training.

Who studies Mechanical Engineering and plays Billiards ?

A. Diksha B. Shreya C. Tanvi D. Akriti

nielit2021dec-scientista logical-reasoning analytical-aptitude

Answer key☟

15.14.5 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 3

Read the following information carefully and answer question given below :

In an engineering college, four students Diksha, Shreya, Tanvi and Akriti exhibit a very strange mix of hobbies and subject interests. One of them studies Computer Science and plays Golf and Lawn Tennis. Diksha and Shreya study Mechanical engineering. Diksha plays Billiards. Both the Mechanical Engineering students play chess. Tanvi is a student of Physics. The physics student play Chess and Badminton. All the friends play two games each and study one subject each. One of the students also does weight training.

Who does not play chess ?

A. Diksha B. Shreya C. Tanvi D. Akriti

nielit2021dec-scientista logical-reasoning analytical-aptitude

Answer key☟

15.14.6 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 4

The day before the day before yesterday is three days after Saturday. What day is it today ?

A. Tuesday B. Wednesday C. Thursday D. Friday

nielit2021dec-scientista logical-reasoning analytical-aptitude

Answer key☟

15.14.7 Logical Reasoning: NIELIT 2021 Dec Scientist A - Section A: 6

In a certain code language, $'493'$ means ‘Friendship Big Challenge’, $'961'$ means ‘Struggle Big Exam’ and $'178'$ means ‘Exam Confidential Subject’. What does ‘Confidential’ stand for ?

A. $7 \text{ or } 8$ B. $7 \text{ or } 9$ C. 8 D. $8 \text{ or } 1$

nielit2021dec-scientista logical-reasoning

Answer key☟

15.14.8 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 11

Directions $(9-13)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

There are five persons $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ and $\mathrm{T}.$ One is football player, one is chess player, one is hockey player. P and S are unmarried ladies and do not participate in any game. None of the ladies plays chess or football. There is a married couple in which T is the husband. Q is the brother of R and is neither a chess player nor a hockey player.

Who is the hockey player ?

A. T B. S C. R D. Q

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.9 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 12

Directions $(9-13)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

There are five persons $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ and $\mathrm{T}.$ One is football player, one is chess player, one is hockey player. P and S are unmarried ladies and do not participate in any game. None of the ladies plays chess or football. There is a married couple in which T is the husband. Q is the brother of R and is neither a chess player nor a hockey player.

Who is the wife of T

A. Q B. R C. S D. None of these

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.10 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 14

Directions $(14-18)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

M and N are good at hockey and volleyball.

O and M are good at hockey and baseball.

P and N are good at cricket and volleyball.

O, P and Q are good at football and basketball.

Who is good at cricket, volleyball and hockey ?

1. Q
2. P
3. O
4. N

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.11 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 15

Directions $(14–18)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

M and N are good at hockey and volleyball.

O and M are good at hockey and baseball.

P and N are good at cricket and volleyball.

O, P and Q are good at football and basketball.

Who is good at baseball, hockey and volleyball ?

A. Q B. P C. O D. M

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.12 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 16

Directions $(14–18)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

M and N are good at hockey and volleyball.

O and M are good at hockey and baseball.

P and N are good at cricket and volleyball.

O, P and Q are good at football and basketball.

Who is good at the largest number of games ?

A. Q B. P C. O D. N

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.13 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 17

Directions $(14 – 18)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

M and N are good at hockey and volleyball.

O and M are good at hockey and baseball.

P and N are good at cricket and volleyball.

O,P and Q are good at football and basketball.

Who is good at cricket, baseball and volleyball $?$

1. Q
2. P
3. O
4. N

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.14 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 18

Directions $(14 – 18)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

M and N are good at hockey and volleyball.

O and M are good at hockey and baseball.

P and N are good at cricket and volleyball.

O, P and Q are good at football and basketball.

Who among the following is good at four games $?$

1. Q
2. P
3. O
4. M

nielit2021dec-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.15 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 20

Directions $(19 \ -21)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

i. Six flats on a floor in two rows facing North and South are allotted to $\text{P}, \text{Q}, \text{R}, \text{S}, \text{T}$ and U.
ii. Q gets a north facing flat and is not next to S.
iii. S and U get diagonally opposite flats.
iv. R next to U, gets a south facing flat and T gets a north facing flat.

If the flats of T and P are interchanged, who's flat will be next to that of U?

A. Q B. T C. P D. R

nielit2021dec-scientistb logical-reasoning spatial-aptitude

Answer key☟

15.14.16 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 21

Directions $(19 - 21)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

i. Six flats on a floor in two rows facing North and South are allotted to $\text{P}, \text{Q}, \text{R}, \text{S}, \text{T}$ and U.
ii. Q gets a north facing flat and is not next to S.
iii. S and U get diagonally opposite flats.
iv. R next to U, gets a south facing flat and T gets a north facing flat.

The flats of which of the other pairs than SU, is diagonally opposite to each other ?

A. PT B. PQ C. QR D. TS

nielit2021dec-scientistb logical-reasoning spatial-aptitude analytical-aptitude

Answer key☟

15.14.17 Logical Reasoning: NIELIT 2021 Dec Scientist B - Section A: 8

Directions $(7 - 8)$: In the given questions below, a statement is given followed by two conclusions numbered I and II. You have to take the statement to be true. Read both the conclusions and decide which of the two or both follow from the given statement. Given answer :

A. If only conclusion I follows.
B. If only conclusion II follows.
C. If either I or II follows.
D. If neither I nor II follows.

A study of planning commission reveals boom in revenues. However, this has been of little avail owing to soaring expenditure. In the event, there has been a high dose of deficit financing, leading to marked rise in prices. Large financial outlays year after year had little impact on level of living.

I. A boom in revenues leads to rise in prices.
II. Large financial outlays should be avoided.

nielit2021dec-scientistb logical-reasoning analytical-aptitude

15.14.18 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 12

Read following information and answer the question.

i. There is a group of five persons, P, Q, R, S and T.
ii. One of them is a Horticulturist, one is Physicist, one is Journalist, one is Industrialist and one is an Advocate.
iii. Three of the P, R and the Advocate prefer tea to coffee and two of them Q and Journalist prefer coffee to tea.
iv. The industrialist, S and P are friends of one another but two of them prefer coffee to tea.
v. The Horticulturist is R's brother.

Who is industrialist ?

A. P
B. Q
C. R
D. S

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.19 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 17

Ten eggs are distributed among ABCD in ratio $1:2:3:4$ randomly. It is known that A gets less eggs than B, and C gets more eggs than D. If A gets half the number of eggs of B, then which of the following is necessarily true ?

A. C gets an even number of eggs
B. D gets an even number of eggs
C. C gets an odd number of eggs
D. D gets an odd number of eggs

nielit2022apr-scientistb probability counting logical-reasoning

Answer key☟

15.14.20 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 24

$\mathrm{P}+\mathrm{Q}$ means P is the brother of Q; P-Q means P is the mother of Q and $\mathrm{P} \times \mathrm{Q}$ means P is the sister of Q. Which of the following means M is the maternal uncle of R ?

A. $\mathrm{M}-\mathrm{R}+\mathrm{K}$
B. $\mathrm{M}+\mathrm{K}$ - R
C. $\mathrm{M}+\mathrm{K}\times\mathrm{R}$
D. $\mathrm{M}+\mathrm{R}+\mathrm{K}$

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.21 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 25

In a code, CORNER is written as GSVRIV. How can CENTRAL be written in that code ?

A. GIRXVEP B. GJRYVEP C. GNFJKER D. DFOUSBM

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.22 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 26

If TOUR is written in a certain code as 1234, CLEAR as 56784 and SPARE as 90847, what will be the 5^{th} Digit for SCULPTURE in the same code ?

A. 3 B. 4 C. 6 D. 0

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.23 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 27

The first republic day of India was celebrated on January $26, 1950$. What day of the week was it ?

A. Wednesday B. Friday C. Thursday D. Tuesday

nielit2022apr-scientistb logical-reasoning analytical-aptitude quantitative-aptitude

Answer key☟

15.14.24 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 29

A told B, "Yesterday I met the only brother of the daughter of my grandmother". Whom did A meet ?

A. Cousin B. Brother C. Father D. Son

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key☟

15.14.25 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 3

In the following question, a statement is followed by two assumptions numbered I and II. Consider the statement and the following assumptions to decide which of the assumptions is implicit in the statement :

Statement : All the employees are notified that the organization will provide transport facilities at half the cost from the nearby railway station to the office except those who have been provided with travelling allowance.

Assumption:

I. Most of the employees will travel by the office transport.
II. Those who are provided with travelling allowance will not read such notice.

A. If only assumption I is implicit;
B. If either I or II is implicit;
C. If only assumption II is implicit;
D. If neither I nor II is implicit.

nielit2022apr-scientistb analytical-aptitude logical-reasoning

Answer key

15.14.26 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 7

Following question contains four arguments of three sentences each. Choose the set in which the third statement is a logical conclusion of the first two.

I. All envelopes are rectangles. All rectangles are rectangular. All envelopes are rectangular.
II. Some things are smart. Some smart things are tiny. All things are tiny.
III. Learneds are well read. Well read know. Learneds know.
IV. Dieting is good for health. Health foods are rare. Dieting is rare.

A. IV only
B. III only
C. Both I and III only
D. All of these

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.27 Logical Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 8

Choose the correct conclusion for the given statements :

Statement I: All athletes in the Asian athletics teams are talented.

II: All Indian athletes are now in the Asian athletics team.

Conclusions:

A. Some Indian athletes are talented.
B. All Indian athletes are talented.
C. India does not have any athletics team.
D. India has more cricketers than athletics.

nielit2022apr-scientistb logical-reasoning analytical-aptitude

Answer key

15.14.28 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 16

Daksh is taller than Manick but not as tall as Rohan. Somesh is shorter than Daksh but taller than Farhan. Who among them is the shortest ?

A. Daksh
B. Manick
C. Farhan
D. Cannot be determined

nielit2023feb-scientistc logical-reasoning

15.14.29 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 19

An application was received by class teacher in the afternoon of a week day. Next day she forwarded it to the Student Coordinator, who was on leave that day. The student Coordinator put up the application to the principal next day in the evening. The Principal studied the application and disposed off the matter on the same day, i.e, Saturday. Which day was the application received by the inward clerk?

A. Monday B. Wednesday C. Tuesday D. Thursday

nielit2023feb-scientistc logical-reasoning analytical-aptitude

15.14.30 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 21

Which letter should be the Tenth letter to the left of the ninth letter from the right, if the first half of the alphabet of English are reversed?

A. D B. F C. E D. I

nielit2023feb-scientistc logical-reasoning analytical-aptitude

15.14.31 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 23

In a family, there are husband-wife, two sons and two daughters. All the ladies were invited to a dinner. Both sons went out to play. Husband did not return from the office. Who was at home?

A. Only wife was at home
B. Nobody was at home
C. Only sons were at home
D. All ladies were at home

nielit2023feb-scientistc logical-reasoning analytical-aptitude

15.14.32 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 27

In a certain code language, OMNIPRESENT is written as QJONPTSMDRD How is CREDIBILITY written in at code?

A. DSFEJDDXSHKH
B. JEFSDCXSHKH
C. JEFSDDXSHKH
D. JEFSDDZUJMJ

nielit2023feb-scientistc logical-reasoning

15.14.33 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 3

Here are some words translated from an artificial language. malgauper means peach cobbler malgaport means peach juice moggagrop means apple jelly moggasport means apple sweet Which word could mean "apple juice"?

A. moggaport B. malgauper C. gropport D. moggagrop

nielit2023feb-scientistc analytical-aptitude logical-reasoning

Answer key☟

15.14.34 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 4

Pointing to a photograph, a man said, "I have no brother or sister, but that man's father is my father's son." Whose photograph was it?

A. His own
B. His Son
C. His Father
D. His Grandfather

nielit2023feb-scientistc logical-reasoning analytical-aptitude

Answer key☟

15.14.35 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 5

In a code language if POSE is coded as OQNPRTDF, then the word TYPE will be coded as :

A. SUXZOQFD B. SUXZQOFD C. SUXZOQDF D. SUXZQODE

nielit2023feb-scientistc logical-reasoning analytical-aptitude

Answer key☟

15.14.36 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 7

Statement (1): Pens cost more than pencils.
Statement (2): Pens cost less than erasers.
Statement (3): Erasers cost more than pencils and pens. If the first two statements are true, the third statement is :

A. True
B. False
C. Uncertain
D. Cannot be determined

nielit2023feb-scientistc logical-reasoning

Answer key☟

15.14.37 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section C | Question: 67

How would you best judge the fruitfulness of research?

A. The fruitfulness of any research can best be judged according to whether or not the results are significant.
B. The fruitfulness of any research could be judged by assessing the impact of the research on the public or other researchers.
C. Fruitfulness of the research is probably best judged in terms of the number of new ideas and insights it offers. This is not easily catalogued; rather it is easy to spot when research lacks novel insight and ideas.
D. All of these.

nielit2023feb-scientistc analytical-aptitude logical-reasoning

15.14.38 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section E | Question: 150

Belief, opinion, knowledge, emotions feelings intentions are the components of

A. OB
B. Job satisfaction
C. Attitude
D. Personality

nielit2023feb-scientistc logical-reasoning analytical-aptitude

15.14.39 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 13

Read the following information carefully and answer the question below.

(i) $\text{M\$ N}$ means M is the mother of N
(ii) $\text{M\# N}$ means M is the father of N
(iii) M@ N means M is the husband of N
(iv) $\text{M\% N}$ means M is the daughter of N

If $\text{G\$M@ K}$, how is K related to G ?

A. Mother
B. Mother-in-law
C. Daughter
D. Daughter-in-law

nielit2023feb-scientistd logical-reasoning analytical-aptitude

15.14.40 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 14

If 'CAMERA' is coded as 'ZIVNZX' in a certain code language, then how would you code 'CHAPRA' in that code language?

A. ZISKZX
B. ZIKSZX
C. ZIKXSZ
D. ZIKZSX

nielit2023feb-scientistd analytical-aptitude logical-reasoning

Answer key

15.14.41 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 16

In the question below, three Statements (I), (II), and (III) are given. You are required to find out which of the given statement (s) is/are sufficient to answer the question. Mark your answer accordingly.

How old was Deepak on 30th July 1996?

(I) Deepak is 6 yr older than his brother Prabir.

(II) Prabir is 29 yr younger than his mother.

(III) Deepak's mother celebrated her 50th birthday on 15^{th} June 1996.

A. (I) and (II)
B. All (I), (II) and (III)
C. (II) and (III)
D. None of these

nielit2023feb-scientistd quantitative-aptitude logical-reasoning

15.14.42 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 23

In a certain code, "Friend" is written as $"\mathrm{EQHDMC}"$. How is "Globe" written in that code?

A. FNKAO B. EJNAD C. FKNBD D. FKNAD

nielit2023feb-scientistd logical-reasoning analytical-aptitude

Answer key

15.14.43 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 26

In a certain code language, $'\mathrm{TAPE}'$ is written as $'\mathrm{SUZBOQDF}'$. How will $'\mathrm{MOCK}'$ be written as in that language?

A. $\mathrm{MOPQDELM}$ B. $\mathrm{LNNPBDJL}$ C. $\mathrm{LOPQBCKL}$ D. $\mathrm{KNNPBEBL}$

nielit2023feb-scientistd analytical-aptitude logical-reasoning

15.14.44 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 27

Four number-pairs have been given, out of which three are alike in some manner and one is different. Select the number-pair that is different from the rest.

A. $76 : 42$ B. $44 : 16$ C. $55 : 25$ D. $28 : 56$

nielit2023feb-scientistd logical-reasoning analytical-aptitude

Answer key

15.14.45 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 28

If Americans are Red, Japanese are Oranges, Indians are Bright then what is the code for a citizen of Tokyo with Indian origin?

A. Bright Red
B. Red Orange
C. Bright Orange
D. None of the options

nielit2023feb-scientistd logical-reasoning analytical-aptitude

Answer key

15.14.46 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 7

In the following question, some groups of letters are given, all of which except one, share a common similarity while one is different. Select the odd one.

A. DFHB
B. KMOJ
C. PRTN
D. XZBV

nielit2023feb-scientistd analytical-aptitude logical-reasoning

Answer key

15.14.47 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 8

If $'+'$ means subtraction, $'-'$ means multiplication, 'symbol of division' means addition and ' x ' means division, then what is the value of the following: $54 \times 9 - 7$ division $8 + 4 =$?

A. 56
B. 46
C. 59
D. 48

nielit2023feb-scientistd analytical-aptitude logical-reasoning

Answer key

15.14.48 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 9

In a code language $'\text{BOMBAY}'$ is written as CQPFFE, then in the same language how will you write $'\text{BHARAT}'$?

A. CJDVEZ
B. CJDVFZ
C. CJEWFZ
D. CIEWEZ

nielit2023feb-scientistd logical-reasoning

Answer key

15.14.49 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 139

________ are the outcome of ________ and ___________, which are always definite.

A. Values, realization, understanding
B. Beliefs, emotions, thoughts
C. Intelligence, values, emotions
D. Thoughts, ideas, beliefs

nielit2023feb-scientistd logical-reasoning

15.14.50 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 140

The green hat of creativity focuses on :

A. Facts and information
B. Creativity
C. Managing thinking process
D. Possibilities

nielit2023feb-scientistd logical-reasoning non-gatecse

Answer key☟

15.14.51 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 141

The blue hat of creativity focuses on :

A. Facts and information
B. Creativity
C. Managing thinking process
D. Possibilities

nielit2023feb-scientistd logical-reasoning analytical-aptitude

Answer key☟

15.14.52 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 142

Dependable and Responsible are part of ___________ Personality Trait as per big five personality trait.

A. Emotional Stability
B. Conscientious
C. Openness to experience
D. Introversion

nielit2023feb-scientistd logical-reasoning verbal-aptitude

15.14.53 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 146

Time Stress can't be managed by :

A. To do lists
B. Action programs
C. Setting priorities
D. Thinking about an action

nielit2023feb-scientistd logical-reasoning

15.14.54 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 150

A Process of receiving, selecting, organizing, interpreting, checking and reacting to sensory stimuli or data so as to form a meaningful and coherent picture of the world is :

A. Attitude
B. Thinking
C. Perception
D. Communication

nielit2023feb-scientistd logical-reasoning analytical-aptitude

15.14.55 Logical Reasoning: NIELIT Scientific Assistant A 2020 November: 21

Ramesh's father is a paediatrician. Ram's father is a trader. Krishan's father is a school teacher. Krishan falls ill. Where should his father take him?

A. to home
B. to school
C. to Ramesh's father
D. to Ram's father

nielit-sta-2020 analytical-aptitude logical-reasoning

Answer key☟

15.14.56 Logical Reasoning: NIELIT Scientific Assistant A 2020 November: 22

Five people are standing in a row. Aman is standing next to Karan but not adjacent to Tanuj. Radhika is standing next to Priyanka who is standing on the extreme left and Tanuj is not standing next to Radhika. Who are Standing adjacent to Aman?

A. Radhika and Karan
B. Karan and Tanuj
C. karan and Priyanka
D. Radhika and Tanuj

nielit-sta-2020 analytical-aptitude logical-reasoning

Answer key☟

15.14.57 Logical Reasoning: NIELIT Scientist B 2020 November: 10

Choose which of the following will be sufficient to find: What time did the bus leave today?

Statements:

I. The bus normally leaves on time.
II. The scheduled departure is at $12:30$

A. I alone is sufficient while II alone is not sufficient
B. II alone is sufficient while I alone is not sufficient
C. Either I or II is sufficient
D. Neither I nor II is sufficient

nielit-scb-2020 analytical-aptitude logical-reasoning

15.14.58 Logical Reasoning: NIELIT Scientist B 2020 November: 13

If X says that his mother is the only daughter of$Y's$ mother, then how is Y related to X?

A. Brother B. Son C. Uncle D. Father

nielit-scb-2020 logical-reasoning

Answer key☟

15.14.59 Logical Reasoning: NIELIT Scientist B 2020 November: 14

Choose the missing terms out of the given alternatives.

$\text{EJO, TYD, INS, XCH, \_\_\_\_?}$

A. NRW B. MRW C. MSX D. NSX

nielit-scb-2020 logical-reasoning analytical-aptitude

Answer key☟

15.14.60 Logical Reasoning: NIELIT Scientist B 2020 November: 25

The admission ticket for an Art Gallery bears a password which is changed after every clock hour based on set of words chosen for each day. The following is an illustration of the code and steps of rearrangement for subsequent clock hours.

The Time is 9 a.m. to 3 p.m. Day's first password:

$\text{First batch} - 9 \text{ a.m. to } 10 \text{ a.m.}$

is not ready cloth simple harmony burning

$\text{Second batch} - 10 \text{ a.m. to } 11 \text{ a.m.}$

ready not is cloth burning harmony simple

$\text{Third batch} - 11 \text{ a.m. to } 12 \text{ noon}$

cloth is not ready simple harmony burning

$\text{Fourth batch} - 12 \text{ noon to } 1 \text{ p.m.}$

not is cloth ready burning harmony simple

$\text{Fifth batch} - 1 \text{ p.m. to } 2 \text{ p.m.}$

ready cloth is not simple harmony burning and so on

If the password for $11\ a.m.$ to $12\ noon$ was – "soap shy miss pen yet the she", what was the password for the First Batch?

A. pen miss shy soap she the yet
B. shy miss pen soap yet the she
C. soap pen miss shy she the yet
D. miss shy soap pen she the yet

nielit-scb-2020 analytical-aptitude logical-reasoning

Answer key☟

15.14.61 Logical Reasoning: NIELIT Scientist B 2020 November: 37

Find the odd one out in the given series:

$ZA, RS, DE, JK, PR, LM, YZ, NO$

A. JK B. LM C. ZA D. PR

nielit-scb-2020 logical-reasoning analytical-aptitude

Answer key

15.14.62 Logical Reasoning: NIELIT Scientist B 2020 November: 8

A university library budget committee must reduce exactly five of eight areas of expenditure -$\text{I, J, K, L, M, N, O and P}$ in accordance with the following conditions:

If both I and O are reduced, P is also reduced.

If L is reduced, Neither N nor O is reduced

If M is reduced, J is not reduced.

Of the three areas J, K and N exactly two are reduced.

If both K and N are reduced, which one of the following is a pair of areas neither of which could be reduced?

A. I, L B. J, L C. J, M D. I, J

nielit-scb-2020 logical-reasoning analytical-aptitude

Answer key

15.15 Logical Sequencing (1)

15.15.1 Logical Sequencing: NIELIT 2023 Feb Scientist D | Section A | Question: 2

In the following question, arrange the words in a logical sequence and choose the correct sequence from the given alternatives :

1. Atomic age,
2. Metallic age,
3. Stone age,
4. Alloy age

A. $1,3,4,2$ B. $3,2,4,1$ C. $2,3,1,4$ D. $4,3,2,1$

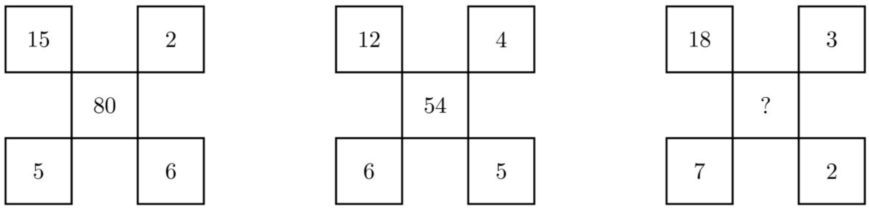
nielit2023feb-scientistd logical-sequencing analytical-aptitude verbal-reasoning

Answer key

15.16 Missing Videos (1)

15.16.1 Missing Videos: NIELIT 2021 Dec Scientist A - Section A: 30

Find the missing number :



A. 46 B. 15 C. 55 D. 32

nielit2021dec-scientista logical-reasoning analytical-aptitude missing-videos

Answer key☟

15.17 Number Series (7)

15.17.1 Number Series: NIELIT 2018-11

Look at this series: $5000, 1001, 201, 41, \ldots$ What number should come next?

A. 9 B. 10 C. 11 D. 42

nielit-2018 general-aptitude analytical-aptitude logical-reasoning sequence-series number-series

Answer key☟

15.17.2 Number Series: NIELIT 2018-9

Look at this series: $25, 25, 37, 37, \ldots, 51, \ldots.$. What number should fill the blank?

A. 51 B. 39 C. 23 D. 25

nielit-2018 general-aptitude analytical-aptitude logical-reasoning sequence-series number-series

Answer key☟

15.17.3 Number Series: NIELIT 2021 Dec Scientist A - Section A: 32

Find the wrong term in the series $5, 11, 29, 83, 245, 765, 2189, 6563$:

A. 245 B. 765 C. 2189 D. 6563

nielit2021dec-scientista number-series analytical-aptitude

Answer key☟

15.17.4 Number Series: NIELIT 2021 Dec Scientist A - Section A: 7

What value should come in place of the question mark $(?)$ in the series given below ?

$\text{BEAG, DGCI, FIEK, ?}$

A. HMIE B. HKGM C. HGKJ D. HKLJ

nielit2021dec-scientista analytical-aptitude number-series logical-reasoning

Answer key☟

15.17.5 Number Series: NIELIT 2021 Dec Scientist B - Section A: 3

Directions $(3-5)$: In each of the following questions, one number is wrong in the series. Find out the wrong number.

$3, 5, 12, 39, 154, 772, 4634:$

A. 5 B. 3 C. 39 D. 154

nielit2021dec-scientistb number-series analytical-aptitude quantitative-aptitude

Answer key☟

15.17.6 Number Series: NIELIT 2023 Feb Scientist D | Section A | Question: 12

In the following number series, only one number is wrong. Find out the wrong number.

$9050, 5675, 3478, 2147, 1418, 1077, 950$

A. 3478 B. 1418 C. 5675 D. 1077

nielit2023feb-scientistd sequence-series analytical-aptitude number-series

15.17.7 Number Series: NIELIT Scientist B 2020 November: 39

What is the next number $16, 30, 54, 88, 132, \ldots$

A. 186 B. 188 C. 190 D. 206

nielit-scb-2020 number-series analytical-aptitude

Answer key☟

15.18 Odd One Out (1)

15.18.1 Odd One Out: NIELIT Scientific Assistant A 2020 November: 10

Find the number which does not fit into the series $8\ 12\ 20\ 32\ 50\ 68.$

A. 20 B. 32 C. 68 D. 50

nielit-sta-2020 analytical-aptitude sequence-series odd-one-out

Answer key☟

15.19 Permutation and Combination (1)

15.19.1 Permutation and Combination: NIELIT Scientist B 2020 November: 31

Five of India's leading models are posing for a photograph promoting "World Peace and Understanding".

But then, Sachin Malhotra the photographer is having a tough time getting them to stand in a straight line, because Natasha refuses to stand in a straight line, because Natasha refuses to stand next to Jessica since Jessica had said something about her in a leading gossip magazine. Rachel and Anna want to stand together because they are good friends. Ria on the other hand cannot get along well with Rachel, because there is some talk about Rachel scheming to get a contract already awarded to Ria. Anna believes her friendly astrologer who has asked her to stand at the extreme right for all group photographs. Finally, Sachin managed to pacify the girls and got a beautiful picture of five beautiful girls smiling beautifully in a straight line, promoting world peace.

If Natasha stands at the extreme left, who is standing second from left?

A. Cannot say B. Jessica C. Rachel D. Ria

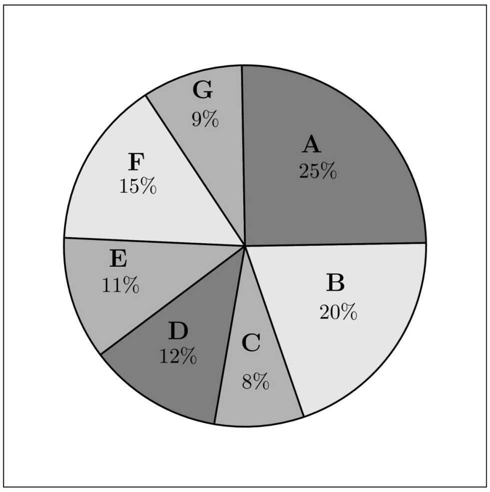
nielit-scb-2020 permutation-and-combination logical-reasoning analytical-aptitude

Answer key☟

15.20 Pie Chart (1)

15.20.1 Pie Chart: NIELIT Scientist B 2020 November: 5

Study the following graph and the table and answer the question given below. (Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

States	Sex and Literacy wise Population Ratio			
	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4

C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

___________ is the percentage of total number of males in F,B and D together to the total population of all the given states.

A. 24% B. 17.5% C. 28.5% D. 29.5%

nielit-scb-2020   data-interpretation   quantitative-aptitude   pie-chart

Answer key☟

15.21 Psychology (2)

15.21.1 Psychology: NIELIT 2023 Feb Scientist C | Section E | Question: 145

Which of the following terms are used to describe the fact that attitudes must be inferred from an individual's overt response to a stimulus?

A. Cognitive consistency
B. Hypothetical mediating variable
C. Selective perception
D. Selective interpretation

nielit2023feb-scientistc   logical-reasoning   psychology   general-aptitude

15.21.2 Psychology: NIELIT 2023 Feb Scientist D | Section E | Question: 137

Which of the following best describes cognitive dissonance?

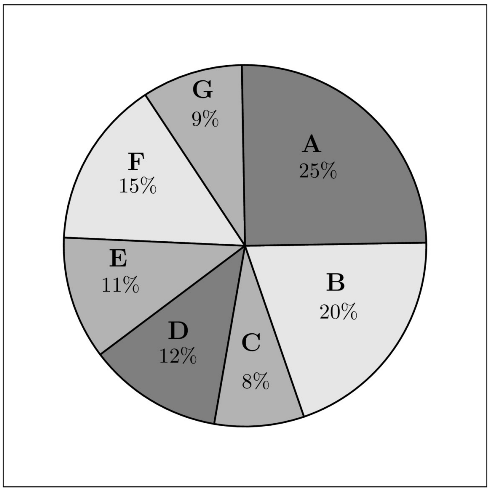
A. An emotional attachment to the organization and a belief in its values.
B. Dissatisfaction expressed through behaviour directed toward leaving the organization.
C. Attitudes used after the fact to make sense out of behaviour.
D. Any incompatibility between two or more attitudes or between behaviour and attitudes.

nielit2023feb-scientistd   logical-reasoning   psychology   analytical-aptitude

15.22 Ratio Proportion (1)

15.22.1 Ratio Proportion: NIELIT Scientist B 2020 November: 4

Study the following graph and the table and answer the question given below. (Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

States	Sex and Literacy wise Population Ratio			
	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

_____________ is the ratio of the number of females in G to the number of females in C.

A. $16:5$ B. $16:7$ C. $15:11$ D. $15:14$

nielit-scb-2020 data-interpretation ratio-proportion analytical-aptitude

Answer key☟

15.23 Round Table Arrangement (2)

15.23.1 Round Table Arrangement: NIELIT Scientific Assistant A 2020 November: 1

Study the following information carefully and answer the question:

Group of girls' gossip with each other. All are sitting surrounding a round table. The name of the girls are Shiksha, radha, Chinu, snigdha and Rani. It is not necessary that they are sitting in the order of the name as mentioned here. Radha is second to the right of Shiksha. Shiksha doesn't sit with Chinu. Rani is second to the right of Radha. Radha sits near snigdha.

Who sits to the left of Shiksha?

A. Rani B. Radha C. Chinu D. Snigdha

nielit-sta-2020 analytical-aptitude round-table-arrangement

Answer key☟

15.23.2 Round Table Arrangement: NIELIT Scientific Assistant A 2020 November: 2

Study the following information carefully and answer the question:

Group of girls' gossip with each other. All are sitting surrounding a round table. The name of the girls are Shiksha, radha, Chinu, snigdha and Rani. It is not necessary that they are sitting in the order of the name as mentioned here. Radha is second to the right of Shiksha. Shiksha doesn't sit with Chinu. Rani is second to the right of Radha. Radha sits near snigdha.

If Radha and Snigdha change their places then who will be second to the left of Rani?

A. Radha B. Snigdha C. Shiksha D. None of the options

nielit-sta-2020 analytical-aptitude round-table-arrangement

Answer key☟

15.24 Seating Arrangement (12)

15.24.1 Seating Arrangement: NIELIT 2021 Dec Scientist A - Section A: 19

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

i. Eleven students $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}, \mathrm{H}, \mathrm{I}, \mathrm{J}$ and K are sitting in a row of the class facing the teacher.
ii. D, who is to the immediate left to F, is second to the right of C.
iii. A, is the second to the right of E, who is at one of the ends.
iv. J is the immediate neighbor of A and B and third to the left of G.
v. H is to the immediate left of D and third to the right of I.

Which of the following statements is true in the context of the above sitting arrangement ?

A. There are three students sitting between D and G.
B. G and C are neighbors sitting to immediate right of H.
C. B is sitting between J and I.
D. K is sitting between A and G.

nielit2021dec-scientista analytical-aptitude seating-arrangement logical-reasoning

Answer key☟

15.24.2 Seating Arrangement: NIELIT 2021 Dec Scientist A - Section A: 20

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

i. Eleven students $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}, \mathrm{H}, \mathrm{I}, \mathrm{J}$ and K are sitting in a row of the class facing the teacher.
ii. D, who is to the immediate left to F, is second to the right of C.

iii. A, is the second to the right of E, who is at one of the ends.
iv. J is the immediate neighbor of A and B and third to the left of G.
v. H is to the immediate left of D and third to the right of I.

Which of the following groups of friends is sitting to the right of G ?

A. $IBJA$ B. $ICHDF$ C. $CHDF$ D. $CHDE$

nielit2021dec-scientista analytical-aptitude seating-arrangement logical-reasoning

Answer key☟

15.24.3 Seating Arrangement: NIELIT 2021 Dec Scientist A - Section A: 21

Read the information given below and on the basis of the information, select the correct alternative for question given after the information.

i. Eleven students $A, B, C, D, E, F, G, H, I, J$ and K are sitting in a row of the class facing the teacher.
ii. D, who is to the immediate left to F, is second to the right of C.
iii. A, is the second to the right of E, who is at one of the ends.
iv. J is the immediate neighbor of A and B and third to the left of G.
v. H is to the immediate left of D and third to the right of I.

Who is sitting in the middle of the row ?

A. C B. I C. B D. G

nielit2021dec-scientista logical-reasoning seating-arrangement analytical-aptitude

Answer key☟

15.24.4 Seating Arrangement: NIELIT 2021 Dec Scientist B - Section A: 19

Directions $(19-21)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

i. Six flats on a floor in two rows facing North and South are allotted to P, Q, R, S, T and U.
ii. Q gets a north facing flat and is not next to S.
iii. S and U get diagonally opposite flats.
iv. R next to U, gets a south facing flat and T gets a north facing flat.

Whose flat is between Q and S?

1. T
2. U

3. R
4. P

nielit2021dec-scientistb analytical-aptitude seating-arrangement logical-reasoning

Answer key☟

15.24.5 Seating Arrangement: NIELIT 2023 Feb Scientist C | Section A | Question: 1

$\mathrm{Q}, \mathrm{R}, \mathrm{S}$, and T are sitting on a bench. P is sitting next to Q, R is sitting next to S, S is not sitting with T who is on the left end of the bench. R is in the second position from the right. P is to the right of Q and T. P and R are sitting together. In which position P is sitting ?

A. Between Q and S
B. Between Q and R
C. Between T and S
D. Between R and T

nielit2023feb-scientistc analytical-aptitude seating-arrangement logical-reasoning

15.24.6 Seating Arrangement: NIELIT 2023 Feb Scientist D | Section A | Question: 17

Directions: $Q(17-21)$ are based on following information :

8 friends named $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}$ and H are sitting in a circle facing each other but not necessarily in the same order. Each of them sings a song but from different movies. The names of the movies are Coolie, Noorie, Sholay, Kranti, Karma, Desh Premi, Tiranga and Satyakama but not necessarily in the same order.

C sings a song from movie Tiranga and sits third to the right of E. Those who sing the songs from movie Desh Premi and Kranti are immediate neighbours of one another but not of E and C. Also, two persons who sing from the movies Noorie and Satyakama are neighbours of each other. F and D do not sing from movie Satyakama. The person who sings from movie Coolie, sits opposite to B, who sings from the movie Sholay. A sings the song from movie Noorie but he is not an immediate neighbour of the person who sings the song from movie Coolie. H sits fourth to the right of the one who sings the song of movie Satyakama G sits third to the right of F, who does not sing song from movie Desh Premi.

Who among the following sits opposite to B?

A. E
B. G
C. F
D. D

nielit2023feb-scientistd logical-reasoning seating-arrangement analytical-aptitude

15.24.7 Seating Arrangement: NIELIT 2023 Feb Scientist D | Section A | Question: 18

Directions : $\mathrm{Q}(17-21)$ are based on following information :

8 friends named $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}$ and H are sitting in a circle facing each other but not necessarily in the same order. Each of them sings a song but from different movies. The names of the movies are Coolie, Noorie, Sholay, Kranti, Karma, Desh Premi, Tiranga and Satyakama but not necessarily in the same order.

C sings a song from movie Tiranga and sits third to the right of E. Those who sing the songs from movie Desh Premi and Kranti are immediate neighbours of one another but not of E and C. Also, two persons who sing from the movies Noorie and Satyakama are neighbours of each other. F and D do not sing from movie Satyakama. The person who sings from movie Coolie, sits opposite to B, who sings from the movie Sholay. A sings the song from movie Noorie but he is not an immediate neighbour of the person who sings the song from movie Coolie. H sits fourth to the right of the one who sings the song of movie Satyakama G sits third to the right of F, who does not sing song from movie Desh Premi.

What is the position of D with respect to H?

A. Immediate right
B. second to the left
C. second to the right
D. third to the left

nielit2023feb-scientistd analytical-aptitude seating-arrangement

15.24.8 Seating Arrangement: NIELIT 2023 Feb Scientist D | Section A | Question: 19

Directions : $\mathrm{Q}(17-21)$ are based on following information :

8 friends named $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}$ and H are sitting in a circle facing each other but not necessarily in the same order. Each of them sings a song but from different movies. The names of the movies are Coolie, Noorie, Sholay, Kranti, Karma, Desh Premi, Tiranga and Satyakama but not necessarily in the same order.

C sings a song from movie Tiranga and sits third to the right of E. Those who sing the songs from movie Desh Premi and Kranti are immediate neighbours of one another but not of E and C. Also, two persons who sing from the movies Noorie and Satyakama are neighbours of each other. F and D do not sing from movie Satyakama. The person who sings from movie Coolie, sits opposite to B, who sings from the movie Sholay. A sings the song from movie Noorie but he is not an immediate neighbour of the person who sings the song from movie Coolie. H sits fourth to the right of the one who sings the song of movie Satyakama G sits third to the right of F, who does not sing song from movie Desh Premi.

How many friends sit between F and C, when counted from the left of C.

A. 4
B. 2
C. 1
D. 3

nielit2023feb-scientistd analytical-aptitude seating-arrangement logical-reasoning

15.24.9 Seating Arrangement: NIELIT 2023 Feb Scientist D | Section A | Question: 20

Directions : $\mathrm{Q}(17-21)$ are based on following information :

8 friends named $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}$ and H are sitting in a circle facing each other but not necessarily in the same order. Each of them sings a song but from different movies. The names of the movies are Coolie, Noorie, Sholay, Kranti, Karma, Desh Premi, Tiranga and Satyakama but not necessarily in the same order.

C sings a song from movie Tiranga and sits third to the right of E. Those who sing the songs from movie Desh Premi and Kranti are immediate neighbours of one another but not of E and C. Also, two persons who sing from the movies Noorie and Satyakama are neighbours of each other. F and D do not sing from movie Satyakama. The person who sings from movie Coolie, sits opposite to B, who sings from the movie Sholay. A sings the song from movie Noorie but he is not an immediate neighbour of the person who sings the song from movie Coolie. H sits fourth to the right of the one who sings the song of movie Satyakama G sits third to the right of F, who does not sing song from movie Desh Premi.

Which of the following movie song G sings?

A. Desh Premi B. Coolie C. Kranti D. Satyakama

nielit2023feb-scientistd analytical-aptitude seating-arrangement

15.24.10 Seating Arrangement: NIELIT 2023 Feb Scientist D | Section A | Question: 21

Directions: $\mathrm{Q}(17-21)$ are based on following information :

8 friends named $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}$ and H are sitting in a circle facing each other but not necessarily in the same order. Each of them sings a song but from different movies. The names of the movies are Coolie, Noorie, Sholay, Kranti, Karma, Desh Premi, Tiranga and Satyakama but not necessarily in the same order.

C sings a song from movie Tiranga and sits third to the right of E. Those who sing the songs from movie Desh Premi and Kranti are immediate neighbours of one another but not of E and C. Also, two persons who sing from the movies Noorie and Satyakama are neighbours of each other. F and D do not sing from movie Satyakama. The person who sings from movie Coolie, sits opposite to B, who sings from the movie Sholay. A sings the song from movie Noorie but he is not an immediate neighbour of the person who sings the song from movie Coolie. H sits fourth to the right of the one who sings the song of movie Satyakama G sits third to the right of F, who does not sing song from movie Desh Premi.

Who among the following sings a song from movie Kranti ?

A. F B. E C. D D. G

nielit2023feb-scientistd analytical-aptitude seating-arrangement logical-reasoning

15.24.11 Seating Arrangement: NIELIT 2023 Feb Scientist D | Section A | Question: 3

Study the following information carefully and answer the questions given below.
Six persons - $\mathrm{K}, \mathrm{L}, \mathrm{M}, \mathrm{N}, \mathrm{O}$ and P - lives on different floor of a building having six floors numbered one to six (the ground floor is numbered 1, the floor above it is numbered 2 and so on and the topmost floor is numbered 6).

L lives on an even numbered floor. L lives on a floor immediately below K's floor and immediately above M 's floor. P lives on a floor immediately above N's floor. P lives on an even numbered floor. O does not live on floor number 4.

Who amongst the following live on the floors exactly between K and P ?

A. O and L
B. L and N
C. L and M
D. M and N

nielit2023feb-scientistd analytical-aptitude seating-arrangement logical-reasoning

Answer key☟

15.24.12 Seating Arrangement: NIELIT Scientific Assistant A 2020 November: 34

Choose the alternative to decide whether the data given in the statements is/are sufficient to answer the question based on the following information.

Five persons $\mathrm{A, B, C, D,}$ and E are sitting in a row. Who is sitting in the middle?

Statements:

I. E is to the left of B.
II. B is in-between C and E.
III. D is in-between E and A.

Choose which of the following will be sufficient to find out who is sitting in the middle?

A. Only (I) and (II)
B. Only (II) and (III)
C. Only (I) and (III)
D. All (I), (II) and (III)

nielit-sta-2020 analytical-aptitude seating-arrangement

Answer key☟

15.25 Sequence Series (15)

15.25.1 Sequence Series: NIELIT 2021 Dec Scientist A - Section A: 8

What value should come in place of the question mark $(?)$ in the series given below ?

$128, 61, Y, 64, 63, S, 32, 65, N, 16, 67, J, 8, 69, G, ?, ?, ? :$

A. $2, 70, J$
B. $3, 70, E$
C. $4, 70, E$
D. $4, 71, E$

nielit2021dec-scientista sequence-series analytical-aptitude

Answer key☟

15.25.2 Sequence Series: NIELIT 2021 Dec Scientist A - Section A: 9

Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

A training college has to conduct a refresher course for teachers of seven different subjects – Mechanics, Psychology, Philosophy, Sociology, Economics, Science and

Engineering from November 22 to November 29.

i. Course should start with Psychology.
ii. November 23, being Sunday, should be a holiday.
iii. Science subject should be on the previous day of the Engineering subject.
iv. Course should end with Mechanics subject.
v. Philosophy should be immediately after holiday.
vi. There should be a gap of one day between Economics and Engineering.
vii. There should be a gap of two days between Sociology and Economics.

Which subject precedes Mechanics?

A. Psychology B. Mechanics C. Economics D. Sociology

nielit2021dec-scientista analytical-aptitude sequence-series

Answer key☟

15.25.3 Sequence Series: NIELIT 2022 April Scientist B | Section A | Question: 33

Find the wrong term in the series $5, 11, 29, 83, 245, 765, 2189, 6563$

A. 245 B. 765 C. 2189 D. 6563

nielit2022apr-scientistb sequence-series analytical-aptitude

Answer key☟

15.25.4 Sequence Series: NIELIT 2023 Feb Scientist C | Section A | Question: 13

Which of the following will come in place of the question mark (?) in the following sequence: $16\text{D}7, 18\text{G}10, 21\text{KI}4, 25\text{P}19, ?$

A. 30 V20
B. 30 V25
C. 30 V24
D. 29 V25

nielit2023feb-scientistc sequence-series analytical-aptitude

Answer key☟

15.25.5 Sequence Series: NIELIT 2023 Feb Scientist C | Section A | Question: 14

Find the next number in the sequence: $16, 30, 54, 88, 132 \ldots$

A. 186 B. 188 C. 190 D. 206

nielit2023feb-scientistc sequence-series analytical-aptitude quantitative-aptitude

Answer key☟

15.25.6 Sequence Series: NIELIT 2023 Feb Scientist C | Section A | Question: 24

$\mathrm{DNN}, \mathrm{FPP}, \mathrm{HRR}, , \mathrm{LVV}.$

A. GRR B. GSS C. JTT D. ITT

nielit2023feb-scientistc analytical-aptitude sequence-series

Answer key

15.25.7 Sequence Series: NIELIT 2023 Feb Scientist C | Section A | Question: 25

Look at this series : $\mathrm{F2}, \mathrm{D8}, \mathrm{C16}, \mathrm{B32}$. What should fill the blank?

A. $\mathrm{A16}$ B. $\mathrm{G4}$ C. $\mathrm{E4}$ D. $\mathrm{E3}$

nielit2023feb-scientistc sequence-series analytical-aptitude

15.25.8 Sequence Series: NIELIT 2023 Feb Scientist C | Section A | Question: 6

Complete the series $\mathrm{E\text{-}5}, \mathrm{G\text{-}7}, \mathrm{I\text{-}9}, \mathrm{K\text{-}11}$,... ?

A. $\mathrm{L\text{-}13}, \mathrm{N\text{-}14}$
B. $\mathrm{M}-13, \mathrm{O}-15$
C. $\mathrm{L\text{-}12}, \mathrm{M\text{-}14}$
D. $\mathrm{K}-12, \mathrm{M}-14$

nielit2023feb-scientistc sequence-series analytical-aptitude

Answer key

15.25.9 Sequence Series: NIELIT 2023 Feb Scientist D | Section A | Question: 15

In the following Question, replace the ? with suitable options: $7, 12, 22, 42, 82, ?$

A. 143 B. 173 C. 183 D. 162

nielit2023feb-scientistd sequence-series analytical-aptitude quantitative-aptitude

Answer key

15.25.10 Sequence Series: NIELIT 2023 Feb Scientist D | Section A | Question: 22

Select the number that will come next in the following series $7, 14, 33, 70, 131,$ __________.

A. 256 B. 222 C. 262 D. 216

nielit2023feb-scientistd sequence-series analytical-aptitude quantitative-aptitude

15.25.11 Sequence Series: NIELIT Scientific Assistant A 2020 November: 11

$5\ 16\ 49\ 104\ 181$ __________

A. 271 B. 298 C. 280 D. 281

nielit-sta-2020 analytical-aptitude sequence-series

Answer key

15.25.12 Sequence Series: NIELIT Scientific Assistant A 2020 November: 12

$14, 28, 20, 40, 32, 64,$ __________

A. 52 B. 56 C. 96 D. 128

nielit-sta-2020 analytical-aptitude sequence-series

Answer key

15.25.13 Sequence Series: NIELIT Scientific Assistant A 2020 November: 20

Complete the following series.
$4, 27, 256, 3125,$ ______

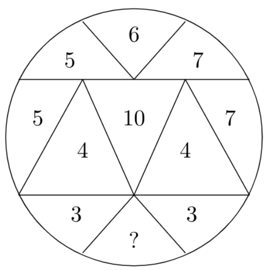
A. 46656 B. 6250 C. 800000 D. 1024

nielit-sta-2020 analytical-aptitude sequence-series

Answer key

15.25.14 Sequence Series: NIELIT Scientific Assistant A 2020 November: 23

Find the missing number.



A. 14 B. 10 C. 9 D. 3

nielit-sta-2020 analytical-aptitude sequence-series

Answer key

15.25.15 Sequence Series: NIELIT Scientific Assistant A 2020 November: 30

Considering 5 as the 1^{st} element in the sequence $5, 11, 23, 47$. What is the 6^{th} element in the sequence?

A. 191 B. 172 C. 342 D. 106

nielit-sta-2020 analytical-aptitude sequence-series

Answer key☟

15.26 Statements Follow (6)

15.26.1 Statements Follow: NIELIT 2022 April Scientist B | Section A | Question: 42

It is 8.00 p.m., when can Hemant get next bus for Ramnagar from Dhanpur ?

I. Buses for Ramnagar leave after every 30 minutes, till 10 p.m.
II. Fifteen minutes ago, one bus has left for Ramnagar.

A. If the data in statement I alone are sufficient to answer the question.
B. If the data in statement II alone are sufficient to answer the question.
C. If the data either in I or II alone are sufficient to answer the question.
D. If the data in both the statements together are needed.

nielit2022apr-scientistb statements-follow logical-reasoning analytical-aptitude

Answer key☟

15.26.2 Statements Follow: NIELIT Scientific Assistant A 2020 November: 5

Two statements followed by four conclusions numbered from (I) to (IV) are given. You have to take the two statements to be true even if these seem to be at variance from the commonly known facts. Read all the conclusions and decide which of the given conclusions logically follow from the two given statements disregarding commonly known facts.

- All Shoes are Socks.
- Some Socks are Gloves.

Conclusions:

I. Some Shoes are Gloves
II. Some Socks are Shoes
III. All Gloves are Shoes
IV. No Shoes are Gloves

A. Only (I) follows
B. Only (II) follows
C. Only (III) follows
D. Only (IV) follows

nielit-sta-2020 analytical-aptitude logical-reasoning statements-follow

Answer key☟

15.26.3 Statements Follow: NIELIT Scientific Assistant A 2020 November: 6

Two statements followed by four conclusions numbered from (I) to (IV) are given. You have to take the two statements to be true even if these seem to be at variance from the commonly known facts. Read all the conclusions and decide which of the given conclusions logically follow from the two given statements disregarding commonly known facts.

- All Boys are Girls.
- No Girl is a Man.

Conclusions:

I. No Boy is a Man
II. Some Boys are Man
III. All Girls are Boys
IV. Some Man are Boys

A. Only (III) follows
B. Only (I) follows
C. All follows
D. None follows

nielit-sta-2020 analytical-aptitude logical-reasoning statements-follow

Answer key☟

15.26.4 Statements Follow: NIELIT Scientific Assistant A 2020 November: 7

Two statements followed by four conclusions numbered from (I) to (IV) are given. You have to take the two statements to be true even if these seem to be at variance from the commonly known facts. Read all the conclusions and decide which of the given conclusions logically follow from the two given statements disregarding commonly known facts.

- All Sentences are Words
- All Words are Alphabets

Conclusions:

I. All Words are sentences
II. All Sentences are alphabets
III. All alphabets are words
IV. Some alphabets are words

A. Only (I) and (III) follows
B. Only (II) , (III) and (IV) follows
C. Only (II) and (IV) follows
D. All follows

nielit-sta-2020 analytical-aptitude logical-reasoning statements-follow

Answer key

15.26.5 Statements Follow: NIELIT Scientific Assistant A 2020 November: 9

Refer the statement and solve the question according to the conclusions.

Statement:

- Some Pigeons are Bird;
- Some Birds are Alive

Conclusion:

I. Some Pigeons are Alive
II. Some Birds are Pigeons

A. Only (I) follows
B. Only (II) follows
C. Both (I) & (II) follows
D. None follows

nielit-sta-2020 analytical-aptitude logical-reasoning statements-follow

Answer key

15.26.6 Statements Follow: NIELIT Scientist B 2020 November: 12

In the following question below are given three statements followed by three conclusions numbered I,II and III

You have to take the two statements to be true even if they seem to be at variance from the commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follow from the two given statements, disregarding commonly known facts.

Statements:

Some pigeons are eagles.

All eagles are sparrows.

Some sparrows are not pigeons.

Conclusions:

I. Some sparrows are pigeons.
II. All pigeons are sparrows.
III. All eagles are pigeons.

A. Only I follows
B. Only II follows
C. Only III follows
D. Both I and III follows

nielit-scb-2020 analytical-aptitude logical-reasoning statements-follow

Answer key

15.27 Trigonometry (1)

15.27.1 Trigonometry: NIELIT 2023 Feb Scientist C | Section D | Question: 104

In triangle $\mathrm{A\ B\ C}$ the value of $\angle \mathrm{A}$ is given by $5 \cos \mathrm{A} - 3 = 0$, then the equation whose roots are $\sin \mathrm{A}$ and $\tan \mathrm{A}$ is :

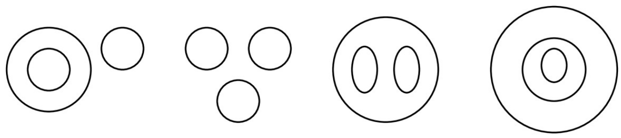
A. $15x^2 - 32x - 16 = 0$
B. $15x^2 + 32x - 16 = 0$
C. $15x^2 + 32x + 16 = 0$
D. $15x^2 - 32x + 16 = 0$

nielit2023feb-scientistc trigonometry geometry

15.28 Venn Diagram (1)

15.28.1 Venn Diagram: NIELIT 2021 Dec Scientist B - Section A: 2

Use following diagrams to answer question number 1 to 2 :



(I) (II) (III) (IV)

Oxygen, Atmosphere, Nitrogen

A. II
B. I
C. IV
D. III

nielit2021dec-scientistb analytical-aptitude venn-diagram

15.29 Verbal Reasoning (8)

15.29.1 Verbal Reasoning: NIELIT 2021 Dec Scientist B - Section A: 7

Directions $(7-8)$: In the given questions below, a statement is given followed by two conclusions numbered I and II. You have to take the statement to be true. Read both the conclusions and decide which of the two or both follow from the given statement. Given answer :

A. If only conclusion I follows.
B. If only conclusion II follows.
C. If either I or II follows.
D. If neither I nor II follows.

The top management has asked the four managers either to resign by tomorrow or face the order of service termination. Three of them have resigned till this very evening.

I. The manager who did not resign yesterday will resign tomorrow.
II. The management will terminate the service of one manager.

nielit2021dec-scientistb logical-reasoning analytical-aptitude verbal-reasoning

Answer key

15.29.2 Verbal Reasoning: NIELIT 2021 Dec Scientist B - Section A: 9

Directions $(9-13)$: Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

There are five persons P, Q, R, S and T. One is football player, one is chess player, one is hockey player. P and S are unmarried ladies and do not participate in any game. None of the ladies plays chess or football. There is a married couple in which T is the husband. Q is the brother of R and is neither a chess player nor a hockey player.

Which of the following is the correct group of ladies ?

A. P, Q and R B. Q, R and S C. P, Q and S D. P, R and S

nielit2021dec-scientistb logical-reasoning verbal-reasoning analytical-aptitude

Answer key

15.29.3 Verbal Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 10

Which of the following will be changed form of the word "OBLIQUE" when the word is written again by substituting each vowel by the second letter following it in the English alphabet and substituting each consonant by the third letter following it in the English alphabet?

A. MEDGTSD B. QEOKTXG C. QEOKTWG D. RDNLSXH

nielit2022apr-scientistb analytical-aptitude verbal-reasoning most-appropriate-word

Answer key

15.29.4 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 22

If in a certain language, MADRAS is coded as NBESBT, how is BOMBAY coded in that code?

A. CPNCBX
B. CPNCBZ
C. CPOCBZ
D. CQOCBZ

nielit2023feb-scientistc verbal-reasoning logical-reasoning

Answer key

15.29.5 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 28

Directions: $(28-29)$ In each of the following letter series, some of the letters are missing which are given in that order as one of the alternatives below it. Choose the correct alternative.

_zy_zxy_yxzx_zyx_xy

A. yxzyz B. zxyzy C. yzxyx D. xyzzy

nielit2023feb-scientistc analytical-aptitude verbal-reasoning sequence-series

Answer key

15.29.6 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 29

Directions **(29-30)**: In each question below is given a statement followed by two course of action numbered (I) and (II). You have to assume everything in the statement to be true and on the basis of the information given in the statement, decide which of the suggested course of action logically follow (s) for pursuing. Give answer as:

(A) if only (I) follows

(B) if only (I) follows

(C) neither (I) nor (II) follows

(D) if both (I) & (II) follows

Statement: There has been large number of cases of Internet hacking in the recent months creating panic among the internet users.
Course of action :

(I): The government machinery should make an all out effort to nab those who are responsible and put them behind the bars.

(II): The internet users should be advised to stay away from using internet till the culprits are caught.

A. only (I) follows
B. only (II) follows
C. neither (I) nor (II) follows
D. both (I) and (II) follows

nielit2023feb-scientistd logical-reasoning verbal-reasoning

Answer key

15.29.7 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 30

Directions **(29-30)**: In each question below is given a statement followed by two course of action numbered (I) and (II). You have to assume everything in the statement to be true and on the basis of the information given in the statement, decide which of the suggested course of action logically follow (s) for pursuing. Give answer as:

(A) if only (I) follows

(B) if only (I) follows

(C) neither (I) nor (II) follows

(D) if both (I) & (II) follows

Statement: The cinema halls are incurring heavy losses these days as people prefer to watch movies in home on TV than to visit cinema hall.

Course of action :

(I): The cinema halls should be demolished and residential multistory buildings should be constructed there.

(II): The cinema halls should be converted into shopping malls.

A. only (I) follows
B. only (II) follows
C. neither (I) nor (II) follows
D. both (I) and (II) follows

nielit2023feb-scientistd logical-reasoning verbal-reasoning analytical-aptitude

15.29.8 Verbal Reasoning: NIELIT Scientist B 2020 November: 22

Number of letter repeated in the given word $'MEASUREMENTS'$ are indicated in front of each alternative. Identify the correct alternative.

A. $M_2E_2A_2S_2U_1R_1N_1T_1$
B. $M_2E_3A_1S_1U_2R_1N_2T_1$
C. $M_2E_2A_1S_2U_1R_1N_1T_1$
D. $M_2E_3A_1S_2U_1R_1N_1T_1$

nielit-scb-2020 logical-reasoning verbal-reasoning analytical-aptitude

Answer key☟

Answer Keys

15.0.1	D	15.0.2	A	15.0.3	C	15.0.4	A	15.0.5	B
15.0.6	B	15.0.7	D	15.0.8	D	15.1.1	A	15.1.2	B
15.1.3	A	15.2.1	A	15.3.1	A	15.4.1	D	15.4.2	C
15.5.1	C	15.6.1	B	15.6.2	B	15.6.3	A	15.7.1	B
15.8.1	B	15.8.2	D	15.8.3	B	15.8.4	C	15.8.5	C
15.8.6	A	15.8.7	D	15.8.8	D	15.8.9	C	15.8.10	B
15.8.11	A	15.8.12	A	15.8.13	A	15.8.14	B	15.8.15	D
15.8.16	C	15.8.17	B	15.8.18	A	15.8.19	A	15.8.20	C
15.8.21	B	15.8.22	D	15.8.23	C	15.9.1	A	15.9.2	C
15.9.3	B	15.9.4	C	15.10.1	A	15.10.2	A	15.10.3	B
15.11.1	A	15.11.2	A	15.11.3	A	15.12.1	B	15.12.2	C
15.12.3	C	15.13.1	D	15.13.2	B	15.13.3	C	15.13.4	A
15.13.5	B	15.14.1	C	15.14.2	A	15.14.3	C	15.14.4	A
15.14.5	D	15.14.6	D	15.14.7	A	15.14.8	C	15.14.9	B
15.14.10	D	15.14.11	D	15.14.12	C	15.14.13	X	15.14.14	C

15.14.15	D	15.14.16	B	15.14.17	D	15.14.18	B	15.14.19	D
15.14.20	B	15.14.21	A	15.14.22	D	15.14.23	C	15.14.24	C
15.14.25	A	15.14.26	C	15.14.27	B	15.14.28	D	15.14.29	D
15.14.30	B	15.14.31	B	15.14.32	C	15.14.33	A	15.14.34	B
15.14.35	C	15.14.36	A	15.14.37	D	15.14.38	C	15.14.39	D
15.14.40	D	15.14.41	B	15.14.42	D	15.14.43	B	15.14.44	D
15.14.45	C	15.14.46	B	15.14.47	B	15.14.48	B	15.14.49	A
15.14.50	D	15.14.51	C	15.14.52	B	15.14.53	D	15.14.54	C
15.14.55	C	15.14.56	A	15.14.57	D	15.14.58	C	15.14.59	B
15.14.60	B	15.14.61	D	15.14.62	B	15.15.1	B	15.16.1	C
15.17.1	A	15.17.2	A	15.17.3	B	15.17.4	B	15.17.5	C
15.17.6	A	15.17.7	A	15.18.1	D	15.19.1	A	15.20.1	C
15.21.1	B	15.21.2	D	15.22.1	D	15.23.1	A	15.23.2	B
15.24.1	C	15.24.2	C	15.24.3	B	15.24.4	A	15.24.5	B
15.24.6	A	15.24.7	C	15.24.8	A	15.24.9	D	15.24.10	A
15.24.11	C	15.24.12	D	15.25.1	D	15.25.2	D	15.25.3	B
15.25.4	D	15.25.5	A	15.25.6	C	15.25.7	C	15.25.8	B
15.25.9	D	15.25.10	B	15.25.11	C	15.25.12	B	15.25.13	A
15.25.14	D	15.25.15	A	15.26.1	D	15.26.2	B	15.26.3	B
15.26.4	C	15.26.5	B	15.26.6	A	15.27.1	D	15.28.1	C
15.29.1	C	15.29.2	D	15.29.3	C	15.29.4	B	15.29.5	A
15.29.6	A	15.29.7	C	15.29.8	D				

16.0.1 NIELIT 2017 Scientist B (CS) - Section A: 5

The present ages of three persons in proportions $4:7:9$. Eight years ago, the sum of their ages was 56. Find their present ages (in years).

A. $8, 20, 28$
B. $16, 28, 36$
C. $20, 35, 45$
D. None of the above options

nielit2017-scientistb-cs general-aptitude

Answer key☟

16.0.2 NIELIT 2017 Scientist B (CS) - Section A: 3

The percentage profit earned by selling an article for $\text{Rs.}1,920$ is equal to the percentage loss incurred by selling the same article for $\text{Rs.}1,280$. At what price should the article be sold to make 25% profit?

A. $\text{Rs.}2,000$
B. $\text{Rs.}2,200$
C. $\text{Rs.}2,400$
D. Data inadequate

nielit2017-scientistb-cs general-aptitude

Answer key☟

16.0.3 NIELIT 2017 Scientist B (CS) - Section A: 1

What is the maximum number of distinct handshakes that can happen in the room with 5 people in it?

A. 15
B. 10
C. 6
D. 5

nielit2017-scientistb-cs general-aptitude

Answer key☟

16.0.4 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 17

A can is filled with 5 paise coins. Another can is filled with 10 paise coins. Another can is filled with 25 paise coins. All the cans are given wrong labels. If the can labeled 25 paise is not having the 10 paise coins, what will the can, labeled 10 paise have?

A. 25 paise
B. 5 paise
C. 10 paise
D. cannot be determined

nielit2017oct-assistanta-cs general-aptitude quantitative-aptitude

Answer key☟

16.0.5 NIELIT Scientist B 2020 November: 28

Gopal went to a fruit market with certain amount of money. With this money he can buy either 50 oranges or 40 mangoes. He retains 10% of the money for taxi fare. If he buys 20 mangoes, then the number of oranges he can buy with balance amount is:

A. 25 B. 20 C. 18 D. 6

nielit-scb-2020 quantitative-aptitude

Answer key☟

16.0.6 NIELIT 2021 Dec Scientist B - Section A: 27

In a fraction, numerator is increased by 25% and the denominator is diminished by 10%. The new fraction obtained is $5/9$. The original fraction is :

A. $2/5$ B. $5/9$ C. $3/5$ D. None of the above

nielit2021dec-scientistb quantitative-aptitude

Answer key☟

16.0.7 NIELIT 2021 Dec Scientist B - Section A: 31

A, B and C rented a pasture by paying ₹ 2160 per month. They put $60, 40$ and 20 sheep respectively. A sells $1/3$ of his sheep to B after 6 months and after 3 months more C sells $2/5$ of his sheep to $\text{A}.$ Find the rent paid by C at the end of the year :

A. ₹ 4355
B. ₹ 3888
C. ₹ 2464
D. ₹ 6224

nielit2021dec-scientistb quantitative-aptitude analytical-aptitude

16.0.8 NIELIT 2022 April Scientist B | Section A | Question: 20

Today it is Thursday. After 132 days, it will be :

A. Monday B. Sunday C. Wednesday D. Thursday

nielit2022apr-scientistb analytical-aptitude quantitative-aptitude

Answer key☟

16.0.9 NIELIT 2022 April Scientist B | Section A | Question: 15

Which largest number of five digits is divisible by 99?

A. 99999 B. 99981 C. 99909 D. 99990

nielit2022apr-scientistb quantitative-aptitude

Answer key☟

16.0.10 NIELIT 2023 Feb Scientist D | Section C | Question: 77

Find the smallest number that must be subtracted from 9100 to make it perfect square?

A. 25 B. 50 C. 75 D. 100

nielit2023feb-scientistd quantitative-aptitude

16.0.11 NIELIT 2023 Feb Scientist D | Section C | Question: 72

Find the least number that must be added to 7900 to obtain a perfect square.

A. 67 B. 87 C. 97 D. 77

nielit2023feb-scientistd quantitative-aptitude

16.0.12 NIELIT 2023 Feb Scientist D | Section C | Question: 69

In a business, B is a sleeping partner and A is a working partner. A invests 5000 and B invests 6000. A receives 12.5% of profit for managing the business and the remaining amount is divided in proportion to their capitals. A's share of profit in a profit of Rs. 880 is :

A. 350 B. 400 C. 420 D. 460

nielit2023feb-scientistd quantitative-aptitude

Answer key☟

16.0.13 NIELIT 2023 Feb Scientist D | Section C | Question: 66

In a Km race A can beat B by $80\ \mathrm{m}$ and B can beat C by $60\ \mathrm{m}$. In the same race, A can beat C by ________.

A. $135.2\ \mathrm{m}$ B. $130.5\ \mathrm{m}$
C. $142\ \mathrm{m}$ D. $132.5\ \mathrm{m}$

nielit2023feb-scientistd quantitative-aptitude analytical-aptitude

16.0.14 NIELIT 2023 Feb Scientist D | Section D | Question: 119

Ram visits recreation club after every five days, Harish visits after every 24 days, while Rajesh goes there after every 9 days. If all three of them met at the club on a Sunday. The days of the week will all three meet again is :

A. Sunday B. Thursday C. Wednesday D. Monday nielit2023feb-scientistd quantitative-aptitude analytical-aptitude

16.0.15 NIELIT 2023 Feb Scientist D | Section D | Question: 118

From a rope of length $38\frac{3}{5}$ m, a piece of length $5\frac{3}{38}$ m is cut off. The length of remaining rope is:

A. $190\frac{99}{33}$ B. $33\frac{90}{190}$ C. $33\frac{99}{190}$ D. 1

nielit2023feb-scientistd quantitative-aptitude

16.0.16 NIELIT 2023 Feb Scientist C | Section C | Question: 89

Find the least number that when divided by $16, 18$ and 20 leaves a remainder 4 in each case, but is completely divisible by 7.

A. 364 B. 2254 C. 2964 D. 2884

nielit2023feb-scientistc quantitative-aptitude

16.0.17 NIELIT 2023 Feb Scientist C | Section C | Question: 87

Ramesh has total of Rs 56000 in four denominations of currency notes namely Rs 100, Rs 200, Rs 500, and Rs $2,000$. How many total currency notes he has, if equal number of currency notes in each denomination are held by him.

A. 28 B. 32 C. 24 D. 20

nielit2023feb-scientistc quantitative-aptitude

16.0.18 NIELIT 2023 Feb Scientist C | Section C | Question: 71

Which is the largest of the following fraction?

A. $3/5$ B. $2/3$ C. $8/11$ D. $11/17$

nielit2023feb-scientistc analytical-aptitude quantitative-aptitude

Answer key☟

16.0.19 NIELIT 2023 Feb Scientist C | Section D | Question: 120

A man invested $\frac{1}{3}$rd of his capital at 7%, $\frac{1}{4}$ th at 8% and remainder at 10%. If his annual income is Rs. 561. Find the capital.

A. Rs. 5000 B. Rs. 6000 C. Rs. 6600 D. Rs. 5600

nielit2023feb-scientistc quantitative-aptitude

16.0.20 NIELIT 2023 Feb Scientist C | Section D | Question: 117

Ram and shyam are two different persons, but when they worked together, they completed a job in 10 days. Had Ram worked at half of his efficiency and shyam at five times of his efficiency, they would have finished the job in 50% of the scheduled time. Shyam can complete the job working alone in y days. Then y is :

A. 20 days B. 30 days C. 40 days D. 35 days

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.21 NIELIT 2023 Feb Scientist C | Section D | Question: 116

The number of integral pair (x, y) satisfying the equation: $(x-y)^2+2y^2=27$ is :

A. 8 B. 7 C. 6 D. 5

nielit2023feb-scientistc analytical-aptitude

16.0.22 NIELIT 2023 Feb Scientist C | Section D | Question: 115

A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate it would have fetched Rs.480 more. Find the principal.

A. Rs. 4000 B. Rs. 5000 C. Rs. 6000 D. Rs. 3000

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.23 NIELIT 2023 Feb Scientist C | Section D | Question: 114

A shopkeeper advertises for selling cloth at 4% loss. However, by using a false meter scale he actually gains 25%. What is the actual length of the scale?

A. 76.8 cm B. 77.8 cm
C. 74.8 cm D. 75.8 cm

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.24 NIELIT 2023 Feb Scientist C | Section D | Question: 102

$\text{P}, \text{Q}, \text{R}$ are three typist who working simultaneously can type 216 pages in 4 hours. In one hour, R can type as many pages more than Q as Q can type more than P. During a period of 5 hours, R can type as many pages as P can during 7 hours. How many pages does each of them type per hour.

A. $14, 17, 20$
B. $15, 17, 22$
C. $15, 18, 21$
D. $16, 18, 22$

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.25 NIELIT 2023 Feb Scientist C | Section D | Question: 101

x, y, z are three positive integers such that x is greater than y & y is greater than z then which of the following is always true:

A. $y\%$ of z is less than $x\%$ of y.
B. $y\%$ of x is less than $z\%$ of y.
C. $z\%$ of x is less than $y\%$ of z.
D. All of the above.

nielit2023feb-scientistc analytical-aptitude quantitative-aptitude

16.1 Age Relation (1)

16.1.1 Age Relation: NIELIT 2018-1

A Professor passed one sixth of his life in childhood, one twelfth in youth, and one seventh more as a bachelor, five years after his marriage a son was born who died four years before his father at half his final age, then what is the age of Professor?

A. 84 years
B. 74 years
C. 64 years
D. 54 years

nielit-2018 general-aptitude quantitative-aptitude age-relation

Answer key☟

16.2 Algebra (1)

16.2.1 Algebra: NIELIT 2023 Feb Scientist D | Section D | Question: 105

If $x = 3 + 3^{\frac{2}{3}} + 3^{\frac{1}{3}}$, then the value of $x^3 - 9x^2 + 18x - 12$ is :

A. 1
B. -1
C. 0
D. 2

nielit2023feb-scientistd algebra functions

Answer key☟

16.3 Alligation Mixture (1)

16.3.1 Alligation Mixture: NIELIT 2023 Feb Scientist D | Section D | Question: 112

The ratio of water and milk in a 30-litre mixture is $7:3$. Find the quantity of water to be added to the mixture in order to make the ratio $6:1$.

A. 30　　B. 32　　C. 35　　D. 33

nielit2023feb-scientistd　quantitative-aptitude　alligation-mixture

Answer key☟

16.4 Approximation (1)

16.4.1 Approximation: NIELIT 2023 Feb Scientist D | Section D | Question: 101

Which is the closest approximation of $0.3333 \times 0.25 \times 0.499 \times 0.125 \times 24$

A. $\frac{1}{8}$　　B. $\frac{3}{4}$　　C. $\frac{3}{8}$　　D. $\frac{2}{5}$

nielit2023feb-scientistd　quantitative-aptitude　approximation　analytical-aptitude

Answer key☟

16.5 Area (1)

16.5.1 Area: NIELIT 2023 Feb Scientist C | Section D | Question: 118

The area of a rhombus is $\frac{x^2}{2} + 2x + \frac{3}{2} \quad (x \geq 0)$ then the smaller diagonal is :

A. $x+2$　　B. $x+3$　　C. $x+1$　　D. 1

nielit2023feb-scientistc　geometry　area　analytical-aptitude

16.6 Arithmetic Series (4)

16.6.1 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section C | Question: 66

The sum of two numbers is 20 and their geometric mean is 20% lower than their arithmetic mean. Find the ratio of the numbers.

A. $4:1$　　B. $9:1$　　C. $1:1$　　D. $17:3$

nielit2023feb-scientistc　quantitative-aptitude　percentage　ratio-proportion　arithmetic-series

Answer key☟

16.6.2 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section C | Question: 88

Find the sum of all two-digit numbers that give a remainder of 3 when they are divided by 7.

A. 686 B. 676 C. 666 D. 656

nielit2023feb-scientistc arithmetic-series quantitative-aptitude

16.6.3 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section D | Question: 107

Find the sum of all three digit numbers that gives a remainder 4 when they are divided by 5.

A. 98270 B. 99270 C. 102090 D. 90270

nielit2023feb-scientistc quantitative-aptitude arithmetic-series

16.6.4 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section D | Question: 108

The value of $5 \times 2 - [3 - \{5 - (7 + 2 \times 4 - 19)\}]$ is :

A. 0 B. 14 C. 16 D. -5

nielit2023feb-scientistc quantitative-aptitude arithmetic-series

Answer key☟

16.7 Average (5)

16.7.1 Average: NIELIT 2022 April Scientist B | Section A | Question: 40

What is the average of first five multiples of 12 ?

A. 36 B. 38 C. 40 D. 42

nielit2022apr-scientistb quantitative-aptitude array average

Answer key☟

16.7.2 Average: NIELIT 2023 Feb Scientist C | Section D | Question: 99

In the month of July of a certain year, the average daily expenditure of an organization was Rs. 68. For the first 15 days of the month, the average daily expenditure was Rs. 85 and for the last 17 days, Rs. 51. Find the amount spent by the organization on the 15^{th} of the month.

A. Rs. 42 B. Rs. 36 C. Rs. 34 D. Rs. 52

nielit2023feb-scientistc average quantitative-aptitude

16.7.3 Average: NIELIT 2023 Feb Scientist D | Section C | Question: 62

It rained as much on Wednesday as on all the others days of the week combined. If the average rainfall for the whole week was 3 cm, then how much did it rain on Wednesday?

A. 2.625 cm B. 3 cm
C. 10.5 cm D. 15 cm

nielit2023feb-scientistd analytical-aptitude average quantitative-aptitude

Answer key☟

16.7.4 Average: NIELIT 2023 Feb Scientist D | Section C | Question: 89

The average of 100 numbers is 60. Numbers $70 \& 80$ are removed What is the new average?

A. 59.69 B. 50.69 C. 11.69 D. 27.27

nielit2023feb-scientistd quantitative-aptitude average

Answer key☟

16.7.5 Average: NIELIT 2023 Feb Scientist D | Section D | Question: 117

A person travels three equal distances at a speed of $x \text{ km/h}, y \text{ km/h}, z \text{ km/h}$ respectively. What will be the average speed during the whole journey?

A. $\frac{xyz}{xy+yz+zx}$ B. $\frac{xy+yz+zx}{xyz}$
C. $\frac{3xyz}{xy+yz+zx}$ D. $\frac{3xyz}{x+y+z}$

nielit2023feb-scientistd average speed-time-distance quantitative-aptitude analytical-aptitude

16.8 Clock Time (1)

16.8.1 Clock Time: NIELIT 2022 April Scientist B | Section A | Question: 22

A clock is set at 4 am. It loses 16 minutes in 24 hours. What will be the correct time when the clock indicates 9 pm on the 4^{th} day ?

A. 8 pm B. 7 pm C. 10 pm D. 11 pm

nielit2022apr-scientistb clock-time analytical-aptitude quantitative-aptitude

Answer key

16.9 Compound Interest (4)

16.9.1 Compound Interest: NIELIT 2022 April Scientist B | Section A | Question: 32

What is the compound interest on $\text{Rs.}2500$ for 2 years at rate of interest 4% per annum ?

A. $\text{Rs.}180$ B. $\text{Rs.}204$ C. $\text{Rs.}210$ D. $\text{Rs.}220$

nielit2022apr-scientistb quantitative-aptitude compound-interest

Answer key

16.9.2 Compound Interest: NIELIT 2023 Feb Scientist C | Section C | Question: 65

A mother divided an amount of ₹$61,000$ between her two daughters aged 18 years and 16 years respectively and deposited their shares in a bond. If the interest rate is 20% compounded annually and if each received the same amount as the other when she attained the age of 20 years, their shares are :

A. ₹$35,600$ and ₹$25,400$
B. ₹ 30500 each
C. ₹ $24,000$ and ₹3700
D. None of these

nielit2023feb-scientistc quantitative-aptitude compound-interest

16.9.3 Compound Interest: NIELIT 2023 Feb Scientist D | Section D | Question: 100

Mr. Kishan invested money in two schemes $\text{A\&B}$ offering compound interest 8% per annum and 9% per annum respectively. If the total amount of interest accrued through two schemes together in two years was Rs. 4818.30 and the total amount invested was Rs. 27000. What was the amount invested in scheme A?

A. Rs. $10000/-$
B. Rs. $13500/-$
C. Rs. $15000/-$
D. Rs. $12000/-$

nielit2023feb-scientistd quantitative-aptitude compound-interest

16.9.4 Compound Interest: NIELIT 2023 Feb Scientist D | Section D | Question: 120

A sum of money doubles itself at compound interest in 15 years. In how many years will it become eight times?

A. 35 years B. 45 years C. 55 years D. 40 years

nielit2023feb-scientistd quantitative-aptitude compound-interest

16.10 Counting (1)

16.10.1 Counting: NIELIT Scientist B 2020 November: 6

A solid cube of each side $8\ cm$, has been painted red, blue and black on pairs of opposite faces. It is then cut into cubical blocks of each side $2\ cm$. How many cubes have no face painted?

A. 0 B. 4 C. 8 D. 12

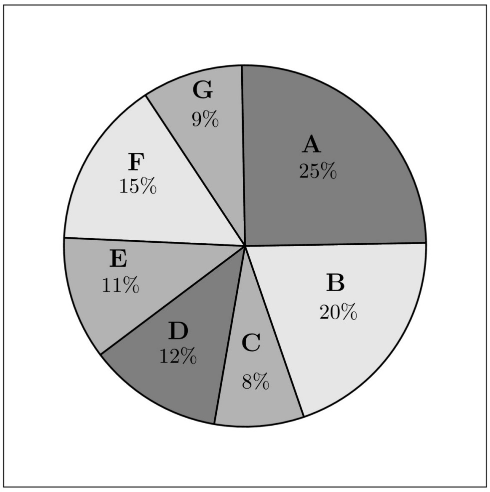
nielit-scb-2020 counting combinatory analytical-aptitude

Answer key☟

16.11 Data Interpretation (2)

16.11.1 Data Interpretation: NIELIT Scientist B 2020 November: 1

Study the following graph and the table and answer the question given below. (Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

States	Sex and Literacy wise Population Ratio			
	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

__________________ is the total number of non-educated people in A and B in 2018.

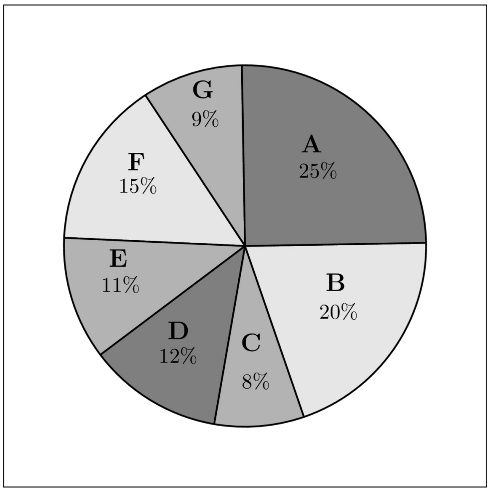
A. 1276040 B. 1032170 C. 1081550 D. 1161160

nielit-scb-2020 data-interpretation quantitative-aptitude analytical-aptitude

Answer key☟

16.11.2 Data Interpretation: NIELIT Scientist B 2020 November: 2

Study the following graph and the table and answer the question given below. (Data of different states regarding population of states in the year 2018)



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B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

The number of males in F in the year 2018 is ______.

A. 294650 B. 294840 C. 301470 D. 301200

nielit-scb-2020 data-interpretation analytical-aptitude

Answer key☟

16.12 Decimal (1)

16.12.1 Decimal: NIELIT 2023 Feb Scientist D | Section D | Question: 107

How many digits will be there to the right of the decimal point in the product of $95.75 \& 0.02554$

A. 5 B. 6 C. 7 D. 8

nielit2023feb-scientistd decimal quantitative-aptitude

16.13 Digit Sum (1)

16.13.1 Digit Sum: NIELIT 2017 DEC Scientist B - Section B: 53

When the sum of all possible two digit numbers formed from three different one digit natural numbers are divided by sum of the original three numbers, the result is

A. 26 B. 24 C. 20 D. 22

nielit2017dec-scientistb general-aptitude quantitative-aptitude digit-sum

Answer key☟

16.14 Divisibility (2)

16.14.1 Divisibility: NIELIT 2023 Feb Scientist C | Section D | Question: 94

Statement (I): 579876543 is divisible by 999

Statement (II): 6800900292 is divisible by 999

A. Both (I) and (II) are true
B. Both (I) and (II) are false
C. (I) is true and (II) is false
D. (I) is false and (II) is true

nielit2023feb-scientistc number-theory divisibility analytical-aptitude quantitative-aptitude

16.14.2 Divisibility: NIELIT 2023 Feb Scientist D | Section D | Question: 97

Statement (I): 579876543 is divisible by 111
Statement (II): 6800900181 is divisible by 111

A. Both (I) and (II) are true
B. Both (II) and (II) are false
C. (II) is true and (II) is false
D. (I) is false and (II) is true

nielit2023feb-scientistd divisibility number-theory analytical-aptitude

16.15 Factors (1)

16.15.1 Factors: NIELIT 2023 Feb Scientist C | Section C | Question: 81

Two friends were discussing their marks in an examination. While doing so they realized that both the numbers had the same prime factors, although Raveesh got a score which had two more factors than Harish. If their marks are represented by one of the options as given below, which of the following options would correctly represent the number of marks they got?

A. $30, 60$ B. $20, 80$ C. $40, 80$ D. $20, 60$

nielit2023feb-scientistc number-theory factors analytical-aptitude

16.16 Geometry (12)

16.16.1 Geometry: NIELIT 2021 Dec Scientist A - Section A: 28

An aeroplane at an altitude of 3000 m observes the angles of depression of opposite points on the two banks of a river to be 45° and 60° respectively. Find the width of the river in metre.

A. 4730 B. 4430 C. 4150 D. 4650

nielit2021dec-scientista geometry analytical-aptitude

16.16.2 Geometry: NIELIT 2021 Dec Scientist A - Section A: 38

The length, breadth and height of a cuboid are in the ratio $3 : 4 : 5$ and its volume is 3840 cm^3, The smallest side has a length of :

A. 12 cm B. 20 cm C. 15 cm D. 18 cm

nielit2021dec-scientista quantitative-aptitude geometry

Answer key☟

16.16.3 Geometry: NIELIT 2021 Dec Scientist B - Section A: 25

Instead of walking along two adjacent sides of a rectangular field, a boy took a short cut along the diagonal and saved a distance equal to half the longer side. Then, the ratio of the shorter side to the longer side is :

A. $1/2$ B. $2/3$ C. $1/4$ D. $3/4$

nielit2021dec-scientistb quantitative-aptitude geometry

Answer key☟

16.16.4 Geometry: NIELIT 2021 Dec Scientist B - Section A: 30

A cylindrical vessel 60 cm in diameter is partially filled with water. A sphere 30 cm in a diameter is dropped into it. The increase in the level of water in the vessel is :

A. 2 cm B. 3 cm C. 4 cm D. 5 cm

nielit2021dec-scientistb geometry quantitative-aptitude

Answer key

16.16.5 Geometry: NIELIT 2021 Dec Scientist B - Section A: 32

A cuboid of dimension $24 \text{ cm} \times 9 \text{ cm} \times 8 \text{ cm}$ is melted and smaller cubes of side 3 cm are formed. Find how many such cubes can be formed :

A. 27 B. 64 C. 54 D. 32

nielit2021dec-scientistb quantitative-aptitude geometry

Answer key

16.16.6 Geometry: NIELIT 2022 April Scientist B | Section A | Question: 1

A water tank is 30 m long, 20 m wide and 12 m deep. It is made of iron sheet which is 3 m wide. The tank is open at the top. If the cost of iron sheet is $\text{Rs. } 10 \text{ per meter}$, what is the total cost of iron sheet required to build the tank?

A. $\text{Rs. } 6000$ B. $\text{Rs. } 8000$ C. $\text{Rs. } 9000$ D. $\text{Rs. } 10,000$

nielit2022apr-scientistb quantitative-aptitude geometry

Answer key

16.16.7 Geometry: NIELIT 2022 April Scientist B | Section A | Question: 28

The tops of two poles are connected by a wire. The heights of the poles are 10 m and 14 m respectively. If the wire makes a 30° angle with the horizontal, find the length of the wire ?

A. 7 m B. 7.5 m C. 8 m D. 9.5 m

nielit2022apr-scientistb geometry analytical-aptitude

16.16.8 Geometry: NIELIT 2023 Feb Scientist C | Section D | Question: 109

Seven equal cubes each of the side 5 cm are joined end to end. Find the surface area of the resultant cuboid.

A. 750 cm^2
B. 1500 cm^2
C. 2250 cm^2
D. 700 cm^2

nielit2023feb-scientistc geometry quantitative-aptitude

16.16.9 Geometry: NIELIT 2023 Feb Scientist D | Section C | Question: 76

The perimeter of a square plot is the same as that of a rectangular plot with sides 35 m and 15 m. The side of the square plot is:

A. 25 m
B. 20 m

C. 100 m D. 50 m

nielit2023feb-scientistd geometry quantitative-aptitude

Answer key☟

16.16.10 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 106

The area of similar triangles $\text{ABC} \& \text{DEF}$ are 144 cm^2 and 81 cm^2 respectively. If the longest side of larger triangle ABC be 36 cm then the longest side of smaller triangle DEF is :

A. 20 cm B. 26 cm
C. 27 cm D. 30 cm

nielit2023feb-scientistd geometry quantitative-aptitude

16.16.11 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 92

A train enters into a tunnel AB at A and exits at B. A jackal is sitting at O in another by passing tunnel AOB, which is connected to AB at A and B, where OA is perpendicular to OB. A cat is sitting at P inside the tunnel AB making the shortest possible distance between O and P, such that $\text{AO} = 30 \text{ km}$ and $\text{PB} = 32 \text{ km}$. When a train before entering into the tunnel AB makes a whistle somewhere before A, the jackal and cat run towards A, they meet with accident at the entrance A. The ratio of speeds of jackal and cat is :

A. $2:3$ B. $4:3$ C. $5:3$ D. $3:2$

nielit2023feb-scientistd analytical-aptitude geometry quantitative-aptitude

16.16.12 Geometry: NIELIT Scientist B 2020 November: 24

If a cube with length, height and width equal to $10 \ cm$, is reduced to a smaller cube of height, length and width of $9 \ cm$ then reduction in volume is :

A. $172 \ cm^3$ B. $729 \ cm^3$ C. $271 \ cm^3$ D. None of the options

nielit-scb-2020 quantitative-aptitude geometry

Answer key☟

16.17 Installments (1)

16.17.1 Installments: NIELIT 2023 Feb Scientist D | Section D | Question: 95

A mobile phone is available for Rs.19650 on cash payment or for Rs.3100 cash down payment and three equal annual installments. If the shopkeeper charges interest at the rate of 10% per annum compounded annually. Calculate the amount of each installment.

A. Rs. 6655 B. Rs. 6555 C. Rs. 6755 D. Rs. 6855

nielit2023feb-scientistd compound-interest quantitative-aptitude installments

16.18 Logarithms (3)

16.18.1 Logarithms: NIELIT 2021 Dec Scientist A - Section A: 35

Which of the following is true ?

A. $\log_{17} 275 = \log_{19} 375$
B. $\log_{17} 275 > \log_{19} 375$
C. $\log_{17} 275 < \log_{19} 375$
D. None of the these

nielit2021dec-scientista logarithms quantitative-aptitude

16.18.2 Logarithms: NIELIT 2021 Dec Scientist A - Section A: 36

$\log(x+3) + \log(x+5) = \log 35,$ solve for x :

A. 1 B. 2 C. 3 D. 4

nielit2021dec-scientista logarithms quantitative-aptitude

Answer key☟

16.18.3 Logarithms: NIELIT 2021 Dec Scientist B - Section A: 28

$\frac{1}{2} \log_{10} 25 – 2 \log_{10} 3 + \log_{10} 18$ equals :

A. 18 B. 1 C. $\log_{10} 3$ D. None of these

nielit2021dec-scientistb logarithms quantitative-aptitude

Answer key☟

16.19 Logical Reasoning (1)

16.19.1 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 9

Four usual dice are thrown on the ground. The total of numbers on the top faces of these four dice is 13 as the top faces showed $4, 3, 1$ and 5 respectively. What is the total of the faces touching the ground?

A. 12
B. 13
C. 15
D. Cannot be determined

nielit2023feb-scientistc probability logical-reasoning

Answer key☟

16.20 Mixtures (1)

16.20.1 Mixtures: NIELIT 2023 Feb Scientist C | Section C | Question: 90

Three vessels having volumes in the ratio of $1 : 2 : 3$ are full of a mixture of coke and soda. In the first vessel, ratio of coke and soda is $2 : 3$, in second, $3 : 7$ and in third, $1 : 4$. If the liquid in all three vessels were mixed in a bigger container, what is the resulting ratio of coke and soda?

A. $4 : 11$ B. $5 : 7$ C. $7 : 11$ D. $7 : 5$

nielit2023feb-scientistc ratio-proportion mixtures quantitative-aptitude

16.21 Modular Arithmetic (1)

16.21.1 Modular Arithmetic: NIELIT 2023 Feb Scientist D | Section C | Question: 70

What is the remainder when 153×698 is divided by 5?

A. 1 B. 2 C. 4 D. 3

nielit2023feb-scientistd modular-arithmetic quantitative-aptitude

Answer key☟

16.22 Number Series (3)

16.22.1 Number Series: NIELIT 2021 Dec Scientist B - Section A: 4

Directions $(3-5)$: In each of the following questions, one number is wrong in the series. Find out the wrong number.

$1, 9, 25, 49, 86, 121$:

A. 25 B. 121 C. 166 D. 86

nielit2021dec-scientistb number-series analytical-aptitude

Answer key☟

16.22.2 Number Series: NIELIT 2021 Dec Scientist B - Section A: 5

Directions $(3-5)$: In each of the following questions, one number is wrong in the series. Find out the wrong number.

$701, 348, 173, 85, 41, 19, 8$:

A. 173 B. 41 C. 19 D. 348

nielit2021dec-scientistb number-series analytical-aptitude quantitative-aptitude

Answer key

16.22.3 Number Series: NIELIT 2022 April Scientist B | Section A | Question: 13

In the following question, a number series is given with one term missing. Choose the correct alternative that will continue the same pattern and fill in the blank spaces.

$97, 86, 73, 58, 45, (\dots)$

A. 24 B. 34 C. 44 D. 54

nielit2022apr-scientistb number-series analytical-aptitude

16.23 Number System (5)

16.23.1 Number System: NIELIT 2018-2

Divide 88 into four parts such as first part known as a, second part b, third part c, and fourth part is d, when 5 is added to the first part, 5 is subtracted from the second part, 3 is multiplied by the third part and the fourth part is divided by 5, then all results value are equal. Find the value of a, b, c and d respectively.

A. $7, 17, 4, 60$ B. $8, 25, 5, 50$ C. $10, 30, 3, 45$ D. $17, 7, 4, 60$

nielit-2018 quantitative-aptitude number-system

Answer key

16.23.2 Number System: NIELIT 2021 Dec Scientist A - Section A: 26

The LCM of two numbers is 45 times their HCF. One number is 125 and the sum of their HCF and LCM is 1150. Find the other number.

A. 275 B. 215 C. 230 D. 225

nielit2021dec-scientista quantitative-aptitude number-system

Answer key

16.23.3 Number System: NIELIT 2022 April Scientist B | Section A | Question: 16

The total number of digits used in numbering the pages of a book having 366 pages is :

A. 730 B. 792 C. 990 D. 1098

nielit2022apr-scientistb analytical-aptitude counting number-system

Answer key

16.23.4 Number System: NIELIT 2022 April Scientist B | Section A | Question: 39

If the number $467\text{X}4$ is divisible by 9, find the value of the digit marked as X.

A. 4 B. 5 C. 6 D. 7

nielit2022apr-scientistb quantitative-aptitude number-system

Answer key☟

16.23.5 Number System: NIELIT 2023 Feb Scientist C | Section C | Question: 83

Find the two-digit number the quotient of whose division by the product of its digits is equal to $8/3$, and the difference between the required number and the number consisting of the digits written in the reverse order is 18?

A. 86
B. 42
C. 75
D. None of these

nielit2023feb-scientistc analytical-aptitude number-system quantitative-aptitude

16.24 Number Theory (8)

16.24.1 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 74

The Product of three consecutive natural numbers, the first of which is an even number, is always divisible by

A. 12
B. 24
C. 6
D. All of these

nielit2023feb-scientistc quantitative-aptitude number-theory

Answer key☟

16.24.2 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 77

Find the number of numbers between 200 and 300, both included, which are not divisible by $2, 3, 4$ and 5.

A. 27 B. 26 C. 25 D. 28

nielit2023feb-scientistc counting number-theory permutation-and-combination quantitative-aptitude

16.24.3 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 79

Find the sum of all-two digit numbers such that the sum of the digits constituting the number is not less than 7; the sum of the squares of the digits is not greater than 30; the number consisting of the same digits written in the reverse order is not larger than half the given number.

A. 52 B. 51 C. 49 D. 53

nielit2023feb-scientistc quantitative-aptitude number-theory

16.24.4 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 86

Which one of the following is a composite number?

A. 101 B. 103 C. 141 D. 137

nielit2023feb-scientistc number-theory

16.24.5 Number Theory: NIELIT 2023 Feb Scientist C | Section D | Question: 97

Find the number of zeroes in the following multiplication :
$5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$

A. 6 B. 7 C. 8 D. 5

nielit2023feb-scientistc counting number-theory quantitative-aptitude

16.24.6 Number Theory: NIELIT 2023 Feb Scientist C | Section D | Question: 98

For the positive integers $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}$, find the minimum value of $(-1)^a+(-1)^b+(-1)^c+(-1)^d$, if $\mathrm{a+b+ c+d} = 1947$.

A. -1 B. -2 C. -3 D. -4

nielit2023feb-scientistc mathematical-logic number-theory analytical-aptitude

16.24.7 Number Theory: NIELIT 2023 Feb Scientist D | Section D | Question: 116

Let n be a natural number such that $\frac{1}{2}+\frac{1}{3}+\frac{1}{7}+\frac{1}{n}$ is also a natural number then which of the following statement is true?

A. 2 divides n B. 3 divides n
C. 7 divides n D. $\mathrm{n} > 84$

nielit2023feb-scientistd number-theory analytical-aptitude

16.24.8 Number Theory: NIELIT 2023 Feb Scientist D | Section D | Question: 91

The digit in the unit place of the number represented by $7^{95}-3^{58}$ is :

A. 0 B. 4 C. 6 D. 7

nielit2023feb-scientistd number-theory quantitative-aptitude

Answer key

16.25 Numerical Computation (1)

16.25.1 Numerical Computation: NIELIT 2017 DEC Scientist B - Section B: 9

Find the smallest number y such that $y \times 162$ (y multiplied by 162) is a perfect cube

A. 24 B. 27 C. 36 D. 38

nielit2017dec-scientistb general-aptitude quantitative-aptitude number-system numerical-computation

Answer key

16.26 Percentage (14)

16.26.1 Percentage: NIELIT 2023 Feb Scientist C | Section C | Question: 69

Arvind spends 75% of his income and saves the rest. His income is increased by 20% and he increased his expenditure by 10%. Then the increase in savings in percentage is

A. 55 B. 52 C. 50 D. 48

nielit2023feb-scientistc percentage quantitative-aptitude

Answer key

16.26.2 Percentage: NIELIT 2023 Feb Scientist C | Section C | Question: 70

Two solutions, the first of which contains 0.8 kg and the second 0.6 kg of salt, were poured together and 10 kg of a new salt solution were obtained. Find the weight of the first and of the second solutions in the mixture if the first solution is known to contain 10 percent more of salt of than the second.

A. $4 \text{ kg}, 6 \text{ kg}$
B. $3 \text{ kg}, 7 \text{ kg}$
C. $4 \text{ kg}, 9 \text{ kg}$
D. $5 \text{ kg}, 9 \text{ kg}$

nielit2023feb-scientistc system-of-equations percentage quantitative-aptitude

16.26.3 Percentage: NIELIT 2023 Feb Scientist C | Section C | Question: 73

If the length of the rectangle is increased by 50% and breadth is decreased by 50% what is the change in the area of the rectangle :

A. Increase by 12.5%
B. Decrease by 12.5%
C. No change
D. Decrease by 25%

nielit2023feb-scientistc   quantitative-aptitude   percentage

16.26.4 Percentage: NIELIT 2023 Feb Scientist C | Section D | Question: 106

Find a single discount equivalent to the discount series of $20\%, 10\%, 5\%$?

A. 30% B. 31.6% C. 68.4% D. 35%

nielit2023feb-scientistc   percentage   quantitative-aptitude

16.26.5 Percentage: NIELIT 2023 Feb Scientist C | Section D | Question: 110

8% of the people eligible to vote are between 20 and 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25, actually voted. In that election the number of persons between 20 and 25, who actually voted, was what percentage of those eligible to vote?

A. 4.2 B. 6.4 C. 6.8 D. 8

nielit2023feb-scientistc   percentage   quantitative-aptitude   analytical-aptitude

16.26.6 Percentage: NIELIT 2023 Feb Scientist C | Section D | Question: 119

When the price of mobile was reduced by 20% the number of mobiles sold is increased by 40%. The effect on the sale was:

A. 12% increase   B. 12% decrease
C. 30% increase   D. 40% decrease

nielit2023feb-scientistc   percentage   quantitative-aptitude

16.26.7 Percentage: NIELIT 2023 Feb Scientist D | Section C | Question: 68

At an election, the candidate who got 56% of the votes cast won by 1440 votes. Find the total number of voters on the voting list if 80% people cast their vote and there were no invalid votes.

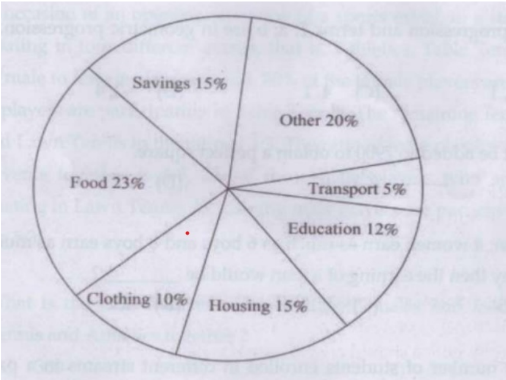
A. 3600 B. 7200 C. 18000 D. 15000

nielit2023feb-scientistd   quantitative-aptitude   percentage

16.26.8 Percentage: NIELIT 2023 Feb Scientist D | Section C | Question: 78

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



If the total expenditure of the family for the year 1999 was Rs. 46000, the family saved during the year:

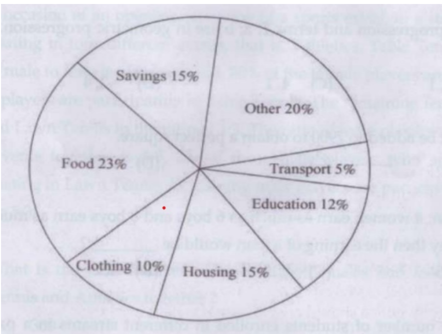
A. Rs.1500
B. Rs.15000
C. Rs.6900
D. Rs.3067 approx.

nielit2023feb-scientistd percentage data-interpretation analytical-aptitude quantitative-aptitude

16.26.9 Percentage: NIELIT 2023 Feb Scientist D | Section C | Question: 80

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



If the total amount spent was Rs. 46000, how much money was spent on clothing and housing together?

A. Rs. 11500
B. Rs. 1150
C. Rs. 10000
D. Rs. 15000

nielit2023feb-scientistd analytical-aptitude percentage

16.26.10 Percentage: NIELIT 2023 Feb Scientist D | Section D | Question: 114

A retailer bought $20\ \mathrm{kg}$ tea at a discount of 10% on market price. Besides $1\ \mathrm{kg}$ tea was freely offered to him by the wholesaler at the purchase of $20\ \mathrm{kg}$ tea. Now he sells all the tea at the marked price to a customer. What is the profit percentage of retailer?

A. 30% B. 12% C. 16.66% D. 10%

nielit2023feb-scientistd quantitative-aptitude percentage

16.26.11 Percentage: NIELIT 2023 Feb Scientist D | Section D | Question: 93

A, B and C went to market to purchase the rings whose costs are same. But each ring is available with two successive discounts. A, B and C avails two successive discounts at the rate of 5% and $20\%, 10\%$ and $15\%, 12\%$ and 13% respectively. Who gets the maximum discounts?

A. A
B. B
C. C
D. All gets the same discount.

nielit2023feb-scientistd percentage analytical-aptitude quantitative-aptitude

16.26.12 Percentage: NIELIT 2023 Feb Scientist D | Section D | Question: 94

The cost of food accounted for 25% of the income of a particular family. If the income gets raised by 20%, then what should be the percentage point decrease in the food expenditure as a percentage of the total income to keep the food expenditure unchanged between two years?

A. 3.5% B. 8.33% C. 4.16% D. 5%

nielit2023feb-scientistd percentage quantitative-aptitude analytical-aptitude

16.26.13 Percentage: NIELIT Scientific Assistant A 2020 November: 24

If 5% income of P is equal to 15% income of Q and 10% income of Q is equal to 20% income of R. If income of R is ₹2000, then what are the incomes of P and Q respectively?

A. ₹4000 and ₹8000
B. ₹12000 and ₹4000
C. ₹15000 and ₹5000
D. ₹18000 and ₹6000

nielit-sta-2020 quantitative-aptitude percentage

Answer key☟

16.26.14 Percentage: NIELIT Scientist B 2020 November: 40

In an office, 30% of the employees were women and 70% of the employees were above the age of 40 years, out of which 60% are men. Find the percentage of women employees who are above 40 years out of the total number of women employees.

A. 96% B. 93.33% C. 70.44% D. 80.66%

nielit-scb-2020 probability percentage analytical-aptitude

Answer key☟

16.27 Permutation and Combination (2)

16.27.1 Permutation and Combination: NIELIT 2021 Dec Scientist B - Section B: 27

The number of 4 digit numbers which contain not more than two different digits is :

A. 576 B. 567 C. 513 D. 504

nielit-2021-it-dec-scientistb counting permutation-and-combination quantitative-aptitude

16.27.2 Permutation and Combination: NIELIT 2021 Dec Scientist B - Section B: 62

The number of 4 digit numbers which contain not more than two different digits is :

A. 576 B. 567 C. 513 D. 504

nielit2021dec-scientistb counting permutation-and-combination number-system

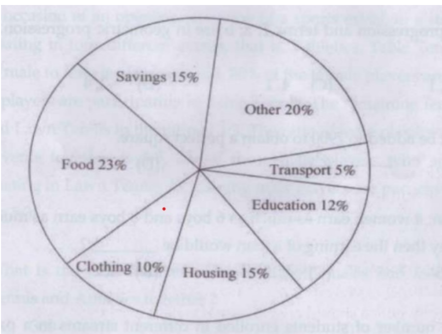
Answer key☟

16.28 Pie Chart (2)

16.28.1 Pie Chart: NIELIT 2023 Feb Scientist D | Section C | Question: 79

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



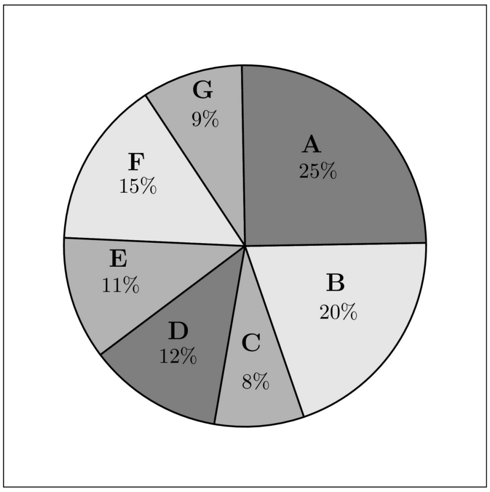
If the total amount spent during the year 1999 was Rs. 46000, the amount spent on food was

A. Rs. 2000 B. Rs. 10580 C. Rs. 23000 D. Rs. 2300

nielit2023feb-scientistd analytical-aptitude quantitative-aptitude pie-chart data-interpretation

16.28.2 Pie Chart: NIELIT Scientist B 2020 November: 3

Study the following graph and the table and answer the question given below. (Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

States	Sex and Literacy wise Population Ratio			
	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

If in the year 2018, population of F is increased by 10% and population of B is increased by 12% as compared to the previous year, then _________ is the ratio of populations of F and B in 2017.

A. $42:55$ B. $62:55$ C. $42:11$ D. $44:5$

nielit-scb-2020 pie-chart data-interpretation quantitative-aptitude

Answer key☟

16.29 Pipe Cistern (1)

16.29.1 Pipe Cistern: NIELIT 2023 Feb Scientist D | Section C | Question: 75

A cistern has two taps which fill it in 12 minutes and 15 minutes, respectively. There is also a waste pipe in the cistern. When all the pipes are opened, the empty cistern is full in 20 minutes. How long will the waste pipe take to empty a full cistern?

A. 8 minutes B. 10 minutes C. 12 minutes D. 16 minutes

nielit2023feb-scientistd quantitative-aptitude work-time pipe-cistern

Answer key☟

16.30 Probability (1)

16.30.1 Probability: NIELIT 2023 Feb Scientist C | Section C | Question: 62

A candidate takes a test and attempts all the 100 questions in it. While any correct answer fetches 1 mark, wrong answers are penalized as follows; one-tenth of the questions carry $1/10$ negative mark each, one-fifth of the questions carry $1/5$ negative marks each and the rest of the questions carry $1/2$ negative mark each. Un-attempted questions carry no marks. What is the difference between the maximum and the minimum marks that he can score?

A. 100
B. 120
C. 140
D. None of these

nielit2023feb-scientistc probability analytical-aptitude quantitative-aptitude

16.31 Profit Loss (6)

16.31.1 Profit Loss: NIELIT 2018-5

Suppose a fraud shopkeeper sells rice to the customer at the cost price, but he uses a false weight of 900 gm for a kg then his percentage gain is ______

A. 5.75% B. 5.56% C. 5.20% D. 5.00%

nielit-2018 general-aptitude quantitative-aptitude profit-loss

Answer key☟

16.31.2 Profit Loss: NIELIT 2023 Feb Scientist C | Section D | Question: 100

Mahesh and Umesh purchase a radio each for the same price and both mark up their respective radios by the same amount. Mahesh gives a discount of Rs.20/- followed by another discount of 20% on the reduced price. While Umesh gives a discount of 20% followed by a discount of Rs $20/-$. If Mahesh's profit percent is equal to thrice Umesh's loss percent. The profit (in Rs) of Mahesh on his radio is:

A. Rs. 10 B. Rs. 3 C. Rs. 15 D. Rs. 7

nielit2023feb-scientistc profit-loss quantitative-aptitude analytical-aptitude

16.31.3 Profit Loss: NIELIT 2023 Feb Scientist C | Section D | Question: 93

A dishonest dealer sells the goods at $6\frac{1}{4}\%$ loss on cost price but uses $12\frac{1}{4}\%$ less weight. What is his percentage profit or loss?

A. $7\frac{1}{4}\%$ B. $4\frac{1}{7}\%$ C. $7\frac{1}{7}\%$ D. $4\frac{1}{4}\%$

nielit2023feb-scientistc percentage profit-loss quantitative-aptitude

Answer key☟

16.31.4 Profit Loss: NIELIT 2023 Feb Scientist D | Section D | Question: 104

A milkman purchases the milk at Rs. x per litre and sells it at Rs. $2x$ per litre still he mixes 2 litres water with every 6 litres of pure milk. What is the profit percentage?

A. 116% B. 166.66% C. 60% D. 100%

nielit2023feb-scientistd quantitative-aptitude percentage profit-loss

Answer key☟

16.31.5 Profit Loss: NIELIT 2023 Feb Scientist D | Section D | Question: 115

A pen manufacturing company produces very fine quality of writing pens. Company knows that on an average 10% of the produced pens are always defective, so are rejected before packing. Company promises to deliver 7200 pens to its wholesaler at Rs 10 each. It estimates the overall profit on all manufactured pens to be 25%. What is the manufacturing cost of each pen?

A. Rs. 6 B. Rs. 5.6 C. Rs. 8 D. Rs. 7.2

nielit2023feb-scientistd percentage profit-loss quantitative-aptitude

16.31.6 Profit Loss: NIELIT Scientific Assistant A 2020 November: 25

A businessman purchases an item at a certain price and marks its price up by 30%. He sells the item at a certain discount on markup price and makes a net profit of 4% on the whole transaction. Find the discount given by businessman on markup price.

A. 10 B. 15 C. 26 D. 20

nielit-sta-2020 quantitative-aptitude profit-loss

Answer key☟

16.32 Quadratic Equations (1)

16.32.1 Quadratic Equations: NIELIT 2018-17

If $2a + 3b + c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$, lies in the interval:

A. $(0, 1)$ B. $(1, 2)$ C. $(2, 3)$ D. $(1, 3)$

nielit-2018 general-aptitude quantitative-aptitude quadratic-equations

Answer key☟

16.33 Ratio Proportion (14)

16.33.1 Ratio Proportion: NIELIT 2021 Dec Scientist A - Section A: 23

The length, breadth and height of a room are in the ratio of $3 : 2 : 1$. If its volume be 1296 m^3, find its breadth.

A. 12 m B. 18 m C. 16 m D. 24 m

nielit2021dec-scientista quantitative-aptitude ratio-proportion

Answer key☟

16.33.2 Ratio Proportion: NIELIT 2021 Dec Scientist B - Section A: 22

A factory employs skilled workers, unskilled workers and clerks in the proportion $8 : 5 : 1$, and the wages of a skilled worker, an unskilled worker and a clerk are in the ratio $5 : 2 : 3$. When 20 unskilled workers are employed, the total daily wages of all (skilled workers, unskilled workers and clerks) amount to ₹ 318. The wages paid to each category of workers are :

A. ₹ 240, ₹ 60, ₹ 18
B. ₹ 200, ₹ 90, ₹ 28
C. ₹ 150, ₹ 108, ₹ 60
D. ₹ 250, ₹ 50, ₹ 18

nielit2021dec-scientistb analytical-aptitude ratio-proportion

Answer key☟

16.33.3 Ratio Proportion: NIELIT 2021 Dec Scientist B - Section A: 24

In a mixture of 60 L, the ratio of milk and water is $2 : 1$. If the ratio of milk and water is to be $1 : 2$, then the amount of water to be further added must be :

A. 40 L B. 30 L C. 20 L D. 60 L

nielit2021dec-scientistb ratio-proportion quantitative-aptitude analytical-aptitude

Answer key☟

16.33.4 Ratio Proportion: NIELIT 2022 April Scientist B | Section A | Question: 2

Ramesh and Suresh enter into a partnership with capitals in the ratio of $10 : 12$. At the end of 8 months, Ramesh withdraws. If they receive profits in the ratio of $10 : 18$. Find how long Suresh's capital was used.

A. 7 months B. 8 months C. 10 months D. 12 months

nielit2022apr-scientistb quantitative-aptitude ratio-proportion

Answer key☟

16.33.5 Ratio Proportion: NIELIT 2022 April Scientist B | Section A | Question: 9

Gold is 19 times as heavy as water and copper 9 times as heavy as water. In what ratio should these metal be mixed so that the mixture may be 15 times as heavy as water ?

A. $3 : 2$ B. $9 : 5$ C. $2 : 3$ D. $7 : 5$

nielit2022apr-scientistb ratio-proportion quantitative-aptitude analytical-aptitude

Answer key☟

16.33.6 Ratio Proportion: NIELIT 2023 Feb Scientist C | Section A | Question: 15

A box contains two dozen of glasses. If it is dropped on a concrete floor then which of the following cannot be the ratio of broken and unbroken glasses.

A. $2 : 1$ B. $5 : 1$ C. $6 : 1$ D. $3 : 1$

nielit2023feb-scientistc probability counting ratio-proportion analytical-aptitude

16.33.7 Ratio Proportion: NIELIT 2023 Feb Scientist C | Section C | Question: 63

A mixture of cement, sand and gravel in the ratio of $1 : 2 : 4$ by volume is required. A person wishes to measure out quantities by weight. He finds that the weight of one cubic foot of cement is $94\,\text{kg}$, of sand $100\,\text{kg}$ and gravel $110\,\text{kg}$. What should be the ratio of cement, sand and gravel by weight in order to give a proper mixture?

A. $47 : 100 : 220$ B. $94 : 100 : 220$
C. $47 : 200 : 440$ D. None of these

nielit2023feb-scientistc ratio-proportion quantitative-aptitude

16.33.8 Ratio Proportion: NIELIT 2023 Feb Scientist C | Section C | Question: 85

A precious stone weighing 35 grams worth ₹$12,500$ is accidentally dropped and gets broken into two pieces having weights in the ratio of $2 : 5$. If the price varies as the square of the weight then find the loss incurred.

A. ₹ 5750 B. ₹ 6000 C. ₹ 5102 D. ₹ 5000

nielit2023feb-scientistc quantitative-aptitude ratio-proportion analytical-aptitude

16.33.9 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 61

The digits of a two-digit number are in the ratio of $2:3$ and the number obtained by interchanging the digits is bigger than the original number by 27. What is the original number?

A. 63 B. 48 C. 96 D. 69

nielit2023feb-scientistd analytical-aptitude ratio-proportion number-system quantitative-aptitude

16.33.10 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 64

Directions for **(64 and 65)**: Read the following information carefully to answer the following questions. On the occasion of an opening ceremony of a sports event, in a stadium, there are 600 players who are participating in four different events, that is, Athletics, Table Tennis, Kho-Kho and Lawn Tennis. The ratio of male to female players is $11:4$. 30% of the female players are participating in Athletics. 10% of the female players are participating in Table Tennis. The remaining female players are participating in KhoKho and Lawn Tennis in the ratio of $1:3$. The ratio of male players who are participating in Athletics and other events together is $3:5.4\%$ of those male players who are not participating in Athletics are participating in Lawn Tennis. Remaining male players are participating in Table Tennis and Kho-Kho in the ratio $5:3$.

What is the total number of players (both males and females together) participating in Table Tennis and Athletics together?

A. 360 B. 358
C. 374 D. None of these

nielit2023feb-scientistd analytical-aptitude ratio-proportion percentage

16.33.11 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 65

Directions for **(64 and 65)**: Read the following information carefully to answer the following questions. On the occasion of an opening ceremony of a sports event, in a stadium, there are 600 players who are participating in four different events, that is, Athletics, Table Tennis, Kho-Kho and Lawn Tennis. The ratio of male to female players is $11:4$. 30% of the female players are participating in Athletics. 10% of the female players are participating in Table Tennis. The remaining female players are participating in KhoKho and Lawn Tennis in the ratio of $1:3$. The ratio of male players who are participating in Athletics and other events together is $3:5.4\%$ of those male players who are not participating in Athletics are participating in Lawn Tennis. Remaining male players are participating in Table Tennis and Kho-Kho in the ratio $5:3$.

What is the ratio of the male players participating in Lawn Tennis to the female players participating in Table Tennis?

A. $11 : 72$
B. $11 : 38$
C. $11 : 16$
D. None of these

nielit2023feb-scientistd analytical-aptitude ratio-proportion percentage

16.33.12 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 67

Two numbers are in the ratio $\mathrm{A}: \mathrm{B}$. When 1 is added to both the numerator and the denominator, the ratio gets changed to C / D. Again, when 1 is added to both the numerator and the denominator, it becomes $1 / 2$. Find the sum of A and B.

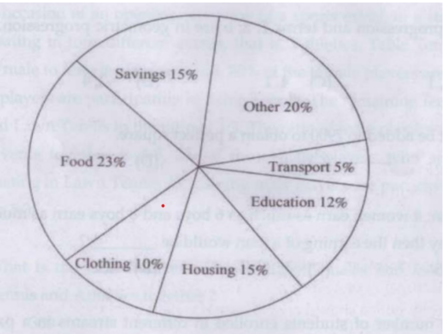
A. 3
B. 4
C. 5
D. 6

nielit2023feb-scientistd ratio-proportion algebra analytical-aptitude quantitative-aptitude

16.33.13 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 82

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



The ratio of the total amount of money spent on housing to that spent on education was:

A. $5 : 2$
B. $2 : 5$
C. $4 : 5$
D. $5 : 4$

nielit2023feb-scientistd analytical-aptitude data-interpretation percentage ratio-proportion

16.33.14 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section D | Question: 102

A cat takes 5 leaps for every 4 leaps of a dog, but 3 leaps of the dog are equal to 4 leaps of the cat. What is the ratio of the speed of the cat to that of dog?

A. $16 : 15$
B. $15 : 11$
C. $11 : 15$
D. $15 : 16$

nielit2023feb-scientistd quantitative-aptitude ratio-proportion

Answer key

16.34 Remainder (1)

16.34.1 Remainder: NIELIT 2023 Feb Scientist C | Section C | Question: 61

What is the greatest number of 4 digits that when divided by any of the numbers $6, 9, 12, 17$ leaves a remainder of 1?

A. 9997 B. 9793 C. 9895 D. 9487

nielit2023feb-scientistc number-theory remainder quantitative-aptitude

16.35 Sequence Series (2)

16.35.1 Sequence Series: NIELIT 2016 MAR Scientist C - Section B: 16

Evaluate the sum $S = 1 + 1 + \frac{3}{2^2} + \frac{3}{2^3} + \frac{5}{2^4} + \dots$

A. 1 B. 2 C. 3 D. 4

nielit2016mar-scientistc sequence-series

Answer key

16.35.2 Sequence Series: NIELIT 2023 Feb Scientist C | Section C | Question: 82

Fourth term of an arithmetic progression is 8. What is the sum of the first 7 terms of the arithmetic progression?

A. 7
B. 64
C. 56
D. Cannot be determined

nielit2023feb-scientistc sequence-series quantitative-aptitude

Answer key

16.36 Set Theory (4)

16.36.1 Set Theory: NIELIT 2022 April Scientist B | Section A | Question: 6

The following question is based on the information given below:

Data of 450 candidates, who took part in an examination in social sciences, Mathematics and Science is given as below :

Number of Students who :

- passed in all subjects are 167

- failed in science $= 191$
- failed in Mathematics $= 199$
- failed in Social sciences $= 175$
- failed in all subjects $= 60$
- passed in Science only $= 52$
- passed in Mathematics only $= 48$
- passed in Social sciences only $= 62$.

How many failed in two subjects only?

A. 162 B. 152 C. 100 D. 52

nielit2022apr-scientistb analytical-aptitude set-theory quantitative-aptitude

Answer key

16.36.2 Set Theory: NIELIT 2023 Feb Scientist D | Section C | Question: 63

In an examination, 30% and 35% students respectively failed in history and geography while 27% students failed in both the subjects. If the number of students passing the examination is 248, find the total number of students who appeared in the examination.

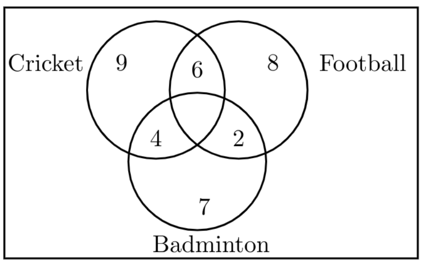
A. 425
B. 380
C. 400
D. None of these

nielit2023feb-scientistd quantitative-aptitude set-theory

Answer key

16.36.3 Set Theory: NIELIT Scientist B 2020 November: 32

In a class there are 40 students who play at least one game out of Football, Cricket and Badminton.



What percentage of students play only one game?

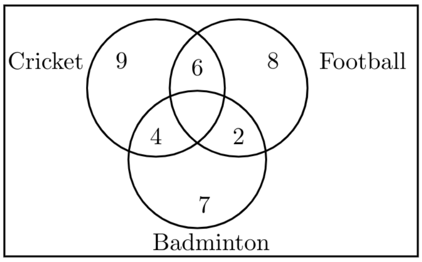
A. 50% B. 60% C. 65% D. 70%

nielit-scb-2020 set-theory quantitative-aptitude counting

Answer key

16.36.4 Set Theory: NIELIT Scientist B 2020 November: 33

In a class there are 40 students who play at least one game out of Football, Cricket and Badminton.



What percentage of students play all the three games?

A. 4% B. 5% C. 8% D. 10%

nielit-scb-2020 set-theory counting probability quantitative-aptitude

Answer key☟

16.37 Simple Compound Interest (1)

16.37.1 Simple Compound Interest: NIELIT Scientific Assistant A 2020 November: 32

₹ 1000 doubled in 10 years when compounded annually. How many more years will it take to get another ₹ 2000 compound interest?

A. 5 years B. 10 years C. 3 years D. 4 years

nielit-sta-2020 quantitative-aptitude simple-compound-interest

Answer key☟

16.38 Simple Interest (1)

16.38.1 Simple Interest: NIELIT 2021 Dec Scientist A - Section A: 22

A certain sum of money amounts to ₹ 6600 in 4 years at a certain rate percent simple interest. If the rate of interest be increased by its 25%, the same sum would amount to ₹ 7000 during the same period. Find the sum.

A. ₹ 6000 B. ₹ 5500 C. ₹ 5000 D. ₹ 7000

nielit2021dec-scientista quantitative-aptitude simple-interest

16.39 Speed Time Distance (6)

16.39.1 Speed Time Distance: NIELIT 2022 April Scientist B | Section A | Question: 19

A train moving at 50 km/hr crosses a bridge in 45 seconds. The length of train is 150 meters. Find the length of the bridge.

A. 525 m B. 545 m C. 475 m D. 575 m

nielit2022apr-scientistb speed-time-distance quantitative-aptitude

Answer key☟

16.39.2 Speed Time Distance: NIELIT 2022 April Scientist B | Section A | Question: 5

A man rows downstream at 20 km/hr and rows upstream at 15 km/hr. At what speed he can row in still water ?

A. 17.5 km/hr B. 18 km/hr C. 20.5 km/hr D. 22 km/hr

nielit2022apr-scientistb quantitative-aptitude speed-time-distance

Answer key☟

16.39.3 Speed Time Distance: NIELIT 2023 Feb Scientist C | Section C | Question: 84

In a 100 m race, Shyam runs at 1.66 m/s. If Shyam gives Sujit a start of 4 m and still beats him by 12 seconds, what is Sujit's speed?

A. 1.11 m/s
B. 0.75 m/s
C. 1.33 m/s
D. 1 km/h

nielit2023feb-scientistc quantitative-aptitude speed-time-distance

16.39.4 Speed Time Distance: NIELIT 2023 Feb Scientist C | Section D | Question: 92

Two trains going on a parallel line in opposite directions take 10 seconds to cross each other. But if they are going in the same direction the longer train crosses the shorter train in 30 seconds. If the length of the longer train decreased by 50%, the time taken to cross the shorter train while going in the same direction decreases by 8 seconds. The time taken by the longer train to cross a tunnel twice its length, if the difference between the length of the trains is 25 m is :

A. 30sec
B. 24sec
C. 36sec
D. 40sec

nielit2023feb-scientistc quantitative-aptitude speed-time-distance

16.39.5 Speed Time Distance: NIELIT 2023 Feb Scientist D | Section C | Question: 90

In a $500 - \text{m}$ race, P runs at 10 m/s. If P gives Q a start of 50 m and still wins the race by 5 seconds, find the speed of Q.

A. 8 m/s
B. 8.18 m/s
C. 9.01 m/s
D. 9.19 m/s

nielit2023feb-scientistd quantitative-aptitude speed-time-distance

16.39.6 Speed Time Distance: NIELIT Scientist B 2020 November: 36

Arjun by car takes double the time taken by bus to travel from Delhi to Agra. What is the Speed of the Bus if the Speed of the Car is $40 \; km/hr$?

A. $40\ km/hr$ B. $60\ km/hr$ C. $80\ km/hr$ D. $30\ km/hr$

nielit-scb-2020 quantitative-aptitude speed-time-distance

Answer key☟

16.40 Statistics (2)

16.40.1 Statistics: NIELIT Scientific Assistant A 2020 November: 61

Find the mode of the following data:

Age	0-6	6-12	12-18	18-24	24-30	30-36	36-42
Frequency	6	11	25	35	18	12	6

A. 20.22 B. 19.47 C. 21.12 D. 20.14

nielit-sta-2020 quantitative-aptitude statistics

Answer key☟

16.40.2 Statistics: NIELIT Scientist B 2020 November: 41

Consists of a question and two statements, labelled (1) and (2), in which certain data are given. You have to decide whether the data given in the statements are sufficient for answering the question. Using the data given in the statements, plus your knowledge of mathematics and everyday facts.

What is the Standard Deviation (SD) of the four numbers A, B, C, D?

Statement (1)

The sum of A, B, C and D is 24.

Statement (2)

The sum of the squares of A, B, C and D is 224.

A. Only Statement (1) is required for answering the question
B. Only Statement (2) is required for answering the question
C. Both statement together are required to answer the question
D. Answer cannot be ascertained with the given information

nielit-scb-2020 statistics analytical-aptitude

Answer key☟

16.41 System of Equations (7)

16.41.1 System of Equations: NIELIT 2021 Dec Scientist A - Section A: 33

If the numerator of a fraction is increased by 2 and the denominator is decreased b y $1,$ then it becomes $2/3.$ If the numerator is increased by 1 and the denominator is increased by $2,$ then it becomes $1/3.$ Find the fraction.

A. $2/9$ B. $2/7$ C. $1/6$ D. $1/5$

nielit2021dec-scientista analytical-aptitude system-of-equations quantitative-aptitude

Answer key

16.41.2 System of Equations: NIELIT 2021 Dec Scientist B - Section A: 29

I f $x = p, y = q$ then which of following are p and q respectively for pair of equations $3x - 7y + 10 = 0$ and $y - 2x - 3 = 0$:

A. $-1,1$ B. $1,1$ C. $1,0$ D. $0,1$

nielit2021dec-scientistb system-of-equations analytical-aptitude

Answer key

16.41.3 System of Equations: NIELIT 2022 April Scientist B | Section A | Question: 35

Four years ago a man was 6 times as old as his son. After 16 years he will be twice as old as his son. What is the present age of man and his son?

A. $34, 9$ B. $33, 7$ C. $35, 5$ D. $36, 6$

nielit2022apr-scientistb system-of-equations analytical-aptitude quantitative-aptitude

Answer key

16.41.4 System of Equations: NIELIT 2022 April Scientist B | Section A | Question: 41

Sum of present age of Suresh and Dinesh is equal to the age of Hema six years back. Five years from now, the ratio of age of Suresh and Dinesh will be $3 : 2$. Rajesh is 5 years older than Hema and his present age is four times the present age of Suresh. What is present age of the Dinesh ?

A. 1 year B. 10 years C. 2 years D. 12 years

nielit2022apr-scientistb analytical-aptitude system-of-equations quantitative-aptitude

Answer key

16.41.5 System of Equations: NIELIT 2023 Feb Scientist C | Section C | Question: 80

The cost of four mangoes, six guavas and sixteen watermelons is ₹500, while the cost of seven mangoes, nine guavas and nineteen watermelons is ₹620. What is the cost of one mango, one guava and one watermelon?

A. 120 B. 40
C. 150 D. Cannot be determined

nielit2023feb-scientistc quantitative-aptitude system-of-equations

16.41.6 System of Equations: NIELIT 2023 Feb Scientist C | Section D | Question: 112

A boat covers 32 km upstream and 36 km downstream in 7hrs. Also, it covers 40 km upstream and $48\ \mathrm{km}$ downstream in $9\mathrm{hrs}$. Find the speed of the stream.

A. $2\ \mathrm{km/hr}$
B. $3\ \mathrm{km/hr}$
C. $1\ \mathrm{km/hr}$
D. $5\ \mathrm{km/hr}$

nielit2023feb-scientistc quantitative-aptitude system-of-equations

16.41.7 System of Equations: NIELIT 2023 Feb Scientist C | Section D | Question: 91

One fourth of a herd of camels was seen in forest. Twice the square root of the herd had gone to mountains and remaining 15 camels were seen on the bank of a river. Find the total number of camels.

A. 24
B. 30
C. 36
D. 42

nielit2023feb-scientistc analytical-aptitude system-of-equations quantitative-aptitude

16.42 Tabular Data (3)

16.42.1 Tabular Data: NIELIT 2021 Dec Scientist B - Section A: 33

Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

YEAR	RURAL		SEMI-URBAN		STATE CAPITAL		METROPOLITAN	
	App	Pass	App	Pass	App	Pass	App	Pass
2015	1652	208	7894	2513	5054	1468	9538	3214
2016	1839	317	8562	2933	7164	3248	10158	4018
2017	2153	932	8139	2468	8258	3159	9695	3038
2018	5032	1798	9432	3528	8529	3628	11247	5158
2019	4915	1668	9784	4015	9015	4311	12518	6328
2020	5628	2392	9969	4263	10725	4526	13624	6449

[*App - Appeared *Pass - Passed]

What approximate value was the percentage drop in the number of semi-urban candidates appeared from 2016 to 2017 ?

A. 15
B. 10
C. 5
D. 8

nielit2021dec-scientistb tabular-data percentage data-interpretation quantitative-aptitude

Answer key☟

16.42.2 Tabular Data: NIELIT 2021 Dec Scientist B - Section A: 34

Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

YEAR	RURAL		SEMI-URBAN		STATE CAPITAL		METROPOLITAN	
	App	Pass	App	Pass	App	Pass	App	Pass
2015	1652	208	7894	2513	5054	1468	9538	3214
2016	1839	317	8562	2933	7164	3248	10158	4018
2017	2153	932	8139	2468	8258	3159	9695	3038
2018	5032	1798	9432	3528	8529	3628	11247	5158
2019	4915	1668	9784	4015	9015	4311	12518	6328
2020	5628	2392	9969	4263	10725	4526	13624	6449

[*App - Appeared *Pass - Passed]

In which of the following years was the percentage passed to appeared candidates from semi-urban area the least ?

A. 2015 B. 2016 C. 2017 D. 2018

nielit2021dec-scientistb tabular-data percentage data-interpretation analytical-aptitude quantitative-aptitude

16.42.3 Tabular Data: NIELIT 2023 Feb Scientist D | Section C | Question: 74

The following table shows the number of students enrolled in different streams in a particular college.

SCIENCE		HUMANITIES		COMMERCE		VOCATIONAL	
Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
32	18	28	45	42	42	13	30

The ratio of the number of girls studying Humanities to the number of girls studying in all other streams is :

A. $1:3$ B. $2:1$ C. $1:2$ D. $3:1$

nielit2023feb-scientistd data-interpretation tabular-data

Answer key☟

16.43 Triangles (2)

16.43.1 Triangles: NIELIT 2018-8

Let us consider the length of the side of a square represented by $2y+3$. The length of the side of an equilateral triangle is $4y$. If the square and the equilateral triangle have equal perimeter, then what is the value of y?

A. 3 B. 4 C. 6 D. 8

nielit-2018 general-aptitude quantitative-aptitude geometry triangles

Answer key☟

16.43.2 Triangles: NIELIT 2023 Feb Scientist C | Section D | Question: 103

The base of a triangle is 15 cm. Two lines are drawn parallel to the base, terminating in the other two sides, and dividing the triangle into three equal areas. The length of the parallel line close to the base is :

A. $5\sqrt{6} \text{ cm}$
B. $4\sqrt{3} \text{ cm}$
C. 10 cm
D. 5.5 cm

nielit2023feb-scientistc geometry area triangles quantitative-aptitude

16.44 Trigonometry (2)

16.44.1 Trigonometry: NIELIT 2023 Feb Scientist C | Section D | Question: 96

The value of the expression $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$ is

A. 2
B. $2\frac{\sin 20^\circ}{\sin 40^\circ}$
C. 4
D. $4\frac{\sin 20^\circ}{\sin 40^\circ}$

nielit2023feb-scientistc trigonometry analytical-aptitude

16.44.2 Trigonometry: NIELIT 2023 Feb Scientist D | Section D | Question: 96

If $\sin x + \sin^2 x = 1$ then $\cos^8 x + 2\cos^6 x + \cos^4 x$ equals to :

A. 0 B. -1 C. 1 D. 2

nielit2023feb-scientistd quantitative-aptitude trigonometry

Answer key☟

16.45 Venn Diagram (6)

16.45.1 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 14

Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- $\text{Football} - 17\%$

- $\text{Handball} - 26\%$
- $\text{Badminton} - 16\%$
- $\text{Table Tennis} - 22\%$
- $\text{Basketball} - 19\%$

Total number of students in the school are $800.$

What is the number of girls who take part in handball, if the ratio of boys to girls is $3:10$ respectively?

A. 48 B. 80 C. 78 D. 160

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key☟

16.45.2 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 15

Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- $\text{Football} - 17\%$
- $\text{Handball} - 26\%$
- $\text{Badminton} - 16\%$
- $\text{Table Tennis} - 22\%$
- $\text{Basketball} - 19\%$

Total number of students in the school are $800.$

What is the respective ratio between the total number of students taking part in Badminton and Table Tennis together and those participating in Basketball and Football together?

A. $11:13$ B. $18:19$ C. $19:18$ D. $29:28$

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key☟

16.45.3 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 16

Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- $\text{Football} - 17\%$
- $\text{Handball} - 26\%$
- $\text{Badminton} - 16\%$
- $\text{Table Tennis} - 22\%$
- $\text{Basketball} - 19\%$

Total number of students in the school are 800.

What is the approximate average of number of participants in Handball, Badminton and Basketball?

A. 162 B. 163 C. 104 D. 169

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key☟

16.45.4 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 17

Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- $\text{Football} – 17\%$
- $\text{Handball} – 26\%$
- $\text{Badminton} – 16\%$
- $\text{Table Tennis} – 22\%$
- $\text{Basketball} – 19\%$

Total number of students in the school are 800.

The number of students taking part in Basketball is approximately what percent more than those taking part in Football?

A. 10.84% B. 9.92% C. 9.32% D. None of the options

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key☟

16.45.5 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 18

Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- $\text{Football} – 17\%$
- $\text{Handball} – 26\%$
- $\text{Badminton} – 16\%$
- $\text{Table Tennis} – 22\%$
- $\text{Basketball} – 19\%$

Total number of students in the school are 800.

If out of the number of students in Basketball, 69 are girls, what is the difference between the number of boys and girls taking part in Basketball?

A. 17 B. 23 C. 86 D. 14

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key

16.45.6 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 31

A Class has 100 students with roll number from 101 to $200.$ All the even numbered students study Physics, whose roll number are divisible by 5 study Chemistry $\&$ students with roll numbers divisible by 7 study Biology. How many students do not study any of the given subject Physics, Chemistry or Biology?

A. 35 B. 45 C. 51 D. 62

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key

16.46 Work Time (12)

16.46.1 Work Time: NIELIT 2018-3

A software engineer has the capability of thinking 200 lines of code in five minutes and can type 200 lines of code in 10 minutes. He takes a break for five minutes after every 10 minutes. How many lines of code will he complete typing after an hour?

A. 200 B. 300 C. 400 D. 500

nielit-2018 general-aptitude quantitative-aptitude work-time

Answer key

16.46.2 Work Time: NIELIT 2018-6

A construction company ready to finish a construction work in 180 days, hired 80 workers each working 8 hours daily. After 90 days, only $2/7$ of the work was completed. How many workers are to be increased to complete the work on time?

Note: If additionally acquired workers do agree to work for 10 hours daily.

A. 90 workers B. 80 workers C. 65 workers D. 85 workers

nielit-2018 general-aptitude quantitative-aptitude work-time

Answer key

16.46.3 Work Time: NIELIT 2021 Dec Scientist A - Section A: 24

Abha can do some work in 10 days, Billu can do it in 20 days and Chintu can do it in 40 days. They start working in turns with Abha starting to work on the first day followed by Billu on the second day and by Chintu on the third day and again by Abha on the fourth day and so on, till the work is completed fully. Find the time taken (approx.) to complete the work fully.

A. 16 days B. 15 days C. 17 days D. 20 days

nielit2021dec-scientista work-time quantitative-aptitude

Answer key

16.46.4 Work Time: NIELIT 2021 Dec Scientist A - Section A: 29

In a company ABC Ltd. a certain number of engineers can develop a design in 40 days. If there were 5 more engineers, it could be finished in 10 days less. How many engineers were there in the beginning ?

A. 18 B. 20 C. 25 D. 15

nielit2021dec-scientista quantitative-aptitude work-time

Answer key☟

16.46.5 Work Time: NIELIT 2022 April Scientist B | Section A | Question: 11

Three pipes A, B and C can fill a cistern in 8 minutes, 12 minutes and 16 minutes respectively. What is the time taken by three pipes to fill the cistern when they are opened together?

A. 3.7 minutes B. 6 minutes C. 4.5 minutes D. 5 minutes

nielit2022apr-scientistb quantitative-aptitude work-time analytical-aptitude

Answer key☟

16.46.6 Work Time: NIELIT 2022 April Scientist B | Section A | Question: 36

In a race of 600 m, A can beat B by 50 m, and in another race of 500 m, B can beat C by 60 m. Then by how many meters will A beat C in a race of 400 meters ?

A. $77\frac{1}{3} \text{ m}$ B. 80 m C. 70 m D. None of these

nielit2022apr-scientistb quantitative-aptitude work-time

Answer key☟

16.46.7 Work Time: NIELIT 2023 Feb Scientist C | Section D | Question: 113

A contractor undertakes to build a wall in 50 days. He employs 50 people for the same. However, after 25 days he finds that the work is only 40% complete. How many more men need to be employed to complete the work 10 days before time.

A. 50 men B. 75 men C. 100 men D. 125 men

nielit2023feb-scientistc quantitative-aptitude work-time

16.46.8 Work Time: NIELIT 2023 Feb Scientist D | Section C | Question: 73

3 men earn as much as 4 women; 4 women earn as much as 6 boys and 8 boys earn as much as 10 girls. If a girl earns Rs. $50/-$ a day then the earning of a man would be?

A. 115 B. 135 C. 125 D. 150

nielit2023feb-scientistd quantitative-aptitude work-time

Answer key☟

16.46.9 Work Time: NIELIT 2023 Feb Scientist D | Section D | Question: 103

Subhash can copy 50 pages in $10 \text{hrs},$, Subhash and Prakash together can copy 300 pages in 40hrs. In how much time can Prakash copy 30 pages.

A. 13hrs
B. 12hrs
C. 11hrs
D. 9hrs

nielit2023feb-scientistd quantitative-aptitude work-time

Answer key☟

16.46.10 Work Time: NIELIT 2023 Feb Scientist D | Section D | Question: 99

A contractor undertakes to build a wall in 50 days. He employs 50 people for the same. However, after 25 days he finds that the work is only 40% complete. How many more men need to be employed to complete the work in time?

A. 50 men B. 75 men C. 100 men D. 25 men

nielit2023feb-scientistd work-time quantitative-aptitude

Answer key☟

16.46.11 Work Time: NIELIT Scientific Assistant A 2020 November: 33

Ram can do a piece of work in 5 days, and Sham can do the same in 10 days. With the help of Karan, they finished the work in 2 days. How many days would it take Karan to do the work?

A. 5 days B. 10 days C. 15 days D. 20 days

nielit-sta-2020 quantitative-aptitude work-time

Answer key☟

16.46.12 Work Time: NIELIT Scientist B 2020 November: 34

Two taps A and B can fill a tank in 12 minutes and 15 minutes respectively. The tank can be emptied by a third tap C in 6 minutes. If A and B are kept open for 5 minutes in the beginning and then C is opened along with A and B being kept open, the time taken to empty the tank is:

A. 60 minutes B. 45 minutes C. 30 minutes D. 75 minutes

nielit-scb-2020 quantitative-aptitude work-time

Answer key☟

Answer Keys

16.0.1	B	16.0.2	A	16.0.3	B	16.0.4	A	16.0.5	B
16.0.6	A	16.0.7	B	16.0.8	C	16.0.9	D	16.0.10	C
16.0.11	TBA	16.0.12	C	16.0.13	A	16.0.14	C	16.0.15	C
16.0.16	D	16.0.17	D	16.0.18	C	16.0.19	C	16.0.20	B
16.0.21	A	16.0.22	A	16.0.23	A	16.0.24	C	16.0.25	A
16.1.1	A	16.2.1	C	16.3.1	D	16.4.1	A	16.5.1	C
16.6.1	A	16.6.2	B	16.6.3	B	16.6.4	C	16.7.1	A
16.7.2	C	16.7.3	C	16.7.4	A	16.7.5	C	16.8.1	C
16.9.1	B	16.9.2	D	16.9.3	D	16.9.4	B	16.10.1	C
16.11.1	D	16.11.2	B	16.12.1	C	16.13.1	D	16.14.1	A
16.14.2	A	16.15.1	C	16.16.1	A	16.16.2	A	16.16.3	D
16.16.4	D	16.16.5	B	16.16.6	A	16.16.7	C	16.16.8	A
16.16.9	A	16.16.10	C	16.16.11	C	16.16.12	C	16.17.1	A
16.18.1	C	16.18.2	B	16.18.3	B	16.19.1	C	16.20.1	A
16.21.1	C	16.22.1	D	16.22.2	D	16.22.3	B	16.23.1	A
16.23.2	B	16.23.3	C	16.23.4	C	16.23.5	D	16.24.1	D
16.24.2	B	16.24.3	A	16.24.4	C	16.24.5	C	16.24.6	B
16.24.7	TBA	16.24.8	B	16.25.1	C	16.26.1	C	16.26.2	A
16.26.3	D	16.26.4	B	16.26.5	C	16.26.6	A	16.26.7	D
16.26.8	C	16.26.9	A	16.26.10	C	16.26.11	A	16.26.12	C
16.26.13	B	16.26.14	B	16.27.1	A	16.27.2	A	16.28.1	B
16.28.2	A	16.29.1	B	16.30.1	A	16.31.1	B	16.31.2	C
16.31.3	TBA	16.31.4	B	16.31.5	A	16.31.6	D	16.32.1	A
16.33.1	A	16.33.2	A	16.33.3	D	16.33.4	D	16.33.5	A
16.33.6	C	16.33.7	A	16.33.8	C	16.33.9	D	16.33.10	D
16.33.11	C	16.33.12	B	16.33.13	D	16.33.14	D	16.34.1	B

16.35.1	D
16.36.4	D
16.39.3	C
16.40.2	C
16.41.5	B
16.42.3	C
16.45.1	D
16.45.6	A
16.46.5	A
16.46.10	D

16.35.2	C
16.37.1	B
16.39.4	B
16.41.1	B
16.41.6	A
16.43.1	A
16.45.2	C
16.46.1	D
16.46.6	D
16.46.11	A

16.36.1	A
16.38.1	C
16.39.5	B
16.41.2	A
16.41.7	C
16.43.2	A
16.45.3	B
16.46.2	B
16.46.7	B
16.46.12	B

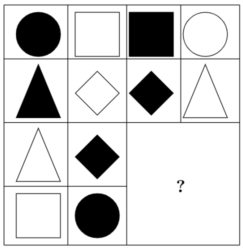
16.36.2	C
16.39.1	C
16.39.6	C
16.41.3	A
16.42.1	C
16.44.1	C
16.45.4	D
16.46.3	A
16.46.8	C

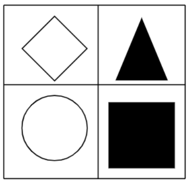
16.36.3	B
16.39.2	A
16.40.1	A
16.41.4	A
16.42.2	C
16.44.2	C
16.45.5	D
16.46.4	D
16.46.9	B

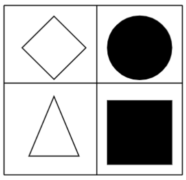
17 General Aptitude: Spatial Aptitude (6)

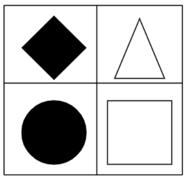
17.0.1 NIELIT Scientist B 2020 November: 11

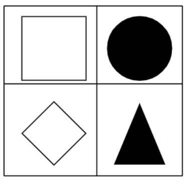
Identify the figure that completes the pattern.



A. 

B. 

C. 

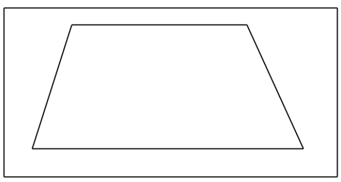
D. 

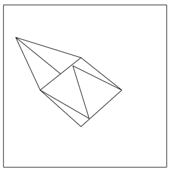
nielit-scb-2020 analytical-aptitude spatial-aptitude

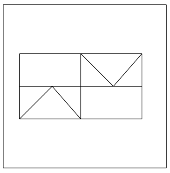
Answer key☟

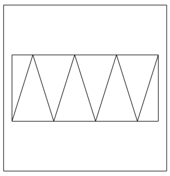
17.0.2 NIELIT Scientist B 2020 November: 7

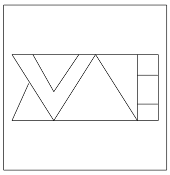
Find out the alternative figure which contains the given figure as its part.



A. 

B. 

C. 

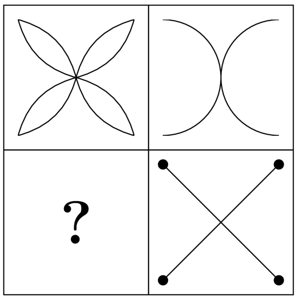
D. 

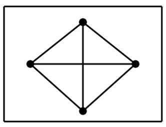
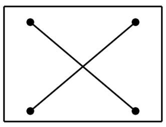
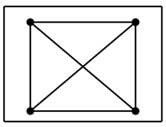
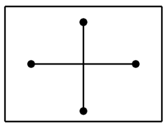
nielit-scb-2020 analytical-aptitude spatial-aptitude

Answer key☟

17.0.3 NIELIT Scientist B 2020 November: 26

Select a suitable figure from the four alternatives that would complete the figure matrix.



A.  B.  C.  D. 

nielit-scb-2020 analytical-aptitude spatial-aptitude

Answer key☟

17.1 Direction Sense (2)

17.1.1 Direction Sense: NIELIT 2023 Feb Scientist C | Section A | Question: 17

A river flows west to east and on the way turns left and goes in a semi-circle round a hillock, and then turns left at right angles. In which direction is the river finally flowing?

A. West B. East C. North D. South

nielit2023feb-scientistc analytical-aptitude direction-sense

17.1.2 Direction Sense: NIELIT 2023 Feb Scientist D | Section A | Question: 10

I was facing East from where I turned to my left and walked 12ft, then I turned towards right and walked 6ft. After that I walked 6ft in South direction and at last, I walked 6ft in the West. Then, in which direction am I standing from the original point?

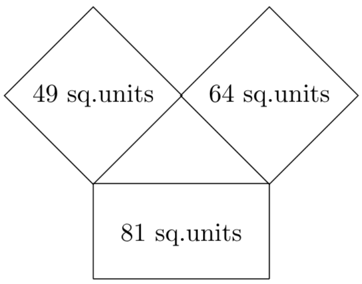
A. West B. East C. South D. North

nielit2023feb-scientistd analytical-aptitude direction-sense

17.2 Geometry (1)

17.2.1 Geometry: NIELIT Scientist B 2020 November: 35

Three squares are there as shown on the three sides of the triangle; find the area of the triangle from the respective areas of the squares.



A. $15\sqrt{5}$ B. $12\sqrt{5}$ C. $2\sqrt{5}$ D. 1

nielit-scb-2020 geometry

Answer key☟

Answer Keys

17.0.1	A	17.0.2	C	17.0.3	C	17.1.1	D	17.1.2	D
17.2.1	B								

18 General Aptitude: Verbal Aptitude (111)

18.0.1 NIELIT 2018-4

Candid is to indirect as honest is to

A. frank B. wicked C. truthful D. untruthful

nielit-2018 general-aptitude verbal-aptitude

Answer key☟

18.0.2 NIELIT 2023 Feb Scientist C | Section E | Question: 136

___________ is not a capability of an employee having Positive attitude

A. Focus B. Creativity C. Pessimism D. Confidence

nielit2023feb-scientistc verbal-aptitude analytical-aptitude

18.0.3 NIELIT 2023 Feb Scientist C | Section E | Question: 130

A pessimist sees the difficulty in every opportunity; an optimist sees the opportunity in every difficulty. " These words were said by :

A. Winston Churchill
B. A. P. J. Abdul Kalam
C. S. Robbins
D. Dressler

nielit2023feb-scientistc verbal-aptitude analytical-aptitude

18.1 English Grammar (56)

18.1.1 English Grammar: NIELIT 2021 Dec Scientist A - Section A: 27

Choose the most appropriate word from the options given below to complete the following sentence: He is _____________ speaker, his discourses are always informative and inspirational.

A. an eloquent B. an amateur C. a novice D. an inarticulate

nielit2021dec-scientista most-appropriate-word english-grammar verbal-aptitude

18.1.2 English Grammar: NIELIT 2021 Dec Scientist A - Section A: 31

“The judge’s standing in the legal community, though shaken by false allegations of wrongdoing, remained ________________ .” The word that best fills the blank in the above sentence is :

A. undiminished B. damaged C. illegal D. uncertain nielit2021dec-scientista english-grammar most-appropriate-word verbal-aptitude

Answer key☟

18.1.3 English Grammar: NIELIT 2021 Dec Scientist A - Section A: 39

The untimely loss of life is a cause of serious global concern as thousands of people get killed _______________ accidents every year while many other die _______________ diseases like cardio vascular disease, cancer, etc.

A. in, of B. from, of C. during, from D. from, from

nielit2021dec-scientista english-grammar verbal-aptitude

Answer key☟

18.1.4 English Grammar: NIELIT 2021 Dec Scientist A - Section A: 40

Given below question has an idiomatic expression followed by four options. Choose the one closest to its meaning :

"To smell a rat"

A. science of plague epidemic
B. bad smell
C. suspect foul dealings
D. to be in a bad mood

nielit2021dec-scientista english-grammar verbal-aptitude

Answer key☟

18.1.5 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 35

He was not only accused of theft _____________ of conspiracy.

A. rather B. but also C. but even D. rather than

nielit2021dec-scientistb english-grammar most-appropriate-word verbal-aptitude

Answer key☟

18.1.6 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 36

"The dress ____________ him so well that she immediately ______________ him on his appearance." The words that best fill the blanks in the above sentence are :

A. complemented, complemented,
B. complimented, complemented,
C. complimented, complimented,
D. complemented, complimented,

nielit2021dec-scientistb english-grammar verbal-aptitude

18.1.7 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 37

Each of these questions $(37 - 38)$ has an idiomatic expression followed by four options. Choose the one closest to its meaning.

Talk shop :

A. Talk about once profession
B. Talk about shopping
C. Ridicule
D. Treat lightly

nielit2021dec-scientistb english-grammar verbal-aptitude

Answer key

18.1.8 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 38

Each of these questions $(37 - 38)$ has an idiomatic expression followed by four options. Choose the one closest to its meaning.

Stick to once guns :

A. Remain faithful to the cause
B. suspect something
C. make something fail
D. be satisfied

nielit2021dec-scientistb english-grammar verbal-aptitude

Answer key

18.1.9 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 39

Identify the correct spelling out of the given options :

A. Managable
B. Manageable
C. Managaeble
D. Managible

nielit2021dec-scientistb english-grammar

Answer key

18.1.10 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 40

“Going by the ____________ that many hands make light work, the school ____________ involved all the students in the task.” The words that best fill the blanks in the above sentence are :

A. principle, principal
B. principal, principle
C. principle, principle
D. principal, principal

nielit2021dec-scientistb english-grammar most-appropriate-word verbal-aptitude

Answer key

18.1.11 English Grammar: NIELIT 2021 Dec Scientist B - Section A: 42

The fisherman, ___________ the flood victims owed their lives, were rewarded by the govt.

A. whom
B. to which
C. to whom
D. to that

nielit2021dec-scientistb english-grammar verbal-aptitude

18.1.12 English Grammar: NIELIT 2023 Feb Scientist C | Section A | Question: 30

According to Wikipedia, 'monologue' is presented by a single character most often to express their mental thoughts aloud, though sometimes also to directly address another character or the audience. Which of the following does **NOT** represent a monologue?

A. A poet reciting a poem on the stage
B. Anshuman hosting an award show
C. A mother narrating a story to her child
D. A teacher taking up student queries in the class

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.13 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 32

Direction: $(31-34)$ In the following questions find the appropriate explanation for idioms/phrases mentioned in bold letters.

They were beginning to scrape the barrels.

A. Forced to use one's last resources
B. Break into a sudden fight after getting drunk
C. To waste one's resources
D. Drink without worrying about the consequences

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.14 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 34

Direction: $(31-34)$ In the following questions find the appropriate explanation for idioms/phrases mentioned in bold letters.

Don't try to boil the ocean.

A. Attack someone who is immensely powerful
B. Try to swindle the high and the mighty
C. Try to do something difficult
D. Pollute the ocean leading to a rise in its temperature

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.15 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 35

Direction: $(35-37)$

Choose the correct spelling of the word from the given options.

A. minascule B. minuscule C. miniiscule D. minnuscule

nielit2023feb-scientistc english-grammar most-appropriate-word verbal-aptitude

Answer key

18.1.16 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 36

Direction: $(35-37)$

Choose the correct spelling of the word from the given options.

A. occasionaly
B. occasionally
C. ocassonally
D. ocassionally

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.17 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 37

Direction: $(35-37)$

Choose the correct spelling of the word from the given options.

A. flabbergested
B. flabberghasted
C. flabbergasted
D. flaberghasted

nielit2023feb-scientistc english-grammar most-appropriate-word

18.1.18 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 38

Direction $(38-40)$**:** Choose the correct substitute for the following phrases/sentences.

One who compiles a dictionary

A. bibliophile
B. cartographer
C. lexicographer
D. linguist

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.19 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 39

Direction $(38-40)$**:** Choose the correct substitute for the following phrases/sentences.

A person who has no money to pay off his debts :

A. penurious
B. insolvent
C. destitute
D. pauper

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.20 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 41

Direction (41-43): Out of the four sentences given below three are incorrect and one correct. The correct one is your answer.

A. I am working really hard, am I?
B. I am working really hard?
C. I am working really hard, isn't it?
D. Am I working hard enough?

nielit2023feb-scientistc english-grammar verbal-aptitude analytical-aptitude

18.1.21 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 42

Direction (41-43): Out of the four sentences given below three are incorrect and one correct. The correct one is your answer.

A. He, you, and I will leave for this secret mission tonight.
B. You, he, and I will leave for this secret mission tonight.
C. I, you, and he will leave for this secret mission tonight.
D. You, I and he will leave for this secret mission tonight.

nielit2023feb-scientistc english-grammar verbal-aptitude

Answer key☟

18.1.22 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 43

Direction (41-43): Out of the four sentences given below three are incorrect and one correct. The correct one is your answer.

A. The only people interested in the book seems to be lawyers.
B. The only people interested in this book are seeming to be lawyers.
C. The only people interested in the book are seemingly to be lawyers.
D. The only people interested in the books seem to be lawyers.

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.23 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 44

Direction $(44-46)$**:** Fill in the blanks with the correct option given below.

This book comprises _________ five chapters.

A. no preposition required
B. of
C. with
D. in

nielit2023feb-scientistc english-grammar

18.1.24 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 45

Direction $(44-46)$**:** Fill in the blanks with the correct option given below.

Everybody sat ________ the floor, but Mr. Sharma slept ________ a cot ________ the corner.

A. over-in-on B. in-over-from C. on-on-in D. about-on-at

nielit2023feb-scientistc english-grammar most-appropriate-word

18.1.25 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 46

Direction $(44-46)$**:** Fill in the blanks with the correct option given below.
_________ such brilliant batting.

A. Never I have seen
B. Rarely I have
C. Never have I seen
D. Never I had seen

nielit2023feb-scientistc english-grammar verbal-aptitude

18.1.26 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 47

Direction $(47-49)$**:** Read the following comprehension carefully and answer the following questions.
Humans are to be blamed for declining bat populations. Bats get poisoned when they consume insects that have been sprayed with pesticides and insecticides. But the biggest problem for bats is the loss of natural habitat. Many bats prefer to roost in dead or dying trees under the loose and peeling bark, or in tree cavities. Some prefer to roost in caverns. Populations have dwindled and diversity has suffered without the protection of these important natural roosts.

Find a word in the passage which means the same as decline.

A. habitat B. suffered C. dwindled D. flourish

nielit2023feb-scientistc english-grammar most-appropriate-word verbal-aptitude

18.1.27 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 49

Direction $(47-49)$**:** Read the following comprehension carefully and answer the following questions.
Humans are to be blamed for declining bat populations. Bats get poisoned when they consume insects that have been sprayed with pesticides and insecticides. But the biggest problem for bats is the loss of natural habitat. Many bats prefer to roost in dead or dying trees under the loose and peeling bark, or in tree cavities. Some prefer to roost in caverns. Populations have dwindled and diversity has suffered without the protection of these important natural roosts.

Find a word in the paragraph which means the opposite of coating.

A. peeling B. protection C. sprayed D. cavern

nielit2023feb-scientistc english-grammar most-appropriate-word verbal-aptitude

18.1.28 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 55

Direction (53-55): Choose the word that is opposite in meaning to the given word.
Object

A. remonstrate B. insult C. cajole D. approve

nielit2023feb-scientistc english-grammar verbal-aptitude

Answer key☟

18.1.29 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 56

Direction $(56-60)$**:** Complete the sentences using most appropriate options given below :

The orator's fiery speech was the for the ensuing

A. Catalyst; revolution
B. Damper; explosion
C. Epilogue; doctrine
D. Barrier; movement

nielit2023feb-scientistc most-appropriate-word english-grammar verbal-aptitude

Answer key☟

18.1.30 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 57

Direction $(56-60)$**:** Complete the sentences using most appropriate options given below :

The millionaire added a __________ to his will just before dying.

A. Prologue B. Codicil C. Formal D. License

nielit2023feb-scientistc english-grammar most-appropriate-word

18.1.31 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 58

Direction $(56-60)$**:** Complete the sentences using most appropriate options given below :

The vicious rumours were untrue but still the politician's

A. Elevated; status
B. Impugned; reputation
C. Interrogated; subordinates
D. Alleviated; difficulties

nielit2023feb-scientistc english-grammar most-appropriate-word verbal-aptitude

18.1.32 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 59

Direction $(56-60)$**:** Complete the sentences using most appropriate options given below :

Restlessness among the students of colleges and universities has from their deep rooted feeling that their views and aspirations are by their elders.

A. arisen, humiliated
B. followed, disregarded
C. stemmed, ignored
D. started, neglected

nielit2023feb-scientistc english-grammar most-appropriate-word verbal-aptitude

18.1.33 English Grammar: NIELIT 2023 Feb Scientist C | Section B | Question: 60

Direction $(56-60)$**:** Complete the sentences using most appropriate options given below :

The ________ of the mountains stood out as craggy________ on the horizon.

A. Colors; shapes
B. Shapes; textures
C. Outline; shapes
D. Textures; lines

nielit2023feb-scientistc english-grammar most-appropriate-word verbal-aptitude

Answer key☟

18.1.34 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 31

Directions: In questions **(31 - 34)** some of the sentences have errors and some have none. Find out which part of a sentence has an error, and the appropriate letter $(\mathrm{A}, \mathrm{B}, \mathrm{C})$ is your answer. If there is no error, D is the answer.

A. The food was excellent and
B. unusually the menu was concise
C. with enough choice to satisfy most tastes.
D. No error

nielit2023feb-scientistd english-grammar verbal-aptitude analytical-aptitude

18.1.35 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 32

Directions: In questions **(31 - 34)** some of the sentences have errors and some have none. Find out which part of a sentence has an error, and the appropriate letter $(\mathrm{A}, \mathrm{B}, \mathrm{C})$ is your answer. If there is no error, D is the answer.

A. As you enter the place,
B. a magnificent setting, a delightful marriage of antique cut stones
C. and the luxuries of modernity, welcoming you.
D. No error

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.36 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 33

Directions: In questions **(31 - 34)** some of the sentences have errors and some have none. Find out which part of a sentence has an error, and the appropriate letter $(\mathrm{A}, \mathrm{B}, \mathrm{C})$ is your answer. If there is no error, D is the answer.

A. Historiography is a meta-level analysis of descriptions of the past
B. usually focuses on the narrative, interpretations, worldview, use of evidence,
C. or method of presentation of other historians.
D. No error

nielit2023feb-scientistd english-grammar verbal-aptitude analytical-aptitude

18.1.37 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 34

Directions: In questions **(31 - 34)** some of the sentences have errors and some have none. Find out which part of a sentence has an error, and the appropriate letter $(\mathrm{A}, \mathrm{B}, \mathrm{C})$ is your answer. If there is no error, D is the answer.

A. To buy a house in the village
B. while lives in the city is a feature of this interconnectivity
C. that blurs in social terms any sharp distinction between city and village or big city and small city.
D. No error

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.38 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 36

Directions: In questions **(35 - 36)**, the sentences have been given in Direct / Indirect speech. From the given alternatives, choose the one which best expresses the given sentence in direct or indirect speech as directed:

Change the following sentence into direct speech :

The moderator told him to sit down and to let others speak.

A. The moderator said, "Sit down and let others speak."
B. The moderator asked him to sit down and let others speak.
C. The moderator commanded him, "to sit down and to let others speak."
D. The moderator commanded him to sit down and to "Let others speak".

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.39 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 37

Directions: In the questions **(37 - 38)**, the given sentences are in Active / Passive voice. From the given alternatives, choose the one which best expresses the given sentence in active or passive voice as directed :

A new book has been published against the hunting of endangered species.

A. A new book has been published against endangered species being hunted.
B. Hunting of endangered species is against the new book that has been published.
C. Hunting of endangered species was against the new book that has been published.
D. A new book has been published against endangered species to be hunted.

nielit2023feb-scientistd english-grammar verbal-aptitude analytical-aptitude

18.1.40 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 39

Directions: In questions **(39-42)** out of the four alternatives choose the one which can be substituted for the given words/sentence.

Art that involves single colour or hue :

A. Contemporary Art
B. Kinetic Art
C. Monochrome Art
D. Motley Art

nielit2023feb-scientistd english-grammar most-appropriate-word verbal-aptitude

18.1.41 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 40

Directions: In questions **(39-42)** out of the four alternatives choose the one which can be substituted for the given words/sentence.

Fiction that deals with nature/environment-oriented subjects :

A. Climate Fiction
B. Eco-fiction
C. Fantasy Novel
D. Historical Fiction

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.42 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 42

Directions: In questions **(39-42)** out of the four alternatives choose the one which can be substituted for the given words/sentence.

The term 'chiasma' means :

A. Point of contact
B. Point of order
C. Point of difference
D. Point of destruction

nielit2023feb-scientistd verbal-aptitude english-grammar analytical-aptitude

18.1.43 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 47

Directions: In questions **(47- 48)** out of the four alternatives choose the most appropriate Synonym to the given word.

Preeminent

A. Sharp
B. Celebrated
C. Guilty
D. Insignificant

nielit2023feb-scientistd english-grammar most-appropriate-word verbal-aptitude

18.1.44 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 48

Directions: In questions **(47- 48)** out of the four alternatives choose the most appropriate Synonym to the given word.

Compelling

A. Captivating
B. Obnoxious
C. Different
D. Comprehensive

nielit2023feb-scientistd english-grammar most-appropriate-word verbal-aptitude

18.1.45 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 49

Directions: In queștions **(49- 51)** out of the four alternatives choose the most appropriate Antonym to the given word.

Compliant

A. Submissive
B. Meek
C. Easygoing
D. Combative

nielit2023feb-scientistd verbal-aptitude most-appropriate-word english-grammar

18.1.46 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 50

Directions: In queștions **(49- 51)** out of the four alternatives choose the most appropriate Antonym to the given word.

Atypical

A. Unique
B. Peculiar
C. Common
D. Special

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.47 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 51

Directions: In queștions **(49- 51)** out of the four alternatives choose the most appropriate Antonym to the given word.

Entrepreneur

A. Employee
B. Founder
C. Businessperson
D. Leader

nielit2023feb-scientistd verbal-aptitude english-grammar

18.1.48 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 52

Choose the most appropriate sentence from the options given below to express the given below sentence in plural.

It is useful for a library to be located away from a noisy building to enable a student to concentrate.

A. It is useful for libraries to be located away from noisy buildings to enable students to concentrate.
B. It is useful for libraries to be located away from a noisy building to enable students to concentrate.
C. It is useful for libraries to be located away from noisy buildings to enable students to concentrates.
D. It is useful for a library to be located away from noisy buildings to enable a student to concentrate.

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.49 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 53

Directions: Without changing the meaning of the given statement, choose the most appropriate option in which the statement is expressed in future tense.

If a cyclist saw an empty stretch, he would speed up.

A. If a cyclist saw an empty stretch, he will speed up.
B. If a cyclist sees an empty stretch, he would speed up.
C. If a cyclist sees an empty stretch, he will speed up.
D. If a cyclist will see an empty stretch, he will speed up.

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.50 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 54

Directions: In questions **(54 - 55)** choose the alternative that is closest to the meaning of the italicized part of the sentence from the options given below.

As a *rule of thumb*, I always shut down my devices rather than leave them in idle mode.

A. Rule-based on personal experience
B. Healthy rule
C. Shortcut
D. National law

nielit2023feb-scientistd english-grammar most-appropriate-word verbal-aptitude

18.1.51 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 57

Directions **(56 to 59)**: Identify which part of the sentence contains error and mark it as your answer.

The principal along with her staff / are practicing very hard / for the / forthcoming Annual Day.

A. The principal along with her staff
B. are practicing very hard
C. for the
D. forthcoming Annual Day

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.52 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 58

Directions **(56 to 59)**: Identify which part of the sentence contains error and mark it as your answer.

Supposing if / there is a thunder storm / what shall / you do?

A. Supposing if
B. there is a thunder storm
C. what shall
D. you do?

nielit2023feb-scientistd english-grammar verbal-aptitude analytical-aptitude

18.1.53 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 59

Directions **(56 to 59)**: Identify which part of the sentence contains error and mark it as your answer.

It was him / who came/running / into the classroom.

A. It was him
B. who came
C. running
D. into the classroom

nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.54 English Grammar: NIELIT 2023 Feb Scientist D | Section B | Question: 60

Choose the most appropriate adjective used in the given below sentence from the given choices : Kavery likes to run along the smooth road to keep fit.

A. Kavery B. run C. smooth D. road

nielit2023feb-scientistd english-grammar most-appropriate-word verbal-aptitude

18.1.55 English Grammar: NIELIT 2023 Feb Scientist D | Section E | Question: 131

Sometimes due to any hard situation _________ of an individual will change but will take a long time.

A. Objective B. Attitude C. Motivation D. Performance nielit2023feb-scientistd english-grammar verbal-aptitude

18.1.56 English Grammar: NIELIT Scientist B 2020 November: 9

Which of these statements reflects a contrast between two flowers?

A. This tulip is as colourful as a rose
B. This tulip does not smell as bad as a daffodil
C. This tulip turns towards light just like sunflower
D. This tulip is grown in bunches, like a lotus

nielit-scb-2020 english-grammar logical-reasoning

18.2 General (3)

18.2.1 General: NIELIT 2023 Feb Scientist C | Section E | Question: 131

Which is NOT true of charismatic leaders?

A. Charismatic leaders have a vision.
B. Charismatic leaders are focused on their personal needs.
C. Charismatic leaders are willing to take high personal risk.
D. Charismatic leaders have behavior that is unconventional.

nielit2023feb-scientistc general logical-reasoning

18.2.2 General: NIELIT 2023 Feb Scientist C | Section E | Question: 139

Leadership is often associated more with

A. Consistency B. Planning C. Paperwork D. Change

nielit2023feb-scientistc general verbal-reasoning

18.2.3 General: NIELIT 2023 Feb Scientist C | Section E | Question: 143

Person perception is :

A. The process of how we manage, experience, and adjust our emotions
B. The process by which we form impressions and process information about people
C. The process by which we adjust to different cultural situations
D. The process by which we learn cultural norms and standards of behaviour

nielit2023feb-scientistc general logical-reasoning

18.3 General Awareness (1)

18.3.1 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 134

Which of the following is not a result of Positive attitude of employee?

A. Improved productivity
B. Successful Organisation
C. Higher performance
D. None of the options

nielit2023feb-scientistc general-awareness

18.4 Is&software Engineering (1)

18.4.1 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 141

Who is a visionary leader?

A. One who has medium term perspective
B. One who has short term perspective
C. One who has long term perspective
D. One who takes initiative for need satisfaction

nielit2023feb-scientistc is&software-engineering logical-reasoning

18.5 Letter Series (1)

18.5.1 Letter Series: NIELIT 2022 April Scientist B | Section A | Question: 37

Which of the following set of letters complete the letter series, when sequentially placed at the gaps ?

$\_bc\_ca\_aba\_c\_ca$

A. abcbb
B. bbbcc
C. baaba
D. abbcc

nielit2022apr-scientistb analytical-aptitude logical-reasoning letter-series

Answer key☟

18.6 Logical Reasoning (6)

18.6.1 Logical Reasoning: NIELIT 2018-7

Here are some words translated from any Language.

dionot means oak tree

blyonot means oak leaf

blycrin means maple leaf

Which word could mean “maple syrup”?

A. blymuth B. hupponot C. patricrin D. crinweel

nielit-2018 general-aptitude verbal-aptitude logical-reasoning

Answer key☟

18.6.2 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section E | Question: 138

Attitude conviction involves which of the following elements?

A. Ego preoccupation
B. Centrality
C. Importance
D. Strength

nielit2023feb-scientistc logical-reasoning

18.6.3 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 122

From the below which is not an outcome from person having good personality?

A. Good Performer
B. Good team player
C. Lead a team well
D. Make Profit

nielit2023feb-scientistd logical-reasoning verbal-aptitude

18.6.4 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 126

_______ are stimulated by events and people external to themselves. They show their feelings, learn by talking and work well in groups.

A. Positive personalities
B. Introverts
C. Extroverts
D. None of the above

nielit2023feb-scientistd logical-reasoning verbal-aptitude

18.6.5 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 129

________ is not a capability of an employee having Positive attitude.

A. Focus B. Creativity C. Pessimism D. Confidence

nielit2023feb-scientistd logical-reasoning verbal-aptitude

18.6.6 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 133

The belief that "discrimination is wrong" is a value statement. Such an opinion is the _________ component of an attitude.

A. Cognitive B. Affective C. Reactive D. Behavioural nielit2023feb-scientistd logical-reasoning

18.7 Most Appropriate Word (8)

18.7.1 Most Appropriate Word: NIELIT 2018-12

If finch related to bird (i.e.,) FINCH : BIRD. Then, find the pair that has a similar relationship.

A. frog : toad
B. elephant : reptile
C. dalmatian : dog
D. collie : marsupial

nielit-2018 verbal-aptitude logical-reasoning most-appropriate-word

Answer key☟

18.7.2 Most Appropriate Word: NIELIT 2018-13

Which word does NOT belong with the others?

A. wing
B. fin
C. beak
D. rudder

nielit-2018 verbal-aptitude logical-reasoning most-appropriate-word

Answer key☟

18.7.3 Most Appropriate Word: NIELIT 2021 Dec Scientist B - Section A: 41

Which of the following options is the closest in meaning to the word below ?

DELETERIOUS

A. delaying
B. glorious
C. harmful
D. graduating

nielit2021dec-scientistb verbal-aptitude most-appropriate-word

Answer key☟

18.7.4 Most Appropriate Word: NIELIT 2023 Feb Scientist C | Section B | Question: 50

Direction $(50-52)$: In the following questions choose the word that is similar in meaning to the given word.

Tireless

A. frugal
B. useless
C. munificent
D. hard-working

nielit2023feb-scientistc verbal-aptitude most-appropriate-word

18.7.5 Most Appropriate Word: NIELIT 2023 Feb Scientist C | Section B | Question: 51

Direction $(50-52)$: In the following questions choose the word that is similar in meaning to the given word.

Accentuate

A. accelerate B. underscore C. mask D. expedite

nielit2023feb-scientistc verbal-aptitude most-appropriate-word

18.7.6 Most Appropriate Word: NIELIT 2023 Feb Scientist C | Section B | Question: 52

Direction $(50-52)$: In the following questions choose the word that is similar in meaning to the given word.

Fastidious

A. meticulous B. vehement C. ludicrous D. generous

nielit2023feb-scientistc verbal-aptitude most-appropriate-word

18.7.7 Most Appropriate Word: NIELIT 2023 Feb Scientist C | Section B | Question: 53

Direction (53-55): Choose the word that is opposite in meaning to the given word.

Abnegation

A. relinquishment
B. eschewal
C. indulgence
D. abstinence

nielit2023feb-scientistc verbal-aptitude most-appropriate-word

18.7.8 Most Appropriate Word: NIELIT 2023 Feb Scientist C | Section B | Question: 54

Direction (53-55): Choose the word that is opposite in meaning to the given word.

Strenuous

A. herculean B. toilsome C. enervated D. elusive

nielit2023feb-scientistc verbal-aptitude most-appropriate-word

18.8 Passage Reading (3)

18.8.1 Passage Reading: NIELIT 2022 April Scientist B | Section A | Question: 18

The first line (P) and last line (U) of the question are fixed. Arrange the other four lines in a logical sequence.

P. First, take five minutes to meditate for peace.
Q. Allow them to radiate from your stillness out into your body.
R. Bring into your mind anyone against whom you have a grievance and let it go.
S. Close your eyes.

T. Put your attention on your heart and inwardly repeat the words: Peace, harmony, laughter, love.
U. Then introduce the intention of peace in your thoughts. After a few moments of silence, repeat this prayer; let me be loved, happy and peaceful; let my friends, my perceived enemies, all beings in the world be happy, loved and peaceful too.

A. RQTS
B. RSTQ
C. TQSR
D. STQR

nielit2022apr-scientistb analytical-aptitude passage-reading

Answer key☟

18.8.2 Passage Reading: NIELIT 2023 Feb Scientist C | Section B | Question: 33

Direction: $(31-34)$ In the following questions find the appropriate explanation for idioms/phrases mentioned in bold letters.

I am beat generally by the time I reach home.

A. Late
B. Drunk
C. Upbeat and energetic
D. Thoroughly fatigued

nielit2023feb-scientistc english-grammar verbal-aptitude passage-reading

18.8.3 Passage Reading: NIELIT 2023 Feb Scientist D | Section B | Question: 45

Directions: In questions **(43-46)**, rearrange the following five sentences $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ and T in the proper sequence to form a meaningful paragraph; then answer the questions given below them.

P. With the death of Queen Anne in August 1714 and the accession of George I, the Tories were a ruined party, and Swift's career in England was at an end.

Q. A Tory ministry headed by Robert Harley (later earl of Oxford) and Henry St. John (later Viscount Bolingbroke) was replacing that of the Whigs.

R. The astute Harley made overtures to Swift and won him over to the Tories.

S. Swift quickly became the Tories' chief pamphleteer and political writer and, by the end of October 1710, had taken over the Tory journal, The Examiner, which he continued to edit until June $14, 1711$.

T. A momentous period began for Swift when in 1710 he once again found himself in London.

Which of the sentences should come first in the sequence?

A. S
B. T
C. P
D. R

nielit2023feb-scientistd verbal-aptitude passage-reading

18.9 Puzzle (1)

18.9.1 Puzzle: NIELIT 2023 Feb Scientist D | Section B | Question: 46

Directions: In questions **(43-46)**, rearrange the following five sentences $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ and T in the proper sequence to form a meaningful paragraph; then answer the questions given below them.

P. With the death of Queen Anne in August 1714 and the accession of George I, the Tories were a ruined party, and Swift's career in England was at an end.

Q. A Tory ministry headed by Robert Harley (later earl of Oxford) and Henry St. John (later Viscount Bolingbroke) was replacing that of the Whigs.

R. The astute Harley made overtures to Swift and won him over to the Tories.

S. Swift quickly became the Tories' chief pamphleteer and political writer and, by the end of October 1710, had taken over the Tory journal, The Examiner, which he continued to edit until June $14, 1711$.

T. A momentous period began for Swift when in 1710 he once again found himself in London.

Which of the following should come second in the sequence?

A. S B. T C. P D. Q

nielit2023feb-scientistd verbal-reasoning puzzle

18.10 Sequence Series (1)

18.10.1 Sequence Series: NIELIT 2018-10

Look at this sequence $SCD\ TEF\ UGH\_\_\_\_WKL$, and find the missing sequence.

A. CMN B. UJI C. VIJ D. IJT

nielit-2018 general-aptitude verbal-aptitude logical-reasoning sequence-series

Answer key☟

18.11 Verbal Reasoning (26)

18.11.1 Verbal Reasoning: NIELIT 2021 Dec Scientist A - Section A: 41

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

$\mathrm{QUISLING : BETRAY}$

A. appreciate : provoke
B. inception : termination
C. juggernaut : crush
D. obstinate : preserve

nielit2021dec-scientista verbal-reasoning analytical-aptitude

Answer key☟

18.11.2 Verbal Reasoning: NIELIT 2021 Dec Scientist A - Section A: 42

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

INTIMATE : CLOSE

A. evanescent : permanency
B. articulate : speech
C. enclose : parentheses
D. obsessed : attracted

nielit2021dec-scientista verbal-reasoning analytical-aptitude

18.11.3 Verbal Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 14

Arrange the given words in alphabetical order and tick the one that comes at the second place.

A. Interfere
B. Interlude
C. Intestine
D. Interview

nielit2022apr-scientistb verbal-reasoning english-grammar

Answer key☟

18.11.4 Verbal Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 21

Choose the word which is least like the other words in the group :

A. Shimmer
B. Simmer
C. Glimmer
D. Glint

nielit2022apr-scientistb verbal-reasoning english-grammar

Answer key☟

18.11.5 Verbal Reasoning: NIELIT 2022 April Scientist B | Section A | Question: 34

Hypsiphobia : Height : : Hylophobia : ?

A. Forests
B. Animals
C. Waters
D. All of the above

nielit2022apr-scientistb verbal-reasoning analytical-aptitude

Answer key☟

18.11.6 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 11

Reluctant: Keen: : Remarkable:?

A. Usual
B. Restrained
C. Striking
D. Evocative

nielit2023feb-scientistc verbal-reasoning analytical-aptitude

18.11.7 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 29

Directions : $(28-29)$ In each of the following letter series, some of the letters are missing which are given in that order as one of the alternatives below it. Choose the correct alternative.

$\text{pqr}\_\_\text{rs}\_\text{rs}\_\_\text{s}\_\text{q}\_$

A. spqpprr B. pqrrppq C. sqppqpr D. sqprrqr

nielit2023feb-scientistc verbal-reasoning analytical-aptitude

18.11.8 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section B | Question: 31

Direction: $(31-34)$ In the following questions find the appropriate explanation for idioms/phrases mentioned in bold letters.

I feel that I have been **left up the creek without a paddle** by my father.

A. Left with a large inheritance
B. Left without an inheritance
C. Left in a pleasant situation
D. Left in an unpleasant situation

nielit2023feb-scientistc english-grammar verbal-reasoning

18.11.9 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section B | Question: 40

Direction $(38-40)$**:** Choose the correct substitute for the following phrases/sentences.

A political leader who seeks support by appealing to popular desires and prejudices rather than by using rational argument :

A. patriarch B. demagogue C. orator D. pedagogue

nielit2023feb-scientistc english-grammar verbal-reasoning

18.11.10 Verbal Reasoning: NIELIT 2023 Feb Scientist C | Section B | Question: 48

Direction $(47-49)$**:** Read the following comprehension carefully and answer the following questions.
Humans are to be blamed for declining bat populations. Bats get poisoned when they consume insects that have been sprayed with pesticides and insecticides. But the biggest problem for bats is the loss of natural habitat. Many bats prefer to roost in dead or dying trees under the loose and peeling bark, or in tree cavities. Some prefer to roost in caverns. Populations have dwindled and diversity has suffered without the protection of these important natural roosts.

Which is an appropriate and accurate explication of this paragraph ?

A. Bats would thrive if humans did not use synthetic pesticides.
B. Responsible citizens should construct friendly habitats for wildlife.
C. Humans have unintentionally contributed to declining bat populations.
D. Bats must seek roosts in caves and caverns when other roosts are unavailable.

nielit2023feb-scientistc passage-reading english-grammar verbal-reasoning

18.11.11 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 4

Rearrange the letters given below to form a meaningful word and select from the given alternatives the word which is almost opposite in meaning to the word so formed :

RBANOEHCER

A. Liking
B. Appreciation
C. Aversion
D. Apprehension

nielit2023feb-scientistd verbal-reasoning analytical-aptitude

Answer key☟

18.11.12 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 5

Directions : **(5-6)** The following questions consist of two words each, that have a certain relationship with each other, followed by four lettered pairs of words. Select the letter pair that has the same relationship as the original pair of words.

Mendacity : Honesty

A. Truth : Beauty
B. Sportsmanship : Fortitude
C. Courageous: Craven
D. Turpitude : Depravity

nielit2023feb-scientistd verbal-reasoning analytical-aptitude

18.11.13 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section A | Question: 6

Directions : **(5-6)** The following questions consist of two words each, that have a certain relationship with each other, followed by four lettered pairs of words. Select the letter pair that has the same relationship as the original pair of words.

Identity: Anonymity

A. Flaw : Perfection
B. Careless : Mistake
C. Truth: Lie
D. Fear: Joy

nielit2023feb-scientistd analytical-aptitude verbal-reasoning

18.11.14 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section B | Question: 35

Directions: In questions **(35 - 36)**, the sentences have been given in Direct / Indirect speech. From the given alternatives, choose the one which best expresses the given sentence in direct or indirect speech as directed:

Change the following sentence into indirect speech :

"Who used the secondary source?"

A. The secondary source was used by me.
B. She asked who is using the secondary source.
C. She enquired who had used the secondary source.
D. She wanted to know who all were reading the secondary source.

nielit2023feb-scientistd english-grammar verbal-reasoning

18.11.15 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section B | Question: 41

Directions: In questions **(39-42)** out of the four alternatives choose the one which can be substituted for the given words/sentence.

An 'extra-terrestrial' is :

A. Originating on earth but capable of travelling into space.
B. Something that remains within the terrafirma.
C. Something that visits the earth or it's atmosphere.
D. Originating, existing, or occurring outside the earth or its atmosphere.

nielit2023feb-scientistd english-grammar verbal-reasoning analytical-aptitude

18.11.16 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section B | Question: 43

Directions: In questions **(43-46)**, rearrange the following five sentences $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ and T in the proper sequence to form a meaningful paragraph; then answer the questions given below them.

P. With the death of Queen Anne in August 1714 and the accession of George I, the Tories were a ruined party, and Swift's career in England was at an end.

Q. A Tory ministry headed by Robert Harley (later earl of Oxford) and Henry St. John (later Viscount Bolingbroke) was replacing that of the Whigs.

R. The astute Harley made overtures to Swift and won him over to the Tories.

S. Swift quickly became the Tories' chief pamphleteer and political writer and, by the end of October 1710, had taken over the Tory journal, The Examiner, which he continued to edit until June $14, 1711$.

T. A momentous period began for Swift when in 1710 he once again found himself in London.

Which of the sentences should come fourth in the sequence?

A. T B. S C. P D. R

nielit2023feb-scientistd verbal-reasoning analytical-aptitude

18.11.17 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section B | Question: 44

Directions: In questions **(43-46)**, rearrange the following five sentences $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$ and T in the proper sequence to form a meaningful paragraph; then answer the questions given below them.

P. With the death of Queen Anne in August 1714 and the accession of George I, the Tories were a ruined party, and Swift's career in England was at an end.

Q. A Tory ministry headed by Robert Harley (later earl of Oxford) and Henry St. John (later Viscount Bolingbroke) was replacing that of the Whigs.

R. The astute Harley made overtures to Swift and won him over to the Tories.

S. Swift quickly became the Tories' chief pamphleteer and political writer and, by the end of October 1710, had taken over the Tory journal, The Examiner, which he continued to edit until June $14, 1711$.

T. A momentous period began for Swift when in 1710 he once again found himself in London.

Which of the sentences should come fifth in the sequence?

A. S B. T C. P D. Q

nielit2023feb-scientistd verbal-reasoning passage-reading

18.11.18 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section B | Question: 55

Directions: In questions **(54 - 55)** choose the alternative that is closest to the meaning of the italicized part of the sentence from the options given below.

Since you didn't.study hard, I am afraid you will have to face the music.

A. Improve your skill
B. Be good in music
C. Face the insult
D. Face the consequences

nielit2023feb-scientistd english-grammar verbal-reasoning

18.11.19 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section B | Question: 56

Directions **(56 to 59)**: Identify which part of the sentence contains error and mark it as your answer.

The capital of Himachal Pradesh / is situating / 2205 meters above / the sea level.

A. The capital of Himachal Pradesh
B. is situating

C. 2190 meters above

D. the sea level

nielit2023feb-scientistd english-grammar verbal-reasoning

18.11.20 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section C | Question: 83

Human technology developed from the first stone tools about two and a half million years ago. At the beginning, the rate of development was slow. Hundreds of thousands of years passed without much change. Today, new technologies are reported daily on television and in newspapers.

This paragraph best supports the statement that:

A. Stone tools were not really technology.
B. Stone tools were in use for two and a half million years.
C. There is no way to know when stone tools first came into use.
D. In today's world, new technologies are constantly being developed.

nielit2023feb-scientistd verbal-reasoning analytical-aptitude

18.11.21 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section C | Question: 84

In the 1966 Supreme Court decision Miranda v. Arizona, the court held that before the police can obtain statements from a person subjected to an interrogation, the person must be given a Miranda warning. This warning means that a person must be told that he or she has the right to remain silent during the police interrogation. Violation of this right means that any statement that the person makes is not admissible in a court hearing.

This paragraph best supports the statement that :

A. Police who do not warn persons of their Miranda rights are guilty of a crime.
B. A Miranda warning must be given before a police interrogation can begin.
C. The police may no longer interrogate persons suspected of a crime unless a lawyer is present.
D. Persons who are interrogated by police should always remain silent until their lawyer comes.

nielit2023feb-scientistd verbal-reasoning analytical-aptitude logical-reasoning

18.11.22 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section C | Question: 85

Yoga has become a very popular type of exercise, but it may not be for everyone. Before you sign yourself up for a yoga class, you need to examine what it is you want from your fitness routine. If you're looking for a high-energy, fast-paced aerobic workout, a yoga class might not be your best choice.

This paragraph best supports the statement that :

A. Yoga is more popular than high-impact aerobics.
B. Before embarking on a new exercise regimen, you should think about your needs and desires.
C. Yoga is changing the world of fitness in major ways.
D. Yoga benefits your body and mind.

nielit2023feb-scientistd verbal-reasoning passage-reading analytical-aptitude

18.11.23 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section C | Question: 86

During colonial times in America, juries were encouraged to ask questions of the parties in the courtroom. The jurors were, in fact, expected to investigate the facts of the case themselves. If jurors conducted an investigation today, we would throw out the case.

This paragraph best supports the statement that :

A. Juries are less important today than they were in colonial times.
B. Courtrooms today are more efficient than they were in colonial times.
C. Jurors in colonial times were more informed than jurors today.
D. The jury system in America has changed since colonial times.

nielit2023feb-scientistd verbal-reasoning analytical-aptitude

18.11.24 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section C | Question: 87

Mathematics allows us to expand our consciousness. Mathematics tells us about economic trends, patterns of disease, and the growth of populations. Math is good at exposing the truth, but it can also perpetuate misunderstandings and untruths. Figures have the power to mislead people. This paragraph best supports the statement that:

A. Figures are sometimes used to deceive people.
B. Words are more truthful than figures.
C. The study of mathematics is more important than other disciplines.
D. The power of numbers is that they cannot lie.

nielit2023feb-scientistd verbal-reasoning analytical-aptitude

18.11.25 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 134

Which of the following statements about behaviour and attitudes is NOT true?

A. Attitudes that are easily remembered are more likely to predict behaviour than attitudes that are not accessible in memory.
B. Attitudes that individuals consider important tend to show a strong relationship to

behaviour.
C. The attitude-behaviour relationship is likely to be much stronger if an attitude refers to something with which the individual has direct personal experience.
D. General attitudes have a greater influence on behaviour than specific attitudes.

nielit2023feb-scientistd logical-reasoning verbal-reasoning analytical-aptitude

18.11.26 Verbal Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 149

In an organization, one can gain valuable perspectives and insights through close association with an experienced person willing to take one under his/her wing. Such an individual is often called a __________.

A. Supervisor B. Facilitator C. Role model D. Mentor

nielit2023feb-scientistd verbal-reasoning english-grammar

18.12 Written Exam (1)

18.12.1 Written Exam: NIELIT 2023 Feb Scientist D | Section B | Question: 38

Directions: In the questions **(37 - 38)**, the given sentences are in Active / Passive voice. From the given alternatives, choose the one which best expresses the given sentence in active or passive voice as directed :

The message would be sorted by Chanda.

A. The message is being sorted by Chanda.
B. Chanda is sorting the message.
C. Chanda will sort the message.
D. The message is sorted by Chanda.

nielit2023feb-scientistd english-grammar verbal-aptitude written-exam

Answer Keys

18.0.1	D	18.0.2	C	18.0.3	A	18.1.1	A	18.1.2	A
18.1.3	A	18.1.4	C	18.1.5	B	18.1.6	D	18.1.7	A
18.1.8	A	18.1.9	B	18.1.10	A	18.1.11	C	18.1.12	D
18.1.13	A	18.1.14	C	18.1.15	B	18.1.16	B	18.1.17	C
18.1.18	C	18.1.19	B	18.1.20	D	18.1.21	B	18.1.22	D
18.1.23	B	18.1.24	C	18.1.25	C	18.1.26	C	18.1.27	A
18.1.28	D	18.1.29	A	18.1.30	B	18.1.31	B	18.1.32	C
18.1.33	C	18.1.34	B	18.1.35	C	18.1.36	B	18.1.37	B

18.1.38	A	18.1.39	A	18.1.40	C	18.1.41	B	18.1.42	A
18.1.43	B	18.1.44	A	18.1.45	D	18.1.46	C	18.1.47	A
18.1.48	A	18.1.49	C	18.1.50	A	18.1.51	B	18.1.52	A
18.1.53	A	18.1.54	C	18.1.55	B	18.1.56	B	18.2.1	B
18.2.2	D	18.2.3	B	18.3.1	D	18.4.1	C	18.5.1	A
18.6.1	D	18.6.2	D	18.6.3	D	18.6.4	C	18.6.5	C
18.6.6	A	18.7.1	C	18.7.2	C	18.7.3	C	18.7.4	D
18.7.5	B	18.7.6	A	18.7.7	C	18.7.8	C	18.8.1	D
18.8.2	D	18.8.3	B	18.9.1	D	18.10.1	C	18.11.1	C
18.11.2	B	18.11.3	A	18.11.4	B	18.11.5	A	18.11.6	C
18.11.7	C	18.11.8	D	18.11.9	B	18.11.10	C	18.11.11	A
18.11.12	C	18.11.13	C	18.11.14	C	18.11.15	D	18.11.16	B
18.11.17	C	18.11.18	D	18.11.19	B	18.11.20	D	18.11.21	B
18.11.22	B	18.11.23	D	18.11.24	A	18.11.25	D	18.11.26	D
18.12.1	B								

19 Non GATE CSE: Big Data Systems (1)

19.1 Big Data Systems (1)

19.1.1 Big Data Systems: NIELIT Scientist B 2020 November: 61

Which of the following is component of Hadoop?

A. $YARN$ B. $HDFS$ C. Map reduce D. All of the options

nielit-scb-2020 big-data-systems data-mining

Answer key☟

Answer Keys

19.1.1	D

20 Non GATE CSE: Cloud Computing (3)

20.1 Cloud Computing (2)

20.1.1 Cloud Computing: NIELIT Scientist B 2020 November: 102

_____________ has a feature of remote access through which a customer can access the data from anywhere and at any time with the help of internet connection.

A. IaaS B. PaaS C. NaaS D. SaaS

nielit-scb-2020 cloud-computing

Answer key☟

20.1.2 Cloud Computing: NIELIT Scientist B 2020 November: 65

Identify the odd one out.

A. Amazon web service
B. Microsoft Azure
C. Google cloud Platform
D. Twitter Platform

nielit-scb-2020 cloud-computing analytical-aptitude

Answer key☟

20.2 Virtual Memory (1)

20.2.1 Virtual Memory: NIELIT Scientist B 2020 November: 67

___________ is a partitioning of single physical server into multiple logical servers.

A. Virtualization
B. Private cloud
C. Hybrid cloud
D. Public cloud

nielit-scb-2020 co-and-architecture virtual-memory

Answer key☟

Answer Keys

20.1.1	C	20.1.2	D	20.2.1	A

21 Non GATE CSE: Computer Graphics (1)

21.0.1 NIELIT 2016 MAR Scientist C - Section C: 6

Memory mapped displays

A. are utilized for high resolution graphics such as maps
B. uses ordinary memory to store the display data in character form
C. stores the display data as individual bits
D. are associated with electromechanical teleprinters

nielit2016mar-scientistc non-gatecse

Answer key☟

Answer Keys

21.0.1	B

22 Non GATE CSE: Computer Peripherals (3)

22.0.1 NIELIT 2016 MAR Scientist B - Section C: 48

The shell

A. accepts command from the user
B. maintains directories of files
C. translates the keyboard's character codes
D. none of these

nielit2016mar-scientistb non-gatecse

Answer key☟

22.1 Computer Peripherals (1)

22.1.1 Computer Peripherals: NIELIT 2016 DEC Scientist B (IT) - Section B: 23

Which of the following is not an input device?

A. Mouse
B. Keyboard
C. Light Pen
D. VDU

nielit2016dec-scientistb-it non-gatecse computer-peripherals

Answer key☟

22.2 Disk (1)

22.2.1 Disk: NIELIT 2022 April Scientist B | Section B | Question: 68

Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is __________

A. 14020
B. 14000
C. 25030
D. 15000

nielit2022apr-scientistb disk output operating-system

Answer Keys

22.0.1	A	22.1.1	D	22.2.1	A

23 Non GATE CSE: Digital Signal Processing (29)

23.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 31

At a room temperature of $300K$, calculate the thermal noise generated by two resistors of $10K\Omega$ and $20K\Omega$ when the bandwidth is $10KHz$.

A. $1.2868 \times 10^{-6}V, 1.819 \times 10^{-6}V$
B. $6.08 \times 10^{-6}V, 15.77 \times 10^{-6}V$
C. $16.66 \times 10^{-6}V, 2.356 \times 10^{-6}V$
D. $1.66 \times 10^{-6}V, 0.23 \times 10^{-6}V$

nielit2017july-scientistb-cs non-gatecse

Answer key☟

23.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 60

The sequence of operation in which PCM is done is

A. Sampling, quantizing, encoding
B. Quantizing, sampling, encoding
C. Quantizing, encoding, sampling
D. None of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key☟

23.0.3 NIELIT 2016 DEC Scientist B (CS) - Section B: 41

The IETF standard documents are called:

A. RFC
B. RCF
C. ID
D. none of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key☟

23.0.4 NIELIT 2016 DEC Scientist B (CS) - Section B: 25

A low pass filter is

A. Passes the frequencies lower than the specified cut off frequency
B. Used to recover signal from sampled signal
C. Rejects higher frequencies
D. All of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key☟

23.0.5 NIELIT 2016 DEC Scientist B (CS) - Section B: 17

$T1$ carrier system is used:

A. For delta modulation
B. Industrial noise
C. For frequency modulated signals
D. None of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key☟

23.0.6 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 27

At 100% modulation, the power in each sideband is _______ of that of carrier.

A. 50% B. 40% C. 60% D. 25%

nielit2017oct-assistanta-cs non-gatecse

23.0.7 NIELIT 2016 MAR Scientist C - Section C: 43

The astable multivibrator has

A. two quasi stable states
B. two stable states
C. one stable and one quasi-stable state
D. none of these

nielit2016mar-scientistc non-gatecse

23.0.8 NIELIT 2016 MAR Scientist C - Section C: 42

A stable multivibrator are used as

A. comparator circuit
B. squaring circuit
C. frequency to voltage converter
D. voltage to frequency converter

nielit2016mar-scientistc non-gatecse

23.0.9 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 12

The open-loop transfer function of a feedback control system is $G(s) \cdot H(s) = 1/(s+1)^3$. The gain margin of the system is

A. 2 B. 4 C. 8 D. 16

nielit2017oct-assistanta-cs non-gatecse

23.0.10 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 11

In MOSFET fabrication, the channel; length is defined during the process of

A. Isolation oxide growth
B. Channel stop implantation
C. Poly-silicon gate patterning
D. Lithography step leading to the contact pad

nielit2017oct-assistanta-cs non-gatecse

23.0.11 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 8

A carrier $Ac\cos(\omega c)t$ is frequency modulated by a signal $Em\cos(\omega m)t$. The modulation index is mf. The expression for the resulting FM signal is

A. $Ac\cos[\omega ct + mf\sin(\omega m)t]$
B. $Ac\cos[\omega ct + mf\cos(\omega m)t]$
C. $Ac\cos[\omega ct + \pi mf\sin\omega mt]$

D. $Ac\cos[\omega ct + 2\pi mfEm\cos(\omega m)t/\omega m]$

nielit2017oct-assistanta-cs non-gatecse

Answer key☟

23.0.12 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 7

If the number of bits per sample in a PCM system is increased from a n to $n+1$, the improvement in signal to quantization nose ratio will be

A. $3\,dB$ B. $6\,dB$ C. $2n\,dB$ D. $n\,dB$

nielit2017oct-assistanta-cs non-gatecse

23.0.13 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 5

For a periodic signal $v(t) = 30\sin 100t + 10\cos 300t + 6\sin(500t + \frac{\pi}{4})$, the fundamental frequency in rad/s

A. 100 B. 300 C. 500 D. None of these

nielit2017oct-assistanta-cs non-gatecse

23.0.14 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 4

For the discrete signal $x[n] = a^n u[n], a > 0$ the z-transform is

A. $\frac{(z+a)}{z}$ B. $\frac{(z-a)}{z}$ C. $\frac{z}{(z-a)}$ D. $\frac{z}{(z+a)}$

nielit2017oct-assistanta-cs non-gatecse

23.0.15 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 3

The final value theorem is used to find the

A. Steady state value of the system output
B. Initial value of the system output
C. Transient behavior of the system output
D. None of these

nielit2017oct-assistanta-cs non-gatecse

23.0.16 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 2

An ideal op-amp is an ideal

A. voltage controlled current source
B. voltage controlled voltage source
C. current controlled current source
D. current controlled voltage source

nielit2017oct-assistanta-cs non-gatecse

Answer key☟

23.0.17 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 1

Twelve $1\,\Omega$ resistances are used as edges to from a cube. The resistance between two diagonally opposite corners of the cube is

A. $\frac{5}{6}\,\Omega$ B. $\frac{1}{6}\,\Omega$ C. $\frac{6}{5}\,\Omega$ D. $\frac{3}{2}\,\Omega$

nielit2017oct-assistanta-cs non-gatecse

23.0.18 NIELIT 2021 Dec Scientist B - Section B: 85

The bit transmission rate in a pulse coded modulation system with number of quantization levels 8 and maximum signal frequency 4000 Hz is given by :

A. 16 kbps B. 24 kbps
C. 32 kbps D. 8 kbps

nielit2021dec-scientistb digital-logic computer-networks

Answer key☟

23.1 Analog Communication (1)

23.1.1 Analog Communication: NIELIT Scientist B 2020 November: 51

Calculate the modulation percentage if the modulating signal is $8\,V$ and carrier is of $12\,V$?

A. 50 B. 67 C. 150 D. 33

nielit-scb-2020 general analog-communication

Answer key☟

23.2 Data Communication (1)

23.2.1 Data Communication: NIELIT 2022 April Scientist B | Section B | Question: 47

In a baseband communication link, frequencies upto 3500 Hz are used for signaling. Using a raised cosine pulse with 75% excess bandwidth and for no inter symbol interference, the maximum possible signaling rate is symbols per sec is:

A. 1750 B. 2625 C. 4000 D. 5250

nielit2022apr-scientistb non-gatecse data-communication probability communication

Answer key☟

23.3 Digital Signal Processing (2)

23.3.1 Digital Signal Processing: NIELIT 2022 April Scientist B | Section B | Question: 55

Flat top sampling of low pass signals ___________

A. gives rise to aperture effect B. implies oversampling

C. lead to aliasing

D. introducing delay distortion

nielit2022apr-scientistb digital-signal-processing analytical-aptitude

Answer key

23.3.2 Digital Signal Processing: NIELIT Scientific Assistant A 2020 November: 60

Term in the MVC architecture that receives events is called _________.

A. Receiver

B. Controller

C. Transmitter

D. Modulator

nielit-sta-2020 digital-signal-processing

Answer key

23.4 Frequency Spectrum (1)

23.4.1 Frequency Spectrum: NIELIT 2022 April Scientist B | Section B | Question: 109

A 1 MHz sinusoidal carrier is amplitude modulated by a symmetrical square wave of period $100 \ \mu\text{sec}$. Which of the following frequencies will not be present in the modulated signal ?

A. 990 MHz

B. 1010 MHz

C. 1020 MHz

D. 1030 MHz

nielit2022apr-scientistb analog-communication signal-processing frequency-spectrum

Answer key

23.5 Microprocessors (2)

23.5.1 Microprocessors: NIELIT 2016 MAR Scientist C - Section C: 48

In a microprocessor, WAIT states are used to

A. make the processor wait during a DMA operation
B. make the processor wait during a power interrupt processing
C. make the processor wait during a power shutdown
D. interface slow peripherals to the processor

nielit2016mar-scientistc non-gatecse microprocessors

Answer key

23.5.2 Microprocessors: NIELIT 2017 DEC Scientific Assistant A - Section B: 8

Microprocessors are used in which generation of computers?

A. I – st Generation

B. II – nd Generation

C. III – rd Generation

D. IV – th Generation

nielit2017dec-assistanta non-gatecse microprocessors

Answer key

23.6 Microprogramming (1)

23.6.1 Microprogramming: NIELIT 2016 MAR Scientist C - Section C: 61

Microprogramming is a technique for

A. writing small programs effectively
B. programming output/input routines
C. programming the microprocessors
D. programming the control steps of a computer

nielit2016mar-scientistc co-and-architecture microprogramming

Answer key☟

23.7 Signal Processing (2)

23.7.1 Signal Processing: NIELIT 2021 Dec Scientist A - Section B: 80

A modulating signal $\mathrm{m}\,(\mathrm{t}) = 10\cos(2\pi \times 10^3\mathrm{t})$ is amplitude modulated with a carrier signal $\mathrm{c}\,(\mathrm{t}) = 50\cos(2\pi \times 10^5\mathrm{t})$. Assume $\mathrm{R} = 1\Omega$. Find the carrier power required for transmitting this AM wave.

A. $1000\ \mathrm{W}$
B. $1250\ \mathrm{W}$
C. $1100\ \mathrm{W}$
D. $50\ \mathrm{W}$

nielit2021dec-scientista analog-communication signal-processing

23.7.2 Signal Processing: NIELIT 2021 Dec Scientist B - Section B: 44

$y = 10\cos(1800\ \pi t) + 20\cos(2000\ \pi t) + 10\cos(220\ \pi t)$. Find the modulation index (μ) of the given wave.

A. 0.3
B. 0.5
C. 0.7
D. 1

nielit2021dec-scientistb analog-communication signal-processing

23.8 Signals and Systems (1)

23.8.1 Signals and Systems: NIELIT 2021 Dec Scientist A - Section B: 64

Energy of power signal is :

A. Finite
B. Zero
C. Infinite
D. 1

nielit2021dec-scientista signals-and-systems analytical-aptitude

Answer key☟

Answer Keys

23.0.1	A	23.0.2	A	23.0.3	A	23.0.4	D	23.0.5	D
23.0.6	D	23.0.7	D	23.0.8	B	23.0.9	C	23.0.10	C
23.0.11	A	23.0.12	B	23.0.13	A	23.0.14	C	23.0.15	A

23.0.16	B
23.3.1	A
23.6.1	D

23.0.17	A
23.3.2	B
23.7.1	B

23.0.18	B
23.4.1	C
23.7.2	D

23.1.1	B
23.5.1	D
23.8.1	C

23.2.1	C
23.5.2	D

24 Non GATE CSE: Geometry (6)

24.1 Area (1)

24.1.1 Area: NIELIT 2023 Feb Scientist C | Section D | Question: 111

A chord AB of a circle of radius 5.25 cm makes an angle 60° at the centre of the circle. Find the area of major segment.

A. 168 cm^2
B. 42 cm^2
C. 84 cm^2
D. 78 cm^2

nielit2023feb-scientistc geometry area quantitative-aptitude

24.2 Geometry (5)

24.2.1 Geometry: NIELIT 2021 Dec Scientist B - Section A: 26

In a parallelogram ABCD, AP and BP are the angle bisectors of $\angle \text{DAB}$ and $\angle \text{ABC}$.Find $\angle \text{APB}$:

A. 85°
B. 90°
C. 94°
D. 86°

nielit2021dec-scientistb geometry analytical-aptitude

Answer key☟

24.2.2 Geometry: NIELIT 2023 Feb Scientist C | Section C | Question: 72

The size of a rhombus is increased by itself, How many times will its perimeter increase?

A. Three times
B. Four times
C. Two times
D. None of the options

nielit2023feb-scientistc geometry analytical-aptitude

24.2.3 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 108

In a triangle $\text{ABC}, 2\text{ac} \sin \left[\frac{1}{2}(\text{A-B+C})\right]$ is equal to :

A. $a^2+b^2-c^2$
B. $c^2+a^2-b^2$
C. $b^2-c^2-a^2$
D. $c^2-a^2-b^2$

nielit2023feb-scientistd geometry

24.2.4 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 109

The vertices of a triangle ABC are $\text{A}(4,6), \text{B}(1,5)$ and $\text{C}(7,2)$. A line is drawn to intersect sides AB and AC at D and E respectively such that

$\mathrm{AD}:\mathrm{AB}=\mathrm{AE}:\mathrm{AC}=:4$. Calculate the ratio of area of triangle ADE and area of triangle ABC.

A. $1:16$ B. $16:1$ C. $15:16$ D. $1:1$

nielit2023feb-scientistd geometry analytical-aptitude

24.2.5 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 110

In a triangle XYZ, P and Q are points on XY, XZ respectively such that $\mathrm{X\,P}{=}2\,\mathrm{P\,Y}, \mathrm{X\,Q}{=}2\,\mathrm{Q\,Z}$, then the ratio, of area of $\triangle\mathrm{XPQ}$ and area of $\triangle\mathrm{XYZ}$ is :

A. $4:9$ B. $2:3$ C. $3:2$ D. $9:4$

nielit2023feb-scientistd geometry analytical-aptitude

Answer key☟

Answer Keys

24.1.1	C	24.2.1	B	24.2.2	C	24.2.3	B	24.2.4	A
24.2.5	A								

25 Non GATE CSE: Information Theory (11)

25.0.1 NIELIT 2016 DEC Scientist B (CS) - Section B: 11

A sinusoidal signal is analog signal, because:

A. it can have a number of values between the negative and positive peaks
B. it is negative for one half cycle
C. it is positive for one half cycle
D. it has positive as well as negative values

nielit2016dec-scientistb-cs non-gatecse

Answer key☟

25.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 10

The noise due to random behavior of charge carriers is:

A. Partition noise
B. Industrial noise
C. Shot noise
D. Flicker noise

nielit2016dec-scientistb-cs non-gatecse

Answer key☟

25.0.3 NIELIT 2016 MAR Scientist B - Section C: 52

Which of the following is possible in a token passing bus network?

A. in-service expansion.
B. unlimited number of stations.
C. both (A) and (B).
D. unlimited distance.

nielit2016mar-scientistb non-gatecse

Answer key☟

25.0.4 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 28

The capacity relationship is given by

A. $C = W \log_2(1 + S/N)$
B. $C = 2W \log_2(1 + S/N)$
C. $C = W \log_2(1 - S/N)$
D. $C = W \log_{10}(1 + S/N)$

nielit2017oct-assistanta-cs non-gatecse

Answer key☟

25.1 Cryptography (1)

25.1.1 Cryptography: NIELIT 2021 Dec Scientist B - Section B: 46

Consider the following two statements:

I. A Hash function, which is used for computing digital signatures, is an injective function
II. An encryption technique such as DES performs a permutation on the elements of its input alphabet

A. Both are false
B. (I) is true (II) is false
C. (II) is true (I) is false
D. Both are true

nielit-2021-it-dec-scientistb cryptography hashing digital-logic

Answer key☟

25.2 Data Compression (1)

25.2.1 Data Compression: NIELIT 2022 April Scientist B | Section B | Question: 97

A message is made up entirely of characters from the set $\mathrm{X}=\{\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}, \mathrm{T}\}$. The table of probabilities for each of the characters is shown below:

Character	**Probability**
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is ______.

A. 225
B. 226
C. 227
D. 228

nielit2022apr-scientistb probability huffman-code encoding analytical-aptitude data-compression

25.3 Information Theory (3)

25.3.1 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 54

An analog signal having $3 \mathrm{~kHz}$ bandwidth is sampled at 1.5 times the Nyquist rate. The successive samples are statistically independent. Each sample is quantized into one of 256 equally likely levels. The information rate of the source is :

A. $3 \mathrm{~kbps}$
B. $72 \mathrm{~kbps}$
C. $256 \mathrm{~kbps}$
D. $9 \mathrm{~kbps}$

nielit2021dec-scientistb information-theory probability

25.3.2 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 67

The channel capacity of a noise free channel having M symbols is given by :

A. M
B. 2^{M}
C. $\log \mathrm{M}$
D. None of these

nielit2021dec-scientistb information-theory probability

Answer key

25.3.3 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 77

A high resolution $\mathrm{B/w\ TV}$ picture contains 3×10^6 picture elements and 16 different brightness levels. Pictures are repeated at a rate of 24 per second. All levels have equal likelihood of occurrence and all picture elements are assumed to be independent. What will be the average rate of information carried by this TV picture source ?

A. 288 Mbps
B. 24 Mbps
C. 132 Mbps
D. 3 Mbps

nielit2021dec-scientistb information-theory probability

25.4 Probability (1)

25.4.1 Probability: NIELIT 2021 Dec Scientist A - Section B: 96

For 8–ary signal or symbol the number of likelihood functions are :

A. 16
B. 8
C. 9
D. 64

nielit2021dec-scientista probability

25.5 Signal Processing (1)

25.5.1 Signal Processing: NIELIT 2022 April Scientist B | Section B | Question: 93

A bandlimited signal is sampled at the Nyquist rate. The signal can be recovered by passing the samples through :

A. an RC filter
B. an envelope detector
C. a PLL
D. an ideal low-pass filter with the appropriate bandwidth

nielit2022apr-scientistb signal-processing analog-communication information-theory

Answer key

Answer Keys

25.0.1	A	25.0.2	C	25.0.3	A	25.0.4	A	25.1.1	C
25.2.1	A	25.3.1	B	25.3.2	C	25.3.3	A	25.4.1	B
25.5.1	D								

26.1 Cyclomatic Complexity (4)

26.1.1 Cyclomatic Complexity: NIELIT 2021 Dec Scientist B - Section B: 39

A particular control flow graph has 15 nodes out of which 7 are decision nodes. The cyclomatic complexity and number of edges respectively can be given by the following :

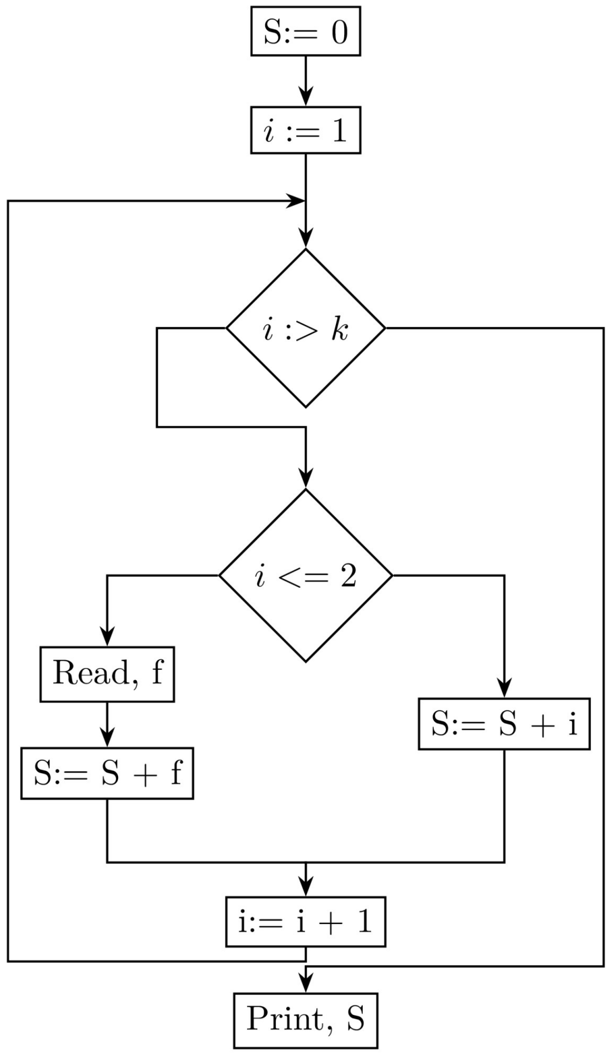
A. $7, 21$ B. $15, 24$ C. $8, 21$ D. $16, 16$

nielit-2021-it-dec-scientistb is&software-engineering cyclomatic-complexity

Answer key☟

26.1.2 Cyclomatic Complexity: NIELIT 2022 April Scientist B | Section B | Question: 84

To carry out white box testing of a program, its flow chart representation is obtained as shown in the figure below:



For basis path based testing of this program, its cyclomatic complexity is

A. 5 B. 4 C. 3 D. 2

nielit2022apr-scientistb is&software-engineering cyclomatic-complexity normal

Answer key

26.1.3 Cyclomatic Complexity: NIELIT Scientific Assistant A 2020 November: 113

Consider the following C program segment.

```
while(first<=last)
{
   if(array[middle]<search)
   first=middle+1;
   else if(array[middle]== search)
   found=True;
   else last=middle-1;
   middle=(first+last)/2;
}
if(first<last)not Present = True;
```

The cyclomatic complexity of the program segment is ______________.

A. 3 B. 4 C. 5 D. 6

nielit-sta-2020 is&software-engineering cyclomatic-complexity

Answer key

26.1.4 Cyclomatic Complexity: NIELIT Scientist B 2020 November: 53

Which one of the following statements is incorrect?

A. The number of regions corresponds to the cyclomatic complexity
B. Cyclomatic complexity for a flow graph G is $V(G) = N - E + 2$, where E is the number of edges and N is the number of nodes in flow graph
C. Cyclomatic complexity for a flow graph G is $V(G) = E - N + 2$, where E is the number of edges and N is the number of nodes in flow graph
D. Cyclomatic complexity for a flow graph G is $V(G) = P + 1$, where P is the number of predicate nodes contained in the flow graph G

nielit-scb-2020 cyclomatic-complexity software-testing is&software-engineering

Answer key

26.2 Data Flow Diagram (1)

26.2.1 Data Flow Diagram: NIELIT 2021 Dec Scientist B - Section B: 53

The ______________ enables the software engineer to develop models of the information domain and functional domain at the same time.

A. data flow diagram
B. state transition diagram
C. control specification
D. activity diagram

nielit2021dec-scientistb is&software-engineering data-flow-diagram

Answer key

26.3 Function Point Metric (1)

26.3.1 Function Point Metric: NIELIT 2017 July Scientist B (IT) - Section B: 27

The Function Points(FP) calculated for software projects are often used to obtain an estimate of Lines of Code(LOC) required for that project. Which of the following statements is FALSE in this context?

A. The relationship between FP and LOC depends on the programming language used to implement the software.
B. LOC requirement for an assembly language implementation will be more for a given FP value, than LOC for implementation in COBOL.
C. On an average, one LOC of C++ provides approximately 1.6 times the functionality of a single LOC of FORTRAN.
D. FP and LOC are not related to each other.

nielit2017july-scientistb-it non-gatecse is&software-engineering function-point-metric

Answer key

26.4 Functions (1)

26.4.1 Functions: NIELIT 2022 April Scientist B | Section B | Question: 98

Consider a software project with the following information domain characteristics for calculation of function point metric.

- Number of external inputs $(\mathrm{I}) = 30$
- Number of external outputs $(\mathrm{O}) = 60$
- Number of external inquiries $(\mathrm{E}) = 23$
- Number of files $(\mathrm{F}) = 08$
- Number of external interfaces (N) = 02$

It is given that the complexity weighting factors for $\mathrm{I}, \mathrm{O}, \mathrm{E}, \mathrm{F}$ and N are $4, 5, 4, 10$ and $7,$ respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors have value $3,$ and each of the remaining factors have value $4.$ The computed value of function point metric is _________.

A. 612.06 B. 212.05 C. 305.09 D. 806.9

nielit2022apr-scientistb is&software-engineering functions

26.5 Is&software Engineering (45)

26.5.1 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 18

Which of the following is not defined in a good Software Requirement Specification(SRS) document?

A. Functional requirement
B. Goals of implementation
C. Nonfunctional requirement
D. Algorithm for software implementation

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.2 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 29

Software Requirement Specification (SRS) is also known as specification of

A. White box testing
B. Integrated testing
C. Acceptance testing
D. Black box testing

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.3 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 32

Line of code(LOC) of the product comes under which type of measures?

A. Indirect measures
B. Coding
C. Direct measures
D. None of the above

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.4 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 38

What is the testing to ensure the WebApp properly interfaces with other applications or databases?

A. Compatibility
B. Interoperability
C. Performance
D. Security

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.5 Is&software Engineering: NIELIT 2016 DEC Scientist B (CS) - Section B: 44

What is described by means of DFDs as studied earlier and represented in algebraic form?

A. Data flow
B. Data storage
C. Data structures
D. Data elements

nielit2016dec-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.6 Is&software Engineering: NIELIT 2016 DEC Scientist B (IT) - Section B: 32

Software projects management comprises of a number of activities, which contains:

A. Project planning
B. Project estimation
C. Scope management
D. All mentioned above

nielit2016dec-scientistb-it non-gatecse is&software-engineering

Answer key☟

26.5.7 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 46

According to Brooks, if n is the number of programmers in a project team, then the number of communication path is

A. $n(n-1)/2$
B. $n \log n$
C. n
D. $n(n+1)/2$

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key☟

26.5.8 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 47

In object oriented design of software, objects have

A. attributes and name only
B. operations and name only
C. attributes name and operations
D. mutation and permutation property

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key☟

26.5.9 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 49

The extent to which the software can control to operate correctly despite the introduction of invalid input is called as

A. reliability
B. robustness
C. fault tolerance
D. portability

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key☟

26.5.10 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 50

On an average, the programmer months is given by $3.6 \times (KLOC)^{1.2}$. If so, a project requiring one thousand source instructions will require

A. 3.6 PM.
B. 0.36 PM.
C. 0.0036 PM.
D. 7.23 PM

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key☟

26.5.11 Is&software Engineering: NIELIT 2016 MAR Scientist B - Section C: 51

Relation of COCOMO model is

A. $E = a * (KLOC)^6$ B. $E = a * (KLOC)^5$ C. $E = a * (KLOC)^7$ D. $E = a * (KLOC)^3$

nielit2016mar-scientistb non-gatecse is&software-engineering

Answer key

26.5.12 Is&software Engineering: NIELIT 2016 MAR Scientist C - Section C: 1

The advantage of better testing in software development is in

A. waterfall model B. prototyping C. iterative D. all of these

nielit2016mar-scientistc non-gatecse is&software-engineering

Answer key

26.5.13 Is&software Engineering: NIELIT 2016 MAR Scientist C - Section C: 23

The graph theoretic concept will be useful in software testing is

A. Cyclomatic number
B. Hamiltonian circuit
C. Eulerian cycle
D. None of these

nielit2016mar-scientistc non-gatecse is&software-engineering

Answer key

26.5.14 Is&software Engineering: NIELIT 2016 MAR Scientist C - Section C: 24

In testing phase, the effort distribution is upto

A. 10% B. 20% C. 40% D. 50%

nielit2016mar-scientistc non-gatecse is&software-engineering

Answer key

26.5.15 Is&software Engineering: NIELIT 2017 DEC Scientist B - Section B: 29

Which of the following is not deliverable of the structured system analysis?

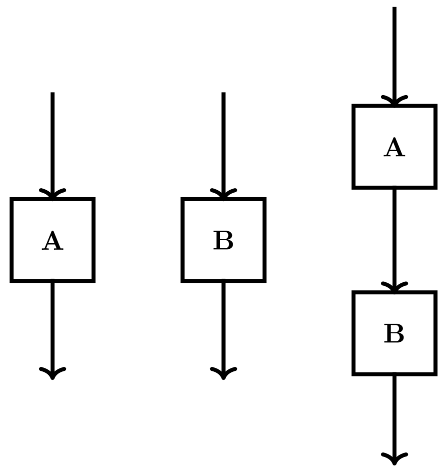
A. Data flow diagram
B. Prototype model
C. Entity Relationship diagram
D. Data dictionary

nielit2017dec-scientistb non-gatecse is&software-engineering

Answer key

26.5.16 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 44

The Cyclomatic complexity of each of the modules A and B shown below is 10. What is the Cyclomatic complexity of the sequential integration shown on the right hand side?



A. 19 B. 21 C. 20 D. 10

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.17 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 45

What is the appropriate pairing of items in the two columns listing various activities encountered in a software life cycle?

P.	Requirements Capture	1.	Module Development and integration
Q.	Design	2.	Domain Analysis
R.	Implementation	3.	Structural and Behavioral Marketing
S.	Maintenance	4.	Performance Tunit

A. $P-3, Q-2, R-4, S-1$
B. $P-2, Q-3, R-1, S-4$
C. $P-3, Q-2, R-1, S-4$
D. $P-2, Q-3, R-4, S-1$

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key

26.5.18 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 51

A company needs to develop a strategy for software product development for which it has a choice of two programming languages $L1$and $L2$. The number of Lines Of Code (LOC) developed using $L2$ is estimated to be twice the LOC developed with $L1$. The product will have to be maintained for five years. Various parameters for the company are given in the table below:

Parameter	Language L1	Language L2
Man years needed for development	LOC/10000	LOC/10000
Development cost per man year	Rs. 10,00,000	Rs. 7,50,000
Maintenance time	5 years	5 years
Cost of maintenance per year	Rs. 1,00,000	Rs. 50,000

Total cost of the project includes cost of development and maintenance. What is the LOC for $L1$ for which the cost of the project using $L1$is equal to the cost of the project using $L2$?

A. 4000 B. 5000 C. 4333 D. 4667

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key☟

26.5.19 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 52

A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 40000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.8 for the software development on embedded systems, while the exponentiation factor is given as 1.20. What is the estimated effort in person-months?

A. 234.25 B. 932.50 C. 287.80 D. 122.40

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key☟

26.5.20 Is&software Engineering: NIELIT 2017 July Scientist B (CS) - Section B: 53

Which one of the following is NOT desired in a good Software Requirement Specifications (SRS) document?

A. Functional Requirements
B. Non-Functional Requirements
C. Goals of Implementation
D. Algorithms for Software Implementation

nielit2017july-scientistb-cs non-gatecse is&software-engineering

Answer key☟

26.5.21 Is&software Engineering: NIELIT 2017 July Scientist B (IT) - Section B: 24

The coupling between different modules of a software is categorized as follows:

I. Content coupling
II. Common coupling
III. Control coupling
IV. Stamp Coupling

V. Data Coupling

Coupling between modules can be ranked in the order of strongest (least desirable) to weakest(most desirable) as follows:

A. I-II-III-IV-V B. V-IV-III-II-I C. I-III-V-II-IV D. IV-II-V-III-I

nielit2017july-scientistb-it non-gatecse is&software-engineering

Answer key☟

26.5.22 Is&software Engineering: NIELIT 2017 July Scientist B (IT) - Section B: 25

Which of the following statements are TRUE?

I. The context diagram should depict the system as a single bubble.
II. External entities should be identified clearly at all levels of DFDs.
III. Control information should not be represented in a DFD.
IV. A data store can be connected either to another data store or to an external entity.

A. II and IV B. II and III C. I and III D. I,II and III

nielit2017july-scientistb-it non-gatecse is&software-engineering

Answer key☟

26.5.23 Is&software Engineering: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 15

By open domain CASE tools we mean

A. tools available in open domain
B. software packages which can be downloaded from the internet
C. software packages to aid each phase of the systems analysis and design which can be downloaded free of cost from the internet
D. source codes of CASE tools

nielit2017oct-assistanta-cs non-gatecse is&software-engineering

Answer key☟

26.5.24 Is&software Engineering: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 3

By open domain CASE tools we mean

A. tools available in open domain
B. software packages which can be downloaded from the internet
C. software packages to aid each phase of the systems analysis and design which can be downloaded free of cost from the internet
D. source codes of CASE tools

nielit2017oct-assistanta-it is&software-engineering

Answer key

26.5.25 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 100

Given the two statements $S1$ and $S2$ for software engineering :

$S1$: Statement coverage cannot guarantee execution of loops in program under test.

$S2$: Use of independent path testing criterion guarantees execution of each loop in a program under test more than once.

Then which among the following is true ?

A. $S1$ is True, $S2$ is True
B. $S1$ is True, $S2$ is False
C. $S1$ is False, $S2$ is True
D. $S1$ is False, $S2$ is False

nielit2021dec-scientista is&software-engineering

26.5.26 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 119

What is the main objective of $\text{ISO } 9001$?

A. Verification
B. Validation
C. S/W Testing
D. H/W Testing

nielit2021dec-scientista is&software-engineering

Answer key

26.5.27 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 69

What is the main focus of Reverse Engineering (RE) ?

A. Data base structure
B. S/W file structure
C. Memory
D. CPU Utilization

nielit2021dec-scientista is&software-engineering

26.5.28 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 81

Elicitation of requirements is a _______________ .

A. $SDLC$ Process
B. Cyclic Process
C. SRS Process
D. Development Process

nielit2021dec-scientista is&software-engineering

Answer key

26.5.29 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 93

Which of the following step is not a part of the requirement engineering process?

A. Feasibility Study
B. Programming Language Requirement Specification
C. Software Requirement Specification
D. Requirement Gathering & Validation

nielit2021dec-scientista is&software-engineering

Answer key☟

26.5.30 Is&software Engineering: NIELIT 2021 Dec Scientist A - Section B: 97

Which one of them is a good software?

A. High cohesion low coupling
B. Low cohesion high coupling
C. High cohesion high coupling
D. Low cohesion low coupling

nielit2021dec-scientista is&software-engineering

Answer key☟

26.5.31 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 100

Delivery of the software product on time is considered as ___________ .

A. SDLC
B. User Satisfaction
C. Planning
D. UI/UX design for software

nielit2021dec-scientistb is&software-engineering

26.5.32 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 14

Efficiency in a software product does not include:

A. responsiveness
B. processing time
C. memory utilization
D. licensing

nielit-2021-it-dec-scientistb is&software-engineering

Answer key☟

26.5.33 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 20

Build & Fix Model is suitable for programming exercises of how many LOC (Line of Code)?

A. $100 – 200$
B. $200 – 400$
C. $400 – 1000$
D. above 1000

nielit-2021-it-dec-scientistb is&software-engineering

Answer key☟

26.5.34 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 61

If developer wants to transform model into source code is also known as _____________ .

A. Backward Testing / Engineering
B. Forward Engineering
C. Forward Testing
D. Reverse Engineering

nielit2021dec-scientistb is&software-engineering

Answer key☟

26.5.35 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 64

Which of these does not account for software failure?

A. Increasing Demand
B. Low expectation
C. Increasing Supply
D. Less reliable and expensive

nielit-2021-it-dec-scientistb is&software-engineering

26.5.36 Is&software Engineering: NIELIT 2022 April Scientist B | Section B | Question: 85

Match the following:

1) Waterfall model	a) Specifications can be developed incrementally
2) Evolutionary model	b) Inflexible partitioning of the project into stages
3) Component-based	c) Explicit recognition of risk software engineering
4) Spiral development	d) Requirement compromises are inevitable

A. $(1)-(a), (2)-(c), (3)-(b), (4)-(d)$
B. $(1)-(b), (2)-(a), (3)-(d), (4)-(c)$
C. $(1)-(d), (2)-(b), (3)-(a), (4)-(c)$
D. $(1)-(c), (2)-(a), (3)-(b), (4)-(d)$

nielit2022apr-scientistb is&software-engineering normal

Answer key☟

26.5.37 Is&software Engineering: NIELIT 2022 April Scientist B | Section B | Question: 89

A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 40000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.8 for the software development on embedded systems, while the exponentiation factor is given as 1.20. What is the estimated effort in person-months?

A. 234.25
B. 932.50
C. 287.80
D. 122.40

nielit2022apr-scientistb is&software-engineering is&software-engineering

26.5.38 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 148

Which can be considered as a difficulty in changing the Attitude of the employees?

A. Escalation of Commitment
B. Providing new information

C. Use of fear

D. Influence by friends and peers

nielit2023feb-scientistc is&software-engineering logical-reasoning

26.5.39 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 145

What are three situational criteria identified in the Fiedler model?

A. Job requirements, position power, and leadership ability
B. Charisma, influence, and leader-member relations
C. Leader-member relations, task structure, and position power
D. Task structure, leadership ability, and group conflict

nielit2023feb-scientistd is&software-engineering

26.5.40 Is&software Engineering: NIELIT Scientific Assistant A 2020 November: 47

Which of the following is not a part of the Test Implementation and Execution Phase?

A. Creating test suites from the test cases
B. Executing test cases either manually or by using test execution tools
C. Comparing actual results
D. Designing the tests

nielit-sta-2020 is&software-engineering

Answer key☟

26.5.41 Is&software Engineering: NIELIT Scientific Assistant A 2020 November: 56

SRD stands for __________.

A. Software Requirements Definition

B. Structured Requirements Definition

C. Software Requirements Diagram

D. Structured Requirements Diagram

nielit-sta-2020 is&software-engineering

Answer key☟

26.5.42 Is&software Engineering: NIELIT Scientific Assistant A 2020 November: 80

A software Requirements Specification (SRS) document should avoid discussing which one of the following?

A. User interface issues

B. Non-functional requirements

C. Design solutions

D. Interfaces with third party software

nielit-sta-2020 is&software-engineering

Answer key☟

26.5.43 Is&software Engineering: NIELIT Scientist B 2020 November: 105

Consider the basic $COCOMO$ model where E is the effort applied in person-months, D is the development time in chronological months, $KLOC$ is the estimated number of delivered lines of code (in thousands) and a_b, b_b, c_b, d_b have their usual meanings. The basic $COCOMO$ equations are of the form.

A. $E = a_b(KLOC)\ exp(b_b), D = c_b(E)\ exp(d_b)$
B. $D = a_b(KLOC)\ exp(b_b), E = c_b(D)\ exp(d_b)$
C. $E = a_b\ exp(b_b), D = c_b(KLOC)\ exp(d_b)$
D. $E = a_b\ exp(d_b), D = c_b(KLOC)\ exp(b_b)$

nielit-scb-2020 is&software-engineering

Answer key☟

26.5.44 Is&software Engineering: NIELIT Scientist B 2020 November: 109

Consider a software project with the following information domain characteristic for calculation of function point metric.

Number of external inputs $(I) = 30$

Number of external output$(O) = 60$

Number of external inquiries $(E) = 23$

Number of files $(F) = 08$

Number of external interfaces $(N) = 02$

It is given that the complexity weighting factors for $\mathrm{I, O, E, F\ and\ N}$ are $4, 5, 4, 10 \text{ and } 7$, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors have value 3, and each of the remaining factors have value 4. The computed value of function point metric is __________

A. 612.06 B. 212.05 C. 305.09 D. 806.9

nielit-scb-2020 is&software-engineering

Answer key☟

26.5.45 Is&software Engineering: NIELIT Scientist B 2020 November: 92

Match the following:

(1)Waterfall model	(a)Specifications can be developed incrementally
(2)Evolutionary model	(b)Re-usability in development
(3)Component based software engineering	(c)Explicit recognition of risk
(4)Spiral development	(d)Inflexible partitioning of the project into stages

A. 1-a, 2-b, 3-c, 4-d
B. 1-d, 2-a, 3-b, 4-c
C. 1-d, 2-b, 3-a, 4-c
D. 1-c, 2-a, 3-b, 4-d

nielit-scb-2020 is&software-engineering match-the-following non-gatecse

Answer key☟

26.6 Productivity (1)

26.6.1 Productivity: NIELIT Scientist B 2020 November: 93

58000 LOC gaming software is developed with effort of 3 person-year. What is the productivity of person-month?

A. 1.9 KLOC
B. 1.6 KLOC
C. 4.8 KLOC
D. 4.2 KLOC

nielit-scb-2020 is&software-engineering productivity

Answer key☟

26.7 Software Design (7)

26.7.1 Software Design: NIELIT 2016 DEC Scientist B (IT) - Section B: 29

RAD software process model stands for:

A. Rapid Application Development.
B. Rapid Application Design.
C. Relative Application Development.
D. Recent Application Development.

nielit2016dec-scientistb-it non-gatecse is&software-engineering software-design

Answer key☟

26.7.2 Software Design: NIELIT 2016 DEC Scientist B (IT) - Section B: 38

What are the characteristics of software?

A. Software is developed or engineered; It is not manufactured in the classical sense.
B. Software can be custom built or custom build.
C. Software doesn't "wear out".
D. All mentioned above.

nielit2016dec-scientistb-it non-gatecse is&software-engineering software-design

Answer key

26.7.3 Software Design: NIELIT 2021 Dec Scientist B - Section B: 68

Which type of illustration lists the functionality of whole project ?

A. $\text{DFD}-0$
B. Class Diagram
C. Use case Diagram
D. State Diagram

nielit2021dec-scientistb is&software-engineering software-design

Answer key

26.7.4 Software Design: NIELIT Scientific Assistant A 2020 November: 104

Software consists of ___________

A. Set of instructions + operating procedures
B. Programs + documentation + operating procedures
C. Programs + hardware manuals
D. Set of programs

nielit-sta-2020 is&software-engineering software-design

Answer key

26.7.5 Software Design: NIELIT Scientific Assistant A 2020 November: 106

Changes are made to the system to reduce the future system failure chances is called __________.

A. Preventive Maintenance
B. Adaptive Maintenance
C. Corrective Maintenance
D. Perfective Maintenance

nielit-sta-2020 is&software-engineering software-design

Answer key

26.7.6 Software Design: NIELIT Scientific Assistant A 2020 November: 116

In the context of modular software design, which one of the following combinations is desirable?

A. High cohesion and high coupling
B. High cohesion and low coupling
C. Low cohesion and high coupling
D. Low cohesion and low coupling

nielit-sta-2020 is&software-engineering software-design

Answer key

26.7.7 Software Design: NIELIT Scientific Assistant A 2020 November: 81

In the Model-View-Controller (MVC) architecture, the model defines the ____________.

A. Data-access layer
B. Presentation layer
C. Business-logic layer
D. Interface layer

nielit-sta-2020 is&software-engineering software-design

Answer key

26.8 Software Development Life Cycle Models (1)

26.8.1 Software Development Life Cycle Models: NIELIT 2022 April Scientist B | Section B | Question: 53

What is the appropriate pairing of items in the two columns listing various activities encountered in a software life cycle?

P.	Requirements Capture	1.	Module Development and integration
Q.	Design	2.	Domain Analysis
R.	Implementation	3.	Structural and Behavioral Marketing
S.	Maintenance	4.	Performance Tunit

A. (P) – (3), (Q) – (2), (R) – (4), (S) – (1)
B. (P) – (2), (Q) – (3), (R) – (1), (S) – (4)
C. (P) – (3), (Q) – (2), (R) – (1), (S) – (4)
D. (P) – (2), (Q) – (3), (R) – (4), (S) – (1)

nielit2022apr-scientistb is&software-engineering software-development-life-cycle-models

26.9 Software Testing (9)

26.9.1 Software Testing: NIELIT 2017 July Scientist B (IT) - Section B: 28

The availability of complex software is 90% its Mean Time Between Failure(MTBF) is 200 days. Because of the critical nature of the usage, the organization deploying the software further enhanced it to obtain an availability of 95%. In the process, the Mean Time To Repair(MTTR) increased by 5 days. What is the MTBF of the enhanced software?(choose the nearest option)

A. 205 days. B. 300 days. C. 500 days. D. 700 days.

nielit2017july-scientistb-it non-gatecse is&software-engineering software-testing

Answer key☟

26.9.2 Software Testing: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 21

A program P calls two subprograms $P1$ and $P2$. $P1$ can fail 50% time and $P2$ can fail 40% times. The program P can fail

A. 50% B. 10% C. 60% D. 70%

nielit2017oct-assistanta-it is&software-engineering software-testing

Answer key☟

26.9.3 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 119

Integration testing, Unit Testing & System Testing are _____________.

A. Fundamental logic of Testing
B. Level Testing
C. Core Testing
D. Testing Suites

nielit2021dec-scientistb software-testing is&software-engineering

Answer key☟

26.9.4 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 44

Boundary value analysis is based on which testing :

A. White Box Testing
B. Black Box Testing
C. White Box & Black Box Testing
D. None of the options

nielit-2021-it-dec-scientistb is&software-engineering software-testing

Answer key☟

26.9.5 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 45

Match the following:

	List-I		**List-II**
W.	Condition coverage	1.	Black-box testing
X.	Equivalence class partitioning	2.	System testing
Y.	Volume testing	3.	White-box testing
Z.	Alpha Testing	4.	Performance testing

A. W-2, X-3, Y-1, Z-4
B. W-3, X-4, Y-2, Z-1
C. W-3, X-1, Y-4, Z-2
D. W-3, X-1, Y-2, Z-4

nielit2021dec-scientistb is&software-engineering software-testing

Answer key☟

26.9.6 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 8

Test levels are performed in which order?

A. Unit, Integration, Acceptance, System.
B. It depends on the nature of a project.
C. Unit, Integration, System, Acceptance.
D. Unit, System, Integration, Acceptance.

nielit-2021-it-dec-scientistb software-testing is&software-engineering

Answer key☟

26.9.7 Software Testing: NIELIT 2022 April Scientist B | Section B | Question: 50

Match the following:

List-I	List-II
(P) Condition coverage	(1) Black-box testing
(Q) Equivalence class partitioning	(2) System testing
(R) Volume testing	(3) White-box testing
(S) Alpha testing	(4) Performance testing

Codes :

	P	Q	R	S
(I)	(2)	(3)	(1)	(4)
(II)	(3)	(4)	(2)	(1)
(III)	(3)	(1)	(4)	(2)
(IV)	(3)	(1)	(2)	(4)

A. (I) B. (II) C. (III) D. (IV)

nielit2022apr-scientistb is&software-engineering software-testing normal non-gatecse

Answer key☟

26.9.8 Software Testing: NIELIT Scientific Assistant A 2020 November: 77

Black Box Software Testing method focuses on the :

A. Boundary condition of the software
B. Control structure of the software
C. Testing of user Interface only
D. Cyclomatic Complexity

nielit-sta-2020 is&software-engineering software-testing

Answer key☟

26.9.9 Software Testing: NIELIT Scientist B 2020 November: 116

Consider a software program that is artificially seeded with 100 faults. While testing this program, 159 faults are detected, out of which 75 faults are from those artificially seeded faults. Assuming that both real and seeded faults are of same nature and have same distribution, the estimated number of undetected real faults is ______________.

A. 28 B. 175 C. 56 D. 84

nielit-scb-2020 probability software-testing is&software-engineering

Answer key☟

Answer Keys

26.1.1	C	26.1.2	C	26.1.3	C	26.1.4	B	26.2.1	A
26.3.1	D	26.4.1	A	26.5.1	D	26.5.2	D	26.5.3	C

26.5.4	B
26.5.9	B
26.5.14	D
26.5.19	A
26.5.24	C
26.5.29	B
26.5.34	B
26.5.39	C
26.5.44	A
26.7.3	C
26.8.1	B
26.9.5	C

26.5.5	A
26.5.10	A
26.5.15	B
26.5.20	D
26.5.25	A
26.5.30	A
26.5.35	C
26.5.40	D
26.5.45	N/A
26.7.4	B
26.9.1	C
26.9.6	B

26.5.6	D
26.5.11	X
26.5.16	A
26.5.21	A
26.5.26	A
26.5.31	B
26.5.36	B
26.5.41	B
26.6.1	B
26.7.5	A
26.9.2	D
26.9.7	C

26.5.7	A
26.5.12	B
26.5.17	B
26.5.22	C
26.5.27	B
26.5.32	D
26.5.37	A
26.5.42	C
26.7.1	A
26.7.6	B
26.9.3	B
26.9.8	C

26.5.8	C
26.5.13	A
26.5.18	B
26.5.23	C
26.5.28	C
26.5.33	A
26.5.38	A
26.5.43	A
26.7.2	D
26.7.7	A
26.9.4	B
26.9.9	A

27 Non GATE CSE: Java (14)

27.1 Java (14)

27.1.1 Java: NIELIT 2016 DEC Scientist B (IT) - Section B: 6

Earlier name of Java programming language was:

A. OAK B. D C. Netbean D. Eclipse

nielit2016dec-scientistb-it non-gatecse java

Answer key

27.1.2 Java: NIELIT 2016 DEC Scientist B (IT) - Section B: 9

Which of the following is a platform free language?

A. JAVA B. C C. Assembly D. Fortran

nielit2016dec-scientistb-it non-gatecse java

Answer key

27.1.3 Java: NIELIT 2017 July Scientist B (IT) - Section B: 41

Which of the following is/are true about packages in Java?

1. Every class is part of some package.
2. All classes in a file are part of the same package.
3. If no package is specified, the classes in the file go into a special unnamed package.
4. If no package is specified, a new package is created with folder name of class and the class is put in this package.

A. Only $1,2$ and 3 B. Only $1,2$ and 4 C. Only 4 D. Only 1 and 3

nielit2017july-scientistb-it non-gatecse java

Answer key

27.1.4 Java: NIELIT 2017 July Scientist B (IT) - Section B: 42

Which of the following is false about abstract classes in Java?

A. If we derive an abstract class and do not implement all the abstract methods, then the derived class should also be marked as abstract using 'abstract' keyword.
B. Abstract classes can have constructors.
C. A class can be made abstract without any abstract method.
D. A class can inherit from multiple abstract classes.

nielit2017july-scientistb-it non-gatecse java

Answer key

27.1.5 Java: NIELIT 2017 July Scientist B (IT) - Section B: 43

Which of the following is true about interfaces in Java?

1. An interface can contain following type of members. . . . public, static, final fields(i.e., constants). . .default and static methods with bodies.
2. An instance of interface can be created.
3. A class can implement multiple interfaces.
4. Many classes can implement the same interface.

A. $1, 3$ and 4 B. $1, 2$ and 4 C. $2, 3$ and 4 D. $1, 2, 3$ and 4

nielit2017july-scientistb-it non-gatecse java

Answer key

27.1.6 Java: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 8

Is null an object?

A. Yes B. No C. Sometimes yes D. None of these

nielit2017oct-assistanta-it non-gatecse java

Answer key

27.1.7 Java: NIELIT Scientific Assistant A 2020 November: 107

In the given program:

```
class Dialog1
{
  public static void main(String args[])
  {
    Frame f1=new Frame("INDIA");
    f1.setSize(300,300);
    f1.setVisible(true);
    FileDialog d=new FileDialog(f1,"MyDialog");
    d.setVisible(true);
    String fname=d.getDirectory()+d.getFile();
    System.out.println("The Selection is"+fname);
  }
}
```

To make the Frame visible, which of the following statements are true?

A. f1.setClear(true); B. f1.setVisible(true);
C. f1.setlook(true); D. f1.setclean(true);

nielit-sta-2020 java

Answer key

27.1.8 Java: NIELIT Scientific Assistant A 2020 November: 108

Which of the following are two main types of overloading in Java?

A. Overloading and linking B. Overriding and linking
C. Reusability and data-hiding D. Overloading and Overriding

nielit-sta-2020 java

Answer key☟

27.1.9 Java: NIELIT Scientific Assistant A 2020 November: 110

In Java, for ensuring the persistence property, the class must implements:

A. Serializable Interface
B. Utilization Interface
C. Threadable Interface
D. Recognizable Interface

nielit-sta-2020 java

Answer key☟

27.1.10 Java: NIELIT Scientific Assistant A 2020 November: 112

In Java, the Dynamic Array are known as :

A. Vectors
B. Cycle
C. Remote
D. Kubernotos

nielit-sta-2020 java

Answer key☟

27.1.11 Java: NIELIT Scientific Assistant A 2020 November: 68

Which of the following construct is not supported by Java Server Pages?

A. JSP Directives
B. JSP Scriptlets
C. JSP Actions
D. JSP Reaction

nielit-sta-2020 java

Answer key☟

27.1.12 Java: NIELIT Scientific Assistant A 2020 November: 74

What is the output of the following program?

```
abstract class sum
{
    public abstract int sumOfTwo(int n1, int n2);
    public abstract int sumOfThree(int n1, int n2, int n3);
    public void disp(){
        System.out.println("Method of class Sum");
    }
}

class DemoAbstract1 extends Sum
{
    public int sumOfTwo(int num1, int num2)
    {
        return num1+num2;
    }
    public int sumOfThree(int num1, int num2, int num3)
    {
        return num1+num2+num3;
    }

    public static void main(String args[]){
        Sum obj=new DemoAbstract1();
```

```
        System.out.println(obj.sumOfTwo(3,7));
        System.out.println(obj.sumOfThree(4,3,19));
        obj.disp();
    }
}
```

A. 10
26
Method of class Sum

B. 26
10
Method of class Sum

C. Method of class Sum
26
10

D. Error

nielit-sta-2020 numerical-answers java

Answer key☟

27.1.13 Java: NIELIT Scientific Assistant A 2020 November: 84

The static keyword is used in public static void main() declaration in Java:

A. To enable the JVM to make call to the main(), as class has not been instantiated
B. To enable the JVM to make call to the main(), as class has not been inherited
C. To enable the JVM to make call to the main(), as class has not been loaded
D. To enable the JVM to make call to the main(), as class has not been finalized

nielit-sta-2020 java

Answer key☟

27.1.14 Java: NIELIT Scientific Assistant A 2020 November: 89

Which of the following Interface is not supported by JDBC for connecting to Database in Java Programming language?

A. Statement Interface
B. Prepared Statement Interface
C. Callable Statement Interface
D. Database Interface

nielit-sta-2020 java

Answer key☟

Answer Keys

27.1.1	A	27.1.2	A	27.1.3	A	27.1.4	D	27.1.5	A

27.1.6	B	27.1.7	B	27.1.8	D	27.1.9	A	27.1.10	A
27.1.11	D	27.1.12	A	27.1.13	A	27.1.14	D		

28 Non GATE CSE: Numerical Methods (8)

28.1 Numerical Methods (6)

28.1.1 Numerical Methods: NIELIT 2016 MAR Scientist B - Section B: 7

In which of the following methods proper choice of initial value is very important?

A. Bisection method
B. False position
C. Newton-Raphson
D. Bairsto method

nielit2016mar-scientistb non-gatecse numerical-methods

Answer key☟

28.1.2 Numerical Methods: NIELIT 2016 MAR Scientist C - Section B: 1

Choose the most appropriate option.

The Newton-Raphson iteration $x_{n+1} = \frac{x_n}{2} + \frac{3}{2x_n}$ can be used to solve the equation

A. $x^2 = 3$
B. $x^3 = 3$
C. $x^2 = 2$
D. $x^3 = 2$

nielit2016mar-scientistc non-gatecse numerical-methods

28.1.3 Numerical Methods: NIELIT 2017 DEC Scientist B - Section B: 3

Using bisection method, one root of $x^4 - x - 1$ lies between 1 and 2. After second iteration the root may lie in interval:

A. $(1.25, 1.5)$
B. $(1, 1.25)$
C. $(1, 1.5)$
D. None of the options.

nielit2017dec-scientistb non-gatecse numerical-methods

Answer key☟

28.1.4 Numerical Methods: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 7

The convergence of the bisection method is

A. Cubic
B. Quadratic
C. Linear
D. None

nielit2017oct-assistanta-cs non-gatecse numerical-methods

Answer key☟

28.1.5 Numerical Methods: NIELIT 2021 Dec Scientist A - Section B: 89

The real root of the equation $x^3 - x - 5 = 0$ lying between 1 and 2 after first iteration by Netwon-Raphson method is ___________, if initial approximation is taken as $x_0 = 2 \in [1, 2]$:

A. 1.909 B. 1.904 C. 1.921 D. 1.940

nielit2021dec-scientista calculus numerical-methods

28.1.6 Numerical Methods: NIELIT 2021 Dec Scientist B - Section B: 60

One root of $x^3 - x - 4 = 0$ lies in $(1, 2)$. In bisection method, after first iteration the root lies in the interval ___________ .

A. $(1, 1.5)$ B. $(1.5, 2)$ C. $(1.25, 1.75)$ D. $(1.75, 2)$

nielit-2021-it-dec-scientistb numerical-methods calculus

28.2 Root Finding (1)

28.2.1 Root Finding: NIELIT 2021 Dec Scientist B - Section B: 106

One root of $x^3 - x - 4 = 0$ lies in $(1, 2)$. In bisection method, after first iteration the root lies in the interval ___________ .

A. $(1, 1.5)$ B. $(1.5, 2)$ C. $(1.25, 1.75)$ D. $(1.75, 2)$

nielit2021dec-scientistb numerical-methods root-finding analytical-aptitude

Answer key☟

28.3 Vector Space (1)

28.3.1 Vector Space: NIELIT 2017 DEC Scientist B - Section B: 20

Let u and v be two vectors in R^2 whose Eucledian norms satisfy $|u| = 2|v|$. What is the value α such that $w = u + \alpha v$ bisects the angle between u and v?

A. 2 B. 1 C. $\frac{1}{2}$ D. -2

nielit2017dec-scientistb non-gatecse vector-space

Answer key☟

Answer Keys

28.1.1	C	28.1.2	A	28.1.3	B	28.1.4	C	28.1.5	A
28.1.6	B	28.2.1	B	28.3.1	A				

29 Non GATE CSE: Object Oriented Programming (15)

29.0.1 NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 24

Which of the following object types are generally autonomous, meaning that they can exhibit some behavior without being operated upon by another object.

A. Passive
B. Active
C. Both (A) and (B)
D. None of the mentioned

nielit2017oct-assistanta-it non-gatecse

Answer key☟

29.1 Object Oriented Programming (13)

29.1.1 Object Oriented Programming: NIELIT 2016 DEC Scientist B (CS) - Section B: 30

The IOS class member function used for formatting IO is

A. width(), precision(), read()
B. width(), precision(), setf()
C. getch(), width(), Io()
D. unsetf() ,setf(), write()

nielit2016dec-scientistb-cs non-gatecse object-oriented-programming

Answer key☟

29.1.2 Object Oriented Programming: NIELIT 2016 DEC Scientist B (IT) - Section B: 1

Given a class named student, which of the following is a valid constructor declaration for the class?

A.
```
Student student(){}
```
B.
```
Private final student(){}
```
C.
```
Student(student s){}
```
D.
```
Void student(){}
```

nielit2016dec-scientistb-it non-gatecse object-oriented-programming

Answer key☟

29.1.3 Object Oriented Programming: NIELIT 2016 DEC Scientist B (IT) - Section B: 24

An operation can be described as:

A. Object behavior
B. Functions
C. Class behavior
D. (A),(B)

nielit2016dec-scientistb-it non-gatecse object-oriented-programming

Answer key☟

29.1.4 Object Oriented Programming: NIELIT 2016 DEC Scientist B (IT) - Section B: 3

An object can have which of the following multiplicities?

A. Zero　B. More than one　C. One　D. All of the above

nielit2016dec-scientistb-it　non-gatecse　object-oriented-programming

Answer key

29.1.5 Object Oriented Programming: NIELIT 2016 DEC Scientist B (IT) - Section B: 46

Object oriented inheritance models:

A. "is a kind of" relationship.
B. "has a" relationship.
C. "want to be" relationship.
D. "contains" of relationship.

nielit2016dec-scientistb-it　non-gatecse　object-oriented-programming

Answer key

29.1.6 Object Oriented Programming: NIELIT 2017 July Scientist B (IT) - Section B: 39

Give the output

```cpp
#include<iostream>
using namespace std;
class Base
{
public:
    int x,y;
public:
    Base(int i, int j){x=i;y=j;}
};
class Derived:public Base
{
public:
    Derived(int i,int j):x(i),y(j){}
    void print(){cout<< x << ""<<y ;}
};
int main(void)
{
    Derived q(10,10);
    q.print();
    return 0;
}
```

A. $10\ 10$　B. Compiler Error　C. $0\ 0$　D. None of the option

nielit2017july-scientistb-it　non-gatecse　object-oriented-programming

Answer key

29.1.7 Object Oriented Programming: NIELIT 2017 July Scientist B (IT) - Section B: 40

Give the output

```cpp
#include<iostream>
```

```
using namespace std;
class Base1{
public:
~Base1() {cout<<"Base1's destructor"<<endl;}
};
class Base2
{
public:
~Base2(){cout<<"Base2's destructor"<<endl;}
};
class Derived : public Base1,public Base2{
public:
~Derived(){cout<<"Derived's destructor"<<endl; }
};
int main()
{
Derived d;
return 0;
}
```

A. Base1's destructor
   Base2's destructor
   Derived's Destructor
B. Derived's Destructor
   Base2's destructor
   Base1's destructor
C. Derived's Destructor
D. Compiler Dependent

nielit2017july-scientistb-it non-gatecse object-oriented-programming

Answer key☟

29.1.8 Object Oriented Programming: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 28

What is 'Basis of Encapsulation'?

A. object
B. class
C. method
D. all of the mentioned

nielit2017oct-assistanta-it non-gatecse object-oriented-programming

Answer key☟

29.1.9 Object Oriented Programming: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 4

If the objects focus on the problem domain, then we are concerned with

A. Object Oriented Analysis
B. Object Oriented Design
C. Object Oriented Analysis and Design
D. None of the above

nielit2017oct-assistanta-it non-gatecse object-oriented-programming

Answer key☟

29.1.10 Object Oriented Programming: NIELIT 2021 Dec Scientist B - Section B: 17

Which is indicating use of virtual functions?

A. Overloading
B. Overriding
C. Static binding
D. Dynamic binding

nielit-2021-it-dec-scientistb object-oriented-programming

Answer key

29.1.11 Object Oriented Programming: NIELIT 2021 Dec Scientist B - Section B: 36

Match the following in from **List-1** and **List-2** :

List-1	List-2
(a) Create new classes from existing class	(i) Polymorphism
(b) Using similar operations to do similar things	(ii) Abstraction
(c) Hiding implementation details	(iii) Inheritance
(d) Wrapping both operator and attributes with operations for model object	(iv) Encapsulation
	(v) Performance

	(a)	(b)	(c)	(d)
(A)	(iii)	(i)	(ii)	(iv)
(B)	(v)	(i)	(iii)	(ii)
(C)	(ii)	(v)	(iv)	(iii)
(D)	(i)	(ii)	(iii)	(iv)

nielit-2021-it-dec-scientistb object-oriented-programming is&software-engineering

Answer key

29.1.12 Object Oriented Programming: NIELIT 2021 Dec Scientist B - Section B: 63

In Software Modeling 'IS A' represents ______________ relationship.

A. Aggregation
B. Over loading
C. Inheritance
D. Design Patterns

nielit2021dec-scientistb object-oriented-programming is&software-engineering

Answer key

29.1.13 Object Oriented Programming: NIELIT Scientific Assistant A 2020 November:

49

During exception handling, which of the following statements hold true?

A. Single try can have multiple associated catch with it
B. A single Catch can have multiple try associated with it
C. Finally block execute only when the class is inherited
D. For a given exception, multiple catch can execute

nielit-sta-2020 object-oriented-programming

Answer key☟

29.2 Operator Overloading (1)

29.2.1 Operator Overloading: NIELIT 2021 Dec Scientist B - Section B: 54

Which of the following operator cannot be overloaded?

A. $++$ B. :: C. $\sim$ D. $()$

nielit-2021-it-dec-scientistb programming-in-c object-oriented-programming operator-overloading

Answer key☟

Answer Keys

29.0.1	B	29.1.1	B	29.1.2	C	29.1.3	D	29.1.4	D
29.1.5	A	29.1.6	B	29.1.7	B	29.1.8	D	29.1.9	A
29.1.10	D	29.1.11	A	29.1.12	C	29.1.13	A	29.2.1	B

30 Non GATE CSE: Optimization (2)

30.0.1 NIELIT 2016 MAR Scientist B - Section C: 30

Bounded minimalization is a technique for

A. proving whether a promotive recursive function is turning computable or not
B. proving whether a primitive recursive function is a total function or not
C. generating primitive recursive functions
D. generating partial recursive functions

nielit2016mar-scientistb non-gatecse

Answer key☟

30.1 Differential Equation (1)

30.1.1 Differential Equation: NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 6

A solution for the differential equation $x'(t) + 2x(t) = \delta(t)$ with initial condition $x(\bar{0}) = 0$

A. $e^{-2t}u(t)$ B. $e^{2t}u(t)$ C. $e^{-t}u(t)$ D. $e^{t}u(t)$

nielit2017oct-assistanta-cs non-gatecse differential-equation

Answer Keys

30.0.1	C	30.1.1	C

31.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 24

How many wires are threaded through the cores in a coincident-current core memory?

A. 2 B. 3 C. 4 D. 6

nielit2017july-scientistb-cs non-gatecse

Answer key

31.0.2 NIELIT 2016 MAR Scientist B - Section C: 33

In what module multiple instances of execution will yield the same result even if one instance has not terminated before the next one has begun?

A. Non reusable module
B. Serially usable
C. Re-enterable module
D. Recursive module

nielit2016mar-scientistb non-gatecse

31.0.3 NIELIT 2016 MAR Scientist C - Section B: 18

Find the volume of the solid obtained by rotating the region bound by the curves $y = x^3 + 1,\ x = 1$, and $y = 0$ about the x-axis

A. $\frac{23\pi}{7}$
B. $\frac{16\pi}{7}$
C. 2π
D. $\frac{19\pi}{7}$

nielit2016mar-scientistc non-gatecse

31.0.4 NIELIT 2016 MAR Scientist C - Section B: 15

The equation of the plane through the point $(-1, 3, 2)$ and perpendicular to each of the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$ is

A. $7x - 8y + 3z + 25 = 0$
B. $7x + 8y + 3z + 25 = 0$
C. $7x - 8y + 3z - 25 = 0$
D. $7x - 8y - 3z - 25 = 0$

nielit2016mar-scientistc non-gatecse

31.0.5 NIELIT 2016 MAR Scientist C - Section B: 14

Find the area bounded by the curve $y = \sqrt{5 - x^2}$ and $y = \mid x - 1 \mid$

A. $\frac{2}{0}(2\sqrt{6} - \sqrt{3}) - \frac{5}{2}$
B. $\frac{2}{3}(6\sqrt{6} + 3\sqrt{3}) + \frac{5}{2}$
C. $2(\sqrt{6} - \sqrt{3}) - 5$
D. $\frac{2}{3}(\sqrt{6} - \sqrt{3}) + 5$

nielit2016mar-scientistc non-gatecse

31.0.6 NIELIT 2016 MAR Scientist C - Section B: 7

The area under the curve $y(x)=3e^{-5x}$ from $x=0$ to $x=\infty$ is

A. $\frac{3}{5}$ B. $\frac{-3}{5}$ C. 5 D. $\frac{5}{3}$

nielit2016mar-scientistc non-gatecse

31.0.7 NIELIT 2016 MAR Scientist C - Section B: 6

If $y=f(x)$, in the interval $[a,b]$ is rotated about the x-axis, the Volume of the solid of revolution is $(f'(x)=dy/dx)$

A. $\int_a^b \pi[f(x)]^2 dx$

B. $\int_a^b [f(x)]^3 dx$

C. $\int_a^b \pi[f'(x)]^2 dx$

D. $\int_a^b \pi^2 f(x) dx$

nielit2016mar-scientistc non-gatecse

31.1 Complex Number (1)

31.1.1 Complex Number: NIELIT 2018-28

For the function $(z)=\frac{1}{z^2(e^z-1)}$, $z=0$ is a pole of order:

A. 1 B. 2 C. 3 D. None of these

nielit-2018 non-gatecse complex-number

Answer key☟

31.2 Differential Equation (4)

31.2.1 Differential Equation: NIELIT 2018-18

If y^a is an integrating factor of the differential equation $2xydx-(3x^2-y^2)dy=0$, then the value of a is

A. -4 B. 4 C. -1 D. 1

nielit-2018 non-gatecse differential-equation

Answer key☟

31.2.2 Differential Equation: NIELIT 2018-20

The general solution of the differential equation $\frac{dy}{dx}=(1+y^2)(e^{-x^2}-2x\tan^{-1}y)$ is:

A. $e^{x^2}\tan^{-1}y=x+c$ B. $e^{-x^2}\tan^{y}=x+c$

C. $e^x \tan y = x^2 + c$

D. $e^{-x} \tan^{-1} y = x^3 + c$

nielit-2018 non-gatecse differential-equation

31.2.3 Differential Equation: NIELIT 2018-22

While solving the differential equation $\frac{d^2y}{dx^2} + 4y = \tan 2x$ by the method of variation of parameters, then value of Wronskion (W) is:

A. 1

B. 2

C. 3

D. 4

nielit-2018 non-gatecse differential-equation

Answer key☟

31.2.4 Differential Equation: NIELIT 2018-25

The general solution of the partial differential equation $(D^2 - D'^2 - 2D + 2D')Z = 0$ where $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$:

A. $f(y+x) + e^{2x} g(y-x)$

B. $e^{2x} f(y+x) + g(y-x)$

C. $e^{-2x} f(y+x) + g(y-x)$

D. $f(y+x) + e^{-2x} g(y-x)$

nielit-2018 non-gatecse differential-equation partial-order

31.3 Double Integration (1)

31.3.1 Double Integration: NIELIT 2018-15

The value of $\int_1^2 \int_0^{x^2} x \, dy \, dx$ is:

A. $\frac{15}{4}$

B. $\frac{5}{4}$

C. $\frac{3}{4}$

D. $\frac{2}{3}$

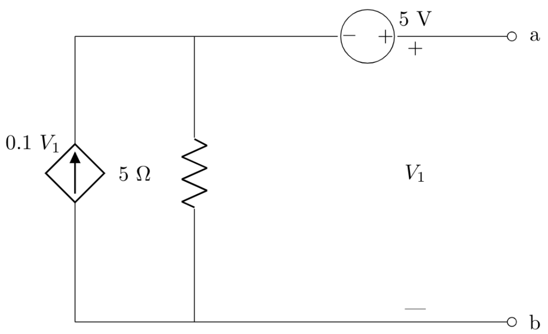
nielit-2018 calculus double-integration

Answer key☟

31.4 Electronic Circuit (1)

31.4.1 Electronic Circuit: NIELIT Scientist B 2020 November: 71

The resistance to be connected across terminal a, b for maximum power transfer to it is:



A. $40\,\Omega$ B. $5\,\Omega$ C. $2.5\,\Omega$ D. $10\,\Omega$

nielit-scb-2020 electronic-circuit non-gatecse

31.5 General (1)

31.5.1 General: NIELIT 2023 Feb Scientist C | Section C | Question: 75

Which of the following is a characteristic of qualitative research?

A. Design flexibility
B. Inductive analysis
C. Context sensitivity
D. All of the above

nielit2023feb-scientistc general

Answer key☟

31.6 Integration (2)

31.6.1 Integration: NIELIT 2018-16

The value of $\int_c \frac{2x^2-5}{(x+2)^2(x^2+4)x^2}dx$, (where c is the square with vertices $1+i, 2+i, 2+2i, i+2i$) is:

A. 0 B. πi C. $2\pi i$ D. $4\pi i$

nielit-2018 non-gatecse integration

31.6.2 Integration: NIELIT 2018-29

Using Green's theorem in plane, evaluate $\int_c (2x-y)dx+(x+y)dy$, where c is the circle $x^2+y^2=4$ in the plane:

A. 2π B. 4π C. -4π D. 8π

nielit-2018 non-gatecse integration

Answer key☟

31.7 Partial Order (3)

31.7.1 Partial Order: NIELIT 2016 MAR Scientist C - Section B: 5

If $f(x,y)=x^3y+e^x$, the partial derivatives, $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are

A. $3x^2y+1,\ x^3+1$
B. $3x^2y+e^x,\ x^3$
C. $x^3y+xe^x,\ x^3+e^x$
D. $2x^2y+\frac{e^x}{x}$

nielit2016mar-scientistc non-gatecse partial-order

Answer key☟

31.7.2 Partial Order: NIELIT 2018-19

If $u = f(y-z,\ z-x,\ x-y)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ is equal to:

A. $x+y+z$ B. $1+x+y+z$ C. 1 D. 0

nielit-2018 non-gatecse differentiation partial-order

31.7.3 Partial Order: NIELIT 2018-23

If $w = f(z) = u(x,y) + i\,v(x,y)$ is an analytic function, then $\frac{dw}{dz}$ is:

A. $\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$
B. $\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial y}$
C. $\frac{\partial u}{\partial x} - i\frac{\partial v}{\partial x}$
D. $\frac{\partial u}{\partial x} + i\frac{\partial u}{\partial y}$

nielit-2018 non-gatecse differentiation partial-order

31.8 Permutation and Combination (1)

31.8.1 Permutation and Combination: NIELIT 2021 Dec Scientist B - Section A: 23

The students in a class are seated according to their marks in the previous examination. Once it so happens that four of these students get equal marks and therefore the same rank. To decide their seating arrangement, the teacher wants to write down all possible arrangements, one in each of separate bits of paper, in order to choose one of these by lots. How many bits of paper are required ?

A. 9 B. 12 C. 15 D. 24

nielit2021dec-scientistb counting permutation-and-combination analytical-aptitude

Answer key☟

Answer Keys

31.0.1	A	31.0.2	C	31.0.3	TBA	31.0.4	A	31.0.5	TBA
31.0.6	A	31.0.7	A	31.1.1	B	31.2.1	A	31.2.2	C
31.2.3	B	31.2.4	A	31.3.1	A	31.4.1	B	31.5.1	D
31.6.1	A	31.6.2	A	31.7.1	B	31.7.2	C	31.7.3	A
31.8.1	D								

32 Non GATE CSE: Web Technologies (23)

32.0.1 NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 11

Which of these events will be generated if we close an applet's window?

A. ActionEvent
B. ComponentEvent
C. AdjustmentEvent
D. WindowEvent

nielit2017oct-assistanta-it non-gatecse

Answer key☟

32.1 Css (1)

32.1.1 Css: NIELIT Scientific Assistant A 2020 November: 92

If we don't want to allow a floating div to the left side of an element, __________ CSS property will we use.

A. margin
B. clear
C. float
D. padding

nielit-sta-2020 web-technologies css

Answer key☟

32.2 Html (7)

32.2.1 Html: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 12

Web links are stored within the page itself and when you wish to 'jump' to the page that is linked, we select the hotspot or anchor. This technique is called

A. Hypertext
B. Hypermedia
C. Both (A) and (B)
D. Anchoring

nielit2017oct-assistanta-cs non-gatecse web-technologies html

Answer key☟

32.2.2 Html: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 3

Web links are stored within the page itself and when you wish to 'jump' to the page that is linked, we select the hotspot or anchor. This technique is called

A. Hypertext
B. Hypermedia
C. Both (A) and (B)
D. Anchoring

nielit2017oct-assistanta-it web-technologies html

Answer key☟

32.2.3 Html: NIELIT 2021 Dec Scientist B - Section B: 29

Which of the following tag is used for inserting the largest heading in HTML?

A. $<\mathrm{h3}>$
B. $<\mathrm{h1}>$
C. $<\mathrm{h5}>$
D. $<\mathrm{h6}>$

nielit-2021-it-dec-scientistb html web-technologies

Answer key☟

32.2.4 Html: NIELIT 2021 Dec Scientist B - Section B: 3

How to add a background color in HTML for marquee Tag?

A. <marquee bgcolor : "red">
B. <marquee bg-color = "red">
C. <marquee bgcolor = "red">
D. <marquee color = "red">

nielit-2021-it-dec-scientistb html

Answer key☟

32.2.5 Html: NIELIT Scientific Assistant A 2020 November: 76

The default character encoding in $\mathrm{HTML5}$ is _________

A. $\mathrm{UTF\text{-}16}$ B. $\mathrm{UTF\text{-}32}$ C. $\mathrm{UTF\text{-}8}$ D. $\mathrm{ISO\text{-}8859\text{-}1}$

nielit-sta-2020 web-technologies html

Answer key☟

32.2.6 Html: NIELIT Scientific Assistant A 2020 November: 90

What does <main> include?

A. Header B. Sidebar C. Article D. Footer

nielit-sta-2020 web-technologies html

Answer key☟

32.2.7 Html: NIELIT Scientist B 2020 November: 94

Which of the following tag is used intended for navigation in $HTML5$?

A. nav B. footer C. section D. navigation tag

nielit-scb-2020 html web-technologies

Answer key☟

32.3 Web Technologies (14)

32.3.1 Web Technologies: NIELIT 2016 DEC Scientist B (IT) - Section B: 20

By which technology do we separate our business logic from the presentation logic?

A. Servlet B. JSP C. Both (A) & (B) D. None of the above

nielit2016dec-scientistb-it non-gatecse web-technologies

Answer key☟

32.3.2 Web Technologies: NIELIT 2017 DEC Scientific Assistant A - Section B: 30

How can we set a Cookie visibility scope to local storage ?

A. $\$$ B. $\%$ C. $/$ D. All of the options

nielit2017dec-assistanta non-gatecse web-technologies

Answer key☟

32.3.3 Web Technologies: NIELIT 2017 July Scientist B (CS) - Section B: 50

Match the problem domains in GROUP I with the solution technologies in GROUP II

	GROUP I		GROUP II
(P)	Service oriented computing	1.	Interoperability
(Q)	Heterogeneous communicating systems	2.	BPMN
(R)	Information representation	3.	Publish-find-bind
(S)	Process description	4.	XML

A. $P-1, Q-2, R-3, S-4$
B. $P-3, Q-4, R-2, S-1$
C. $P-3, Q-1, R-4, S-2$
D. $P-4, Q-3, R-2, S-1$

nielit2017july-scientistb-cs non-gatecse web-technologies

Answer key☟

32.3.4 Web Technologies: NIELIT 2017 July Scientist B (IT) - Section B: 29

HTML(Hypertext Markup language) has language elements which permit certain actions other than describing the structure of the web document. Which of the following actions is NOT supported by pure HTML(without any server or client side scripting) pages?

A. Embed web objects from different sites into the same page.
B. Refresh the page automatically after a specified interval.
C. Automatically redirect to another page upon download.
D. Display the client time as part of the page.

nielit2017july-scientistb-it non-gatecse web-technologies

Answer key☟

32.3.5 Web Technologies: NIELIT 2017 July Scientist B (IT) - Section B: 30

Consider the HTML table definition given below:

```
<table border=1>
<tr>
<td rowspan=2>ab</td>
<td colspan=2>cd</td>
</tr>
<tr>
<td>ef</td>
<td rowspan=2>gh</td>
</tr>
<tr><td colspan=2>ik</td>
</tr>
</table>
```

The number of rows in each column and the number of columns in each row are

A. $(2, 2, 3)$ and $(2, 3, 2)$
B. $(2, 2, 3)$ and $(2, 2, 3)$
C. $(2, 3, 2)$ and $(2, 3, 2)$
D. $(2, 3, 2)$ and $(2, 2, 3)$

nielit2017july-scientistb-it non-gatecse web-technologies

Answer key

32.3.6 Web Technologies: NIELIT 2017 July Scientist B (IT) - Section B: 31

Which of the following statements is/are False?

1. XML overcomes the limitations in HTML to support a structured way of organizing content .
2. XML specification is not case sensitive while HTML specification is case sensitive.
3. XML supports user defined tags while HTML uses pre-defined tags.
4. XML tags need not be closed while HTML tags must be closed.

A. 2 only.
B. 1 only.
C. 2 and 4 only.
D. 3 and 4 only.

nielit2017july-scientistb-it non-gatecse web-technologies

Answer key

32.3.7 Web Technologies: NIELIT 2021 Dec Scientist B - Section B: 33

What is WordPress?

A. It is a software used to press text
B. It is a text formatting software
C. It is a CMS (Content Management System)
D. It is mail service

nielit-2021-it-dec-scientistb web-technologies

Answer key

32.3.8 Web Technologies: NIELIT 2021 Dec Scientist B - Section B: 43

____________ is used to write a comment in CSS.

A. $/^*$ a comment $^*/$
B. $//$ a comment $//$
C. $/$ a comment $/$
D. $<$ 'a comment' $>$

nielit-2021-it-dec-scientistb web-technologies

Answer key

32.3.9 Web Technologies: NIELIT Scientific Assistant A 2020 November: 59

Adding the style attributes in HTML elements, is known to be ___________.

A. Internal B. Inline C. Outline D. External

nielit-sta-2020 web-technologies

Answer key

32.3.10 Web Technologies: NIELIT Scientific Assistant A 2020 November: 71

___________ CSS property allow to wrap a block of text around an image.

A. wrap B. push C. float D. align

nielit-sta-2020 web-technologies

Answer key

32.3.11 Web Technologies: NIELIT Scientist B 2020 November: 119

____________________ is automatically loaded and operates as part of browser.

A. Add-ons B. Plug-ins C. Utilities D. Widgets

nielit-scb-2020 web-technologies

Answer key

32.3.12 Web Technologies: NIELIT Scientist B 2020 November: 56

Which element is used to define discrete unit of content such as a blogpost, comment and so on?

A. section B. class C. article D. none of the options

nielit-scb-2020 web-technologies

Answer key

32.3.13 Web Technologies: NIELIT Scientist B 2020 November: 89

PI in XML specification stands for ___________.

A. priceless instruction
B. processing instruction
C. polymorphic inheritance
D. primary instruction

nielit-scb-2020 web-technologies

Answer key

32.3.14 Web Technologies: NIELIT Scientist B 2020 November: 95

Which of the following is correct Content-Type header that a server side script should send for SSE in $HTML5$?

A. Content-Type: text/event-stream
B. Content-Type: text/application-stream
C. Content-Type: text/data-stream
D. None of the options

nielit-scb-2020 web-technologies

Answer key☟

Answer Keys

32.0.1	D	32.1.1	B	32.2.1	C	32.2.2	C	32.2.3	B
32.2.4	C	32.2.5	C	32.2.6	C	32.2.7	A	32.3.1	B
32.3.2	C	32.3.3	C	32.3.4	D	32.3.5	C	32.3.6	C
32.3.7	C	32.3.8	A	32.3.9	B	32.3.10	C	32.3.11	B
32.3.12	C	32.3.13	B	32.3.14	A				

33.0.1 NIELIT 2018-50

The process executes the following code and after execution _______ number of child process get created

```
fork();
fork();
fork();
fork();
```

A. 4 B. 1 C. 15 D. 16

nielit-2018 operating-system

Answer key

33.0.2 NIELIT 2017 DEC Scientific Assistant A - Section B: 58

Page fault frequency in an operating system is reduced when the :

A. Process tend to be I/O bound
B. Locality of reference is applicable to the process
C. Size of pages is reduced
D. Processes tend to be CPU bound

nielit2017dec-assistanta operating-system page-fault

Answer key

33.0.3 NIELIT 2017 DEC Scientific Assistant A - Section B: 48

The process of loading the operating system into memory is called :

A. Booting B. Spooling C. Thrashing D. Formatting

nielit2017dec-assistanta operating-system

Answer key

33.0.4 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 33

The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows. What is the number of Page Faults?
$$0100, 0200, 0430, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370.$$

A. $15,4$ B. $6,4$ C. $7,2$ D. $4,6$

nielit2017oct-assistanta-cs operating-system demand-paging page-fault

Answer key

33.0.5 NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 10

The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows. What is the number of Page Faults?

$0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370.$

A. $15, 4$ B. $6, 4$ C. $7, 2$ D. $4, 6$

nielit2017oct-assistanta-it operating-system demand-paging page-fault

Answer key☟

33.0.6 NIELIT Scientist B 2020 November: 70

On computers where there are multiple operating system, the decision to load a particular one is done by ________.

A. PCB B. Inode C. File Control Block D. Boot Loader

nielit-scb-2020 operating-system

Answer key☟

33.1 Access Control (1)

33.1.1 Access Control: NIELIT 2021 Dec Scientist B - Section B: 64

The operating system stores an ____________ in order to decide to which user to grant which access rights to which file ?

A. File allocation table
B. Process control block
C. Access control matrix
D. File control matrix

nielit2021dec-scientistb operating-system access-control matrix

33.2 Bankers Algorithm (2)

33.2.1 Bankers Algorithm: NIELIT 2016 DEC Scientist B (IT) - Section B: 48

The Banker's algorithm is used:

A. to rectify deadlock.
B. to prevent deadlock.
C. to detect deadlock.
D. to detect and solve deadlock.

nielit2016dec-scientistb-it operating-system deadlock-prevention-avoidance-detection bankers-algorithm

Answer key☟

33.2.2 Bankers Algorithm: NIELIT 2022 April Scientist B | Section B | Question: 100

An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types $\mathrm{X}, \mathrm{Y},$ and Z to three processes $\mathrm{P0}, \mathrm{P1},$ and $\mathrm{P2}$. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution.

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

There are 3 units of type $X, 2$ units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state:

REQ1: P0 requests 0 units of $\text{X}, 0$ units of Y and 2 units of Z

REQ2: P1 requests 2 units of $\text{X}, 0$ units of Y and 0 units of Z

Which one of the following is TRUE ?

A. Only REQ1 can be permitted.
B. Only REQ2 can be permitted.
C. Both REQ1 and REQ2 can be permitted.
D. Neither REQ1 nor REQ2 can be permitted.

nielit2022apr-scientistb operating-system deadlock-prevention-avoidance-detection bankers-algorithm

33.3 CO and Architecture (2)

33.3.1 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 110

In ______________ machine is executing operating system instruction :

A. System mode
B. User mode
C. Normal mode
D. Safe mode

nielit2021dec-scientistb operating-system co-and-architecture

Answer key☟

33.3.2 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 45

Which of the following is true?

A. Unless enabled, a CPU will not be able to process interrupts
B. Loop instructions cannot be interrupted till they complete
C. A processor need not checks for interrupts before executing a new instruction
D. Only level triggered interrupts are possible on microprocessors

nielit-2021-it-dec-scientistb co-and-architecture operating-system

Answer key☟

33.4 Cache Memory (2)

33.4.1 Cache Memory: NIELIT 2016 DEC Scientist B (CS) - Section B: 37

The principle of locality of reference justifies the use of

A. Non reusable
B. Cache memory
C. Virtual memory
D. None of the above

nielit2016dec-scientistb-cs operating-system cache-memory

Answer key☟

33.4.2 Cache Memory: NIELIT 2017 DEC Scientific Assistant A - Section B: 56

In a particular system it is observed that, the cache performance gets improved as a result of increasing the block size of the cache. The primary reason behind this is :

A. Programs exhibits temporal locality
B. Programs have small working set
C. Read operation is frequently required rather than write operation
D. Programs exhibits spatial locality

nielit2017dec-assistanta operating-system cache-memory

Answer key☟

33.5 Context Switch (1)

33.5.1 Context Switch: NIELIT 2017 DEC Scientist B - Section B: 17

The time taken to switch between user and kernel modes of execution be $t1$ while the time taken to switch between two processes be $t2$. Which of the following is TRUE?

A. $t1 > t2$
B. $t1 = t2$
C. $t1 < t2$
D. nothing can be said about the relation between $t1$ and $t2$

nielit2017dec-scientistb operating-system context-switch

Answer key☟

33.6 Disk (7)

33.6.1 Disk: NIELIT 2016 MAR Scientist B - Section C: 36

What is the elapsed time of P if records of F are organized using a blocking factor of 2(i.e. each block on D contains two records of F) and P uses one buffer?

A. 12 sec.
B. 14 sec.
C. 17 sec.
D. 21 sec.

nielit2016mar-scientistb operating-system disk

33.6.2 Disk: NIELIT 2016 MAR Scientist B - Section C: 8

When we move from the outermost track to the innermost track in a magnetic disk, then density(bits per linear inch)

A. increases.
B. decreases.
C. remains the same.
D. either remains constant or decreases.

nielit2016mar-scientistb operating-system disk

Answer key

33.6.3 Disk: NIELIT 2016 MAR Scientist C - Section C: 26

The seek time of a disk is 30 ms. It rotates at the rate of 30 rotations/second. The capacity of each track is 300 words. The access time is (approximately)

A. 62 ms
B. 60 ms
C. 50 ms
D. 47 ms

nielit2016mar-scientistc operating-system disk

Answer key

33.6.4 Disk: NIELIT 2017 July Scientist B (CS) - Section B: 25

Which access method is used for obtaining a record from cassette tape?

A. Direct
B. Sequential
C. Random
D. Parallel

nielit2017july-scientistb-cs operating-system disk

Answer key

33.6.5 Disk: NIELIT 2018-52

In the disk, swap space is used to _______.

A. save XML files
B. save process data
C. save drivers
D. save HTML files

nielit-2018 operating-system disk

Answer key

33.6.6 Disk: NIELIT 2022 April Scientist B | Section B | Question: 118

An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected)

A. 0.50 s
B. 1.50 s
C. 1.25 s
D. 1.00 s

nielit2022apr-scientistb operating-system disk co-and-architecture

33.6.7 Disk: NIELIT 2022 April Scientist B | Section B | Question: 45

Consider a hard disk with 16 recording surfaces $(0-15)$ having 16384 cylinders $(0-16383)$ and each cylinder contains 64 sectors $(0-63)$. Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder number, surface number, sector number>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is $< 1200, 9, 40 >$. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

A. 1281 B. 1282 C. 1283 D. 1284

nielit2022apr-scientistb operating-system disk normal

33.7 Disk Scheduling (4)

33.7.1 Disk Scheduling: NIELIT 2017 July Scientist B (CS) - Section B: 35

A disk has 200 tracks (numbered 0 through 199). At a given time, it was servicing the request of reading data from track 120, and at the previous request, service was for track 90. The pending requests (in order of their arrival) are for track numbers.

$30\ 70\ 115\ 130\ 110\ 80\ 20\ 25$

How many times will the head change its direction for the disk scheduling policies SSTF(Shortest Seek Time First) and FCFS (First Come First Serve)?

A. 2 and 3 B. 3 and 3 C. 3 and 4 D. 4 and 4

nielit2017july-scientistb-cs operating-system disk-scheduling

Answer key ☟

33.7.2 Disk Scheduling: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 35

Disk request come to a disk driver for cylinders in the order $10, 22, 20, 2, 40, 6$ and 38, at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms per cylinder. The total seek time, if the disk arm scheduling algorithm is first-come-first-served is

A. 900 ms B. 850 ms C. 360 ms D. 876 ms

nielit2017oct-assistanta-cs operating-system disk-scheduling

Answer key ☟

33.7.3 Disk Scheduling: NIELIT 2021 Dec Scientist B - Section B: 70

Consider a memory system where a request comes for disk driver for cylinders $10, 22, 20, 2, 40, 6$ and 38 (head start at 20). The seek time is 6 ms per cylinder. The total seek time if the disk arm scheduling algorithm is FCFS :

A. 850 ms B. 906 ms C. 400 ms D. 876 ms

nielit-2021-it-dec-scientistb operating-system disk-scheduling

33.7.4 Disk Scheduling: NIELIT Scientist B 2020 November: 72

One disk queue with requests for I/O to blocks on cylinders. The Request are in the following manner:

$98\ 183\ 37\ 122\ 14\ 124\ 65\ 67$

Considering $SSTF$ (shortest seek time first) scheduling, the total number of head movements is, if the disk head is initially at 53 is:

A. 236 B. 246 C. 220 D. 240

nielit-scb-2020 process-scheduling operating-system disk-scheduling

Answer key☟

33.8 Effective Memory Access (1)

33.8.1 Effective Memory Access: NIELIT Scientist B 2020 November: 85

If main memory access time is $400\ \mu s$, TLB access time $50\ \mu s$, considering TLB hit 90%, what wil be the overall access time?

A. $800\ \mu s$ B. $490\ \mu s$ C. $485\ \mu s$ D. $450\ \mu s$

nielit-scb-2020 virtual-memory effective-memory-access operating-system

Answer key☟

33.9 File Organization (2)

33.9.1 File Organization: NIELIT 2016 MAR Scientist B - Section C: 35

The file structure that redefines its first record at a base of zero uses the term

A. relative organization. B. key fielding.
C. dynamic reallocation. D. all of these.

nielit2016mar-scientistb operating-system file-system file-organization

Answer key☟

33.9.2 File Organization: NIELIT 2017 DEC Scientific Assistant A - Section B: 47

The open file table has a/an __________ associated with each file.

A. file content B. file permission C. open count D. close count

nielit2017dec-assistanta operating-system file-organization

Answer key☟

33.10 File System (3)

33.10.1 File System: NIELIT 2016 MAR Scientist C - Section C: 2

The file manager is responsible for

A. naming files
B. saving files
C. deleting files
D. all of these

nielit2016mar-scientistc operating-system file-system

Answer key☟

33.10.2 File System: NIELIT 2017 DEC Scientific Assistant A - Section B: 7

If file size is large and if it is to be accessed randomly then which of the following allocation strategy should be best to use in a system?

A. Linked allocation
B. Indexed allocation
C. Contiguous allocation
D. None of the options

nielit2017dec-assistanta operating-system file-system

Answer key☟

33.10.3 File System: NIELIT Scientist B 2020 November: 104

Which table is used in MS DOS for linked list allocation?

A. TLB
B. Page Table
C. FAT
D. Index Table

nielit-scb-2020 file-system linked-list

Answer key☟

33.11 Fragmentation (2)

33.11.1 Fragmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 29

What is Compaction ?

A. a technique for overcoming internal fragmentation
B. a paging technique
C. a technique for overcoming external fragmentation
D. a technique for overcoming fatal error

nielit2017dec-assistanta operating-system fragmentation

Answer key☟

33.11.2 Fragmentation: NIELIT Scientist B 2020 November: 49

Contiguous memory allocation having variable size partition suffers from:

A. External Fragmentation
B. Internal Fragmentation
C. Both External and Internal Fragmentation
D. None of the options

nielit-scb-2020 operating-system fragmentation memory-management

Answer key☟

33.12 Memory Management (9)

33.12.1 Memory Management: NIELIT 2016 DEC Scientist B (IT) - Section B: 51

Copying a process from memory to disk to allow space for other processes is called:

A. Swapping B. Demand Paging C. Deadlock D. Page Fault

nielit2016dec-scientistb-it operating-system memory-management

Answer key☟

33.12.2 Memory Management: NIELIT 2016 MAR Scientist C - Section C: 8

If there are 32 segments, each of size 1 K byte, then the logical address should have

A. 13 bits
B. 14 bits
C. 15 bits
D. 16 bits

nielit2016mar-scientistc operating-system memory-management

Answer key☟

33.12.3 Memory Management: NIELIT 2017 DEC Scientist B - Section B: 16

Which of the following is added to the page table in order to track whether a page of cache has been modified since it was read from the memory?

A. Reference bit B. Dirty bit C. Tag bit D. Valid bit

nielit2017dec-scientistb operating-system memory-management paging cache-memory

Answer key☟

33.12.4 Memory Management: NIELIT 2017 July Scientist B (CS) - Section B: 32

A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a Translation Look-aside Buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

A. 11 bits B. 13 bits C. 15 bits D. 20 bits

nielit2017july-scientistb-cs operating-system memory-management paging translation-lookaside-buffer

Answer key☟

33.12.5 Memory Management: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 4

If a processor has 32-bit virtual address, 28-bit physical address, 2 kb pages. How many bits are required for the virtual, physical page number?

A. $17, 21$ B. $21, 17$ C. $6, 10$ D. None

nielit2017oct-assistanta-cs operating-system memory-management

Answer key☟

33.12.6 Memory Management: NIELIT 2018-38

Find the effective access time for the memory for the given data.

Page fault service time $= 8$ ms

Average memory access time $= 20$ ns

One page fault is generated for every memory access $= 10^6$

A. 29 ns B. 33 ns C. 28 ns D. 30 ns

nielit-2018 operating-system memory-management

Answer key☟

33.12.7 Memory Management: NIELIT 2021 Dec Scientist B - Section B: 69

If the size of the logical address space is 2^m and the page size is 2^n addressing units, then the high order m-n bits of a logical address designate the ____________ .

A. offset B. page no C. frame no D. physical address

nielit2021dec-scientistb operating-system memory-management

Answer key☟

33.12.8 Memory Management: NIELIT 2021 Dec Scientist B - Section B: 76

In a paging scheme if page size is of 2048 bytes, then while accommodating a process of $72, 766 \text{ bytes}$, how much internal fragmentation occurs ?

A. 962 bytes
B. 2048 bytes
C. 1024 bytes
D. 1086 bytes

nielit2021dec-scientistb operating-system memory-management

Answer key☟

33.12.9 Memory Management: NIELIT Scientific Assistant A 2020 November: 70

A computer has a single cache (off-chip) with a 3 ns hit time and a 95% hit rate. Main memory has a 50 ns access time. If we add an on-chip cache with a 0.6 ns hit time and a 98% hit rate, the computer's effective access time:

A. 2.8 ns
B. 5.5 ns
C. 0.7 ns
D. None of the options

nielit-sta-2020 operating-system memory-management

Answer key☟

33.13 Monitors (1)

33.13.1 Monitors: NIELIT 2016 MAR Scientist B - Section C: 37

A long-term monitor

A. should show any immediate performance problems
B. should show I/O, paging and processor activity
C. need show only the I/O and processor activity
D. usually reports only on terminal displays

nielit2016mar-scientistb operating-system monitors

Answer key☟

33.14 Page Replacement (9)

33.14.1 Page Replacement: NIELIT 2016 DEC Scientist B (CS) - Section B: 48

In which one of the following pages replacement policies, Belady's anomaly may occur?

A. FIFO B. LRU C. Optimal D. MRU

nielit2016dec-scientistb-cs operating-system page-replacement

Answer key☟

33.14.2 Page Replacement: NIELIT 2016 MAR Scientist B - Section C: 38

Determine the number of page faults when references to pages occur in the following order: $1, 2, 4, 5, 2, 1, 2, 4$. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 having been brought earlier than page 2.(LRU algorithm is used).

A. 3 B. 5 C. 4 D. None of these.

nielit2016mar-scientistb operating-system page-replacement page-fault least-recently-used

Answer key☟

33.14.3 Page Replacement: NIELIT 2016 MAR Scientist B - Section C: 39

Working set(t, k) at an instant of time, t,is

A. The set of k future references that the operating system will make
B. The set of future references that the operating system will make in the next $'k'$ time units
C. The set of k references with high frequency
D. The set of pages that have been referenced in the last k time units

nielit2016mar-scientistb operating-system page-replacement

Answer key

33.14.4 Page Replacement: NIELIT 2017 DEC Scientist B - Section B: 51

The total number of page faults for the reference string $1, 2, 3, 4, 5, 6, 7, 8, 9, 10$ using FIFO page replacement policy for a process, if 3 frames are allocated to it are

A. 9 B. 10 C. 8 D. 11

nielit2017dec-scientistb operating-system page-replacement page-fault

Answer key

33.14.5 Page Replacement: NIELIT 2021 Dec Scientist A - Section B: 102

Consider a system with page size p and average process size m and size of each page table entry is e. What is the amount of space required by page table ?

A. $\mathrm{me/p}$ B. $\mathrm{mp/e}$ C. mpe D. $\mathrm{pe/m}$

nielit2021dec-scientista operating-system page-replacement memory-management

Answer key

33.14.6 Page Replacement: NIELIT 2021 Dec Scientist A - Section B: 107

Consider a system with three frames in memory and following memory references in the working set

$$2 \quad 1 \quad 2 \quad 3 \quad 5 \quad 4 \quad 1 \quad 3 \quad 4 \quad 2 \quad 1$$

How many page fault will be there if we use second chance page replacement algorithm?

A. 7 B. 8 C. 9 D. 10

nielit2021dec-scientista operating-system page-replacement memory-management

33.14.7 Page Replacement: NIELIT 2021 Dec Scientist A - Section B: 114

Which of the following Page Replacement Algorithm suffers from the Belady's anomaly ?

A. LRU
B. Optimal page Replacement
C. FIFO
D. Both LRU and FIFO

nielit2021dec-scientista operating-system page-replacement

Answer key

33.14.8 Page Replacement: NIELIT 2022 April Scientist B | Section B | Question: 78

Consider the virtual page reference string

$1, 2, 3, 2, 4, 1, 3, 2, 4, 1$

On a demand paged virtual memory system running on a computer system that main memory size of 3 pages frames which are initially empty. Let $\mathrm{LRU}, \mathrm{FIFO}$ and

$\mathrm{OPTIMAL}$ denote the number of page faults under the corresponding page replacement policy. Then

A. $\mathrm{OPTIMAL} < \mathrm{LRU} < \mathrm{FIFO}$
B. $\mathrm{OPTIMAL} < \mathrm{FIFO} < \mathrm{LRU}$
C. $\mathrm{OPTIMAL} = \mathrm{LRU}$
D. $\mathrm{OPTIMAL} = \mathrm{FIFO}$

nielit2022apr-scientistb operating-system page-replacement virtual-memory

33.14.9 Page Replacement: NIELIT Scientist B 2020 November: 46

Which of the following Page Replacement Algorithm suffers from the belady's anomaly?

A. LRU
B. Optimal Page Replacement
C. $FIFO$
D. Both LRU and $FIFO$

nielit-scb-2020 operating-system page-replacement

Answer key☟

33.15 Process (2)

33.15.1 Process: NIELIT 2022 April Scientist B | Section B | Question: 61

Consider three processes, all arriving at time zero, with total execution time of $10, 20$ and 30 units, respectively. Each process spends the first 20% of execution time doing $\mathrm{I/O}$, the next 70% of time doing computation, and the last 10% of time doing $\mathrm{I/O}$ again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on $\mathrm{I/O}$ or when the running process finishes its compute burst. Assume that all $\mathrm{I/O}$ operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

A. 0%
B. 10.6%
C. 30.0%
D. 89.4%

nielit2022apr-scientistb operating-system process-scheduling process shortest-path

33.15.2 Process: NIELIT Scientist B 2020 November: 86

In an operating system, processes that are terminated but, for some reason must have its task structure in the process table are referred as __________.

A. Zombies
B. Orphans
C. Parent Process
D. Child Process

nielit-scb-2020 operating-system process

Answer key☟

33.16 Process Scheduling (13)

33.16.1 Process Scheduling: NIELIT 2016 DEC Scientist B (IT) - Section B: 56

The degree of multi programming is controlled by:

A. CPU Scheduler
B. Long-term Scheduler
C. Context Switching
D. Medium term Scheduler

nielit2016dec-scientistb-it operating-system process-scheduling

Answer key

33.16.2 Process Scheduling: NIELIT 2016 MAR Scientist C - Section C: 7

In real-time operating systems, which of the following is the most suitable scheduling scheme?

A. round-robin
B. first-come-first-served
C. preemptive
D. random scheduling

nielit2016mar-scientistc operating-system process-scheduling

Answer key

33.16.3 Process Scheduling: NIELIT 2017 DEC Scientific Assistant A - Section B: 34

Process is in a ready state _______ .

A. when process is scheduled to run after some execution
B. when process is unable to run until some task has been completed
C. when process is using the CPU
D. none of the options

nielit2017dec-assistanta operating-system process-scheduling

Answer key

33.16.4 Process Scheduling: NIELIT 2017 DEC Scientific Assistant A - Section B: 57

Starvation can be avoided by which of the following statements:

i. By using shortest job first resource allocation policy .
ii. By using first come first serve resources allocation policy.

A. (i) only
B. (i) and (ii) only
C. (ii) only
D. None of the options

nielit2017dec-assistanta operating-system process-scheduling

Answer key

33.16.5 Process Scheduling: NIELIT 2017 DEC Scientist B - Section B: 12

Consider the following four processes with their corresponding arrival time and burst time:

Process	Arrival time	Burst time(in ms)
P1	0.0	8
P2	0.6	6
P3	3.8	4
P4	4.4	2

What is the average turn around time (in ms) for these processes using FCFS scheduling algorithm?

A. 15　　B. 12.8　　C. 13　　D. none of the options

nielit2017dec-scientistb　operating-system　process-scheduling

Answer key☟

33.16.6 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 44

Consider three processes (process id $0, 1, 2$ respectively) with compute time bursts $2, 4$ and 8 time units. All processes arrive at time zero. Consider the Longest Remaining Time First (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is

A. 13 units　　B. 14 units　　C. 15 units　　D. 16 units

nielit2017july-scientistb-it　operating-system　process-scheduling

Answer key☟

33.16.7 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 45

Consider three processes, all arriving at time zero, with total execution time of $10, 20$ and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

A. 0%　　B. 10.6%　　C. 30.0%　　D. 89.4%

nielit2017july-scientistb-it　operating-system　process-scheduling

Answer key☟

33.16.8 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 46

Consider three CPU-intensive processes, which require $10, 20$ and 30 time units and arrive at times $0, 2$ and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

A. 1 B. 2 C. 3 D. 4

nielit2017july-scientistb-it operating-system process-scheduling

Answer key☟

33.16.9 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 47

Which of the following process scheduling algorithm may lead to starvation?

A. FIFO
B. Round Robin
C. Shortest Job Next
D. None of the option

nielit2017july-scientistb-it operating-system process-scheduling

Answer key☟

33.16.10 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 48

A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

A. This algorithm is equivalent to the first come first serve algorithm.
B. This algorithm is equivalent to the round-robin algorithm.
C. This algorithm is equivalent to the shortest-job-first algorithm.
D. This algorithm is equivalent to the shortest-remaining time-first algorithm.

nielit2017july-scientistb-it operating-system process-scheduling

Answer key☟

33.16.11 Process Scheduling: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 9

Which of the following scheduling algorithms could result in starvation?

A. Priority B. Round Robin C. FCFS D. None of the above

nielit2017oct-assistanta-it operating-system process-scheduling

Answer key☟

33.16.12 Process Scheduling: NIELIT 2018-78

Identify the true statement from the given statements.

1. FIFO is non-preemptive
2. Round robin is non-preemptive
3. Multilevel Queue Scheduling is non-preemptive

A. 1 B. 1 and 2 C. $1, 2$ and 3 D. 2

nielit-2018 operating-system process-scheduling

Answer key

33.16.13 Process Scheduling: NIELIT 2021 Dec Scientist B - Section B: 57

___________ is the elapsed time between the time a program or job is submitted and the time when it is completed.

A. Response time
B. Turnaround time
C. Waiting time
D. Throughput

nielit2021dec-scientistb operating-system process-scheduling

Answer key

33.17 Process Synchronization (2)

33.17.1 Process Synchronization: NIELIT 2016 DEC Scientist B (CS) - Section B: 51

A process that is based on IPC mechanism which executes on different systems and can communicate with other processes using message based communication is called _______.

A. Local Procedure Call
B. Remote Procedure Call
C. Inter Process Communication
D. Remote Machine Invocation

nielit2016dec-scientistb-cs operating-system process-synchronization

Answer key

33.17.2 Process Synchronization: NIELIT 2016 DEC Scientist B (IT) - Section B: 50

Which of the following statements about semaphores is true?

A. P and V operations should be indivisible operations.
B. A semaphore implementation should guarantee that threads do not suffer indefinite postponement.
C. If several threads attempt a $P(S)$ operation simultaneously, only one thread should be allowed to proceed.
D. All of the above.

nielit2016dec-scientistb-it operating-system process-synchronization

Answer key

33.18 Process and Threads (2)

33.18.1 Process and Threads: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 23

Which two are valid constructions for Thread?

1. Thread(Runnable r, String name)
2. Thread()
3. Thread(int priority)

4. Thread(Runnable r, ThreadGroup g)
5. Thread(Runnable r, int priority)

A. 1 and 3 B. 2 and 4 C. 1 and 2 D. 2 and 5

nielit2017oct-assistanta-it operating-system process-and-threads threads

Answer key☟

33.18.2 Process and Threads: NIELIT 2021 Dec Scientist A - Section B: 120

Which one of the following cannot be scheduled by the kernel ?

A. Kernel level thread
B. User level thread
C. Process
D. None of the option

nielit2021dec-scientista operating-system process-scheduling process-and-threads

Answer key☟

33.19 Resource Allocation (6)

33.19.1 Resource Allocation: NIELIT 2016 DEC Scientist B (CS) - Section B: 56

Consider a system with m resources of same type being shared by n processes. Resources can be requested and released by processes only one at a time. The system is deadlock free if and only if

A. The sum of all max needs is $< m+n$
B. The sum of all max needs is $> m+n$
C. Both of above
D. None

nielit2016dec-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key☟

33.19.2 Resource Allocation: NIELIT 2017 DEC Scientist B - Section B: 21

A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units then, deadlock

A. Can never occur
B. Has to occur
C. May occur
D. None of the options

nielit2017dec-scientistb operating-system resource-allocation

Answer key☟

33.19.3 Resource Allocation: NIELIT 2017 July Scientist B (CS) - Section B: 36

Consider the following snapshot of a system running n processes. Process i is holding X_i instances of a resource R, $1 \leq i \leq n$. Currently, all instances of R are occupied. Further, for all i, process i has placed a request for an additional Y_i instances while holding the X_i instances it already has. There are exactly two processes p and q and such that $Y_p = Y_q = 0$. Which one of the following can serve as a necessary condition to guarantee that the system is not approaching a deadlock?

A. $\min(X_p, X_q) < \max(Y_i)$ where $i! = p$ and $i! = q$
B. $X_q \geq \min(Y_i)$ where $i! = p$ and $i! = q$
C. $\max(X_p, X_q) > 1$
D. $\min(X_p, X_q) > 1$

nielit2017july-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.19.4 Resource Allocation: NIELIT 2017 July Scientist B (CS) - Section B: 37

A system has n resources $R_0, \ldots, R_{n-1}$, and k processes $P_0, \ldots, P_{k-1}$. The implementation of the resource request logic of each process P_i is as follows:

```
if(i%2==0){
   if(i<n) request Ri;
   if(i+2<n) request Ri+2;
}
else{
   if(i<n) request Rn-i;
   if(i+2<n) request Rn-i-2;
}
```

In which of the following situations is a deadlock possible?

A. $n = 40, k = 26$ B. $n = 21, k = 12$ C. $n = 20, k = 10$ D. $n = 41, k = 19$

nielit2017july-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.19.5 Resource Allocation: NIELIT 2017 July Scientist B (CS) - Section B: 38

A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is _________.

A. 6 B. 7 C. 8 D. 9

nielit2017july-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.19.6 Resource Allocation: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 15

Consider a system having $'m'$ resources of the same type. These resources are shared by 3 processes A, B, C; which have peak time demands of $3, 4, 6$ respectively. The minimum value of $'m'$ that ensures that deadlock will never occur is

A. 11 B. 12 C. 13 D. 14

nielit2017oct-assistanta-it operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.20 Segmentation (1)

33.20.1 Segmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 17

Non-contiguous memory allocation splits program into blocks of memory called ________ that can be loaded in non-adjacent holes in main memory.

A. Pages B. Frames C. Partition D. Segments

nielit2017dec-assistanta operating-system page-replacement segmentation

Answer key

33.21 Semaphore (3)

33.21.1 Semaphore: NIELIT 2018-57

In a system, counting semaphore was initialized to 10, then $6P$(wait) operations and $4V$ (signal) operations were completed on this semaphore. So _____ is the final value of the semaphore.

A. 7 B. 8 C. 13 D. 12

nielit-2018 operating-system semaphore

Answer key

33.21.2 Semaphore: NIELIT 2021 Dec Scientist B - Section B: 79

A protected variable which can be accessed and changed by particular set of operation is called **:**

A. interrupt B. monitor C. semaphore D. IPC

nielit2021dec-scientistb operating-system process-synchronization semaphore

Answer key

33.21.3 Semaphore: NIELIT 2022 April Scientist B | Section B | Question: 110

The atomic *fetch-and-set* x, y instruction unconditionally sets the memory location x to 1 and fetches the old value of x in y without allowing any intervening access to the memory location x. Consider the following implementation of P and V functions on a binary semaphore S.

```
void P (binary_semaphore *s) {
   unsigned y;
   unsigned *x = &(s->value);
   do {
      fetch-and-set x, y;
   } while (y);
}

void V (binary_semaphore *s) {
   S->value = 0;
}
```

Which one of the following is true?

A. The implementation may not work if context switching is disabled in P
B. Instead of using *fetch-and –set*, a pair of normal load/store can be used
C. The implementation of V is wrong
D. The code does not implement a binary semaphore

nielit2022apr-scientistb operating-system process-synchronization semaphore

33.22 System Calls (1)

33.22.1 System Calls: NIELIT 2017 DEC Scientific Assistant A - Section B: 37

How many times the word "PROCESS" will be printed when executing the following program ?

```
main(){
    printf("PROCESS");
    fflush();
    fork();
    fork();
}
```

A. 8 B. 4 C. 6 D. 7

nielit2017dec-assistanta operating-system system-calls

Answer key☟

33.23 Threads (2)

33.23.1 Threads: NIELIT 2017 DEC Scientific Assistant A - Section B: 9

Operating System maintains the page table for :

A. each process
B. each thread
C. each instruction
D. each address

nielit2017dec-assistanta operating-system paging threads

Answer key☟

33.23.2 Threads: NIELIT 2021 Dec Scientist B - Section B: 40

Choose the correct sequence of items mentioned in **List-1** out of items mentioned in **List-2** :

List-1	**List-2**
(a) Threads	(i) operating system
(b) Non pre-emptive	(ii) first come first serve
(c) Pre-emptive	(iii) round robin algorithm

(d) Scheduling	(iv) Context switch
(e) Process switch	(v) light weight process with in reduced time

	(a)	(b)	(c)	(d)	(e)
(A)	(v)	(iii)	(iv)	(ii)	(i)
(B)	(ii)	(iii)	(v)	(iv)	(i)
(C)	(v)	(ii)	(iii)	(i)	(iv)
(D)	(ii)	(iv)	(v)	(i)	(ii)

nielit-2021-it-dec-scientistb nielit2021dec-scientistb operating-system process-scheduling threads process-synchronization

Answer key☟

33.24 Virtual Memory (6)

33.24.1 Virtual Memory: NIELIT 2017 July Scientist B (CS) - Section B: 33

A computer uses 46-bit virtual address, 32-bit physical address, and a three–level paged page table organization. The page table base register stores the base address of the first-level table $(T1)$, which occupies exactly one page. Each entry of $T1$ stores the base address of a page of the second-level table $(T2)$. Each entry of $T2$ stores the base address of a page of the third-level table $(T3)$. Each entry of $T3$ stores a Page Table Entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16 way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

What is the size of a page in KB in this computer?

A. 2 B. 4 C. 8 D. 16

nielit2017july-scientistb-cs operating-system virtual-memory

Answer key☟

33.24.2 Virtual Memory: NIELIT 2017 July Scientist B (CS) - Section B: 34

A computer uses $46-bit$ virtual address, $32-bit$ physical address, and a three–level paged page table organization. The page table base register stores the base address of the first-level table ($T1$), which occupies exactly one page. Each entry of $T1$ stores the base address of a page of the second-level table ($T2$). Each entry of $T2$ stores the base address of a page of the third-level table ($T3$). Each entry of $T3$ stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16 way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

What is the minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?

A. 2 B. 4 C. 8 D. 16

nielit2017july-scientistb-cs operating-system virtual-memory

33.24.3 Virtual Memory: NIELIT 2018-48

Decreasing the RAM causes

A. fewer page faults
B. more page faults
C. virtual memory gets increased
D. virtual memory gets decreased

nielit-2018 operating-system virtual-memory

Answer key☟

33.24.4 Virtual Memory: NIELIT 2021 Dec Scientist B - Section B: 42

What is virtual memory?

A. An illusion of an extremely large memory, that is physically not present
B. A type of memory used in nano computers
C. A memory that is in direct touch of CPU
D. A memory that works on the principle of set associative mapping

nielit-2021-it-dec-scientistb operating-system virtual-memory

Answer key☟

33.24.5 Virtual Memory: NIELIT 2021 Dec Scientist B - Section B: 53

A page fault occurs when:

A. the Deadlock happens
B. the Segmentation starts
C. the page is found in the memory
D. the page is not found in the memory

nielit-2021-it-dec-scientistb operating-system virtual-memory

Answer key☟

33.24.6 Virtual Memory: NIELIT 2021 Dec Scientist B - Section B: 77

Consider a memory system in which the size of page is 8 KB. A CPU that generates the 32 bit virtual address. What will be minimum size of TLB (translation look aside buffer) tag, if TLB has total 256 page table and is 8 way set associative?

A. 12 bit B. 13 bit C. 14 bit D. 15 bit

nielit-2021-it-dec-scientistb co-and-architecture memory-management virtual-memory

Answer key☟

Answer Keys

33.0.1	C	33.0.2	B	33.0.3	A	33.0.4	C	33.0.5	C
33.0.6	D	33.1.1	C	33.2.1	B	33.2.2	B	33.3.1	A
33.3.2	A	33.4.1	B	33.4.2	D	33.5.1	C	33.6.1	TBA
33.6.2	A	33.6.3	D	33.6.4	B	33.6.5	B	33.6.6	B
33.6.7	D	33.7.1	C	33.7.2	D	33.7.3	D	33.7.4	A
33.8.1	B	33.9.1	A	33.9.2	C	33.10.1	D	33.10.2	B
33.10.3	C	33.11.1	C	33.11.2	A	33.12.1	A	33.12.2	C
33.12.3	B	33.12.4	C	33.12.5	B	33.12.6	C	33.12.7	B
33.12.8	A	33.12.9	C	33.13.1	B	33.14.1	A	33.14.2	C
33.14.3	D	33.14.4	B	33.14.5	A	33.14.6	C	33.14.7	C
33.14.8	B	33.14.9	C	33.15.1	B	33.15.2	A	33.16.1	B
33.16.2	C	33.16.3	A	33.16.4	C	33.16.5	B	33.16.6	A
33.16.7	B	33.16.8	B	33.16.9	C	33.16.10	B	33.16.11	A
33.16.12	A	33.16.13	B	33.17.1	B	33.17.2	D	33.18.1	C
33.18.2	B	33.19.1	A	33.19.2	A	33.19.3	B	33.19.4	B
33.19.5	A	33.19.6	A	33.20.1	D	33.21.1	B	33.21.2	C
33.21.3	A	33.22.1	X	33.23.1	A	33.23.2	C	33.24.1	C
33.24.2	C	33.24.3	B	33.24.4	A	33.24.5	D	33.24.6	D

34.0.1 NIELIT Scientific Assistant June 2025 | Question: 30

30. Which chemical element is represented as "Hg" in the periodic table ?
(A) Silver (B) Tin (C) Copper (D) Mercury

30. Which chemical element is represented as " Hg " in the periodic table ?

A. Silver B. Tin C. Copper D. Mercury

nielit-sta-2025

34.0.2 NIELIT Scientific Assistant June 2025 | Question: 29

29. If you are facing west and turn 90° clockwise, then 180° anti-clockwise, which direction will you be facing ?
(A) North (B) East (C) South (D) West

29. If you are facing west and turn 90° clockwise, then 180° anti-clockwise, which direction will you be facing ?
(A) North
(B) East
(e) South
(D) West

nielit-sta-2025

34.0.3 NIELIT Scientific Assistant June 2025 | Question: 28

28. Maya is younger than Priya. Priya is elder than Rani but younger than Arjun. Arjun is elder than Komal. Who is the eldest ?
(A) Arjun (B) Maya (C) Priya (D) Rani

28. Maya is younger than Priya. Priya is elder than Rani but younger than Arjun. Arjun is elder than Komal. Who is the eldest ?

A. Arjun B. Maya C. Priya D. Rani

nielit-sta-2025

34.0.4 NIELIT Scientific Assistant June 2025 | Question: 27

27. The simple interest on Rs. 1000 for 2 years at 5% per annum is :
(A) Rs. 50 (B) Rs. 100 (C) Rs. 75 (D) Rs. 60

27. The simple interest on Rs. 1000 for 2 years at 5% per annum is : 1

A. Rs. 50 B. Rs. 100 C. Rs. 75 D. Rs. 60

nielit-sta-2025

34.0.5 NIELIT Scientific Assistant June 2025 | Question: 26

26. If a train travels 360 km at a speed of 90 km/h, how long does it take to complete the journey ?
(A) 5 hours (B) 4 hours (C) 3 hours (D) 6 hours

26. If a train travels 360 km at a speed of $90 \mathrm{~km/h}$, how long does it take to complete

the journey ?

A. 5 hours B. 4 hours C. 3 hours D. 6 hours

nielit-sta-2025

34.0.6 NIELIT Scientific Assistant June 2025 | Question: 25

25. The area of a triangle with base 10 cm and height 8 cm is :

A. 30 sq cm B. 40 sq cm C. 60 sq cm D. 80 sq cm

nielit-sta-2025

34.0.7 NIELIT Scientific Assistant June 2025 | Question: 24

24. The difference between compound interest and simple interest on Rs. 8000 at 5\% per annum for 2 years is:

A. Rs. 10 B. Rs. 15 C. Rs. 20 D. Rs .25

nielit-sta-2025

34.0.8 NIELIT Scientific Assistant June 2025 | Question: 23

23. If the perimeter of a rectangle is 40 cm and its length is 12 cm , find its breadth.

A. 8 cm B. 10 cm C. 9 cm D. 11 cm

nielit-sta-2025

34.0.9 NIELIT Scientific Assistant June 2025 | Question: 22

22. Who won the 2024 Formula 1 Drivers' Championship ?

A. George Russell B. Max Verstappen
C. Charles Leclerc D. Sergio Pérez

nielit-sta-2025

34.0.10 NIELIT Scientific Assistant June 2025 | Question: 21

21. Find the missing number: $12, 24, 48, 96$?

A. 190 B. 192 C. 194 D. 196

34.0.11 NIELIT Scientific Assistant June 2025 | Question: 20

20. If there are 8 teams in a tournament and each team plays with every other team once, how many matches are played ?
(A) 36 (B) 30 (C) 32 (D) 28

20. If there are 8 teams in a tournament and each team plays with every other team once, how many matches are played ?

A. 36 B. 30 C. 32 D. 28

nielit-sta-2025

34.0.12 NIELIT Scientific Assistant June 2025 | Question: 19

19. If the simple interest on a sum for 3 years at 10% per annum is Rs. 600, find the principal amount?
(A) Rs. 3000 (B) Rs. 5000 (C) Rs. 2000 (D) Rs. 2500

19. If the simple interest on a sum for 3 years at 10% per annum is Rs. 600 , find the principal amount?

A. Rs. 3000 B. Rs. 5000 C. Rs. 2000 D. Rs .2500

nielit-sta-2025

34.0.13 NIELIT Scientific Assistant June 2025 | Question: 18

18. Find the missing number :
2, 6, 12, 20, ?
(A) 28 (B) 30 (C) 32 (D) 40

18. Find the missing number: 2, 6, 12, 20, ?
(A) 28
(B) 30°

32
(D) 40

nielit-sta-2025

34.0.14 NIELIT Scientific Assistant June 2025 | Question: 17

17. Select the correct option to improve the sentence :
He has been working here since four years.
(A) working here for (B) working here from
(C) working here while (D) No improvement

17. Select the correct option to improve the sentence:

He has been working here since four years.
(A) working here for
(B) working here from
(C) working here while
(B) No improvement

34.0.15 NIELIT Scientific Assistant June 2025 | Question: 16

16. What is the sum of the first 20 natural numbers ?
(A) 200 (B) 210 (C) 220 (D) 420

16. What is the sum of the first 20 natural numbers?

A. 200 B. 210 C. 220 D. 420

nielit-sta-2025

34.0.16 NIELIT Scientific Assistant June 2025 | Question: 15

15. Nayab Saini was sworn in as the 11th Chief Minister of which State ?
(A) Punjab (B) Uttarakhand (C) Rajasthan (D) Haryana

15. Nayab Saini was swom in as the 11^{th} Chief Minister of which State?

A. Punjab B. Uttarakhand C. Rajasthan D. Haryana

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34.0.17 NIELIT Scientific Assistant June 2025 | Question: 14

14. Which pair is the odd one out ?
(A) Apple Fruit
(B) Dog Animal
(C) Rose Flower
(D) Car Road

14. Which pair is the odd one out?

A. Apple Fruit B. Dog Animal C. Rose Flower D. Car Road

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34.0.18 NIELIT Scientific Assistant June 2025 | Question: 13

13. A man walks 5 km north, then turns right and walks 3 km, then turns right again and walks 5 km. Which direction is he facing ?
(A) East (B) South (C) West (D) North

13. A man walks 5 km north, then turns right and walks 3 km , then turns right again and walks 5 km . Which direction is he facing ?
(算) East
(B) South
(C) West
(D) North

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34.0.19 NIELIT Scientific Assistant June 2025 | Question: 12

12. In a certain code language, LION is written as 50, then TIGER is written as ______
(A) 52 (B) 59 (C) 58 (D) 61

12. In a certain code language, LION is written as 50 , then TIGER is written as

A. 52 B. 59 C. 58 D. 61

34.0.20 NIELIT Scientific Assistant June 2025 | Question: 11

11. In the given question, select the related word from the given alternatives.
Moon : Satellite : : Earth : ?
(A) Sun (B) Solar system (C) Planet (D) Round

11. In the given question, select the related word from the given alternatives.

Moon : Satellite :: Earth : ?

A. Sun
B. Solar system
C. Planet
D. Round

34.0.21 NIELIT Scientific Assistant June 2025 | Question: 10

10. Who wrote the novel "War and Peace" ?
(A) Anton Chekhov (B) Fyodor Dostoevsky
(C) Leo Tolstoy (D) Ivan Turgenev

10. Who wrote the novel "War and Peace" ?

A. Anton Chekhov
B. Fyodor Dostoevsky
C. Leo Tolstoy
D. Ivan Turgenev

34.0.22 NIELIT Scientific Assistant June 2025 | Question: 9

9. What is the capital of Canada ?
(A) Toronto (B) Vancouver (C) Ottawa (D) Montreal

9. What is the capital of Canada ?

A. Toronto
B. Vancouver
C. Ottawa
D. Montreal

34.0.23 NIELIT Scientific Assistant June 2025 | Question: 8

8. Who was declared as the winner of the Pritzker Architecture Prize 2024 ?
(A) Riken Yamamoto (B) David Chipperfield
(C) B.V. Doshi (D) Yvonne Farrell

8. Who was declared as the winner of the Pritzker Architecture Prize 2024 ?

A. Riken Yamamoto
B. David Chipperfield
C. B.V. Doshi
D. Yvonne Farrell

34.0.24 NIELIT Scientific Assistant June 2025 | Question: 7

7. Amit is elder than Rohit, but younger than Suman. Suman is younger than Neha. Rohit is elder than Kabir. Who is the youngest among them ?
(A) Rohit (B) Suman (C) Kabir (D) None of the options

7. Amit is elder than Rohit, but younger than Suman. Suman is younger than Neha. Rohit is elder than Kabir. Who is the youngest among them ?

A. Rohit B. Suman C. Kabir D. None of the options

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34.0.25 NIELIT Scientific Assistant June 2025 | Question: 6

6. Find the missing term :

BDF, CEG, DFH, ____ FIK

A. EGI B. EGH C. EHI D. DGI

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34.0.26 NIELIT Scientific Assistant June 2025 | Question: 5

5. Find the next term in the series: $7, 14, 28, 56, 4$

A. 64 B. 110 C. 115 D. 112

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34.0.27 NIELIT Scientific Assistant June 2025 | Question: 4

4. Which is the largest lake in the World ?

A. Caspian Sea B. Baikal C. Lake Superior D. Ontario

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34.0.28 NIELIT Scientific Assistant June 2025 | Question: 3

3. What is the next term in the series: $3, 9, 27, 81$, ?

A. 162 B. 243 C. 99 D. 150

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34.0.29 NIELIT Scientific Assistant June 2025 | Question: 2

2. If $\mathrm{M} * \mathrm{~N}+\mathrm{O}$ means " M is the father of N and N is the mother of O ," then how is M related to O ?

A. Uncle B. Father C. Brother D. Grandfather

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34.0.30 NIELIT Scientific Assistant June 2025 | Question: 1

1. A company sells a product of Rs. 1500 with a 20% discount and has no profit and no loss. If the company wants to make a 20% profit, at what price should the company sell that product ?
(A) Rs. 1400
(B) Rs. 1450
(G) Rs, 1440
(D) Rs. 1300

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34.0.31 NIELIT Scientific Assistant June 2025 | Question: 60

60. Which type of feasibility focuses on whether the hardware and software available can support the proposed system ?

A. Operational Feasibility
B. Technical Feasibility
C. Legal Feasibility
D. Economic Feasibility

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34.0.32 NIELIT Scientific Assistant June 2025 | Question: 59

59. Which of the following is the primary goal of information gathering in system development ?
(A) To design the system architecture
B) To perform system testing
(C) To identify the available hardware and software
(D) To collect detailed user requirements and system needs

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34.0.33 NIELIT Scientific Assistant June 2025 | Question: 58

58. Which of the following is the main purpose of the AM demodulator (envelope detector) ?

A. To convert the modulated signal into the carrier frequency
B. To filter out noise from the modulated signal
C. To generate a new carrier signal
D. To extract the message signal from the modulated signal

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34.0.34 NIELIT Scientific Assistant June 2025 | Question: 57

57. Consider a complete graph G with 4 vertices. The graph G has ________ spanning trees.
(A) 15 (B) 8 (C) 16 (D) 13

57. Consider a complete graph G with 4 vertices. The graph G has ________ spanning trees.

A. 15
B. 8
C. 16
D. 13

nielit-sta-2025

34.0.35 NIELIT Scientific Assistant June 2025 | Question: 56

56. The interval in which the function $f(x)=\sin x+\cos x,\ 0 \le x \le 2\pi$ is strictly increasing or strictly decreasing is :
(A) Strictly increasing for $x\in\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
(B) Strictly increasing for $x\in[0, \pi)$ and strictly decreasing for $x\in(\pi, 2\pi)$
(C) Strictly increasing for $x\in\left[0, \frac{\pi}{4}\right] \cap \left(\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
(D) Strictly decreasing for $x\in\left(0, \frac{\pi}{4}\right) \cup \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly increasing for $x \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

56. The interval in which the function $f(x) = \sin x + \cos x, 0 \leq x \leq 2\pi$ is strictly increasing or strictly decreasing is :

A. Strictly increasing for $x \in \left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
B. Strictly increasing for $x \in [0, \pi)$ and strictly decreasing for $x \in (\pi, 2\pi)$
C. Strictly increasing for $x \in \left[0, \frac{\pi}{4}\right] \cap \left(\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
D. Strictly decreasing for $x \in \left(0, \frac{\pi}{4}\right) \cup \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly increasing for $x \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

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34.0.36 NIELIT Scientific Assistant June 2025 | Question: 55

55. Which of the following is false in the case of a spanning tree of a graph G ?
(A) It is tree that spans G (B) It is a subgraph of the G
(C) It includes every vertex of the G (D) It can be either cyclic or acyclic

55. Which of the following is false in the case of a spanning tree of a graph G ?

A. It is tree that spans G
B. It is a subgraph of the G
C. It includes every vertex of the G
D. It can be either cyclic or acyclic

nielit-sta-2025

34.0.37 NIELIT Scientific Assistant June 2025 | Question: 54

54. A bag contains 'x' red, '$x+5$' blue and '$x+7$' grey balls. If two balls are randomly drawn from the bag and the probability that both the balls are of same colour is $\frac{148}{435}$, then the total number of balls in the bag is :
(A) 36 (B) 64 (C) 58 (D) 30

54. A bag contains 'x' red, '$x+5$' blue and '$x+7$' grey balls. If two balls are randomly drawn from the bag and the probability that both the balls are of same colour is $\frac{148}{435}$, then the total number of balls in the bag is :

A. 36 B. 64 C. 58 D. 30

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34.0.38 NIELIT Scientific Assistant June 2025 | Question: 53

53. $\lim_{x \to \frac{\pi}{2}} \left(\frac{\tan 2x}{x - \frac{\pi}{2}} \right) =$ ________.

(A) 0 (B) 1 (C) 2 (D) Undefined

53. $\lim_{x \to \frac{\pi}{2}} \left(\frac{\tan 2x}{x - \frac{\pi}{2}} \right) =$

A. 0 B. 1 C. 2 D. Undefined

nielit-sta-2025

34.0.39 NIELIT Scientific Assistant June 2025 | Question: 52

52. If $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$ is an eigen vector of $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, the eigen value corresponding to the given eigen vector is :

(A) 1 (B) 4 (C) 6 (D) -4.5

52. If $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$ is an eigen vector of $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, the eigen value corresponding to the given eigen vector is:

A. 1 B. 4 C. 6 D. -4.5

nielit-sta-2025

34.0.40 NIELIT Scientific Assistant June 2025 | Question: 51

51. The approximate value of integral $\int_0^1 (3x+7)dx$ by trapezoidal rule of the step size 0.0001 is ________.

(A) $\frac{3}{2}$ (B) 7 (C) $\frac{17}{2}$ (D) None of the options

51. The approximate value of integral $\int_0^1 (3x+7)dx$ by trapezoidal rule of the step size 0.0001 is

A. $\frac{3}{2}$ B. 7 C. $\frac{17}{2}$ D. None of the options

nielit-sta-2025

34.0.41 NIELIT Scientific Assistant June 2025 | Question: 50

50. Which of the following is simplified sum of product form for the Boolean expression ?
(P+Q'+R') (P+Q'+R) (P+Q+R')

(A) P+QR' (B) P+Q'R (C) P+Q'R' (D) P'+Q'R'

50. Which of the following is simplified sum of product form for the Boolean expression ?
$\left(\mathrm{P}+\mathrm{Q}'+\mathrm{R}'\right)\left(\mathrm{P}+\mathrm{Q}'+\mathrm{R}\right)\left(\mathrm{P}+\mathrm{Q}+\mathrm{R}'\right)$

A. $\mathrm{P}+\mathrm{QR}'$
B. $\mathrm{P}+\mathrm{Q}'\mathrm{R}$
C. $\mathrm{P}+\mathrm{Q}'\mathrm{R}'$
D. $P'+Q'R'$

nielit-sta-2025

34.0.42 NIELIT Scientific Assistant June 2025 | Question: 49

49. If A=Baby is eating, B=Baby is hungry, then the correct propositional logic to the English sentence "Baby is eating and not hungry" is :
(A) $A \Rightarrow \neg B$ (B) $A \wedge \neg B$ (C) $\neg(A \Rightarrow \neg B)$ (D) $A \vee \neg B$

49. If $\mathrm{A}=\mathrm{Baby}$ is eating, $\mathrm{B}=\mathrm{Baby}$ is hungry, then the correct propositional logic to the English sentence

A. $\mathrm{A} \Rightarrow \neg \mathrm{B}$ and not hungry * is :
B. $\mathrm{A} \wedge \neg \mathrm{B}$
C. $\neg(A \Rightarrow -B)$
D. $A \vee \neg B$

nielit-sta-2025

34.0.43 NIELIT Scientific Assistant June 2025 | Question: 48

48. For what values of x :
$[1\,2\,1]\begin{bmatrix}1 & 2 & 0\\ 2 & 0 & 1\\ 1 & 0 & 2\end{bmatrix}\begin{bmatrix}0\\2\\x\end{bmatrix}=0$
(A) 1 (B) 2 (C) 3 (D) -1

48. For what values of x :

$$\begin{bmatrix}1 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 0\\ 2 & 0 & 1\\ 1 & 0 & 2\end{bmatrix}\begin{bmatrix}0\\ 2\\ x\end{bmatrix}=0$$

A. 1
B. 2
C. 3
D. -1

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34.0.44 NIELIT Scientific Assistant June 2025 | Question: 47

47. For two events A and B, we have the following probabilities.
$P(A)=P\left(\frac{A}{B}\right)=\frac{1}{4}, P\left(\frac{B}{A}\right)=\frac{1}{2}$
Consider the statements :
(1) A and B are independent. (2) $P(B)=\frac{1}{2}$
Which of the following is correct ?
(A) (1) is True; (2) is False
(B) (1) is True; (2) is True
(C) (1) is False; (2) is False
(D) (1) is False; (2) is True

47. For two events A and B, we have the following probabilities.

$$P(A)=P\left(\frac{A}{B}\right)=\frac{1}{4}, P\left(\frac{B}{A}\right)=\frac{1}{2}$$

Consider the statements :
(1) A and B are independent.
(2) $P(B) = \frac{1}{2}$

Which of the following is correct ?

A. (1) is True; (2) is False
B. (1) is True; (2) is True
C. (1) is False; (2) is False
D. (1) is False; (2) is True

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34.0.45 NIELIT Scientific Assistant June 2025 | Question: 46

46. The sum of all the 4 digit numbers that can be formed with the digits 3, 4, 5 and 6 is :
(A) 119988 (B) 11988
(C) 191988 (D) None of the options

46. The sum of all the 4 digit numbers that can be formed with the digits $3, 4, 5$ and 6 is :

A. 119988
B. 11988
C. 191988
D. None of the options

nielit-sta-2025

34.0.46 NIELIT Scientific Assistant June 2025 | Question: 45

45. Let X = {1, 2, 3, 4}. If
A = {<x, y> | x ∈ X ∧ y ∈ X ∧ (x − y) is a nonzero multiple of 2}
B = {<x, y> | x ∈ X ∧ y ∈ X ∧ (x − y) is a nonzero multiple of 3}
Then A ∩ B is :
(A) {(1, 4), (4, 1)}
(B) {(1, 3), (3, 1), (2, 4), (4, 2)}
(C) {(1, 3), (3, 1), (2, 4), (4, 2), (1, 4), (4, 1)}
(D) ϕ

45. Let $X = \{1, 2, 3, 4\}$. If
$\mathrm{A} = \{\langle x, y\rangle \mid x \in X \wedge y \in X \wedge (x - y)$ is a nonzero multiple of 2$\}$
$\mathrm{B} = \{\langle x, y\rangle \mid x \in X \wedge y \in X \wedge (x - y)$ is a nonzero multiple of 3$\}$ Then $A \cap B$ is :

A. $\{(1,4),(4,1)\}$
B. $\{(1,3),(3,1),(2,4),(4,2)\}$
C. $\{(1,3),(3,1),(2,4),(4,2),(1,4),(4,1)\}$
D. ϕ

nielit-sta-2025

34.0.47 NIELIT Scientific Assistant June 2025 | Question: 44

44. The LU factorization requires :
(A) $\frac{n^3}{6} - \frac{n}{5}$ multiplication/division
(B) $\frac{n^3}{3} - \frac{n^2}{2} + \frac{n}{6}$ addition/subtraction
(C) $\frac{n^3}{5} - \frac{n^2}{3} + \frac{n}{6}$ addition/subtraction
(D) None of the above

44. The LU factorization requires:

A. $\frac{n^3}{6} - \frac{n}{5}$ multiplication/division
B. $\frac{n^3}{3} - \frac{n^2}{2} + \frac{n}{6}$ addition/subtraction
C. $\frac{n^3}{5} - \frac{n^2}{3} + \frac{n}{6}$ addition/subtraction
D. None of the above

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34.0.48 NIELIT Scientific Assistant June 2025 | Question: 43

43. The number of straight lines that can be drawn out of 12 points of which 8 are collinear is
(A) 39 (B) 29 (C) 49 (D) 59

43. The number of straight lines that can be drawn out of 12 points of which 8 are collinear is:

A. 39 B. 29 C. 49 D. 59

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34.0.49 NIELIT Scientific Assistant June 2025 | Question: 42

42. The product of two numbers is 120, and their sum is 22. What are the numbers ?
(A) 12, 11 (B) 6, 20 (C) 8, 15 (D) 10, 12

42. The product of two numbers is 120 , and their sum is 22 . What are the numbers ?

A. 12,11 B. 6,20 C. 8,15 D. 10,12

nielit-sta-2025

34.0.50 NIELIT Scientific Assistant June 2025 | Question: 41

41. First Bank in India to introduce talking ATMs for disabled persons :
(A) SBI (B) Punjab National Bank
(C) ICICI Bank (D) Union Bank of India

41. First Bank in India to introduce talking ATMs for disabled persons :

A. SBI
B. Punjab National Bank
C. ICICI Bank
D. Union Bank of India

nielit-sta-2025

34.0.51 NIELIT Scientific Assistant June 2025 | Question: 40

40. If the diameter of a circle is reduced by 50%, what happens to its area ?
(A) Reduced by 25% (B) Reduced by 50%
(C) Reduced by 75% (D) Reduced by 100%

40. If the diameter of a circle is reduced by 50%, what happens to its area ?

A. Reduced by 25%
B. Reduced by 50\%
C. Reduced by 75%
D. Reduced by 100%

nielit-sta-2025

34.0.52 NIELIT Scientific Assistant June 2025 | Question: 39

39. What is the probability of rolling a prime number on a fair six-sided die ?
(A) 1/3 (B) 1/5 (C) 1/2 (D) 1/6

39. What is the probability of rolling a prime number on a fair six-sided die ?

A. $1/3$ B. $1/5$ C. $1/2$ D. $1/6$

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Answer key☟

34.0.53 NIELIT Scientific Assistant June 2025 | Question: 38

38. A person walks 10 km north, then turns right and walks 2 km, then turns right again and walks 10 km. Finally, he turns left and walks 4 km. How far is he from his starting point ?
(A) 6 km (B) 4 km (C) 10 km (D) 8 km

38. A person walks 10 km north, then turns right and walks 2 km , then turns right again and walks 10 km . Finally, he turns left and walks 4 km . How far is he from his starting point ?

A. 6 km B. 4 km C. 10 km D. 8 km

nielit-sta-2025

34.0.54 NIELIT Scientific Assistant June 2025 | Question: 37

37. Who was the Governor of Reserve Bank of India as on 5th March 2025 ?
(A) Sanjay Malhotra (B) Shaktikant Das
(C) S. Jaishankar (D) R.N. Malhotra

37. Who was the Governor of Reserve Bank of India as on 5^{th} March 2025 ?

A. Sanjay Malhotra
B. Shaktikant Das
C. S. Jaishankar
D. R.N. Malhotra

nielit-sta-2025

34.0.55 NIELIT Scientific Assistant June 2025 | Question: 36

36. If a and b are positive integers, such that a*a=4 b*b = 36, then what is a*b ?
(A) 19 (B) 12 (C) 10 (D) 20

36. If a and b are positive integers, such that $a^*a = 4 \quad b^*b = 36$, then what is a^*b ?

A. 19 B. 12 C. 10 D. 20

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Answer key

34.0.56 NIELIT Scientific Assistant June 2025 | Question: 35

35. The new manager is known for his ________ nature.
(A) Ambiguous (B) Animosity (C) Hostile (D) Amiable

35. The new manager is known for his nature.

A. Ambiguous B. Animosity C. Hostile D. Amiable

nielit-sta-2025

34.0.57 NIELIT Scientific Assistant June 2025 | Question: 34

34. Study of insects is known as ________.
(A) Astrology (B) Emetology (C) Entomology (D) Geology

34. Study of insects is known as

A. Astrology B. Emetology C. Entomology D. Geology

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Answer key

34.0.58 NIELIT Scientific Assistant June 2025 | Question: 33

33. If STAR is coded as VWDU, how is MOON coded ?

A. PRRQ B. QPQP C. ORRP D. PPRQ

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Answer key

34.0.59 NIELIT Scientific Assistant June 2025 | Question: 32

32. Find the next term :

$$\mathrm{AZ}, \mathrm{BY}, \mathrm{CX}, ?$$

A. DW B. EV C. DU D. FW

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Answer key

34.0.60 NIELIT Scientific Assistant June 2025 | Question: 31

31. Statements :

All cats are dogs.
Some dogs are lions.
Conclusions:
I. Some lions are cats.
II. All lions are dogs.
(A) Only Conclusion I follows
(C) Both follow
(B) Only Conclusion II follows
(D). Neither follows

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34.0.61 NIELIT Scientific Assistant June 2025 | Question: 90

90. Which OSI layer is responsible for data compression and encryption ?
(A) Transport (B) Application (C) Presentation (D) Session
11

90. Which OSI layer is responsible for data compression and encryption ?

A. Transport
B. Application
C. Presentation
D. Session

nielit-sta-2025

34.0.62 NIELIT Scientific Assistant June 2025 | Question: 89

89. Which normal form deals with removing partial dependencies ?
(A) 1st Normal Form (1NF)
(B) 2nd Normal Form (2NF)
(C) 3rd Normal Form (3NF)
(D) Boyce-Codd Normal Form (BCNF)

89. Which normal form deals with removing partial dependencies ?

A. 1st Normal Form (INF)
B. 2nd Normal Form (2NF)
C. 3 rd Normal Form (3NF)
D. Boyce-Codd Normal Form (BCNF)

nielit-sta-2025

34.0.63 NIELIT Scientific Assistant June 2025 | Question: 88

88. Which of the following is true for the LL(1) parsing method ?
(A) It uses left recursion in its grammar
(B) It can be used for all context-free languages
(C) It requires a look-ahead of 1 symbol
(D) It is a type of shift-reduce parsing

88. Which of the following is true for the LL(1) parsing method?
(A) It uses left recursion in its grammar
B) It can be used for all context-free languages
(C) It requires a look-ahead of 1 symbol
(D) It is a type of shift-reduce parsing

nielit-sta-2025

34.0.64 NIELIT Scientific Assistant June 2025 | Question: 87

87. What is the main purpose of a device driver ?
(A) To control the physical hardware of the device
(B) To manage network communication
(C) To control memory allocation
(D) To execute system calls

87. What is the main purpose of a device driver ?

A. To control the physical hardware of the device
B. To manage network communication
C. To control memory allocation
D. To execute system calls

nielit-sta-2025

34.0.65 NIELIT Scientific Assistant June 2025 | Question: 86

86. What does a Push-Down Automaton (PDA) use to make decisions ?
(A) Stack (B) Queue (C) Tape (D) Register

86. What does a Push-Down Automaton (PDA) use to make decisions ?

A. Stack B. Queue C. Tape D. Register

nielit-sta-2025

34.0.66 NIELIT Scientific Assistant June 2025 | Question: 85

85. Which of the following is a feature of a Real-Time Operating System (RTOS) ?

A. Processes are executed in a sequential manner.
B. Tasks are executed without strict timing constraints.
C. It guarantees that tasks will be executed within a predefined time limit.
D. It does not support multitasking.

nielit-sta-2025

34.0.67 NIELIT Scientific Assistant June 2025 | Question: 84

84. What does the size of a Page Table Entry (PTE) depend on in a paging system ?

A. The number of bits required to represent the frame number
B. The size of the virtual memory address
C. The size of the physical memory address
D. The size of the page

nielit-sta-2025

34.0.68 NIELIT Scientific Assistant June 2025 | Question: 83

83. Which of the following is an example of passing a parameter by reference in C ?
(A) void func(int x)
(C) void func(int x, int y)
(B) void func(int \& x)
(D) void func(int * x)

nielit-sta-2025

34.0.69 NIELIT Scientific Assistant June 2025 | Question: 82

82. The modulation index of AM is defined as :
(A) The ratio of carrier frequency to message signal frequency
(B) The ratio of the peak amplitude of the modulated signal to the peak amplitude of the carrier
(C) The ratio of the carrier power to the total power in the AM signal
(D) The ratio of the amplitude of the modulating signal to the amplitude of the carrier signal

82. The modulation index of AM is defined as :
(A) The ratio of carrier frequency to message signal frequency
(B) The ratio of the peak amplitude of the modulated signal to the peak amplitude of the carrier
(D) The ratio of the carrier power to the total power in the AM signal

nielit-sta-2025

34.0.70 NIELIT Scientific Assistant June 2025 | Question: 81

81. If the input to an LTI system is a random signal with a known mean and variance, the output will have :
(A) The same mean and variance as the input signal
(B) The same mean, but a different variance
(C) The same variance, but a different mean
(D) A completely random mean and variance

81. If the input to an LTI system is a random signal with a known mean and variance, the output will have:

A. The same mean and variance as the input signal
B. The same mean, but a different variance
C. The same variance, but a different mean
D. A completely random mean and variance

nielit-sta-2025

34.0.71 NIELIT Scientific Assistant June 2025 | Question: 80

80. What is the purpose of public key cryptography ?
(A) To encrypt data using a single shared key
(B) To replace the need for encryption in secure communication
(C) To ensure confidentiality by using two keys: a public key and a private key
(D) To decrypt digital signatures

80. What is the purpose of public key cryptography ?

A. To encrypt data using a single shared key
B. To replace the need for encryption in secure communication
C. To ensure confidentiality by using two keys: a public key and a private key
D. To decrypt digital signatures

nielit-sta-2025

34.0.72 NIELIT Scientific Assistant June 2025 | Question: 79

79. Which isolation level in database systems allows transactions to be executed concurrently but without allowing dirty reads ?
(A) Read Uncommitted (B) Read Committed
(C) Repeatable Read (D) Serializable

79. Which isolation level in database systems allows transactions to be executed concurrently but without allowing dirty reads ?

(A) Read Uncommitted
(C) Repeatable Read
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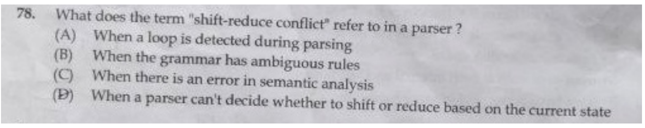
(D) Serializable

nielit-sta-2025

34.0.73 NIELIT Scientific Assistant June 2025 | Question: 78


78. What does the term "shift-reduce conflict" refer to in a parser ?
(A) When a loop is detected during parsing
(B) When the grammar has ambiguous rules
(C) When there is an error in semantic analysis
(D) When a parser can't decide whether to shift or reduce based on the current state

78. What does the term "shift-reduce conflict" refer to in a parser ?
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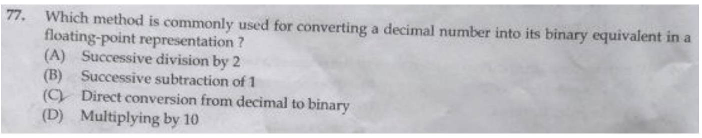
34.0.74 NIELIT Scientific Assistant June 2025 | Question: 77


77. Which method is commonly used for converting a decimal number into its binary equivalent in a floating-point representation ?
(A) Successive division by 2
(B) Successive subtraction of 1
(C) Direct conversion from decimal to binary
(D) Multiplying by 10

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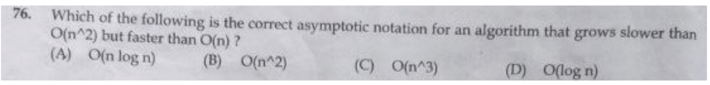
34.0.75 NIELIT Scientific Assistant June 2025 | Question: 76


76. Which of the following is the correct asymptotic notation for an algorithm that grows slower than O(n^2) but faster than O(n) ?
(A) O(n log n) (B) O(n^2) (C) O(n^3) (D) O(log n)

76. Which of the following is the correct asymptotic notation for an algorithm that grows slower than $\mathrm{O}\left(n^\wedge 2\right)$ but faster than $\mathrm{O}(n)$?

A. $O(n \log n)$
B. $\mathrm{O}\left(\mathrm{n}^\wedge 2\right)$
C. $\mathrm{O}\left(\mathrm{n}^\wedge 3\right)$
D. $O(\log n)$

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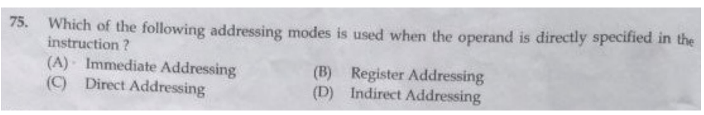
34.0.76 NIELIT Scientific Assistant June 2025 | Question: 75


75. Which of the following addressing modes is used when the operand is directly specified in the instruction ?
(A) Immediate Addressing (B) Register Addressing
(C) Direct Addressing (D) Indirect Addressing

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34.0.77 NIELIT Scientific Assistant June 2025 | Question: 74

74. What is the advantage of using a full adder over a half adder in binary addition ?

A. A full adder does not require a carry-in input.
B. A full adder can add two 4 -bit numbers.
C. A full adder handles carry inputs as well as carry outputs.
D. A full adder is used in sequential circuits only.

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34.0.78 NIELIT Scientific Assistant June 2025 | Question: 73

73. Which of the following phases in the Software Development Life Cycle (SDLC) involves gathering and analyzing user requirements ?

A. Design
B. Implementation
C. Requirement Analysis
D. Maintenance

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34.0.79 NIELIT Scientific Assistant June 2025 | Question: 72

72. Which of the following problems can be solved using a greedy approach ?

A. Knapsack problem (0/1 version)
B. Job Scheduling with deadlines and profits
C. Fibonacci sequence calculation
D. Finding the shortest path in a weighted graph

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34.0.80 NIELIT Scientific Assistant June 2025 | Question: 71

71. What is the purpose of the sliding window protocol in data communication ?
(A) To manage the amount of data that can be sent before receiving an acknowledgment
(B) To check for errors in received data
(C) To determine the routing path in a network
(D) To initiate a connection between devices

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34.0.81 NIELIT Scientific Assistant June 2025 | Question: 70

70. What is a key advantage of B+ trees over B− trees ?
(A) B+ trees allow multiple key values per node
(B) B+ trees use more disk space
(C) B+ trees store data at both internal and leaf nodes
(D) B+ trees have better performance for range queries

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nielit-sta-2025

34.0.82 NIELIT Scientific Assistant June 2025 | Question: 69

69. Which traversal method can be used to print the nodes of a binary tree in ascending order ?
(A) Pre-order traversal
(B) Post-order traversal
(C) In-order traversal
(D) Level-order traversal

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nielit-sta-2025

34.0.83 NIELIT Scientific Assistant June 2025 | Question: 68

68. Which of the following describes binding in C ?
(A) The process of assigning a value to a variable
(B) The process of declaring the type of a variable
(C) The process of associating a function with a return type
(D) The process of associating a function name with a memory address

68. Which of the following describes binding in C ?

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D. The process of associating a function name with a memory address

nielit-sta-2025

Answer key☟

34.0.84 NIELIT Scientific Assistant June 2025 | Question: 67

67. Which of the following algorithms is used to find the minimum spanning tree in a graph ?
(A) Dijkstra's Algorithm
(C) Bellman-Ford Algorithm
(B) Kruskal's Algorithm
(D) Floyd-Warshall Algorithm

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34.0.85 NIELIT Scientific Assistant June 2025 | Question: 66

66. In which addressing mode, the address of the operand is given by the sum of a constant value and the content of a register ?
(A) Indexed Addressing
(C) Direct Addressing
(B) Base Register Addressing
(D) Register Indirect Addressing

nielit-sta-2025

Answer key☟

34.0.86 NIELIT Scientific Assistant June 2025 | Question: 65

65. In PCM, the quantization error is caused by :

A. Sampling the signal
B. The difference between the input signal and the quantized value
C. The modulation of the signal
D. The carrier frequency variation

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34.0.87 NIELIT Scientific Assistant June 2025 | Question: 64

64. What is the key difference between a DFA (Deterministic Finite Automaton) and an NFA (Nondeterministic Finite Automaton) ?
(A) A DFA has a single state transition for each symbol, while an NFA can have multiple transitions.
(B) An NFA has a single state transition for each symbol, while a DFA can have multiple transitions.
(C) DFAs are less powerful than NFAs.
(D) NFAs can recognize more languages than DFAs.

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34.0.88 NIELIT Scientific Assistant June 2025 | Question: 63

63. What is the time complexity for searching an element in a balanced Binary Search Tree ?
(A) O(1) (B) O(log n) (C) O(n) (D) O(n log n)

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nielit-sta-2025

34.0.89 NIELIT Scientific Assistant June 2025 | Question: 62

62. Which of the following is the primary operation for adding an element to a queue in C ?
(A) Enqueue (B) Dequeue (C) Push (D) Pop

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nielit-sta-2025

34.0.90 NIELIT Scientific Assistant June 2025 | Question: 61

61. What is the main purpose of the synthesis of combinational circuits ?
(A) To optimize the circuit speed
(B) To find the minimal sum of products
(C) To design the flip-flops for sequential circuits
(D) To convert the truth table into a Boolean expression

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34.0.91 NIELIT Scientific Assistant June 2025 | Question: 120

20. Which of the following is the main characteristic of sequential files ?
(A) Data is stored in an unordered fashion
(B) Data is stored in sequential order
(C) Data is indexed for fast access
(D) Data can be accessed randomly

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nielit-sta-2025

34.0.92 NIELIT Scientific Assistant June 2025 | Question: 119

19. Which of the following problems is solved using dynamic programming ?
(A) Merge sort (B) Depth-first search
(C) Breadth-first search (D) Fibonacci sequence calculation

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34.0.93 NIELIT Scientific Assistant June 2025 | Question: 118

18. Which of the following is true about 'Level 0' DFD (Context Diagram) ?
(A) It shows the complete system with all its processes and data flows
(B) It shows a high-level view of the system with one process and its interactions with external entities
(C) It is used to detail the internal components of the system
(D) It shows only data stores and external entities

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34.0.94 NIELIT Scientific Assistant June 2025 | Question: 117

17. Which data structure is used by the lexical analyzer in a compiler ?
(A) Stack (B) Queue (C) Symbol table (D) Tree

17. Which data structure is used by the lexical analyzer in a compiler ?

A. Stack B. Queue C. Symbol table D. Tree

34.0.95 NIELIT Scientific Assistant June 2025 | Question: 116

6. Which of the following is the most powerful computational model ?
(A) Finite Automaton
(C) Turing Machine
(B) Push-Down Automaton
(D) Linear Bounded Automaton

34.0.96 NIELIT Scientific Assistant June 2025 | Question: 115

115. In CPU control design, which of the following is responsible for generating control signals ?

A. ALU B. Control Unit C. Registers D. Data Bus

34.0.97 NIELIT Scientific Assistant June 2025 | Question: 114

114. What is the primary disadvantage of using indirect addressing mode ?

A. It eliminates the need for instructions
B. It is faster than direct addressing
C. It reduces the need for registers
D. It requires extra memory access

34.0.98 NIELIT Scientific Assistant June 2025 | Question: 113

113. What is the main advantage of using a DFA over an NFA ?

A. DFAs are easier to construct.
B. DFAs always have fewer states than NFAs.
C. DFAs have a deterministic behavior, making them more predictable.
D. DFAs can recognize more complex languages than NFAs.

34.0.99 NIELIT Scientific Assistant June 2025 | Question: 112

112. Which of the following best describes the purpose of the "backpatching" technique in code generation?

A. It is used to resolve forward jumps by filling in missing addresses.
B. It is used to optimize code by evaluating constant expressions at compile-time.
C. It is used to detect and correct syntax errors in the source code.
D. It is used for constant folding.

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34.0.100 NIELIT Scientific Assistant June 2025 | Question: 111

111. Which of the following is the primary objective of the "round-robin" CPU scheduling algorithm ?

A. Minimize the average waiting time
B. Avoid starvation
C. Minimize the turnaround time
D. Provide each process with a fair share of CPU time

nielit-sta-2025

Answer key☟

34.0.101 NIELIT Scientific Assistant June 2025 | Question: 110

110. Which of the following is a characteristic of Boyce-Codd Normal Form (BCNF) ?

A. Every determinant is a candidate key
B. There are no transitive dependencies
C. There are no partial dependencies
D. All of the above

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34.0.102 NIELIT Scientific Assistant June 2025 | Question: 109

109. The Power Spectral Density (PSD) of a random signal is modified by an LTI system in what way ?
(A) It is completely filtered out
(B) It is multiplied by the square of the system's frequency response
(C) It is unaffected
(D) It is multiplied by the impulse response

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34.0.103 NIELIT Scientific Assistant June 2025 | Question: 108

108. Which type of cryptography uses the same key for both encryption and decryption ?
(A) Digital signatures (B) Asymmetric key cryptography
(C) Hash functions (D) Symmetric key cryptography

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34.0.104 NIELIT Scientific Assistant June 2025 | Question: 107

107. Which of the following operations can be performed in O(log n) time in a binary heap ?
(A) Deleting the root element (B) Inserting an element
(C) Both (A) and (B) (D) Searching for an element

107. Which of the following operations can be performed in $\mathrm{O}(\log n)$ time in a binary heap ?
(A) Deleting the root element
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34.0.105 NIELIT Scientific Assistant June 2025 | Question: 106

106. Which of the following statements is true for a relation in 3rd Normal Form (3NF) ?
(A) It is in 2nd Normal Form (2NF)
(B) It does not contain transitive dependencies
(C) It does not contain partial dependencies
(D) All of the above

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dependencies

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Answer key

34.0.106 NIELIT Scientific Assistant June 2025 | Question: 105

105. The number of bits used for quantization in PCM determines:

A. The transmission speed
B. The power efficiency
C. The signal-to-noise ratio (SNR)
D. The bandwidth of the system

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34.0.107 NIELIT Scientific Assistant June 2025 | Question: 104

104. Which of the following operations does a floating-point representation support that a fixed-point representation cannot?

A. Arithmetic operations on integers
B. Operations on very large or very small numbers
C. Simple addition and subtraction
D. Operations on binary fractions

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34.0.108 NIELIT Scientific Assistant June 2025 | Question: 103

103. Which of the following is a property of a regular language ?

A. It cannot be represented by a finite automaton.
B. It cannot be represented by a regular expression.
C. It can only be represented by a context-free grammar.
D. It is closed under union, concatenation, and Kleene star operations.

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Answer key

34.0.109 NIELIT Scientific Assistant June 2025 | Question: 102

102. Which of the following is true about Turing Machines ?

A. Turing machines are equivalent to finite automata
B. Turing machines can simulate any computation that can be described algorithmically
C. Turing machines are not capable of recognizing context-free languages
D. Turing machines cannot halt

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34.0.110 NIELIT Scientific Assistant June 2025 | Question: 101

101. What does a class diagram in UML represent ?
(A) The sequence of operations
B) The data flow between entities
(C) The structure and relationships between classes
(D) The physical storage of data

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34.0.111 NIELIT Scientific Assistant June 2025 | Question: 100

100. Which of the following is NOT a key consideration in input/output design ?

A. User interface consistency
B. Efficiency of data retrieval
C. Ensuring security of system processes
D. Creating an intuitive user experience

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34.0.112 NIELIT Scientific Assistant June 2025 | Question: 99

99. Which of the following is the primary property of a binary heap ?

A. The parent node is greater than its children (max-heap).
B. The parent node is less than its children (min-heap).
C. All nodes are arranged in a sorted order.
D. It is always a balanced tree.

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34.0.113 NIELIT Scientific Assistant June 2025 | Question: 98

98. Which of the following is an example of a combinational circuit ?

A. Counter B. Register C. Multiplexer D. Shift Register

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Answer key☟

34.0.114 NIELIT Scientific Assistant June 2025 | Question: 97

97. Which of the following is the main objective of memory management in an operating system ?

A. Efficient allocation of memory
B. Protection of memory
C. Virtual memory management
D. All of the options

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Answer key☟

34.0.115 NIELIT Scientific Assistant June 2025 | Question: 96

96. What is the primary use of a Karnaugh Map (K-map) in Boolean algebra ?

A. To simplify logic circuits
B. To find the truth table
C. To convert a Boolean expression into its canonical form
D. To convert between logic gates

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Answer key☟

34.0.116 NIELIT Scientific Assistant June 2025 | Question: 95

95. Which of the following is NOT a component of syntax analysis in a compiler ?

A. Syntax tree B. Parse tree C. Token D. Grammar rules

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34.0.117 NIELIT Scientific Assistant June 2025 | Question: 94

94. What is the main advantage of a Link-State Routing algorithm over a Distance-Vector Routing algorithm ?
(A) It is more efficient in terms of bandwidth usage
(B) It has a faster convergence time
(C) It requires less memory
(D) It is simpler to implement

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nielit-sta-2025

Answer key

34.0.118 NIELIT Scientific Assistant June 2025 | Question: 93

93. What is the main purpose of a file system ?
(A) To protect the system from malware
(B) To manage the data stored on a storage device
(C) To provide networking services
(D) To execute user programs

93. What is the main purpose of a file system ?

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nielit-sta-2025

Answer key

34.0.119 NIELIT Scientific Assistant June 2025 | Question: 92

92. What happens in TCP when congestion is detected in the network ?
(A) The sender reduces its congestion window
(B) The sender increases its congestion window
(C) The receiver sends an acknowledgment for all unacknowledged packets
(D) The sender stops sending data

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34.0.120 NIELIT Scientific Assistant June 2025 | Question: 91

91. Which of the following is an intermediate representation used by compilers ?
(A) Abstract syntax tree (B) Parse tree
(C) Machine code (D) Source code

91. Which of the following is an intermediate representation used by compilers ?

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D. Source code

nielit-sta-2025

Answer key

34.1 General Awareness (4)

34.1.1 General Awareness: NIELIT 2022 April Scientist B | Section A | Question: 23

Assertion (A): Increase in carbon dioxide would melt polar ice.

Reason (R): Global temperature would rise.

A. Both (A) and (R) are true but (R) is not the correct explanation of (A)
B. Both (A) and (R) are true and (R) is the correct explanation of (A)
C. (A) is true but (R) is false.
D. (A) is false but (R) is true.

nielit2022apr-scientistb general-awareness logical-reasoning

Answer key

34.1.2 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 135

The Ohio State Leadership Studies revealed _________ and initiating structure as two major dimensions of leadership behaviour:

A. Control
B. Communication
C. Collaboration
D. Consideration

nielit2023feb-scientistc general-awareness

34.1.3 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 144

Which style of leadership focuses on goals, standards, and organization?

A. Task leadership
B. Social leadership
C. Semantic leadership
D. Transformational leadership

nielit2023feb-scientistc general-awareness

34.1.4 General Awareness: NIELIT 2023 Feb Scientist D | Section A | Question: 24

Directions: Answer questions **(24-25)** on the basis of following instructions :

Instructions: For the Assertions (A) and Reason (R) below, choose the correct alternative from the following :

(I) Both (A) and (R) are true and (R) is the correct explanation of (A).

(II) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(III) (A) is true but (R) is false.

(IV) (A) is false but (R) is true.

Assertion (A): The filament in an electric bulb does not burn up although its temperature is about $2700^\circ \mathrm{C}$ when it glows.

Reason (R): Bulb filament is made of tungsten.

A. (I) B. (II) C. (III) D. (IV)

nielit2023feb-scientistd logical-reasoning general-awareness analytical-aptitude

34.2 Is&software Engineering (14)

34.2.1 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 132

You are asked by your instructor to discuss about Contingency theory. What should you be included in your discussion?

A. It is a theory in leadership that views the pattern of leader behaviour as dependent on the interaction of the personality and the needs of the situation.
B. It emphasizes that both leaders and followers should act on one another to raise their motivation.
C. It states that leadership qualities inspire followers to be motivated by what they do.
D. None of the options

nielit2023feb-scientistc is&software-engineering logical-reasoning

34.2.2 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 137

A Charismatic leader's ___________ is the key to follower acceptance.

A. Energy
B. History with the organization
C. Credentials
D. Vision

nielit2023feb-scientistc is&software-engineering

34.2.3 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 140

Participative leadership has which of the following characteristics?

A. Believe success arises form leaders and staff working together
B. Employs a clear chain of command
C. Takes the view that rewards and punishment motivate staff
D. Seeks to involve staff in the decision making process

nielit2023feb-scientistc is&software-engineering logical-reasoning

34.2.4 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 123

This style is a one side approach where the leader-manager pays much attention to the security and comfort of the employees but he has low concern for production.

A. Impoverished
B. Country club
C. Dictatorial or Perish
D. The Team or Sound

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.5 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 124

________ are the theories which gives an idea about what employees wants or needs and what are the key factors the managers can utilize to motivate the employees.

A. Maslow theory
B. Herzberg Theory
C. Process Theory
D. Content Theory

nielit2023feb-scientistd is&software-engineering

Answer key☟

34.2.6 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 125

In addition to basic managerial functions of planning, organizing, staffing, directing, and controlling, leaders are ascribed :

A. Procedural and external roles.
B. Procedural and internal roles.
C. Strategic and internal roles.
D. Strategic and external roles.

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.7 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 128

_________ commitment refers to an employee's obligation to remain with an organization for moral or ethical reasons.

A. Affective
B. Continuance
C. Theoretical
D. Normative

nielit2023feb-scientistd is&software-engineering

34.2.8 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question:

132

The process by which an older and more experienced person helps to socialize and encourage younger organizational colleagues is called __________.

A. Evaluating B. Consulting C. Mentoring D. Networking

nielit2023feb-scientistd is&software-engineering

34.2.9 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 135

In managerial grid, the style of management depicts the style of a leader who is neither concerned about the people nor does he care about the tasks to be performed is _________.

A. task management
B. impoverished style
C. country club
D. team management style

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.10 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 136

A range of management behaviors based on the various ways that task-oriented and, employee-oriented styles (each expressed as a continuum on a scale of 1 to 9) is known as :

A. Bi-dimensional managerial grid
B. Rectangular managerial grid
C. Diagonal managerial grid
D. BCG Matrix

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.11 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 138

The trait approach to leadership suggests that:

A. Leaders have special innate qualities.
B. Leadership traits are clearly visible.
C. Traits are based on social class.
D. Traits cannot be measured.

nielit2023feb-scientistd is&software-engineering

34.2.12 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 143

All decision-making power is centralized in the leader is under ____________.

A. autocratic style
B. liberal leader

C. democratic leader D. institutional leader

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.13 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 147

Which path-goal leadership style leads to greater satisfaction when tasks are ambiguous or stressful?

A. Directive B. Supportive C. Participative D. Mixed

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.14 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 148

Which role focuses on bringing about order and consistency by drawing up formal plans?

A. Leadership
B. Management
C. Task structure
D. Initiating structure

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.3 Logical Reasoning (1)

34.3.1 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 144

Groups created by managerial decision in order to accomplish stated goals of the organization are called ___________.

A. Formal groups
B. Informal groups
C. Task groups
D. Interest groups

nielit2023feb-scientistd logical-reasoning

Answer Keys

34.0.1	D	34.0.2	C	34.0.3	A	34.0.4	B	34.0.5	B
34.0.6	B	34.0.7	C	34.0.8	A	34.0.9	B	34.0.10	B
34.0.11	D	34.0.12	C	34.0.13	B	34.0.14	A	34.0.15	B
34.0.16	D	34.0.17	D	34.0.18	B	34.0.19	B	34.0.20	C
34.0.21	C	34.0.22	C	34.0.23	A	34.0.24	C	34.0.25	A

34.0.26	D	34.0.27	A	34.0.28	B	34.0.29	D	34.0.30	C
34.0.31	B	34.0.32	D	34.0.33	D	34.0.34	C	34.0.35	A
34.0.36	D	34.0.37	D	34.0.38	C	34.0.39	B	34.0.40	C
34.0.41	C	34.0.42	B	34.0.43	D	34.0.44	B	34.0.45	A
34.0.46	D	34.0.47	B	34.0.48	A	34.0.49	D	34.0.50	D
34.0.51	C	34.0.52	C	34.0.53	A	34.0.54	A	34.0.55	B
34.0.56	D	34.0.57	C	34.0.58	A	34.0.59	A	34.0.60	D
34.0.61	C	34.0.62	B	34.0.63	C	34.0.64	A	34.0.65	A
34.0.66	C	34.0.67	A	34.0.68	D	34.0.69	D	34.0.70	B
34.0.71	C	34.0.72	B	34.0.73	D	34.0.74	A	34.0.75	A
34.0.76	A	34.0.77	C	34.0.78	C	34.0.79	B	34.0.80	A
34.0.81	D	34.0.82	C	34.0.83	D	34.0.84	B	34.0.85	A
34.0.86	B	34.0.87	A	34.0.88	B	34.0.89	A	34.0.90	D
34.0.91	B	34.0.92	D	34.0.93	B	34.0.94	C	34.0.95	C
34.0.96	B	34.0.97	D	34.0.98	C	34.0.99	A	34.0.100	D
34.0.101	A	34.0.102	B	34.0.103	D	34.0.104	C	34.0.105	D
34.0.106	C	34.0.107	B	34.0.108	D	34.0.109	B	34.0.110	C
34.0.111	C	34.0.112	D	34.0.113	C	34.0.114	D	34.0.115	A
34.0.116	C	34.0.117	B	34.0.118	B	34.0.119	A	34.0.120	A
34.1.1	A	34.1.2	D	34.1.3	A	34.1.4	A	34.2.1	A
34.2.2	D	34.2.3	D	34.2.4	B	34.2.5	D	34.2.6	D
34.2.7	D	34.2.8	C	34.2.9	B	34.2.10	A	34.2.11	A
34.2.12	A	34.2.13	B	34.2.14	B	34.3.1	A		

35 Others: Others (11)

35.0.1 NIELIT 2022 April Scientist B | Section B | Question: 117

Consider sinusoidal modulation in an AM system. Assuming no over modulation, the modulation index (μ) when the maximum and minimum values of the envelope, respectively, are $3 \mathrm{~V}$ and $1 \mathrm{~V}$ is ___________.

A. 0.1 B. 0 C. 0.3 D. 0.5

nielit2022apr-scientistb analytical-aptitude

35.1 Cache Memory (1)

35.1.1 Cache Memory: NIELIT 2021 Dec Scientist B - Section B: 22

Cache memory works on the principle of:

A. Locality of data
B. Locality of memory
C. Locality of reference
D. Locality of reference & memory

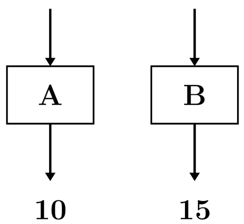
nielit-2021-it-dec-scientistb cache-memory co-and-architecture

Answer key☟

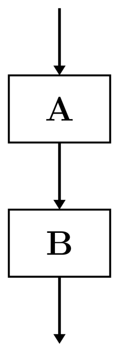
35.2 Cyclomatic Complexity (2)

35.2.1 Cyclomatic Complexity: NIELIT 2021 Dec Scientist A - Section B: 63

The Cyclomatic complexity of two modules A and B are 10 and 15 respectively :



What is the cyclomatic complexity of sequential integration of A and B ?



A. 19 B. 21 C. 24 D. 25

nielit2021dec-scientista is&software-engineering cyclomatic-complexity logical-reasoning

35.2.2 Cyclomatic Complexity: NIELIT 2021 Dec Scientist B - Section B: 48

Which testing is equivalent to Cyclomatic complexity?

A. Black box testing
B. Green box testing
C. Yellow box testing
D. White box testing

nielit-2021-it-dec-scientistb software-testing cyclomatic-complexity

Answer key☟

35.3 Effective Memory Access (2)

35.3.1 Effective Memory Access: NIELIT 2021 Dec Scientist B - Section B: 113

When a cache is 10 times faster than main memory, and the cache can be used 70% of the time, how much speed can be gained using cache ?

A. $\simeq 10$
B. $\simeq 0.3$
C. $\simeq 0.7$
D. $\simeq 2.7$

nielit2021dec-scientistb cache-memory effective-memory-access co-and-architecture

35.3.2 Effective Memory Access: NIELIT 2022 April Scientist B | Section B | Question: 102

Consider a machine with 40 MHz processor which has run a benchmark program. The executed program consists of $100,000$ instruction executions, with the following instructions mix and clock cycle count. What will be the effective CPI, MIPS rate, and execution time.

Instruction Type	Instruction Count	Cycles/Instructions
Integer arithmetic	45000	1
Data Transfer	32000	2
Floating point	15000	2
Control transfer	8000	2

A. CPI : 3.55; MIPS : 30; Execution time : 1.87 ms
B. CPI : 1.55; MIPS : 25.8; Execution time : 3.87 ms
C. CPI : 5.60; MIPS : 45.8; Execution time : 2.87 ms
D. CPI : 2.55; MIPS : 35.8; Execution time : 4.87 ms

nielit2022apr-scientistb co-and-architecture effective-memory-access

Answer key☟

35.4 Group Theory (1)

35.4.1 Group Theory: NIELIT 2022 April Scientist B | Section B | Question: 65

Let G be a multiplicative group and $a \in \mathrm{G}$. If the order of a is 6, then the order of a^5 is equal to :

A. 1 B. 5 C. 6 D. 30

nielit2022apr-scientistb group-theory functions

Answer key

35.5 Is&software Engineering (1)

35.5.1 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 24

Consider a program with following data:

Unique operator $= 10$, unique operands $= 15$

Total operator $= 30$, Total operands $= 40$

What is the estimated length of program?

A. 132 B. 92 C. 32 D. 82

nielit-2021-it-dec-scientistb is&software-engineering

Answer key

35.6 Percentage (1)

35.6.1 Percentage: NIELIT 2021 Dec Scientist A - Section B: 62

What is the annual change in traffic of software with 1 million lines of code with 30% lines added and 10% lines are deleted ?

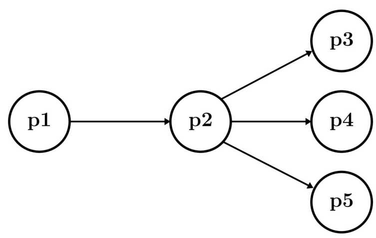
A. 0.25 B. 0.15 C. 0.4 D. 0.6

nielit2021dec-scientista is&software-engineering percentage quantitative-aptitude

35.7 Software Design (1)

35.7.1 Software Design: NIELIT 2022 April Scientist B | Section B | Question: 79

In a data flow diagram, the segment shown below is identified as having transaction flow characteristics, with $p2$ identified as the transaction center



A first level architectural design of this segment will result in a set of process modules with an associated invocation sequence. The most appropriate architecture is

A. $p1$ invokes $p2, p2$ invokes either $p3$, or $p4$, or $p5$
B. $p2$ invokes $p1$ and then invokes $p3$, or $p4$, or $p5$
C. A new module Tc is defined to control the transaction flow. This module Tc first invokes $p1$ and then invokes $p2$. $p2$ then invokes $p3$, or $p4$, or $p5$
D. A new module Tc is defined to control the transaction flow. This module Tc invokes $p2$. $p2$ invokes $p1$, and then invokes $p3$, or $p4$, or $p5$

nielit2022apr-scientistb is&software-engineering software-design

35.8 Software Testing (1)

35.8.1 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 47

Maintenance testing is performed using which methodology?

A. Breadth test and depth test
B. Sanity testing
C. Retesting
D. Confirmation testing

nielit-2021-it-dec-scientistb is&software-engineering software-testing

Answer key☟

Answer Keys

35.0.1	D	35.1.1	C	35.2.1	C	35.2.2	D	35.3.1	D
35.3.2	B	35.4.1	C	35.5.1	B	35.6.1	C	35.7.1	C
35.8.1	A								

36 Programming and DS (12)

36.1 Algorithm Design (1)

36.1.1 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 109

What does the following code do ?

```
int a, b;
a = a + b;
b = a - b;
a = a - b;
```

A. leaves a and b unchanged
B. a doubles and stores in a
C. b doubles and stores in a
D. Exchanges a and b

nielit2021dec-scientistb programming-in-c algorithm-design output

Answer key

36.2 Compiler tokenization (2)

36.2.1 Compiler tokenization: NIELIT 2022 April Scientist B | Section B | Question: 70

What does the following C-statement declare?

```
int (*f) (int * );
```

A. A function that takes an integer pointer as argument and returns an integer.
B. A function that takes an integer as argument and returns an integer pointer.
C. A pointer to a function that takes an integer pointer as argument and returns an integer.
D. A function that takes an integer pointer as argument and returns a function pointer.

nielit2022apr-scientistb programming-in-c pointers compiler-tokenization

36.2.2 Compiler tokenization: NIELIT Scientist B 2020 November: 47

The number of tokens in the following C statement is

```
printf("i=%d, &i=%x", i, &i);
```

A. 8
B. 4
C. 7
D. 10

nielit-scb-2020 compiler-tokenization programming-in-c

Answer key

36.3 Functions (1)

36.3.1 Functions: NIELIT 2021 Dec Scientist B - Section B: 67

Choose correct statement for code segment

```
int multiply (int a, int b = 5),
```

A. variable a and b are of int types and the initial value of a and b are 5
B. variable b is of int type and will always have value 5
C. variable b is of global scope and will have value 5
D. variable b will have value 5 if not specified when calling a function in further program

nielit-2021-it-dec-scientistb programming-in-c functions

36.4 Memory Management (1)

36.4.1 Memory Management: NIELIT 2021 Dec Scientist B - Section B: 26

In $C++$, dynamic memory allocation is accomplished with which of the following operator:

A. this B. new C. delete D. malloc()

nielit-2021-it-dec-scientistb programming-in-c data-structures memory-management

Answer key☟

36.5 Output (1)

36.5.1 Output: NIELIT 2021 Dec Scientist B - Section B: 34

Output of following code

```
main(){printf ("\n%%%%");}
```

A. Error message
B. $\%\%$
C. 00
D. $\&\&\&\&$

nielit-2021-it-dec-scientistb programming-in-c output

Answer key☟

36.6 Parameter Passing (1)

36.6.1 Parameter Passing: NIELIT 2022 April Scientist B | Section B | Question: 76

What is printed by the print statements in the program $P1$ assuming call by reference parameter passing ?

```
Program P1()
{
   x = 10;
   y = 3;
   func1(y, x, x);
```

```
    print x;
    print y;
}
func1(x, y, z)
{
    y = y+4;
    z = x+ y+ z;
}
```

A. $10, 3$ B. $31, 3$ C. $27, 7$ D. $33, 5$

nielit2022apr-scientistb programming-in-c parameter-passing data-structures

Answer key☟

36.7 Pointers (1)

36.7.1 Pointers: NIELIT 2022 April Scientist B | Section B | Question: 95

Consider the following C code:

```
#include<stdio.h>
int *assignval (int *x, int val) {
    *x = val;
    return x;
}

int main () {
    int *x = malloc(sizeof(int));
    if (NULL == x) return;
    x = assignval (x,0);
    if (x) {
        x = (int *)malloc(sizeof(int));
        if (NULL == x) return;
        x = assignval (x,10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

A. compiler error as the return of malloc is not typecast appropriately.
B. compiler error because the comparison should be made as $x == \text{NULL}$ and not as shown.
C. compiles successfully but execution may result in dangling pointer.
D. compiles successfully but execution may result in memory leak.

nielit2022apr-scientistb programming-in-c memory-management pointers

36.8 Programming In C (4)

36.8.1 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 49

The best statement to find out that value of a variable lies in the range 7 to 10 or 12 to 15 in C language is :

A. switch B. while C. for D. continue

nielit-2021-it-dec-scientistb programming-in-c

Answer key

36.8.2 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 55

In C language value of $\mathrm{ix}+\mathrm{j}$, if j is of integer type and ix long type would be:

A. Integer B. Float C. Long integer D. Double precision

nielit-2021-it-dec-scientistb programming-in-c

36.8.3 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 65

If $\mathrm{a}{+}{+}$ is replaced with ${+}{+}\mathrm{a}$, which statement does not get affected?

A.
```
printf("% d % d", ++a, a++);
```
B.
```
a = 20; a++;
```
C.
```
while(a++ = 20) cout<<a;
```
D.
```
d = a++;
```

nielit-2021-it-dec-scientistb nielit2021dec-scientistb programming-in-c

36.8.4 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 91

Following are implicitly provided in C programming language :

A. Output facility
B. Input facility
C. Both Input and Output facility
D. No Input and Output facility

nielit2021dec-scientistb programming-in-c

Answer Keys

36.1.1	D	36.2.1	C	36.2.2	D	36.3.1	D	36.4.1	B
36.5.1	B	36.6.1	B	36.7.1	D	36.8.1	A	36.8.2	C
36.8.3	B	36.8.4	C						

37 Programming and DS: Data Structures (77)

37.1 AVL Tree (1)

37.1.1 AVL Tree: NIELIT 2017 July Scientist B (CS) - Section B: 7

A balance factor in AVL tree is used to check

A. what rotation to make
B. if all child nodes are at same level
C. when the last rotation occurred
D. if the tree is unbalanced

nielit2017july-scientistb-cs data-structures avl-tree easy

Answer key☟

37.2 Algorithm Design (1)

37.2.1 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 37

Linked lists are not suitable data structures for which one of the following problems?

A. Insertion sort
B. Binary search
C. Radix sort
D. Polynomial manipulation

nielit-2021-it-dec-scientistb data-structures linked-list algorithm-design

Answer key☟

37.3 Array (4)

37.3.1 Array: NIELIT 2016 MAR Scientist C - Section C: 55

To sort many large objects or structures, it would be most efficient to place

A. them in an array and sort the array
B. pointers to them in an array and sort the array
C. them in a linked list and sort the linked list
D. references to them in an array and sort the array

nielit2016mar-scientistc data-structures array

Answer key☟

37.3.2 Array: NIELIT 2017 July Scientist B (CS) - Section B: 4

Let A be a square matrix of size $n \times n$. Consider the following program. What is the expected output?

```
C=100
for i=1 to n do
for j=1 to n do
{
 Temp=A[i][j]+C
 A[i][j]=A[j][i]
 A[j][i]=Temp-C
}
for i=1 to n do
```

```
for j=1 to n do
 Output(A[i][j]);
```

A. The matrix A itself.
B. Transpose of matrix A.
C. Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A.
D. None of the option.

nielit2017july-scientistb-cs data-structures array

Answer key

37.3.3 Array: NIELIT 2017 July Scientist B (CS) - Section B: 40

Kadane algorithm is used to find

A. Maximum sum subsequence in an array
B. Maximum sum subarray in an array
C. Maximum product subsequence in an array
D. Maximum product subarray in an array

nielit2017july-scientistb-cs data-structures array

Answer key

37.3.4 Array: NIELIT Scientific Assistant A 2020 November: 109

A program P reads in 500 integers in the range $[0 \ldots 100]$ representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for P to store the frequencies?

A. An array of 50 numbers
B. An array of 100 numbers
C. An array of 500 numbers
D. A dynamically allocated array of 550 numbers

nielit-sta-2020 data-structures array easy

Answer key

37.4 Binary Heap (5)

37.4.1 Binary Heap: NIELIT 2016 MAR Scientist C - Section C: 49

We have a binary heap on n elements and wish to insert n more elements (not necessarily one after another) into this heap. Total time required for this is

A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. $\Theta(n^2)$

nielit2016mar-scientistc data-structures binary-heap

Answer key

37.4.2 Binary Heap: NIELIT 2017 DEC Scientist B - Section B: 31

Consider a complete binary tree where the left and the right sub trees of the root are max-heaps. The lower bound for the number of operations to convert the tree to a heap is:

A. $\Omega(\log n)$ B. $\Omega(n \log n)$ C. $\Omega(n)$ D. $\Omega(n^2)$

nielit2017dec-scientistb data-structures binary-tree binary-heap

Answer key☟

37.4.3 Binary Heap: NIELIT 2018-37

The smallest element that can be found in time ____ in a binary max heap.

A. $O(n \log n)$ B. $O(\log n)$ C. $O(n)$ D. $O(n^2)$

nielit-2018 data-structures binary-heap

Answer key☟

37.4.4 Binary Heap: NIELIT 2018-84

For the given nodes:

$89, 19, 50, 17, 12, 15, 2, 5, 7, 11, 6, 9, 100$

minimum ______ number of interchanges are required to convert it into a max-heap.

A. 3 B. 4 C. 5 D. 6

nielit-2018 data-structures binary-heap

Answer key☟

37.4.5 Binary Heap: NIELIT 2022 April Scientist B | Section B | Question: 107

Consider a max heap, represented by the array: $40, 30, 20, 10, 15, 16, 17, 8, 4$. Now consider that a value 35 is inserted into this heap. After insertion, the new heap is

A. $40, 30, 20, 10, 15, 16, 17, 8, 4, 35$
B. $40, 35, 20, 10, 30, 16, 17, 8, 4, 15$
C. $40, 30, 20, 10, 35, 16, 17, 8, 4, 15$
D. $40, 35, 20, 10, 15, 16, 17, 8, 4, 30$

nielit2022apr-scientistb data-structures binary-heap algorithms

37.5 Binary Search (1)

37.5.1 Binary Search: NIELIT 2016 MAR Scientist C - Section C: 52

Consider the process of inserting an element into a $Max\ Heap$, where the $Max\ Heap$ is represented by an $array$. Suppose we perform a binary search on the path from the new leaf to the root to find the position for the newly inserted element, the number of $comparisons$ performed is

A. $\Theta(\log_2 n)$ B. $\Theta(n \log_2 \log_2 n)$

C. $\Theta(n)$

D. $\Theta(n \log_2 n)$

nielit2016mar-scientistc data-structures binary-search time-complexity binary-heap

Answer key☟

37.6 Binary Search Tree (3)

37.6.1 Binary Search Tree: NIELIT 2016 MAR Scientist C - Section C: 50

You are given the postorder traversal, P, of a binary search tree on the n elements $1, 2, \ldots, n$. You have to determine the unique binary search tree that has P as its postorder traversal. What is the time complexity of the most efficient algorithm for doing this?

A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. None of the above, as the tree cannot be uniquely determined.

nielit2016mar-scientistc data-structures binary-search-tree

Answer key☟

37.6.2 Binary Search Tree: NIELIT 2018-67

____ is the worst-case time complexity for all operations (i.e.,) search, update and delete) in a general Binary Search tree

A. $O(n)$

B. $O(n \log n)$

C. $O(\log n)$ for search and insert, and $O(n)$ for delete

D. None of these

nielit-2018 data-structures binary-search-tree

Answer key☟

37.6.3 Binary Search Tree: NIELIT Scientific Assistant A 2020 November: 93

When we perform in order traversal on a binary tree, we get the ascending order array. The tree is:

A. Heap tree

B. almost complete binary tree

C. Binary search tree

D. Cannot be determined

nielit-sta-2020 data-structures binary-search-tree

Answer key☟

37.7 Binary Tree (11)

37.7.1 Binary Tree: NIELIT 2016 DEC Scientist B (CS) - Section B: 15

Which of the following need not be a binary tree?

A. Search tree B. Heap C. AVL tree D. B tree nielit2016dec-scientistb-cs data-structures binary-tree

Answer key☟

37.7.2 Binary Tree: NIELIT 2016 DEC Scientist B (IT) - Section B: 13

The number of unused pointers in a complete binary tree of depth 5 is:

A. 4 B. 8 C. 16 D. 32

nielit2016dec-scientistb-it data-structures binary-tree

Answer key☟

37.7.3 Binary Tree: NIELIT 2016 MAR Scientist C - Section C: 31

A full binary tree with n non-leaf nodes contains

A. $\log_2 n$ nodes B. $n+1$ nodes
C. $2n$ nodes D. $2n+1$ nodes

nielit2016mar-scientistc data-structures binary-tree

Answer key☟

37.7.4 Binary Tree: NIELIT 2017 DEC Scientific Assistant A - Section B: 18

In a full binary tree number of nodes is 63 then the height of the tree is :

A. 2 B. 4 C. 3 D. 6

nielit2017dec-assistanta data-structures binary-tree

Answer key☟

37.7.5 Binary Tree: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 2

The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number number of nodes in a binary tree of height h is

A. 2^h B. $2^{h-1}-1$ C. $2^{h+1}-1$ D. 2^{h+1}

nielit2017oct-assistanta-it data-structures binary-tree

Answer key☟

37.7.6 Binary Tree: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 20

The number of possible binary trees with 4 nodes is

A. 12 B. 13 C. 14 D. 15

nielit2017oct-assistanta-it data-structures binary-tree

Answer key☟

37.7.7 Binary Tree: NIELIT 2021 Dec Scientist A - Section B: 76

When the left sub-tree of the tree is one level higher than that of the right sub-tree, then the balance factor is ____________ .

A. 0 B. 1 C. -1 D. 2

nielit2021dec-scientista data-structures binary-tree

Answer key

37.7.8 Binary Tree: NIELIT 2021 Dec Scientist A - Section B: 78

Total number of nodes at the n^{th} level of a full binary tree can be given as ______________ .

A. 2n+1 B. 2n^2 C. 2^n D. 2n-1

nielit2021dec-scientista binary-tree data-structures

Answer key

37.7.9 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 38

The maximum number of binary trees that can be formed with three unlabeled nodes is :

A. 1 B. 5 C. 4 D. 3

nielit-2021-it-dec-scientistb binary-tree combinatory data-structures algorithms

Answer key

37.7.10 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 51

In a binary tree, the number of internal nodes of degree 1 is 5, and the number of nodes of degree 2 is 10. The number of leaf nodes in binary tree is:

A. 10 B. 11 C. 12 D. 15

nielit-2021-it-dec-scientistb data-structures binary-tree

Answer key

37.7.11 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 90

A binary tree of depth K is called a full binary tree of depth K, if it has exactly ____________ nodes.

A. K B. 2^k C. $2^\text{k}-1$ D. $2^\text{k}+1$

nielit2021dec-scientistb binary-tree data-structures

37.8 Data Structures (1)

37.8.1 Data Structures: NIELIT 2018-70

____ number of underlined graphs can be constructed using $V = (v1, v2, \ldots, vn)$.

A. n^3 B. $2^{n(n-1)}/2$ C. $n - 1/2$ D. $2^{(n-1)}/2$

nielit-2018 data-structures graph-theory

Answer key

37.9 Depth First Search (3)

37.9.1 Depth First Search: NIELIT 2016 DEC Scientist B (IT) - Section B: 30

What data structures is used for depth first traversal of a graph?

A. Queue B. Stack C. List D. None of the above

nielit2016dec-scientistb-it data-structures stack depth-first-search

Answer key

37.9.2 Depth First Search: NIELIT 2016 DEC Scientist B (IT) - Section B: 42

In the ________ traversal we process all of a vertex's descendants before we move to an adjacent vertex.

A. Depth First B. Breadth First C. Width First D. Depth Limited

nielit2016dec-scientistb-it data-structures stack depth-first-search

Answer key

37.9.3 Depth First Search: NIELIT 2017 July Scientist B (IT) - Section B: 4

What are the appropriate data structures for graph traversal using Breadth First Search(BFS) and Depth First Search(DFS) algorithms?

A. Stack for BFS and Queue for DFS
B. Queue for BFS and Stack for DFS
C. Stack for BFS and Stack for DFS
D. Queue for BFS and Queue for DFS

nielit2017july-scientistb-it data-structures graph-algorithms breadth-first-search depth-first-search

Answer key

37.10 Expression Evaluation (2)

37.10.1 Expression Evaluation: NIELIT 2018-54

______ to evaluate an expression without any embedded function calls.

A. Two stacks are required
B. one stack is needed
C. Three stacks are required
D. More than three stacks are required

nielit-2018 data-structures stack expression-evaluation

Answer key

37.10.2 Expression Evaluation: NIELIT 2018-85

Evaluation of the given postfix expression $10\ 10 + 60\ 6/^{*}8-$ is

A. 192 B. 190 C. 110 D. 92

nielit-2018 data-structures expression-evaluation

Answer key

37.11 Hashing (5)

37.11.1 Hashing: NIELIT 2016 MAR Scientist C - Section C: 30

A hash table with 10 buckets with one slot per bucket is depicted. The symbols, $S1$ to $S7$ are initially emerged using a hashing function with linear probing. Maximum number of comparisons needed in searching an item that is not present is

0	S7
1	S1
2	
3	S4
4	S2
5	
6	S5
7	
8	S6
9	S3

A. 6 B. 5 C. 4 D. 3

nielit2016mar-scientistc data-structures hashing

Answer key

37.11.2 Hashing: NIELIT 2016 MAR Scientist C - Section C: 56

The average search time of hashing, with linear probing will be less if the load factor

A. is far less than one
B. equals one
C. is far greater than one
D. none of these

nielit2016mar-scientistc data-structures hashing

Answer key

37.11.3 Hashing: NIELIT 2018-65

For a given hash table T with 10 slots that stores 1000 elements, the load factor α for T is

A. 100 B. 0.01 C. 200 D. 1.05

nielit-2018 data-structures hashing

Answer key

37.11.4 Hashing: NIELIT 2021 Dec Scientist A - Section B: 51

Which open addressing technique is free from Clustering problems?

A. Linear probing
B. Quadratic probing
C. Double hashing
D. Rehashing

nielit2021dec-scientista hashing data-structures algorithms

Answer key

37.11.5 Hashing: NIELIT Scientist B 2020 November: 73

Which open addressing technique is free from Clustering problems?

A. Linear probing
B. Quadratic probing
C. Double hashing
D. Rehashing

nielit-scb-2020 hashing data-structures

Answer key

37.12 Infix Prefix (1)

37.12.1 Infix Prefix: NIELIT 2017 July Scientist B (CS) - Section B: 6

Assume that the operators $+, -, \times$ are left associative and $\wedge$ is right associative. The order of precedence(from highest to lowest) is $\wedge, \times, +, -$. The postfix expression corresponding to the infix expression $a + b \times c - d \wedge e \wedge f$ is

A. $abc \times +def \wedge \wedge -$
B. $abc \times +de \wedge f \wedge -$
C. $ab + c \times d - e \wedge f \wedge$
D. $- + a \times bc \wedge \wedge def$

nielit2017july-scientistb-cs data-structures stack infix-prefix

Answer key

37.13 Linked List (7)

37.13.1 Linked List: NIELIT 2016 MAR Scientist C - Section C: 54

In a circularly linked list organization, insertion of a record involves the modification of

A. no pointer
B. 1 pointer
C. 2 pointers
D. 3 pointers

nielit2016mar-scientistc data-structures linked-list

Answer key

37.13.2 Linked List: NIELIT 2017 DEC Scientific Assistant A - Section B: 52

The address field of linked list :

A. Contain address of next node
B. May contain null character
C. Contain address of next pointer
D. Both (A) and (B)

nielit2017dec-assistanta data-structures linked-list

Answer key

37.13.3 Linked List: NIELIT 2017 July Scientist B (CS) - Section B: 1

What does the following function do for a given Linked List with first node as head?

```
void fun1(struct node* head)
{
if(head==NULL)
return;
fun1(head->next);
printf("%d",head->data);
}
```

A. Prints all nodes of linked lists
B. Prints all nodes of linked list in reverse order
C. Prints alternate nodes of Linked List
D. Prints alternate nodes in reverse order

nielit2017july-scientistb-cs data-structures linked-list

37.13.4 Linked List: NIELIT 2017 July Scientist B (CS) - Section B: 3

Consider the following function that takes reference to head of a Doubly Linked List as parameter. Assume that a node of doubly linked list has previous pointer as $prev$ and next pointer as $next$.

```
void fun(struct node **head_ref)
{
struct node *temp=NULL;
struct node *current=*head_ref;
while(current!=NULL)
{
 temp=current->prev;
 current->prev=current->next;
 current->next=temp;
 current=current->prev;
}
if(temp!=NULL)
*head_ref=temp->prev;
}
```

Assume that reference of head of following doubly linked list is passed to above function $1 \leftrightarrow 2 \leftrightarrow 3 \leftrightarrow 4 \leftrightarrow 5 \leftrightarrow 6$. What should be the modified linked list after the function call?

A. $2 \leftrightarrow 1 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 6 \leftrightarrow 5$
B. $5 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 2 \leftrightarrow 1 \leftrightarrow 6$
C. $6 \leftrightarrow 5 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 2 \leftrightarrow 1$
D. $6 \leftrightarrow 5 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 1 \leftrightarrow 2$

nielit2017july-scientistb-cs data-structures linked-list

37.13.5 Linked List: NIELIT 2021 Dec Scientist A - Section B: 104

Which type of linked list stores the address of the header node in the next field of the last node ?

A. Singly linked list
B. Circular linked list
C. Doubly linked list
D. Circular header linked list

nielit2021dec-scientista data-structures linked-list

Answer key☟

37.13.6 Linked List: NIELIT 2021 Dec Scientist B - Section B: 48

The following type definition is for a ____________ .

type pointer $= \uparrow$ node

node = record

data : integer

link : pointer

end;

A. Structure
B. Link List
C. Stack
D. Doubly link list

nielit2021dec-scientistb data-structures linked-list programming-in-c

Answer key☟

37.13.7 Linked List: NIELIT Scientist B 2020 November: 90

Which type of linked list stores the address of the header node in the next field of the last node?

A. Singly linked list
B. Circular linked list
C. Doubly linked list
D. Hashed list

nielit-scb-2020 linked-list data-structures

Answer key☟

37.14 Operator Precedence (2)

37.14.1 Operator Precedence: NIELIT 2016 DEC Scientist B (IT) - Section B: 17

Which of the following is the correct order of evaluation for the below expression?

$z = x + y^* z/4\%2 - 1$

A. $^*/\% + - =$
B. $=^* /\% + -$
C. $/^*\% - + =$
D. $^*\%/ - + =$

nielit2016dec-scientistb-it data-structures operator-precedence stack

Answer key☟

37.14.2 Operator Precedence: NIELIT 2017 DEC Scientific Assistant A - Section B: 59

The expression $5 - 2 - 3 * 5 - 2$ will evaluate to 18, if :

A. '$-$' is left associative and '$*$' has precedence over '$-$'
B. '$-$' is right associative and '$*$' has precedence over '$-$'
C. '$-$' is right associative and '$-$' has precedence over '$*$'
D. '$-$' is left associative and '$-$' has precedence over '$*$'

nielit2017dec-assistanta data-structures operator-precedence

Answer key☟

37.15 Pre Order Traversal (1)

37.15.1 Pre Order Traversal: NIELIT Scientist B 2020 November: 63

Binary search tree contains the values $1, 2, 3, 4, 5, 6, 7, 8$. The tree is traversed in pre-order and the values are printed out. Which of the following sequences is a valid output?

A. 53124786 B. 53126487 C. 53241678 D. 53124768

nielit-scb-2020 binary-search-tree tree data-structures pre-order-traversal

Answer key☟

37.16 Priority Queue (1)

37.16.1 Priority Queue: NIELIT 2017 July Scientist B (CS) - Section B: 8

A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: $10, 8, 5, 3, 2$. Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the elements is

A. $10,8,7,3,2,1,5$
B. $10,8,7,2,3,1,5$
C. $10,8,7,1,2,3,5$
D. $10,8,7,5,3,2,1$

nielit2017july-scientistb-cs data-structures priority-queue binary-heap

Answer key☟

37.17 Queue (10)

37.17.1 Queue: NIELIT 2016 DEC Scientist B (IT) - Section B: 15

A ________ is a linear list in which insertions and deletions are made to from either end of the structure.

A. Circular queue. B. Priority queue. C. Stack. D. Dequeue.

nielit2016dec-scientistb-it data-structures queue

Answer key☟

37.17.2 Queue: NIELIT 2017 July Scientist B (CS) - Section B: 10

A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT(n refers to the number of items in the queue)?

A. Both operations can be performed in $O(1)$ time.
B. At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$.
C. The worst case time complexity for both operations will be $\Omega(n)$.
D. Worst case time complexity for both operations will be $\Omega(\log n)$.

nielit2017july-scientistb-cs data-structures queue

Answer key☟

37.17.3 Queue: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 12

If queue is implemented using arrays, what would be the worst run time complexity of queue and dequeue operations?

A. $O(n), O(n)$ B. $O(n), O(1)$ C. $O(1), O(n)$ D. $O(1), O(1)$

nielit2017oct-assistanta-it data-structures queue

Answer key☟

37.17.4 Queue: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 6

Which of the following is useful in traversing a given graph by breadth first search?

A. Stack B. Set C. List D. Queue

nielit2017oct-assistanta-it data-structures queue

Answer key☟

37.17.5 Queue: NIELIT 2018-69

____ number of queues are needed to implement a stack

A. 1 B. 2 C. 3 D. 4

nielit-2018 data-structures queue

Answer key☟

37.17.6 Queue: NIELIT 2021 Dec Scientist B - Section B: 19

In which types of following data structure, an element is inserted at one end called Rear and deleted at other end called Front?

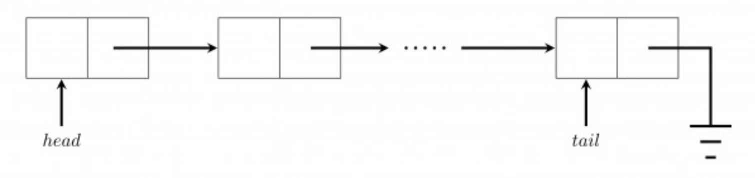
A. Stack B. Queue
C. Height balanced Avl tree D. Binary tree

nielit-2021-it-dec-scientistb data-structures queue

Answer key☟

37.17.7 Queue: NIELIT 2022 April Scientist B | Section B | Question: 90

A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let 'enqueue' be implemented by inserting a new node at the head, and 'dequeue' be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of 'enqueue' and 'dequeue, respectively, for this data structure?

A. $\Theta(1), \Theta(1)$
B. $\Theta(1), \Theta(n)$
C. $\Theta(n), \Theta(1)$
D. $\Theta(n), \Theta(n)$

nielit2022apr-scientistb data-structures linked-list queue time-complexity

37.17.8 Queue: NIELIT Scientific Assistant A 2020 November: 114

Priority queue is implemented by :

A. Doubly link list
B. Graph
C. Heap
D. Stack

nielit-sta-2020 data-structures queue

Answer key☟

37.17.9 Queue: NIELIT Scientific Assistant A 2020 November: 58

A stack can be implemented using queue, but then we need to use atleast :

A. 3 queues
B. 2 queues
C. only one queue is sufficient
D. none of the options

nielit-sta-2020 data-structures queue

Answer key☟

37.17.10 Queue: NIELIT Scientist B 2020 November: 58

Typical time requirement for operations on queues is :

A. $O(1)$
B. $O(n)$
C. $O(logn)$
D. $O(n^2)$

nielit-scb-2020 data-structures time-complexity queue

Answer key☟

37.18 Recurrence Relation (1)

37.18.1 Recurrence Relation: NIELIT 2016 DEC Scientist B (IT) - Section B: 16

The recurrence relation capturing the optimal execution time of the Towers of Hanoi problem with n discs is:

A. $T(n) = 2T(n-2) + 2$
B. $T(n) = 2T(n/2) + 1$
C. $T(n) = 2T(n-1) + n$
D. $T(n) = 2T(n-1) + 1$

nielit2016dec-scientistb-it data-structures recurrence-relation

Answer key☟

37.19 Searching (1)

37.19.1 Searching: NIELIT 2021 Dec Scientist A - Section B: 55

If a hash table is implemented as a search tree, the expected time required to enter n names and make m searches is proportional to :

A. $(\mathrm{n}+\mathrm{m}) \log_2 \mathrm{n}$
B. $(\mathrm{n}+\mathrm{m}) \log_2 \mathrm{m}$
C. $\mathrm{mn} \log_2 \mathrm{n}$
D. $\mathrm{mn} \log_2 \mathrm{m}$

nielit2021dec-scientista data-structures searching time-complexity algorithm-design

Answer key☟

37.20 Sorting (1)

37.20.1 Sorting: NIELIT Scientist B 2020 November: 82

Which of the following techniques deals with sorting the data stored in the computer's memory?

A. Distribution sort
B. Internal sort
C. External sort
D. Radix sort

nielit-scb-2020 sorting data-structures

Answer key☟

37.21 Stack (2)

37.21.1 Stack: NIELIT 2022 April Scientist B | Section B | Question: 106

A single array $A[1 \ldots \mathrm{MAXSIZE}]$ is used to implement two stacks. The two stacks grow from opposite ends of the array. Variables $\text{top } 1$ and $\text{top } 2$ $(\text{top } 1 < \text{top } 2)$ point to the location of the topmost element in each of the stacks. If the space is to be used efficiently, the condition for "stack full" is

A. $(\text{top } 1 = \mathrm{MAXSIZE}/2)$ and $(\text{top } 2 = \mathrm{MAXSIZE}/2 + 1)$
B. $\text{top } 1 + \text{top } 2 = \mathrm{MAXSIZE}$
C. $(\text{top } 1 = \mathrm{MAXSIZE}/2)$ or $(\text{top } 2 = \mathrm{MAXSIZE})$
D. $\text{top } 1 = \text{top } 2 - 1$

nielit2022apr-scientistb data-structures stack array

37.21.2 Stack: NIELIT Scientific Assistant A 2020 November: 115

A recursive problem like tower of hanoi can be rewritten without recursion using:

A. stack B. priority queue C. graph D. cycles

nielit-sta-2020 data-structures stack

Answer key☟

37.22 Tree (6)

37.22.1 Tree: NIELIT 2016 DEC Scientist B (CS) - Section B: 42

The maximum number of nodes in a binary tree of level $k, k \geq 1$ is:

A. $2^k + 1$ B. 2^{k-1} C. $2^k - 1$ D. $2^{k-1} - 1$

nielit2016dec-scientistb-cs data-structures tree binary-tree

Answer key☟

37.22.2 Tree: NIELIT 2017 July Scientist B (CS) - Section B: 54

In a complete k-ary tree, every internal node has exactly k children. The number of leaves in such a tree with n internal nodes is

A. nk B. $(n-1)k+1$ C. $n(k-1)+1$ D. $n(k-1)$

nielit2017july-scientistb-cs data-structures tree

Answer key☟

37.22.3 Tree: NIELIT 2018-72

_____ number of leaf nodes in a rooted tree of n nodes, where each node is having 0 or 3 children

A. $\frac{n}{2}$ B. $\frac{(2n+1)}{3}$ C. $\frac{(n-1)}{n}$ D. $(n-1)$

nielit-2018 data-structures tree

Answer key☟

37.22.4 Tree: NIELIT 2021 Dec Scientist A - Section B: 47

Suppose a binary search tree has been constructed from the following sequence of numbers in the order in which they arrive : $6, 2, 10, 1, 5, 7, 11, 3, 9, 4, 8$. Consider the following piece of code :

Show(root) $\{$ if $(\text{root} \mathrel{!}= \text{NULL})$

$\{\text{printf}(\text{"}\%d\text{"}, \text{root} \rightarrow \text{key});$

show $(\text{root} \rightarrow \text{right});$

show $(\text{root} \rightarrow \text{left});$

$\}$

else

return ;

}

The sequence printed will be :

A. $6, 11, 10, 7, 8, 9, 2, 4, 3, 5, 1$
B. $6, 11, 7, 9, 8, 10, 2, 5, 1, 3, 4$
C. $6, 10, 11, 7, 9, 8, 2, 5, 3, 4, 1$
D. $6, 10, 2, 11, 7, 9, 8, 5, 3, 4, 1$

nielit2021dec-scientista binary-search-tree tree binary-tree data-structures programming-in-c

Answer key☟

37.22.5 Tree: NIELIT 2022 April Scientist B | Section B | Question: 59

Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:

The height of a tree with a single node is 0.

A. 4 and 15 respectively
B. 3 and 14 respectively
C. 4 and 14 respectively
D. 3 and 15 respectively

nielit2022apr-scientistb binary-search-tree data-structures tree

Answer key☟

37.22.6 Tree: NIELIT Scientist B 2020 November: 106

Suppose we have to insert the following sequence of keys into an empty binary search tree:

$5, 7, 45, 60, 50, 23, 15, 54$

What would be the height of binary search tree?

A. 3
B. 4
C. 5
D. 6

nielit-scb-2020 binary-search-tree data-structures tree

Answer key☟

37.23 Tree Traversal (7)

37.23.1 Tree Traversal: NIELIT 2016 MAR Scientist B - Section C: 18

Traversing a binary tree first root and then left and right subtrees called _______ traversal.

A. postorder.
B. preorder.
C. inorder.
D. none of these.

nielit2016mar-scientistb data-structures binary-tree tree-traversal

Answer key☟

37.23.2 Tree Traversal: NIELIT 2017 DEC Scientific Assistant A - Section B: 2

In binary search tree which traversal is used for getting ascending order values ?

A. Inorder B. Preorder C. Postorder D. None of the options

nielit2017dec-assistanta data-structures binary-search-tree tree-traversal

Answer key☟

37.23.3 Tree Traversal: NIELIT 2017 DEC Scientist B - Section B: 30

If for a given Binary Search Tree (BST) the pre-order traversal is $41, 23, 11, 31, 62, 50, 73$. Then which of the following is its post-order traversal?

A. $11, 31, 23, 50, 73, 62, 41$
B. $31, 11, 23, 50, 41, 62, 73$
C. $11, 31, 50, 23, 73, 62, 41$
D. $11, 31, 23, 50, 62, 73, 41$

nielit2017dec-scientistb data-structures binary-search-tree tree-traversal

Answer key☟

37.23.4 Tree Traversal: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 5

A binary search tree contains the values-$1, 2, 3, 4, 5, 6, 7$ and 8. The tree is traversed in preorder and the values are printed out. Which of the following sequences is a valid output?

A. $5\ 3\ 1\ 2\ 4\ 7\ 8\ 6$
B. $5\ 3\ 1\ 2\ 6\ 4\ 9\ 7$
C. $5\ 3\ 2\ 4\ 1\ 6\ 7\ 8$
D. $5\ 3\ 1\ 2\ 4\ 7\ 6\ 8$

nielit2017oct-assistanta-cs data-structures binary-search-tree tree-traversal

Answer key☟

37.23.5 Tree Traversal: NIELIT 2018-80

_____ traversals are not sufficient to build a binary tree.

A. Preorder and Inorder
B. Postorder and Inorder
C. Postorder and Preorder
D. None of these

nielit-2018 data-structures tree-traversal

Answer key☟

37.23.6 Tree Traversal: NIELIT 2021 Dec Scientist B - Section B: 74

The preorder traversal of a binary search tree is given by $12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20$.

Then postorder traversal will be:

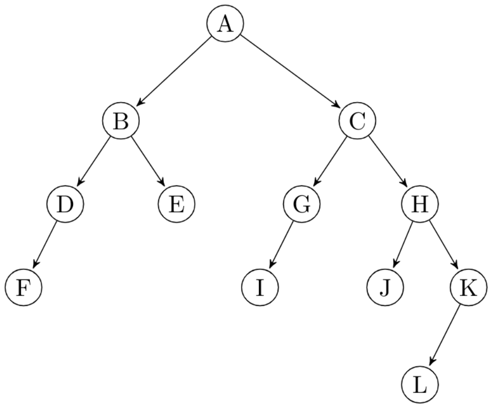
A. $2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20$
B. $2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12$
C. $7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12$
D. $7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12$

nielit-2021-it-dec-scientistb binary-search-tree binary-tree tree-traversal data-structures algorithm-design

Answer key☟

37.23.7 Tree Traversal: NIELIT Scientist B 2020 November: 45

The Preorder traversal of a tree given below is:



A. A B D F E C G I H J K L
B. A B C D E G H F I J K L
C. A B E D F C G H I J K L
D. A B D F E C G I J H K L

nielit-scb-2020 tree-traversal tree data-structures

Answer key☟

Answer Keys

37.1.1	D	37.2.1	B	37.3.1	B	37.3.2	A	37.3.3	B
37.3.4	A	37.4.1	B	37.4.2	A	37.4.3	C	37.4.4	A
37.4.5	B	37.5.1	B	37.6.1	B	37.6.2	A	37.6.3	C
37.7.1	D	37.7.2	D	37.7.3	D	37.7.4	X	37.7.5	C
37.7.6	C	37.7.7	B	37.7.8	C	37.7.9	B	37.7.10	B
37.7.11	C	37.8.1	B	37.9.1	B	37.9.2	A	37.9.3	B
37.10.1	B	37.10.2	A	37.11.1	X	37.11.2	A	37.11.3	A
37.11.4	C	37.11.5	C	37.12.1	A	37.13.1	C	37.13.2	D
37.13.3	B	37.13.4	C	37.13.5	B	37.13.6	B	37.13.7	B
37.14.1	A	37.14.2	A	37.15.1	D	37.16.1	A	37.17.1	D
37.17.2	A	37.17.3	D	37.17.4	D	37.17.5	B	37.17.6	B
37.17.7	B	37.17.8	C	37.17.9	B	37.17.10	A	37.18.1	D
37.19.1	A	37.20.1	B	37.21.1	D	37.21.2	A	37.22.1	C
37.22.2	C	37.22.3	B	37.22.4	C	37.22.5	B	37.22.6	D
37.23.1	B	37.23.2	A	37.23.3	A	37.23.4	D	37.23.5	C
37.23.6	B	37.23.7	A						

38.1 Recursion (1)

38.1.1 Recursion: NIELIT 2022 April Scientist B | Section B | Question: 77

Consider the following two functions.

```
void fun1(int n) {
    if(n == 0) return;
    printf("%d", n);
    fun2(n - 2);
    printf("%d", n);
}
void fun2(int n) {
    if(n == 0) return;
    printf("%d", n);
    fun1(++n);
    printf("%d", n);
}
```

The output printed when $\text{fun1}(5)$ is called is

A. 53423122233445
B. 53423120112233
C. 53423122132435
D. 53423120213243

nielit2022apr-scientistb programming normal tricky recursion

Answer Keys

38.1.1	C

39 Programming: Programming in C (39)

39.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 58

Bug means

A. A logical error in a program
B. A difficult syntax error in a program
C. Documenting programs using an efficient documentation tool
D. All of the above

nielit2017july-scientistb-cs programming

Answer key

39.0.2 NIELIT 2017 July Scientist B (IT) - Section B: 37

Consider the following C declaration

```
struct
{
    short s[5];
    union
    {
        float y;
        long z;
    }u;
}t;
```

Assume that objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes respectively. The memory requirement for variable t ignoring alignment considerations, is

A. 22 bytes
B. 14 bytes
C. 18 bytes
D. 10 bytes

nielit2017july-scientistb-it programming

Answer key

39.0.3 NIELIT 2017 July Scientist B (IT) - Section B: 26

The following program is to be tested for statement coverage:

```
begin
if(a==b){S1;exit;}
else if (c==d){S2;}
else{S3;exit;}
S4;
end
```

The test cases $T1, T2, T3$ and $T4$ given below are expressed in terms of the properties satisfied by the values of variables a, b, c and d. The exact values are not given.

$T1 : a, b, c$ and d are all equal
$T1 : a, b, c$ and d are all distinct
$T3 : a = b$ and $c! = d$
$T4 : a! = b$ and $c = d$

Which of the test suites given below ensures coverage of statements $S1, S2, S3$ and

$S4$?

A. T_1, T_2, T_3 B. T_2, T_4 C. T_3, T_4 D. T_1, T_2, T_4

nielit2017july-scientistb-it programming

Answer key

39.0.4 NIELIT 2016 MAR Scientist B - Section C: 45

Control structures include

A. iteration
B. rendezvous statements
C. exception statements
D. all of these

nielit2016mar-scientistb programming

Answer key

39.1 Array of Pointers (4)

39.1.1 Array of Pointers: NIELIT 2016 DEC Scientist B (IT) - Section B: 11

What is the meaning of following declaration?

```
int(*p[7])();
```

A. p is pointer to function
B. p is pointer to such function which return type is array
C. p is array of pointer to function
D. p is pointer to array of function

nielit2016dec-scientistb-it programming-in-c pointers array-of-pointers

Answer key

39.1.2 Array of Pointers: NIELIT 2017 DEC Scientific Assistant A - Section B: 42

Which of the following is illegal declaration in C language?

A.
```
char*str ="Raj is a Research scholar";
```
B.
```
char str[25] ="Raj is a Research scholar";
```
C.
```
char str[40] ="Raj is a Research scholar";
```
D.
```
char[]str ="Raj is a Research scholar";
```

nielit2017dec-assistanta programming-in-c pointers array-of-pointers

Answer key

39.1.3 Array of Pointers: NIELIT 2017 DEC Scientist B - Section B: 44

If x is a one dimensional array, then

A. $*(x+i)$ is same as $*(\&x[i])$
B. $\&x[i]$ is same as $x+i-1$
C. $*(x+i)$ is same as $*x[i]$
D. $*(x+i)$ is same as $*x+i$

nielit2017dec-scientistb programming-in-c array-of-pointers

Answer key☟

39.1.4 Array of Pointers: NIELIT 2017 July Scientist B (IT) - Section B: 35

Assume that float takes 4 bytes, predict the output of following program.

```
#include<stdio.h>
int main()
{
float arr[5]={12.5,10.0,13.5,90.5,0.5};
float *ptr1=&arr[0];
float *ptr2=ptr1+3;
printf("%f",*ptr2);
printf("%d",ptr2-ptr1);
return 0;
}
```

A. $90.500000 \;\; 3$
B. $90.500000 \;\; 12$
C. $10.000000 \;\; 12$
D. $0.500000 \;\; 3$

nielit2017july-scientistb-it programming-in-c pointers array-of-pointers

Answer key☟

39.2 Data Structures (1)

39.2.1 Data Structures: NIELIT 2018-41

_____ is used to convert from recursive to iterative implementation of an algorithm

A. Array
B. Tree
C. Stack
D. Queue

nielit-2018 data-structures

Answer key☟

39.3 Dynamic Scoping (1)

39.3.1 Dynamic Scoping: NIELIT 2016 DEC Scientist B (IT) - Section B: 60

How will you free the allocated memory?

A. remove(var-name)
B. free(var-name)
C. delete(var-name)
D. dalloc(var-name)

nielit2016dec-scientistb-it programming-in-c dynamic-scoping

Answer key☟

39.4 Functions (1)

39.4.1 Functions: NIELIT 2016 DEC Scientist B (IT) - Section B: 25

The keyboard used to transfer control from a function back to the calling function is:

A. switch B. go to C. go back D. return

nielit2016dec-scientistb-it programming functions

Answer key

39.5 Output (13)

39.5.1 Output: NIELIT 2016 DEC Scientist B (CS) - Section B: 13

What will be output if you will compile and execute the following C code?

```
void main()
{
printf("%d",sizeof(5.2));
}
```

A. 4 B. 8 C. 2 D. 16

nielit2016dec-scientistb-cs programming-in-c output

Answer key

39.5.2 Output: NIELIT 2016 DEC Scientist B (CS) - Section B: 8

What will be output if you will compile and execute the following C code?

```
void main(){
char c=125;
c=c+10;
printf("%d",c);
}
```

A. 135 B. 115 C. -121 D. -8

nielit2016dec-scientistb-cs programming-in-c output

Answer key

39.5.3 Output: NIELIT 2016 MAR Scientist B - Section C: 15

Output of the following loop is

```
for(putchar('c');putchar
('a');putchar('r'))
putchar('t');
```

A. a syntax error. B. cartrt. C. catrat. D. catratratratrat...

nielit2016mar-scientistb programming-in-c output

Answer key

39.5.4 Output: NIELIT 2017 July Scientist B (IT) - Section B: 32

Output of following program

```
#include<stdio.h>
int main()
```

```
{
int i=5;
printf("%d %d %d", i++,i++,i++);
return 0;
}
```

A. 7 6 5 B. 5 6 7 C. 7 7 7 D. Compiler Dependent

nielit2017july-scientistb-it programming-in-c output

Answer key

39.5.5 Output: NIELIT 2017 July Scientist B (IT) - Section B: 33

Output of following program?

```
#include<stdio.h>
void dynamic(int s,...)
{
printf("%d",s);
}
int main()
{
dynamic(2,4,6,8);
dynamic(3,6,9);
return 0;
}
```

A. 2 3 B. Compiler Error C. 4 3 D. 3 2

nielit2017july-scientistb-it programming-in-c output

Answer key

39.5.6 Output: NIELIT 2017 July Scientist B (IT) - Section B: 36

Assume that size of an integer is 32 bit. What is the output of following ANSI C program?

```
#include<stdio.h>
struct st
{
int x;
static int y;
};
int main()
{
printf(%d",sizeof(struct st));
return 0;
}
```

A. 4 B. 8 C. Compiler Error D. Runtime Error

nielit2017july-scientistb-it programming-in-c output

Answer key

39.5.7 Output: NIELIT 2017 July Scientist B (IT) - Section B: 38

Output of the following program?

```c
#include<stdio.h>
struct st
{
    int x;
    struct st next;
};
int main()
{
    struct st temp;
    temp.x=10;
    temp.next=temp;
    printf("%d",temp.next,x);
    return 0;
}
```

A. Compiler Error B. 10 C. Runtime Error D. Garbage Value

nielit2017july-scientistb-it programming-in-c output

Answer key

39.5.8 Output: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 6

The question is based on the following program fragment.

```c
f(intY[10],int x){

int u,j,k;

i=0;j=9;

do{

        k=(i+j)/2;

        if(Y[k] < x) i=k; else j=k;

    } while(Y[k]!=x) && (i<j));

     if (Y[k]==x) printf("x is in the array.");

      else printf("x is not in the array.");

}
```

On which of the following contents of ‘Y’ and ‘x’ does the program fail?

A. Y is $[1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10]$ and $x < 10$
B. Y is $[1\ 3\ 5\ 7\ 9\ 11\ 13\ 15\ 17\ 19]$ and $x < 1$
C. Y is $[2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2]$ and $x > 2$
D. Y is $[2\ 4\ 6\ 8\ 10\ 12\ 14\ 16\ 18\ 20]$ and $2 < x < 20$ and $'x'$ is even

nielit2017oct-assistanta-cs programming-in-c output

Answer key

39.5.9 Output: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 11

What will be the output of following?

```
main()

{

    Static int a = 3;

    Printf("%d",a--);

    If(a)

    main();

}
```

A. 3
B. $3\ 2\ 1$
C. $3\ 3\ 3$
D. Program will fall in continuous loop and print 3

nielit2017oct-assistanta-cs programming-in-c output

Answer key☟

39.5.10 Output: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 3

The following program fragment prints

```
int i = 5;
do { putchar(i+100); printf("%d", i--;) }
while(i);
```

A. i5h4g3f2el
B. 14h3g2f1e0
C. An error message
D. None of the above

nielit2017oct-assistanta-cs programming-in-c output

Answer key☟

39.5.11 Output: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 1

What will be the output of following?

```
main()

{

    Static int a = 3;

    Printf("%d",a--);

    If(a)

    main();

}
```

A. 3
B. $3\ 2\ 1$
C. $3\ 3\ 3$
D. Program will fall in continuous loop and print 3

nielit2017oct-assistanta-it programming-in-c output

Answer key

39.5.12 Output: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 1

The question is based on the following program fragment.

```
f(intY[10],int x){

int u,j,k;

i=0;j=9;

do{

        k=(i+j)/2;

        if(Y[k] < x) i=k; else j=k;

    } while((Y[k]!=x) && (i<j));

    if (Y[k]==x) printf("x is in the array.");

     else printf("x is not in the array.");

}
```

On which of the following contents of ‘Y’ and ‘x’ does the program fail?

A. Y is $[1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10]$ and $x < 10$
B. Y is $[1\ 3\ 5\ 7\ 9\ 11\ 13\ 15\ 17\ 19]$ and $x < 1$
C. Y is $[2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2\ 2]$ and $x > 2$
D. Y is $[2\ 4\ 6\ 8\ 10\ 12\ 14\ 16\ 18\ 20]$ and $2 < x < 20$ and $'x'$ is even

nielit2017oct-assistanta-it programming-in-c output

Answer key

39.5.13 Output: NIELIT 2021 Dec Scientist B - Section B: 21

What will be the output of the following ‘C’ program?

```
void count(int n)
{
static int d = 1;
printf(" %d", n);
printf(" %d", d);
d++;
if (n>1) count (n - 1);
printf(" %d", d);
}
void main()
{
count(3);
}
```

A. $3\ 1\ 2\ 2\ 1\ 3\ 4\ 4\ 4$ B. $3\ 1\ 2\ 1\ 1\ 1\ 2\ 2\ 2$

C. 3 1 2 2 1 3 4 D. 3 1 2 1 1 1 2

nielit-2021-it-dec-scientistb programming-in-c output

Answer key☟

39.6 Pointers (3)

39.6.1 Pointers: NIELIT 2016 MAR Scientist B - Section C: 14

Prior to using a pointer variable it should be

A. declared.
B. initialized.
C. both declared and initialized.
D. none of these.

nielit2016mar-scientistb programming-in-c pointers

Answer key☟

39.6.2 Pointers: NIELIT 2017 July Scientist B (IT) - Section B: 34

Output of following program?

```
#include<stdio.h>
int main()
{
int *ptr;
int x;
ptr=&x;
*ptr=0;
printf("x=%d\n",x);
printf("*ptr=%d\n",*ptr);
*ptr+=5;
printf("x=%d\n",x);
printf("*ptr=%d\n",*ptr);
(*ptr)++;
printf(“x=%d\n",x);
printf("*ptr=%d\n",*ptr);
return 0;
}
```

A. $x = 0$
$*ptr = 0$
$x = 5$
$*ptr = 5$
$x = 6$
$*ptr = 6$

B. $x =$garbage value
$*ptr = 0$
$x =$garbage value
$*ptr = 5$
$x =$garbage value
$*ptr = 6$

C. $x = 0$
$*ptr = 0$
$x = 5$
$*ptr = 5$

x =garbage value
$*ptr$ =garbage value

D. $x = 0$
$*ptr = 0$
$x = 0$
$*ptr = 0$
$x = 0$
$*ptr = 0$

nielit2017july-scientistb-it programming-in-c pointers

Answer key

39.6.3 Pointers: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 29

Consider the following declaration.

```
int a, *b = &a, **c = &b;

a = 4;

**c = 5;
```

If the statement

```
b = (int *)**c
```

Is appended to the above program fragment then

A. Value of b becomes 5
B. Value of b will be the address of c
C. Value of b is unaffected
D. None of these

nielit2017oct-assistanta-it programming pointers

Answer key

39.7 Programming In C (10)

39.7.1 Programming In C: NIELIT 2016 DEC Scientist B (IT) - Section B: 10

The maximum combined length of the command-line arguments including the spaces between adjacent arguments is:

A. 128 characters
B. 256 characters
C. 67 characters
D. It may very from one Operating System to another

nielit2016dec-scientistb-it programming-in-c

Answer key

39.7.2 Programming In C: NIELIT 2016 MAR Scientist B - Section C: 11

What is the correct way to round off x, a float to an int value?

A. $y = (\text{int})(x + 0.5)$
B. $y = \text{int}(x + 0.5)$
C. $y = (\text{int})x + 0.5$
D. $y = (\text{int})(\text{int})x + 0.5)$

nielit2016mar-scientistb programming-in-c

Answer key

39.7.3 Programming In C: NIELIT 2016 MAR Scientist B - Section C: 13

What error would the following function give on compilation?

```
f(int a, int b)
{
int a;
a=20;
return a;
}
```

A. Missing parenthesis is $return$ statement.
B. Function should be defined as $\text{int f(int a, int b)}$
C. Redeclaration of a.
D. None of these.

nielit2016mar-scientistb programming-in-c

Answer key

39.7.4 Programming In C: NIELIT 2016 MAR Scientist B - Section C: 16

If space occupied by two strings s_1 and s_2 in 'C' are respectively m and n, then space occupied by string obtained by concatenating s_1 and s_2 is always

A. less than $m+n$
B. equal to $m+n$
C. greater than $m+n$
D. none of these

nielit2016mar-scientistb programming-in-c

Answer key

39.7.5 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 29

An external variable

A. is globally accessible by all functions
B. has a declaration “extern” associated with it when declared within a function
C. will be initialized to 0 if not initialized
D. all of these

nielit2016mar-scientistc programming-in-c

Answer key

39.7.6 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 57

If initialization is a part of declaration of a structure, then storage class can be

A. automatic
B. register
C. static
D. anything

nielit2016mar-scientistc programming-in-c

Answer key

39.7.7 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 58

For x and y are variables as declared below $double\, x = 0.005,\ y = -0.01;$ What is the value of $\text{ceil}\ (x + y)$, where ceil is a function to compute ceiling of a number?

A. 1 B. 0 C. 0.005 D. 0.5

nielit2016mar-scientistc programming-in-c

Answer key☟

39.7.8 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 59

In C programming language, if the first and the second operands of operator $+$ are of types int and float, respectively, the result will be of type

A. int B. float C. char D. long int

nielit2016mar-scientistc programming-in-c

Answer key☟

39.7.9 Programming In C: NIELIT 2016 MAR Scientist C - Section C: 60

What will be the value of x and y after execution of the following statement(C language)

```
n = 5; x = n++; y = -x;
```

A. $5, -4$ B. $6, -5$ C. $6, -6$ D. $5, -5$

nielit2016mar-scientistc programming-in-c

Answer key☟

39.7.10 Programming In C: NIELIT 2017 July Scientist B (CS) - Section B: 5

Following is C like Pseudo code of a function that takes a number as an argument, and uses a stack S to do processing.

```
void fun(int n)
{
Stack S;//Say it creates an empty stack S
while(n>0)
{
 // This line pushes the value of n%2 to stack S;
 Push(&S,n%2);
 n=n/2;
}
// Run while Stack S is not empty
while(!is Empty(&S))
printf("%d",pop(&S));//pop an element from S and print it
}
```

What does the above function do in general ?

A. Prints binary representation of n in reverse order.

B. Prints binary representation of n.
C. Prints the value of $\log n$.
D. Prints the value of $\log n$ in reverse order.

nielit2017july-scientistb-cs programming-in-c

Answer key

39.8 Recursion (2)

39.8.1 Recursion: NIELIT 2021 Dec Scientist B - Section B: 32

Consider the following C function :

```
int f(int n) {
   static int i = 1;
   if (n > = 5) return n;
   n=n + i;i++;
   return f(n); }
```

The value returned by $\mathrm{f}(1)$ is :

A. 5 B. 6 C. 7 D. 8

nielit-2021-it-dec-scientistb programming-in-c recursion functions

Answer key

39.8.2 Recursion: NIELIT 2021 Dec Scientist B - Section B: 72

What is the value of $f(4)$ using the following C code:

```
int f(int k){
   if(k<3)
      return k;
   else
      return f(k-1) * f(k-2) + f(k-3);
}
```

A. 5 B. 6 C. 7 D. 8

nielit2021dec-scientistb programming-in-c recursion recurrence-relation data-structures

Answer Keys

39.0.1	A	39.0.2	C	39.0.3	D	39.0.4	D	39.1.1	C
39.1.2	D	39.1.3	A	39.1.4	A	39.2.1	C	39.3.1	B
39.4.1	D	39.5.1	B	39.5.2	C	39.5.3	D	39.5.4	D
39.5.5	A	39.5.6	C	39.5.7	A	39.5.8	C	39.5.9	B
39.5.10	A	39.5.11	B	39.5.12	C	39.5.13	A	39.6.1	C

39.6.2	A	39.6.3	A	39.7.1	D	39.7.2	A	39.7.3	C
39.7.4	A	39.7.5	D	39.7.6	C	39.7.7	B	39.7.8	B
39.7.9	D	39.7.10	B	39.8.1	C	39.8.2	A		

40 Theory of Computation (80)

40.1 Binary Tree (1)

40.1.1 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 120

Let B_n denote the number of full binary trees with n vertices. Then a recurrence relation for B_n is :

A. $\mathrm{B}_n = \mathrm{B}_{n-1} + \mathrm{O}(1)$
B. $\mathrm{B}_n = 2\mathrm{B}_{n-1} + \mathrm{O}(1)$
C. $\mathrm{B}_n = \mathrm{B}_{n-1} + \mathrm{O}(n)$
D. $\mathrm{B}_n = 2\mathrm{B}_{n-1} + \mathrm{O}(n)$

nielit2021dec-scientistb recurrence-relation binary-tree data-structures algorithms

40.2 Complexity Classes (2)

40.2.1 Complexity Classes: NIELIT 2021 Dec Scientist A - Section B: 74

Assume that P and NP are different i.e. $\mathrm{P}! = \mathrm{NP}$ then for the expression $\text{NP-Complete} \cap \mathrm{P} = ?$ Which among the following is correct ?

A. NP-Hard
B. $\oslash$
C. P
D. NP-Complete

nielit2021dec-scientista theory-of-computation complexity-classes

Answer key☟

40.2.2 Complexity Classes: NIELIT 2021 Dec Scientist B - Section B: 46

__________ is the class of decision problems that can be solved by non-deterministic polynomial algorithms.

A. NP
B. P
C. Hard
D. Complete

nielit2021dec-scientistb theory-of-computation p-np-npc-nph complexity-classes

Answer key☟

40.3 Complexity Theory (1)

40.3.1 Complexity Theory: NIELIT 2021 Dec Scientist A - Section B: 109

Given a graph with n vertices, deciding if there exists a clique of size ≥ 195 is :

A. Solvable in polynomial time
B. NP
C. NP-Complete
D. None of the above

nielit2021dec-scientista graph-theory complexity-theory p-np-npc-nph algorithm-design

Answer key☟

40.4 Context Free Grammar (3)

40.4.1 Context Free Grammar: NIELIT 2016 MAR Scientist B - Section C: 29

The CFG $S \rightarrow aS \mid bS \mid a \mid b$ is equivalent to

A. $(a+b)$ B. $(a+b)(a+b)^*$ C. $(a+b)(a+b)$ D. all of these

nielit2016mar-scientistb theory-of-computation context-free-grammar

Answer key☟

40.4.2 Context Free Grammar: NIELIT 2017 DEC Scientist B - Section B: 7

Let G be a grammar in CFG and let $W_1, W_2 \in L(G)$ such that $|W_1 |=| W_2|$ then which of the following statements is true?

A. Any derivation of W_1 has exactly the same number of steps as any derivation of W_2.
B. Different derivation have different length.
C. Some derivation of W_1 may be shorter than the derivation of W_2
D. None of the options

nielit2017dec-scientistb theory-of-computation context-free-grammar

Answer key☟

40.4.3 Context Free Grammar: NIELIT 2021 Dec Scientist A - Section B: 98

Which of the following problems is undecidable ?

A. Membership problem for CFGs.
B. Ambiguity problem for CFGs.
C. Finiteness problem for FSAs.
D. Equivalence problem for FSAs.

nielit2021dec-scientista theory-of-computation decidability context-free-grammar finite-automata

Answer key☟

40.5 Context Free Language (5)

40.5.1 Context Free Language: NIELIT 2016 DEC Scientist B (CS) - Section B: 3

If $L1$ is CFL and $L2$ is regular language which of the following is false?

A. $L1 - L2$ is not Context free
B. $L1$ intersection $L2$ is Context free
C. $\sim L1$ is Context free
D. Both (A) and (C)

nielit2016dec-scientistb-cs theory-of-computation context-free-language

Answer key☟

40.5.2 Context Free Language: NIELIT 2016 MAR Scientist B - Section C: 26

If L_1 and L_2 are context free language and R a regular set, then which one of the languages below is not necessarily a context free language?

A. $L_1 L_2$ B. $L_1 \cap L_2$ C. $L_1 \cap R$ D. $L_1 \cup L_2$

nielit2016mar-scientistb theory-of-computation context-free-language

Answer key

40.5.3 Context Free Language: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 31

Which of the following definitions generates the same languages as L, where $L = \{x^n y^n, n \geq 1\}$

i. $E \rightarrow xEy \mid xy$
ii. $xy \mid x^+ xyy^+$
iii. $x^+ y^+$

A. (i)
B. (i) and (ii) only
C. (ii) and (iii) only
D. (ii) only

nielit2017oct-assistanta-cs theory-of-computation context-free-language

Answer key

40.5.4 Context Free Language: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 9

Which of the following definitions generates the same languages as L, where $L = \{x^n y^n, n \geq 1\}$

i. $E \rightarrow xEy \mid xy$
ii. $xy \mid x^+ xyy^+$
iii. $x^+ y^+$

A. (i) Only
B. (i) and (ii) only
C. (ii) and (iii) only
D. (ii) only

nielit2017oct-assistanta-it theory-of-computation context-free-language

Answer key

40.5.5 Context Free Language: NIELIT 2018-33

The language $\{W^a X^b Y^{a+b} \mid a, b, > 1\}$ is

A. Regular
B. Context-free but not regular
C. Context sensitive but not context free
D. Type$= 0$ but not context sensitive

nielit-2018 theory-of-computation context-free-language

Answer key

40.6 Context Sensitive (1)

40.6.1 Context Sensitive: NIELIT 2021 Dec Scientist B - Section B: 80

The following grammer is an example of _____________ .

$\mathrm{A} \rightarrow \mathrm{a\ A\ B\ C}$

$\mathrm{CB} \rightarrow \mathrm{B\ c}$

$\mathrm{A} \rightarrow \mathrm{a\ b\ c}$

$\mathrm{b\ B} \rightarrow \mathrm{b\ b}$

$\mathrm{B} \rightarrow \mathrm{c\ b\ a}$

A. Type 0 Grammer
B. Type 1 Grammer
C. Type 2 Grammer
D. Type 3 Grammer

nielit2021dec-scientistb grammar theory-of-computation context-sensitive

40.7 Cyk Algorithm (1)

40.7.1 Cyk Algorithm: NIELIT 2017 DEC Scientist B - Section B: 50

Let n is the length of string to test for membership, then the number of table entry in CYK algorithm is

A. $n(n+1)$ B. n^2+1 C. n^2-1 D. $n(n+1)/2$

nielit2017dec-scientistb theory-of-computation cyk-algorithm

Answer key☟

40.8 Decidability (2)

40.8.1 Decidability: NIELIT 2017 DEC Scientist B - Section B: 11

Which of the following are undecidable?

$P1$: The language generated by some CFG contains any words of length less than some given number n.

$P2$: Let $L1$ be CFL and $L2$ be regular, to determine whether $L1$ and $L2$ have common elements

$P3$: Any given CFG is ambiguous or not.

$P4$: For any given CFG G, to determine whether epsilon belongs to $L(G)$

A. $P2$ only B. $P1$ and $P2$ only C. $P2$ and $P3$ only D. $P3$ only

nielit2017dec-scientistb theory-of-computation decidability

Answer key☟

40.8.2 Decidability: NIELIT 2022 April Scientist B | Section B | Question: 108

Consider the following problem X.
Given a Turing machine M over the input alphabet Σ, any state q of M and a word $w \in \Sigma^*$, does the computation of M on w visit the state of q ?
Which of the following statements about X is correct?

A. X is decidable

B. X is undecidable but partially decidable

C. X is undecidable and not even partially decidable

D. X is not a decision problem

nielit2022apr-scientistb theory-of-computation turing-machine decidability

40.9 Dpda (2)

40.9.1 Dpda: NIELIT 2017 DEC Scientific Assistant A - Section B: 6

Which of the following statements is true ?

A. Melay and Moore machines are language acceptors.
B. Finite State automata is language translator.
C. NPDA is more powerful than DPDA.
D. Melay machine is more powerful than Moore machine.

nielit2017dec-assistanta theory-of-computation dpda npda

Answer key☟

40.9.2 Dpda: NIELIT 2017 DEC Scientist B - Section B: 39

Which of the following is **true**?

A. Mealy and Moore machine are language acceptors.
B. Finite State automata is language translator.
C. NPDA is more powerful than DPDA.
D. Melay machine is more powerful than Moore machine.

nielit2017dec-scientistb theory-of-computation finite-automata npda dpda

Answer key☟

40.10 Expression Evaluation (2)

40.10.1 Expression Evaluation: NIELIT 2021 Dec Scientist A - Section B: 87

The postfix equivalent of the infix expression $(a+b)^*(c^*d-e)^*f/g$ is :

A. $ab+cd^*e–fg^*/^*$

B. $ab+cd^*e-fg/^{**}$

C. $ab+cde^*–fg/^{**}$

D. $abcd+e^*fg–/^{**}$

nielit2021dec-scientista expression-evaluation stack data-structures compiler-design

Answer key☟

40.10.2 Expression Evaluation: NIELIT 2021 Dec Scientist B - Section B: 60

$\mathrm{ABC}{*}{+}$ is the postfix form of :

A. $\mathrm{A}*\mathrm{B}+\mathrm{C}$

B. $\mathrm{A}*+\mathrm{BC}$

C. $\mathrm{A}+\mathrm{B}*\mathrm{C}$

D. none of these

nielit2021dec-scientistb expression-evaluation data-structures infix-prefix stack

Answer key

40.11 Finite Automata (10)

40.11.1 Finite Automata: NIELIT 2016 DEC Scientist B (CS) - Section B: 1

Palindromes can't be recognized by any Finite State Automata because:

A. FSA cannot remember arbitrarily large amount of information.
B. FSA cannot deterministically fix the midpoint.
C. Even if the mid-Point is known an FSA cannot find whether the second half of the string matches the first half.
D. All of the above.

nielit2016dec-scientistb-cs theory-of-computation finite-automata

Answer key

40.11.2 Finite Automata: NIELIT 2016 DEC Scientist B (CS) - Section B: 24

Given two DFA's $M1$ and $M2$. They are equivalent if

A. $M1$ and $M2$ has the same number of states
B. $M1$ and $M2$ accepts the same language i.e $L(M1) = L(M2)$
C. $M1$ and $M2$ has the same number of final states
D. None of the above

nielit2016dec-scientistb-cs theory-of-computation finite-automata

Answer key

40.11.3 Finite Automata: NIELIT 2016 DEC Scientist B (CS) - Section B: 46

$(00 + 01 + 10)(0 + 1)^*$ represents

A. Strings not starting with 11
B. Strings of odd length
C. Strings starting with 00
D. Strings of even length

nielit2016dec-scientistb-cs theory-of-computation finite-automata

Answer key

40.11.4 Finite Automata: NIELIT 2016 MAR Scientist C - Section C: 32

Two finite state machines are said to be equivalent if they

A. have same number of states
B. have same number of edges
C. have same number of states and edges
D. recognize same set of tokens

nielit2016mar-scientistc theory-of-computation finite-automata

Answer key

40.11.5 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 3

The automaton which allows transformation to a new state without consuming any input symbols :

A. NFA B. DFA C. $NFA - 1$ D. All of the options

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key☟

40.11.6 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 4

Complement of a DFA can be obtained by :

A. making starting state as final state.
B. make final as a starting state.
C. making final states non-final and non-final as final.
D. None of the options

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key☟

40.11.7 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 40

What is the relation between DFA and NFA on the basis of computational power ?

A. $DFA > NFA$ B. $NFA > DFA$ C. Equal D. Can't be said

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key☟

40.11.8 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 5

Concatenation Operation refers to which of the following set operations :

A. Union B. Dot C. Kleene D. None of the options

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key☟

40.11.9 Finite Automata: NIELIT 2017 DEC Scientific Assistant A - Section B: 60

Finite automata requires minimum _____________ number of stacks.

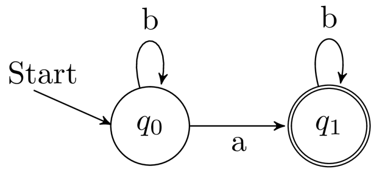
A. 1 B. 0 C. 2 D. None of the options

nielit2017dec-assistanta theory-of-computation finite-automata

Answer key☟

40.11.10 Finite Automata: NIELIT 2018-40

Consider the following Finite State Automaton. The language accepted by the automaton is given by the regular expression.



A. ab^*b^* B. a^*b^* C. b^*b D. b^*ab^*

nielit-2018 theory-of-computation finite-automata

Answer key

40.12 First Order Logic (1)

40.12.1 First Order Logic: NIELIT 2021 Dec Scientist B - Section B: 49

Let $\mathrm{P}(x)$ be "x is perfect", $\mathrm{F}(x)$ be "x is your friend" and the domain be all people. The statement, "At least one of your friends is perfect" is :

A. $\forall x\ (\mathrm{F}(x) \rightarrow \mathrm{P}(x))$
B. $\forall x\ (\mathrm{F}(x) \wedge \mathrm{P}(x))$
C. $\exists x\ (\mathrm{F}(x) \wedge \mathrm{P}(x))$
D. $\exists x\ (\mathrm{F}(x) \rightarrow \mathrm{P}(x))$

nielit2021dec-scientistb propositional-logic first-order-logic logical-reasoning

Answer key

40.13 Grammar (2)

40.13.1 Grammar: NIELIT 2021 Dec Scientist A - Section B: 67

_______________ is the most general phase structured grammar.

A. Regular B. Context free
C. Context sensitive D. All of the above

nielit2021dec-scientista grammar theory-of-computation

Answer key

40.13.2 Grammar: NIELIT 2021 Dec Scientist B - Section B: 112

$(\mathrm{a+b})^2$ corresponds to the language :

A. $\{\mathrm{a+b, a+b}\}$ B. $\{\mathrm{aa, ab, ba, bb}\}$
C. $\{\mathrm{abab, baba}\}$ D. $\{\mathrm{a+b}, (\mathrm{a+b})^2\}$

nielit2021dec-scientistb theory-of-computation regular-expression grammar

Answer key

40.14 Identify Class Language (7)

40.14.1 Identify Class Language: NIELIT 2016 MAR Scientist B - Section C: 25

Regarding power of recognition of language, which of the following statements is false?

A. Non deterministic finite-state automata are equivalent to deterministic finite-state automata.
B. Non-deterministic push-down automata are equivalent to deterministic push-down automata.
C. Non-deterministic Turing Machines are equivalent to deterministic push-down automata.
D. Multi-tape Turing Machines are equivalent to Single-tape Turing Machines.

nielit2016mar-scientistb theory-of-computation identify-class-language

Answer key

40.14.2 Identify Class Language: NIELIT 2016 MAR Scientist C - Section C: 14

If every string of a language can be determined, whether it is legal or illegal in finite time, the language is called

A. decidable B. undecidable C. interpretive D. non-deterministic

nielit2016mar-scientistc theory-of-computation identify-class-language

Answer key

40.14.3 Identify Class Language: NIELIT 2016 MAR Scientist C - Section C: 15

The defining language for developing a formalism in which language definitions can be stated, is called

A. syntactic meta language
B. decidable language
C. intermediate language
D. high level language

nielit2016mar-scientistc theory-of-computation identify-class-language non-gatecse

Answer key

40.14.4 Identify Class Language: NIELIT 2017 DEC Scientific Assistant A - Section B: 31

A finite automaton accepts which type of language :

A. Type 0 B. Type 1 C. Type 2 D. Type 3

nielit2017dec-assistanta theory-of-computation identify-class-language finite-automata

Answer key

40.14.5 Identify Class Language: NIELIT 2017 DEC Scientist B - Section B: 56

Which of the following statement is **true**?

A. Deterministic context free language are closed under complement.
B. Deterministic context free language are not closed under Union.
C. Deterministic context free language are closed under intersection with regular set.
D. All of the options

nielit2017dec-scientistb theory-of-computation identify-class-language context-free-language

Answer key☟

40.14.6 Identify Class Language: NIELIT 2017 DEC Scientist B - Section B: 58

Which of the following is a correct hierarchical relationships of the following where

L_1: set of languages accepted by NFA

L_2: set of languages accepted by DFA

L_3: set of languages accepted by DPDA

L_4: set of languages accepted by NPDA

L_5: set of recursive language

L_6: set of recursive enumerable languages?

A. $L_1, L_2 \subset L_3 \subset L_4 \subset L_5 \subset L_6$
B. $L_1 \subset L_2 \subset L_3 \subset L_4 \subset L_5 \subset L_6$
C. $L_2 \subset L_1 \subset L_3 \subset L_4 \subset L_5 \subset L_6$
D. $L_1 \subset L_2 \subset L_3 \subset L_4 \subset L_6 \subset L_5$

nielit2017dec-scientistb theory-of-computation identify-class-language recursive-and-recursively-enumerable-languages

Answer key☟

40.14.7 Identify Class Language: NIELIT 2022 April Scientist B | Section B | Question: 92

L1 is a recursively enumerable language over Σ. An algorithm A effectively enumerates its words as $w_1, w_2, w_3, \dots$ Define another language L2 over Σ Union $\{\#\}$ as $\{wi\#wj : wi, wj \in \text{L1}, i < j\}$. Here # is new symbol. Consider the following assertions.

- S1 : L1 is recursive implies L2 is recursive
- S2 : L2 is recursive implies L1 is recursive

Which of the following statements is true?

A. Both S1 and S2 are true
B. S1 is true but S2 is not necessarily true
C. S2 is true but S1 is not necessarily true
D. Neither is necessarily true

nielit2022apr-scientistb theory-of-computation recursive-and-recursively-enumerable-languages decidability identify-class-language

40.15 Injective Function (1)

40.15.1 Injective Function: NIELIT 2022 April Scientist B | Section B | Question: 82

Consider the following two statements:

i. A hash function (these are often used for computing digital signatures) is an injective function.
ii. encryption technique such as DES performs a permutation on the elements of its input alphabet.

Which one of the following options is valid for the above two statements?

A. Both are false
B. Statement (i) is true and the other is false
C. Statement (ii) is true and the other is false
D. Both are true

nielit2022apr-scientistb hashing encryption-decryption injective-function digital-signature combinatory cryptography

40.16 Language Classes (1)

40.16.1 Language Classes: NIELIT 2021 Dec Scientist A - Section B: 99

Let $\mathrm{L} = \mathrm{L}_1 \cap \mathrm{L}_2$ where L_1 and L_2 are language defined below :

$\mathrm{L}_1 = \{a^m b^m c\, a^n b^n \quad m, n > 0\}$

$\mathrm{L}_2 = \{a^i\, b^j\, c^k \quad i, j, k > 0\}$

Then L is

A. Not Recursive
B. Regular
C. Context free but not regular
D. Recursively enumerable but not context free

nielit2021dec-scientista theory-of-computation context-free-language recursively-enumerable-languages language-classes

40.17 Lexical Analysis (1)

40.17.1 Lexical Analysis: NIELIT Scientist B 2020 November: 57

Which of the following machine model can be used in a necessary and sufficient sense for lexical analysis in modern computer language?

A. Deterministic Push down Automata
B. Finite Automata
C. Non-Deterministic Finite Automata
D. Turing Machine

nielit-scb-2020 finite-automata lexical-analysis compiler-design

Answer key☟

40.18 Number of Dfa (1)

40.18.1 Number of Dfa: NIELIT 2017 DEC Scientific Assistant A - Section B: 50

How many DFA's exits with two states over input alphabet $\{0, 1\}$

A. 16 B. 26 C. 32 D. 64

nielit2017dec-assistanta theory-of-computation finite-automata number-of-dfa

Answer key

40.19 P NP NPC NPH (2)

40.19.1 P NP NPC NPH: NIELIT 2016 MAR Scientist B - Section C: 19

If there is in NP-Complete language L whose complement is in NP, then complement of any language in NP is in

A. P B. NP C. both (A) and (B) D. None of these

nielit2016mar-scientistb theory-of-computation p-np-npc-nph

Answer key

40.19.2 P NP NPC NPH: NIELIT Scientific Assistant A 2020 November: 57

Which of the following problem is not NP complete but undecidable?

A. Partition Problem
B. Halting Problem
C. Hamiltonian Circuit
D. Bin Packing

nielit-sta-2020 theory-of-computation p-np-npc-nph

Answer key

40.20 Pumping Lemma (1)

40.20.1 Pumping Lemma: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 12

The logic of pumping lemma is a good example of

A. the pigeon-hole principle
B. the divide and conquer technique
C. recursion
D. iteration

nielit2017oct-assistanta-cs theory-of-computation pumping-lemma

Answer key

40.21 Pushdown Automata (1)

40.21.1 Pushdown Automata: NIELIT 2017 DEC Scientist B - Section B: 57

Which machine is equally powerful in both deterministic and non-deterministic form?

A. Push Down Automata
B. Turing machine
C. Linear Bounded Automata
D. None of the options

nielit2017dec-scientistb pushdown-automata turing-machine

Answer key

40.22 Quantifiers (1)

40.22.1 Quantifiers: NIELIT 2021 Dec Scientist A - Section B: 43

Let the predicates $D(x, y)$ mean "team x defeated team y" and $P(x, y)$ mean "team x has played team y". The quantified formula for the statement that there is a team that has beaten every team it has played is:

A. $\exists x \forall y (P(x, y) \rightarrow D(x, y))$
B. $\forall x \exists y (P(x, y) \rightarrow D(x, y))$
C. $\forall y \exists x (P(x, y) \rightarrow D(x, y))$
D. $\exists x \forall y (D(x, y) \rightarrow P(x, y))$

nielit2021dec-scientista first-order-logic logical-reasoning quantifiers

Answer key

40.23 Recursion (1)

40.23.1 Recursion: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 32

Choose the correct statements.

A. A total recursive function is also a partial recursive function
B. A partial recursive function is also a total recursive function
C. A partial recursive function is also a primitive recursive function
D. None of the above

nielit2017oct-assistanta-cs theory-of-computation recursion

Answer key

40.24 Recursive and Recursively Enumerable Languages (9)

40.24.1 Recursive and Recursively Enumerable Languages: NIELIT 2017 DEC Scientist B - Section B: 19

Recursive enumerable languages are not closed under _________.

A. Set difference
B. Complement
C. Both (A) and (B)
D. None of the options

nielit2017dec-scientistb theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key

40.24.2 Recursive and Recursively Enumerable Languages: NIELIT 2017 DEC Scientist B - Section B: 27

If any string of a language L can be effectively enumerated by an enumerator in a lexicographic order then language L is _______.

A. Regular
B. Context free but not necessarily regular
C. Recursive but not necessarily context free
D. Recursively enumerable but not necessarily recursive

nielit2017dec-scientistb theory-of-computation recursive-and-recursively-enumerable-languages

Answer key☟

40.24.3 Recursive and Recursively Enumerable Languages: NIELIT 2017 DEC Scientist B - Section B: 32

The collection of Turing recognizable languages are closed under:

i. Union
ii. Intersection
iii. Complement
iv. Concatenation
v. Star Closure

A. (i) Only
B. Both (i),(iv)
C. (i),(ii),(iv)and(v)
D. All of the options

nielit2017dec-scientistb theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key☟

40.24.4 Recursive and Recursively Enumerable Languages: NIELIT 2017 July Scientist B (CS) - Section B: 59

Let L be a language and L' be its complement. Which one of the following is NOT a viable possibility?

A. Neither L nor L' is RE.
B. One of the L and L' is RE but not recursive;the other is not RE.
C. Both L and L' are RE but not recursive.
D. Both L and L' are recursive.

nielit2017july-scientistb-cs theory-of-computation recursive-and-recursively-enumerable-languages

Answer key☟

40.24.5 Recursive and Recursively Enumerable Languages: NIELIT 2017 July Scientist B (CS) - Section B: 60

Let $L1$ be a recursive language, and let $L2$ be a recursively enumerable but not recursive language. Which one of the following is TRUE?

A. $L1'$ is recursive and $L2'$is recursively enumerable.
B. $L1'$ is recursive and $L2'$is not recursively enumerable.
C. $L1'$ and $L2'$is recursively enumerable.

D. $L1'$ is recursively enumerable and $L2'$is recursive.

nielit2017july-scientistb-cs theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.24.6 Recursive and Recursively Enumerable Languages: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 9

A language L for which there exists a $TM \ \ 'T'$, that accepts every word in L and either rejects or loops for every word that is not in L, is said to be

A. Recursive
B. Recursively enumerable
C. NP-HARD
D. None of the above

nielit2017oct-assistanta-cs theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.24.7 Recursive and Recursively Enumerable Languages: NIELIT 2018-81

Identify the true statement from the given statements:

1. A recursive formal language is a recursive subset in the set of all possible words over the alphabet of the language.
2. The complement of a recursive languages is recursive
3. The complement of a context-free language is context-free

A. Only 1
B. 1 and 2
C. $1, 2$ and 3
D. 2 and 3

nielit-2018 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

40.24.8 Recursive and Recursively Enumerable Languages: NIELIT 2021 Dec Scientist B - Section B: 52

A language L is recognizable by a turing machine M if and only if L is a ______________ language.

A. Type 0
B. Type 1
C. Type 2
D. Type 3

nielit2021dec-scientistb theory-of-computation turing-machine recursive-and-recursively-enumerable-languages

Answer key

40.24.9 Recursive and Recursively Enumerable Languages: NIELIT 2022 April Scientist B | Section B | Question: 43

Consider the following types of languages:

$\mathrm{L1}$: Regular,

$\mathrm{L2}$: Context-free,

$\mathrm{L3}$: Recursive,

$\mathrm{L4}$: Recursively enumerable.

Which of the following is/are TRUE ?

I. $\mathrm{L3}' \cup \mathrm{L4}$ is recursively enumerable
II. $\mathrm{L2} \cup \mathrm{L3}$ is recursive
III. $\mathrm{L1}^* \cup \mathrm{L2}$ is context-free
IV. $\mathrm{L1} \cup \mathrm{L2}'$ is context-free

A. I only
B. I and III only
C. I and IV only
D. $\mathrm{I, II}$ and III only

nielit2022apr-scientistb theory-of-computation recursive-and-recursively-enumerable-languages

Answer key☟

40.25 Regular Expression (13)

40.25.1 Regular Expression: NIELIT 2016 MAR Scientist B - Section C: 24

Which of the following regular expressions denotes a language comprising all possible strings over the alphabet $\{a, b\}$?

A. a^*b^*
B. $(a \mid b)^*$
C. $(ab)^+$
D. $(a \mid b^*)$

nielit2016mar-scientistb theory-of-computation regular-expression

Answer key☟

40.25.2 Regular Expression: NIELIT 2016 MAR Scientist C - Section C: 17

Regular expression $(a \mid b)(a \mid b)$ denotes the set

A. $\{a, b, ab, aa\}$
B. $\{a, b, ba, bb\}$
C. $\{a, b\}$
D. $\{aa, ab, ba, bb\}$

nielit2016mar-scientistc theory-of-computation regular-expression

Answer key☟

40.25.3 Regular Expression: NIELIT 2017 DEC Scientific Assistant A - Section B: 19

Let P, Q, R be a regular expression over Σ. If P does not contain null string, then $R = Q + RP$ has a unique solution ___________ .

A. Q^*P
B. QP^*
C. Q^*P^*
D. $(P^*O^*)^*$

nielit2017dec-assistanta theory-of-computation regular-expression

Answer key☟

40.25.4 Regular Expression: NIELIT 2017 DEC Scientific Assistant A - Section B: 35

$(0 + \varepsilon)(1 + \varepsilon)$ represents :

A. $\{0, 1, 01, \varepsilon\}$
B. $\{0, 1, \varepsilon\}$

C. $\{0,1,01,11,00,10,\varepsilon\}$

D. $\{0,1,\}$

nielit2017dec-assistanta theory-of-computation regular-expression

Answer key

40.25.5 Regular Expression: NIELIT 2017 DEC Scientific Assistant A - Section B: 54

Complement of $(a+b)^*$ will be :

A. $Phi\,(\Phi)$

B. Null

C. a

D. b

nielit2017dec-assistanta theory-of-computation regular-expression

Answer key

40.25.6 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 10

A regular expression is $(a+b^*c)$ is equivalent to

A. set of strings with either a or one or more occurrence of b followed by c.

B. (b^*c+a)

C. set of strings with either a or zero or more occurrence of b followed by c.

D. Both (B) and (C)

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.7 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 2

According to the given language, which among the following expressions does it correspond to ?

Language $L = \{x \in \{0,1\} \mid x \text{ is of length 4 or less}\}$.

A. $(0+1+0+1+0+1+0+1)^4$

B. $(0+1)^4$

C. $(01)^4$

D. $(0+1+\varepsilon)^4$

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.8 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 22

What is the meaning of regular expression $\Sigma^*001\Sigma^*$?

A. Any string containing '1' as substring

B. Any string containing '01' as substring

C. Any string containing '011' as substring

D. All string containing '001' as substring

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.9 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 35

Which of the following regular expression is equal to $(r_1 + r_2)^*$?

A. $r_1^* r_2^*$
B. $(r_1 r_2)^*$
C. $r_1^* r_2^* + r_1 r_2$
D. $(r_1^* r_2^*)^*$

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.10 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 41

Which of the following is equivalent regular expressions?

i. $((01)^*(10)^*)^*$
ii. $(10 + 01)^*$
iii. $(01)^* + (11)^*$
iv. $(0^* + (11)^* + 0^*)^*)$

A. (i) and (ii)
B. (ii) and (iii)
C. (iii) and (iv)
D. (iv) and (i)

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.11 Regular Expression: NIELIT 2017 DEC Scientist B - Section B: 45

The string 1101 does not belong to the set represented by

A. $(00 + (11)^*0)$
B. $1(0 + 1)^*101$
C. $(10)^*(01)^*(00 + 11)^*$
D. $110^*(0 + 1)$

nielit2017dec-scientistb theory-of-computation regular-expression

Answer key

40.25.12 Regular Expression: NIELIT 2018-43

Which of the following languages over the alphabet $[0, 1]$ is described by the given regular expression: $(0 + 1)^*1(0 + 1)^*1$?

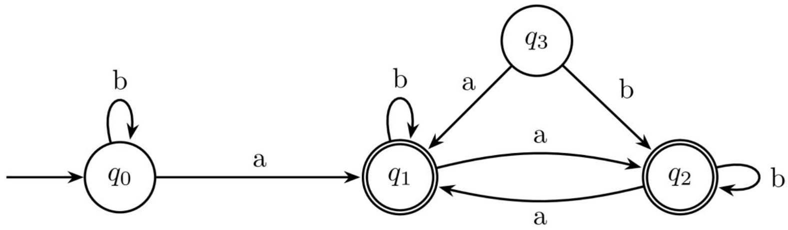
A. The set of all strings containing the substring 11
B. The set of all strings containing at most two 1's
C. The set of all strings containing at least two 1's
D. The set of all strings that begins and ends with only 0

nielit-2018 theory-of-computation regular-expression

Answer key

40.25.13 Regular Expression: NIELIT 2021 Dec Scientist A - Section B: 103

Consider the finite automata given below



The language b accepted by this automata is a given by the regular expression :

A. b* ab * a b * a b *
B. (a+b)*
C. b*a (a+b)*
D. b* ab* ab*

nielit2021dec-scientista   finite-automata   regular-expression   theory-of-computation

Answer key☟

40.26 Regular Language (5)

40.26.1 Regular Language: NIELIT 2016 MAR Scientist B - Section C: 27

If L be a language recognizable by a finite automaton, then language from $\{L\} = \{w$ such that w is a prefix of v where $v \in L\}$, is a

A. regular language.
B. context-free language.
C. context-sensitive language.
D. recursive enumeration language

nielit2016mar-scientistb   theory-of-computation   regular-language

Answer key☟

40.26.2 Regular Language: NIELIT 2016 MAR Scientist B - Section C: 28

Which of the following statements is correct?

A. $A = \{a^n b^n \mid n = 0, 1, 2, 3 \ldots\}$ is regular language
B. Set B of all strings of equal number of a's and b's defines a regular language
C. $L(A^* B^*) \cap B$ gives the set A
D. None of these.

nielit2016mar-scientistb   theory-of-computation   regular-language

Answer key☟

40.26.3 Regular Language: NIELIT 2017 DEC Scientific Assistant A - Section B: 15

If $L1$ and $L2$ are regular sets then intersection of these two will be :

A. Regular
B. Non Regular
C. Recursive
D. Non Recursive

nielit2017dec-assistanta   theory-of-computation   regular-language

Answer key☟

40.26.4 Regular Language: NIELIT 2018-83

In the Given language $L = \{ab, aa, baaa\}$, ____ number of strings are in L^*

1. baaaba
2. aabaaaa
3. baaabaaaabaa
4. baaabaaa

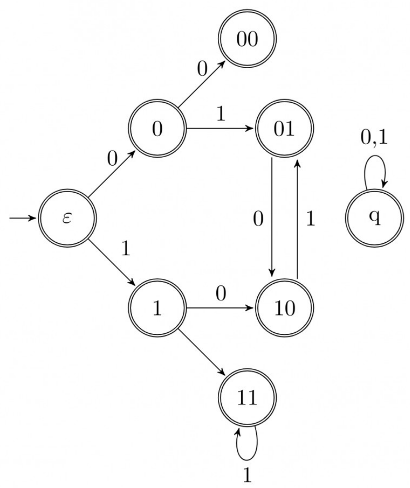
A. 1 B. 2 C. 3 D. 4

nielit-2018 theory-of-computation regular-language

Answer key☟

40.26.5 Regular Language: NIELIT 2022 April Scientist B | Section B | Question: 87

Consider the set of strings on $\{0, 1\}$ in which, every substring of 3 symbols has at most two zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.



The missing arcs in the DFA are:

A.

	00	**01**	**10**	**11**	**q**
00	1	0			
01				1	
10	0				
11			0		

B.

	00	01	10	11	q
00		0			1
01		1			
10				0	
11		0			

C.

	00	01	10	11	q
00		1			0
01		1			
10			0		
11		0			

D.

	00	01	10	11	q
00		1			0
01				1	
10	0				
11			0		

nielit2022apr-scientistb theory-of-computation finite-automata regular-language

40.27 Turing Machine (3)

40.27.1 Turing Machine: NIELIT 2016 DEC Scientist B (CS) - Section B: 6

Which of the following is wrong?

A. Turing machine is a simple mathematical model of general purpose computer
B. Turing machine is more powerful than finite automata
C. Turing Machine can be simulated by a general purpose computer
D. All of these

nielit2016dec-scientistb-cs theory-of-computation turing-machine

Answer key☟

40.27.2 Turing Machine: NIELIT 2017 DEC Scientist B - Section B: 37

Which of the following statement is true?

$S1$: The power of a multi-tape Turing machine is greater than the power of a single tape Turing machine.

$S2$: Every non-deterministic Turing machine has an equivalent deterministic Turing machine.

A. $S1$ B. $S2$ C. Both $S1$ and $S2$ D. None of the options nielit2017dec-scientistb theory-of-computation turing-machine

Answer key☟

40.27.3 Turing Machine: NIELIT 2021 Dec Scientist A - Section B: 91

If T1 and T2 are two Turing machines. The composite can be represented using the expression :

A. T1T2

B. $\text{T1} \cup \text{T2}$

C. $\text{T1} \times \text{T2}$

D. None of the options

nielit2021dec-scientista theory-of-computation turing-machine

Answer key☟

Answer Keys

40.1.1	B	40.2.1	B	40.2.2	A	40.3.1	C	40.4.1	B
40.4.2	C	40.4.3	B	40.5.1	D	40.5.2	B	40.5.3	A
40.5.4	A	40.5.5	B	40.6.1	B	40.7.1	D	40.8.1	D
40.8.2	B	40.9.1	C	40.9.2	C	40.10.1	TBA	40.10.2	C
40.11.1	D	40.11.2	B	40.11.3	A	40.11.4	D	40.11.5	C
40.11.6	C	40.11.7	C	40.11.8	B	40.11.9	B	40.11.10	D
40.12.1	C	40.13.1	C	40.13.2	B	40.14.1	B	40.14.2	A
40.14.3	A	40.14.4	D	40.14.5	D	40.14.6	A	40.14.7	A
40.15.1	C	40.16.1	B	40.17.1	B	40.18.1	D	40.19.1	B
40.19.2	B	40.20.1	A	40.21.1	B	40.22.1	A	40.23.1	A
40.24.1	C	40.24.2	C	40.24.3	C	40.24.4	C	40.24.5	B
40.24.6	B	40.24.7	B	40.24.8	A	40.24.9	D	40.25.1	B
40.25.2	D	40.25.3	B	40.25.4	A	40.25.5	A	40.25.6	D
40.25.7	D	40.25.8	D	40.25.9	D	40.25.10	A	40.25.11	C
40.25.12	X	40.25.13	C	40.26.1	A	40.26.2	C	40.26.3	A
40.26.4	B	40.26.5	D	40.27.1	D	40.27.2	B	40.27.3	A